FOOT & ANKLE — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations · 48 of 48 topics –

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Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer

5TH METATARSAL BASE FRACTURE

– Foot & Ankle · 1 / 48 –

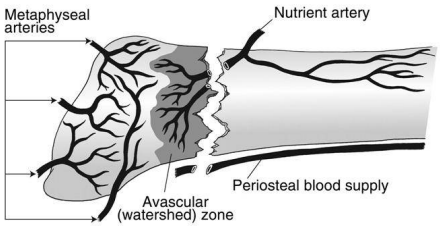
1 Basics

Epidemiology

Frequency	among the **commonest foot fractures** · base fractures = **25% of all metatarsal fractures** · **90% are zone 1**
Who	**athletes, military recruits, manual labourers** · runners, dancers

Anatomy & blood supply

Regions	tuberosity → base → metadiaphysis → diaphysis → neck → head
Base + tuberosity	**cancellous, highly vascular** · tuberosity = insertion of **peroneus brevis + lateral band of plantar fascia**
Metadiaphysis	**no tendon attachments** · **vascular watershed** between **metaphyseal vessels** and **diaphyseal nutrient artery** (terminal capillaries)
Diaphysis	dorsal curve in distal ⅓ · **peroneus tertius** inserts dorsally
Shear force (OB)	fracture lies in the **hypovascular zone between peroneus brevis and peroneus tertius insertions** → tendons shear across fracture site
Function	lateral border of forefoot · **lever during push-off**
Mimics	**open apophysis** or **os peroneum** → comparison radiographs



Watershed zone between metaphyseal arteries and the nutrient artery

Genes & lab markers

CMT (cavovarus → 5th MT stress fracture)	**duplication on chromosome 17** ✗ Wrong options: absent dystrophin (muscular dystrophy, equinovarus) · PITX1 (clubfoot) · PTT insufficiency (flatfoot) · tarsal coalition
Nonunion / refracture labs	**vitamin D deficiency (hypovitaminosis D)** · **thyroid** (hormonal) deficiency
Female athlete triad (TO)	disordered eating + **menstrual irregularities** + ↓ BMD → ↑ metatarsal stress fractures (TO)

– – Active space – –

2 Mechanism

Zone	Mechanism
1	**Plantarflexion + hindfoot inversion** → avulsion by **peroneus brevis**, **lateral band of plantar fascia** or **long plantar ligament**
2	**Forefoot adduction** (Orthobullets, OB) · **strong abduction force to forefoot → bending moment at metaphyseal–diaphyseal junction** (TO) – read the options · acute inversion injury (OB)
3	**Repetitive microtrauma → stress fracture**

Associated conditions **Lisfranc / midfoot injury** · **lateral ankle ligament laxity** · **cavus foot + varus hindfoot**

Hindfoot varus / cavus **overloads lateral column** → stress fracture distal to 4/5 articulation

3 Presentation

Symptom **pain over lateral border of forefoot**, worse with WB · stress fracture: **antecedent pain**, rapid ↑ workload / training change

Exam point tenderness at base · rare **skin tenting (zone 1)** · **varus hindfoot**, **cavus**, **lateral shoe-tread wear**, **5th MT head callosity**

Check lateral ligament instability · is varus hindfoot correctable · **pain with resisted eversion** (peroneal weakness)

4 Diagnosis & Classification

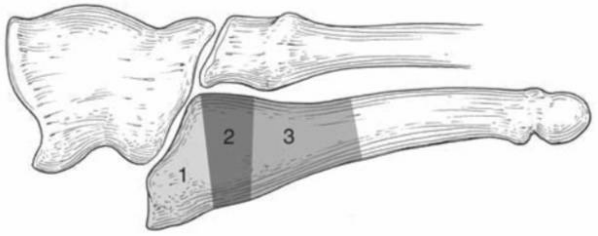
Key radiology findings

Study	Use / finding
X-ray AP, lateral, oblique foot	**Diagnosis**; location + displacement decide fixation
X-ray chronicity	**Intramedullary sclerosis + no periosteal callus** = chronic · callus forms **medially first → laterally**
Plantar fracture gap	**Poor prognosis for nonop union**
Cavovarus signs	**Plantarflexed 1st MT**, **high Meary angle**
Bone scan	Suspected stress fracture, equivocal X-ray → **diaphyseal uptake**
MRI	Equivocal X-ray / scan → **high-signal stress reaction + oedema**
CT	**Healing in delayed / nonunion** or after fixation · **not at time of injury** (X-ray suffices)
Screw planning (TO)	**AP = screw diameter** · **lateral = screw length** (TO)

Lawrence & Botte anatomic zones (also called Torg anatomic zones)

– – Active space – –

Zone	Name	Location	Key facts
1	**Pseudo-Jones**	**Proximal tubercle avulsion**; may enter **cubometatarsal (5th TMT) joint**; **proximal to 4–5 intermetatarsal joint**	Well vascularised; nonunion rare, **usually asymptomatic (fibrous)**
2	**Jones**	**Metaphyseal–diaphyseal junction**; **enters 4th–5th intermetatarsal articulation**	**Watershed**; acute; **nonunion 15–30%**
3	**Proximal diaphyseal**	**Distal to 4–5 articulation**; **proximal 1.5 cm of diaphysis (TO)**	**Stress fracture in athletes**; cavovarus, sensory neuropathy; ↑ nonunion



Anatomic zones 1–3 of the proximal 5th metatarsal

Jones location — **1.5–3 cm** from proximal tip (OB) · **~2.5 cm** distal to base (OB) · intra-articular at 4–5 **intermetatarsal** facet; does **not** extend distal to it or into metatarsocuboid joint

Joint entered — **Jones → intermetatarsal (4–5) joint** · **avulsion → tarsometatarsal joint** (OB) · one OB explanation calls the Jones joint "4th–5th tarsometatarsal" — read the options

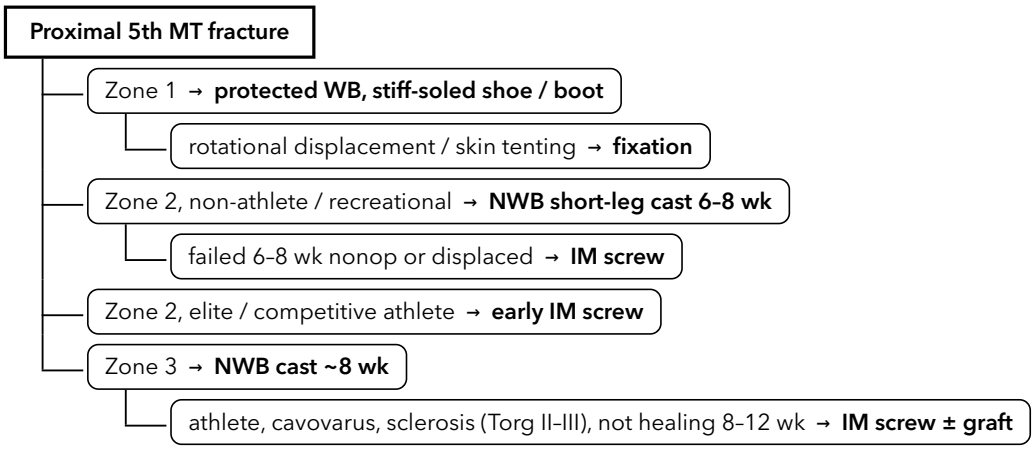
Zone 2 vs 3 (TO) — some studies: **no distinction needed**, treatment + outcome the same (TO)

Torg radiographic classification (stress fractures, zone 3)

Type	Orthobullets	TO	Age
I	**Narrow line, no IM sclerosis**	**Acute fracture**	**Acute**
II	**Widened line + IM sclerosis**	**Periosteal reaction, widening, some IM sclerosis**	**Delayed union**
III	**Widened (obliterated) canal, no callus**	**IM sclerosis + blunted fracture edges**	**Nonunion**

– – Active space – –

5 Management



Operative criteria

Nonoperative	Operative (IM screw)
Zone 1 without rotational displacement – **even elite athletes / dancers**	**Zone 1 with rotational displacement or skin tenting** · gross displacement / articular involvement in young active (TO)
Zone 2 in non-athlete / recreational athlete (e.g. 45-yo recreational soccer)	**Zone 2 in elite / competitive athlete** (collegiate, Olympic, professional) – **even acute + nondisplaced**
High-school / recreational athlete until **6–8 wk nonop failed**	**Non-elite after failed 6–8 wk** nonop or **unacceptable displacement**
Zone 3 acute, low demand	**Zone 3 in athletes, cavovarus, sclerosis / nonunion (Torg II–III)** · **not healing at 8–12 wk (TO)** · occupation can't tolerate NWB (TO)

Zone 1 avulsion – weight bearing in a hard-soled shoe

Treatment	**protected / immediate WB in stiff- (hard-) soled shoe, postoperative shoe, walking boot or short-leg walking cast**, advance as pain allows
Outcome	**union by 8 wk** · fibrous unions rarely symptomatic · early RTW, symptoms up to **6 months**
Soft dressing (TO)	compressive dressing + WB in cast boot → **shorter recuperation, better foot score than cast** (TO)
Elite dancer, zone 1	**protected WB, stiff-soled shoe, gradual return** ✗ Wrong options: immediate return to competition · NWB short-leg cast (unnecessary) · IM screw (true Jones only) · resection + peroneus brevis advancement
57-yo, inversion off curb, zone 1	**immediate WB in postoperative shoe** ✗ Wrong options: acute fragment excision · IM screw · NWB cast · K-wires
Which pattern → immediate WB in boot?	**zone 1** ✗ Wrong options: **Lisfranc** (unstable) · **displaced 1st MT** fracture (no intermetatarsal ligament → surgery) · zone 2 (NWB if nonop)
Symptomatic zone 1 nonunion	**fragment excision + peroneus brevis reattachment** ✗ Wrong options: acutely
Zone 1 ORIF (TO)	rarely; **displaced intra-articular in highly competitive** · plate, screw or **tension-band** (TO)

Acute Jones (zone 2) – IM screw in athletes vs casting

– – Active space – –

Nonop — **NWB short-leg cast 6–8 wk** (Orthobullets, OB) · then WB cast until union (TO) · advance WB with callus (~4–6 wk) · **NWB cast > walking cast** for healing

Zone 2 outcome	Value
Nonunion, nonop	**15–30%** (Orthobullets) · **10–30%** (OB) · **25%** (OB) · **up to 33%** (OB) · **> 30%** in some studies (OB) – read the options
Union with cast	**72–93%** (OB) · **healing rate 75%** (TO)
Refracture after nonop	**33%** (Orthobullets) · **~30%** (OB)
Union with IM screw	**Approaching 100%**
Reoperation after ORIF (Yates)	**8.5%** – refracture, persistent nonunion, symptomatic hardware

What IM screw achieves — **improved union rate** (primary advantage) · **faster / decreased time to union** · **earlier return to play / work** · minimises nonunion + prolonged activity restriction · better patient-reported scores in athletes

What it does not do — ↓ pain · ↓ HO · ↑ ROM · ↓ cost · **eliminate refracture** · avoid hardware complications · ↓ sural irritation (**more** with surgery) · prevent midfoot arthritis (not a sequela)

✗ Casting traps — **> 95% union** with cast (false) · **asymptomatic fibrous nonunion** with cast (that's zone 1; zone 2 nonunion usually symptomatic) · "casting unacceptable" (false – still valid if avoiding surgery) · "ORIF ≈ casting" (false) · "ORIF high hardware failure" (false)

Bone graft — **not required** in primary fixation · use in **revision / nonunion / refracture** (± BMAC)

✗ Other wrong answers for athlete Jones — **K-wires** (not supported) · **tubular plate** (long lines into zone 3 / multifragmentary diaphyseal) · **MRI** · **taping + immediate RTS** · **Brostrom** · **Lisfranc repair** · **midfoot fusion**

TO answer key (TO) — painful 5th MT base stress fracture in runner: **"All correct"** = **internal fixation always required** + **more in females with menstrual abnormalities** + **RICE is first treatment** (TO) · conflicts with OB/Orthobullets (cast for non-athletes) – read the options

RICE (TO) — rest, ice, compression, elevation = **primary treatment in all fractures** (TO)

Competitive athletes (TO) — **early ORIF to ↓ disability time** (TO)

Stress fractures, delayed union & nonunion – fixation

Zone 3 nonop (TO) — chronic (canal obliteration, sclerosis) → **NWB casting ~8 wk** may work (TO) · Orthobullets: NWB 6–8 wk (often 6–7 wk) · **ESWT** union ≈ internal fixation (reports)

Refracture common in zone 3 (TO) — risk factors: **cavovarus**, **hypovitaminosis D**, **metatarsus adductus**, **curved 5th MT with prominent base** (TO)

Two proven operations, zones 2–3 (TO) — **IM screw (5.5 mm)** · **corticocancellous inlay bone graft** after clearing canal of sclerotic bone (TO)

– – Active space – –

Delayed union after 6 wk cast, widened line (15-yo)	**percutaneous IM screw** ✗ Wrong options: ORIF (not needed) · **electrical stimulation alone** · more casting · hard-sole shoe
Zone 3, 11 wk, sclerotic edges, minimal callus	**ORIF ± bone grafting** ✗ Wrong options: MRI · bone stimulator (controversial) · continued cast / boot
HS quarterback, refracture after 8 wk NWB boot	**IM compression screw** ✗ Wrong options: plate · fragment excision (zone 1) · NWB + stimulator · midfoot fusion
Zone 3, triathlete wants running at 6 wk	**IM screw** ✗ Wrong options: boot / hard-soled shoe / early PT (zone 1) · NWB cast (non-athletes)
Nonunion risk factor (dancer, 4 months)	**fracture location** (zone 2–3) ✗ Wrong options: NSAIDs · high calcium · early WB · female sex
Nonunion treatment	**revision IM screw + bone graft** · plate = salvage after failed IM screw

IM screw technique & failure

Position	supine, bump under hip, fluoroscopy
Approach	short incision **proximal to tuberosity**, parallel to sole · blunt dissection past **sural branches** · between **peroneus longus + brevis**
Start point	**"high and inside"** = **superior + medial** on tuberosity · K-wire not past MT curvature
Screw	**> 4 mm diameter** · **4.5, 5.5 or 6.5 mm partially threaded** (solid or cannulated) · **largest diameter the canal accepts** (max fill + compression, ↓ refracture) · **5.5 mm** (TO)
Check	**all threads cross fracture** · no distal cortical perforation · tap measures length

Screw error	Result
< 4.5 mm (not including 4.5)	**Delayed / nonunion**
Too long	**Straightens curved shaft → fracture gapping + distal / medial cortical perforation** → failure, nonunion
Too short	**No compression**
Prominent head	**Sural nerve irritation**



Too-long screw straightening the shaft and gapping the fracture

Rehab	**NWB 1–3 wk** → protected WB (from ~2 wk) → **running + impact at 6 wk** if pain-free + **radiographic callus** · RTP average **4–6 wk**
Failure risk factors	**elite athletes** · **return to sport before radiographic union** · **early unprotected WB in athletes** · **too-long screw** ✗ Wrong options: female · age < 20 or > 40 · solid vs cannulated · **4.5 mm** or **> 4.5 mm** screw · no bone graft · no ultrasound stimulator · partially threaded
Bent 5.0 mm screw at 12 wk	cause = **return to play before radiographic union** · fix = **remove + larger IM screw**
NFL QB, failure at 8 wk	**too long partially threaded screw** ✗ Wrong options: WB at 2 wk · impact at 6 wk · no BMAC · largest screw
IM screw vs plate	IM **more rigid** + **less hardware prominence** (OB) · plate **equivalent strength** in early data (Orthobullets) – read the options
Plate (3 mm, plantar-lateral)	same indications; **salvage after failed IM screw**; reduces plantar gap / rotation

Association with cavovarus foot (CMT)

Cavus foot	**↑ refracture** + ↑ nonunion even after successful treatment ✗ Wrong options: ↓ nonunion · earlier / later RTS
Varus hindfoot	overloads lateral column → **zone 3 stress fracture** distal to 4/5 articulation
Refracture treatment	internal fixation **+ correct cavovarus** · **leave screw in until end of athletic career** (removal ↑ refracture)

Iselin disease (apophysitis) (TO)

Definition (TO)	**apophysitis (osteochondritis) of 5th metatarsal base** · benign, **self-limiting** · **young adolescent sportspersons** (TO)

Osteochondritis site (TO)	Eponym
5th MT base	Iselin
2nd MT head	Freiberg
Navicular	Köhler
Calcaneal tuberosity	Sever
Lower pole patella	Sinding-Larsen–Johansson
Tibial tubercle	Osgood–Schlatter
Lateral part of medial femoral condyle	Osteochondritis dissecans
Femoral head	Perthes
Lunate	Kienböck
Capitulum	Panner
Vertebral ring epiphysis	Scheuermann
Central bony nucleus of vertebral body	Calvé

– – Active space – –

6 Complications

Complication	Key facts
Nonunion	Zone 2 **15–30%**; zones 2–3 (watershed); **screw < 4.5 mm**; vitamin D / thyroid deficiency → **revision IM screw + graft**
Failure of fixation	**Elite athlete**, **RTS before radiographic union**, **screw length** (too long → gapping) → revision fixation
Refracture	**33% zone 2 nonop**; cavovarus, stress fracture, vitamin D, **screw removal** → fix + correct cavovarus
Painful hardware	Rare; prominent head → **modified shoewear**
Sural nerve injury	**Dorsolateral branch within 2–3 mm of tuberosity** · prevent: **tissue protector + countersink head**
Chronic pain	Zone 1 nonunion after nonop → **fragment excision + peroneus brevis reattachment**

X vs Y

	Zone 1 (pseudo-Jones)	Zone 2 (Jones)	Zone 3 (diaphyseal stress)
Site	Tuberosity	**Metadiaphysis**	**Proximal diaphysis**
Joint	5th TMT (cubometatarsal)	**4–5 intermetatarsal**	Distal to 4–5 joint
Mechanism	**PF + inversion**	**Forefoot adduction** (abduction – TO)	**Repetitive stress**
Blood supply	Good	**Watershed**	Watershed
Nonop	**WB hard-soled shoe**	**NWB cast 6–8 wk**	**NWB cast ~8 wk**
Athlete	Still nonop	**Early IM screw**	**IM screw**

How it's asked

"Elite dancer, tuberosity avulsion – management?"	**→ protected WB in stiff-soled shoe, gradual return**
"Zone 1 avulsion in 57-yo – treatment?"	**→ immediate WB in postoperative shoe**
"Which pattern → immediate WB in walking boot?"	**→ zone 1 (pseudo-Jones)**
"Zone 1 vs zone 2 – true?"	**→ zone 1 (pseudo-Jones) → hard-soled shoe, allowed to weight bear as tolerated**
"Fastest union + return to sport, collegiate lineman Jones?"	**→ intramedullary screw fixation**
"Professional basketball player, Jones – quickest return?"	**→ IM screw** ✗ Wrong options: MRI · taping

"Primary advantage of fixing a Jones fracture?"	→ **improved union rate**
"45-yo recreational soccer, Jones?"	→ **NWB short-leg cast**
"Collegiate runner, acute metadiaphyseal fracture?"	→ **acute meta-diaphyseal fracture → IM screw**
"Which patient needs early surgery?"	→ **collegiate soccer player, acute nondisplaced zone 2**
"IM screw vs cast for Jones – true?"	→ **decreased time to union**
"Operative vs casting in Jones – true?"	→ **shorter time to union, quicker return to work**
"Proposed benefit of IM screw?"	→ **faster radiographic union + earlier return to play**
"Zone III in triathlete wanting to run at 6 wk?"	→ **IM screw**
"Delayed union after 6 wk cast in 15-yo?"	→ **percutaneous screw fixation**
"Collegiate varsity athlete, Jones on X-ray?"	→ **internal fixation**
"Recurrent pain after nonop Jones in HS quarterback?"	→ **IM compression screw**
"Jones in cavus vs planus foot?"	→ **higher risk of refracture**
"Zone III, 11 wk, sclerotic edges?"	→ **ORIF ± bone grafting**
"Risk factor for nonunion?"	→ **fracture location**
"Bent screw after release to play?"	→ **return to play before radiographic union**
"Risk factor for failure after fixation?"	→ **return to sport before radiographic union**
"Higher failure proportion after IM screw?"	→ **early unprotected WB in athletes**
"IM screw for zone 3 – true?"	→ **screw diameter as large as possible**
"Failure at 8 wk in NFL QB?"	→ **too long a partially threaded screw**
"Screw diameter linked to nonunion?"	→ **< 4.5 mm**
"Best view for screw diameter / length?"	→ **AP / lateral** (TO)
"Associated with 5th MT stress fracture + cavus foot?"	→ **duplication on chromosome 17 (CMT)**
"Painful 5th MT base stress fracture in runner – correct?"	→ **all correct** (fixation, female menstrual abnormality, RICE) (TO)
"Iselin disease is?"	→ **osteochondritis of 5th metatarsal base** (TO)
"Freiberg disease is?"	→ **osteochondritis of 2nd metatarsal head** (TO)

– – Active space – –

ACHILLES TENDON RUPTURE

– Foot & Ankle · 2 / 48 –

1 Basics

Epidemiology & risk factors

Incidence **18 : 100,000 / yr · missed in up to 25%**

Demographics **men** · **30–40 yr** (Orthobullets) · **males 30–39** (TO)

Classic patient episodic athlete, **"weekend warrior"** (basketball, tennis, pickleball, volleyball)

Risk factor	Source
Fluoroquinolones	Orthobullets, TO
Steroids – local injections + systemic	Orthobullets, TO
Hypercholesterolaemia, **gout**, **RA**	TO
Long-term dialysis, **renal transplantation**	TO
Chronic tendinopathy with Haglund disease	TO
Foot / ankle malalignment, training errors, degenerative tendon	TO

Haglund deformity ↑ TA rupture true (TO)

"Antithrombolytics have a role" the **incorrect statement** in the TO list (TO)

Anatomy

Achilles **largest** (Orthobullets) / **strongest** (OB) tendon in body · confluence of **soleus + medial and lateral gastrocnemius**

Tendon blood supply (TO)

Fact	TO answer
Three sources	**Musculotendinous junction**, **osteotendinous junction**, **paratenon** (**paratenon = most important**)
Major artery	**Posterior tibial artery** (also Orthobullets)
Minor	**Peroneal artery** – small contribution to mid-section
Most abundant zone	**At the tendon insertion** (✗ Wrong options: mid-section between MTJ and bone–tendon junction)
Least vascularised zone	**2–6 cm above calcaneal insertion**, at all ages → poor repair
Main surface of supply	**Anterior (deep) surface** (✗ Wrong options: posterior · medial · lateral) – branches loop round / pass through to posterior surface

Sural nerve

Course	**lateral to the TA** · accompanied by **small saphenous vein**, which lies **just lateral** to the nerve
Protect	**skin incision slightly medial to TA** (TO, Orthobullets) · injury → **heel + lateral foot numbness** (TO) ✗ Wrong options: saphenous · tibial · lateral plantar nerve



Axial T1 MRI – A peroneus longus, B peroneus brevis, C small saphenous vein, D sural nerve, E FHL

2 Mechanism

Mechanism	**sudden forced plantarflexion** · **violent dorsiflexion of a plantarflexed foot** · **push-off with weight-bearing forefoot while extending knee** (TO) · **sudden unexpected dorsiflexion** (TO) · **eccentric load** (OB)
Site	**4–6 cm above calcaneal insertion** (Orthobullets, OB) · **3–6 cm above os calcis**, watershed (TO) · least vascular zone **2–6 cm** (TO) – read the options

3 Presentation

History	**"pop"** · weakness, difficulty walking, heel pain · ecchymosis
Inspection	**↑ resting ankle dorsiflexion prone with knees bent** (loss of resting tension) · calf atrophy (chronic)
Palpation	**palpable gap**
Motion	**weak plantarflexion** · **↑ passive dorsiflexion** · chronic: **unable to toe off / toe walk**



Loss of resting tension – affected foot rests in more dorsiflexion

– – Active space – –

Diagnosis – Thompson test & ultrasound

Test (TO)	Method	Positive = rupture
Thompson (Simmonds) squeeze	Prone, feet over couch edge; squeeze calf **just distal to maximal girth**	**Little / no plantarflexion** vs other side · **sensitivity 0.89–0.93**
Matles	Prone, **actively flex knees to 90°**	Foot falls to **neutral or slight dorsiflexion**
O'Brien needle	**25-gauge needle** in midline of proximal tendon, passive ankle ROM	**Needle doesn't move** (moves = continuity)

"Thomas test is a reliable test" — TO uses "Thomas / Simmonds" for the calf squeeze (true statement, TO) · **Thomas test = hip flexion contracture**, **Thessaly = meniscus** (OB) – read the options

Movement tested against resistance (TO) — options inversion + PF / dorsiflexion / plantarflexion / eversion + DF – **answer key not provided by the source** · Orthobullets: **weakness to ankle plantarflexion**

Differential – xanthomas (TO)

Bilateral swellings over Achilles (TO) — options rheumatoid nodules / tophaceous gout / **tendoachilles xanthomas** / ganglion – **answer key not provided by the source** (module filed as xanthomas) · hypercholesterolaemia = rupture risk factor (TO)

4 Diagnosis & Classification

Key radiology findings

Study	Key facts
X-ray	**Rule out other pathology** (✗ Wrong options: IOC)
Ultrasound	**Investigation of choice** after + Thompson (TO) · **complete vs partial** rupture (Orthobullets) · as good as MRI for **tendinopathy + full-thickness tear** (TO)
MRI	**Highest accuracy** (TO) · **superior for partial tears + types of tendinopathy** (TO) · **equivocal exam**, **chronic ruptures**, gauging **gap size** · retracted tendon edges
Complete tear	**US = MRI** (TO)
No	**Venography**, **CT** (TO)

Diagnostic criteria – ultrasound criteria for nonoperative treatment (TO)

Position	Criterion
Maximal plantarflexion	**Gap < 5 mm**
Neutral	**Gap < 10 mm**
20° plantarflexion	**> 75% tendon apposition**

Acute vs chronic – definition

Chronic — **> 3–4 wk** (OB) · **> 4 wk** (OB, JAAOS algorithm) · **> 6 wk** (other OBs) · Orthobullets: acute = **< 6 wk** · TO: chronic **4–6 wk** – read the options

Chronic sleeve avulsion – diagnosis

Chronic Achilles avulsion (sleeve) rupture	old cast-treated rupture + new fall · X-ray: **calcified / ossified distal tendon moved away from insertion** · MRI: **chronic avulsion from insertion** ✗ Wrong options: HO of proximal tendon · untreated midsubstance · gout · musculotendinous junction

Medial gastrocnemius tear (tennis leg)

Mechanism	**simultaneous knee extension + ankle dorsiflexion** · athletes
Presentation	sudden **medial proximal calf** pain + ecchymosis · US / MRI confirm (often unnecessary)
Treatment	**conservative first line → excellent outcomes, most return to activity within 3 months** ✗ Wrong options: surgery · quarter need surgery (nonop failure rare)
Myotendinous Achilles rupture	nonop boot + PT from 3 wk → complete healing (Orthobullets: more proximal injuries → nonop)

5 Management

Acute Achilles rupture

- Sedentary, frail, proximal / myotendinous, or preference → **functional bracing in equinus + early rehab**
- Active, wants surgery → **open end-to-end or percutaneous repair + early functional rehab**
- Deep infection after repair → **debride necrotic tendon, no reconstruction, IV antibiotics 6 wk**

Chronic rupture – gap after debridement

- < 2 cm → **end-to-end repair**
- 2–5 cm → **V-Y advancement ± FHL transfer**
- > 5 cm → **FHL transfer ± V-Y / turndown**
  - Defect too big for local tissue (> 10 cm) → **allograft**

Chronic / neglected rupture > 4 wk (JAAOS, OB)

- Avoid sacrificing tendons / donor morbidity → **synthetics**
- Gap 3–5 cm, viable ends → **V-Y or local turndown ± FHL**
- Gap > 5 cm or unhealthy tissue → **FHL / FDL transfer ± dermal matrix**

– – Active space – –

Operative criteria

Indication	Treatment
Acute, preference for nonop · **sedentary** · **medically frail** · **proximal (musculature) injuries**	**Functional bracing / cast in resting equinus (20° PF)** + early functional rehab
Acute (< ~6 wk)	**Open end-to-end repair**
Cosmesis concern	**Percutaneous repair**
Chronic, defect < 3 cm	**V-Y advancement** (Orthobullets)
Chronic, defect > 3 cm	**FHL transfer ± V-Y advancement** – **requires functioning tibial nerve** (Orthobullets)
Defect < 3 cm after debridement + < 3 months	**Direct end-to-end repair** (TO, OB)

Operative vs nonoperative acute treatment

Outcome	Surgery vs nonop with functional rehab
Re-rupture	**Lower with surgery vs cast 6 wk** (classic) · **equal with early functional rehab** (Willits, Soroceanu) · mixed (OB)
Re-rupture rates	**~2% op vs 10–40% nonop** (Orthobullets) · 2.3% vs 3.9% (meta-analysis) · op **3–4%** (TO) · **1–2%** (OB)
Complications	**↑ with surgery** (~2×; 4.9% vs 1.6%) – wound, infection, sural
Strength, ROM, calf girth, SMFA at 12 mo	**No difference** · **equivalent PF strength** (some ↑ PF strength with surgery; no ↑ DF strength)
DF ROM at 6 mo	Not improved – some studies ↓ in op group
Functional outcomes	**No significant difference**
DVT	**No difference** (7–10%)
Return to work	**Quicker with surgery**
Return to sport	**Quicker with surgery** (one OB) · **similar** (another OB) – read the options · TO: ~60% after surgery
Nerve injury	**Higher with surgery**
Cost	**Higher with surgery**
Satisfaction, ultimate strength	Same (even vs cast)

Cast 6 wk nonop vs surgery — **higher risk of re-rupture**
✗ Wrong options: ↓ satisfaction · less return to sport · ↓ strength · ↑ skin problems

Re-rupture: greater in conservatively managed cases — true (TO)

↓ Re-rupture after repair — **early functional rehabilitation protocol** (WB in boot ~2 wk + ROM)
✗ Wrong options: delayed WB > 6 wk (stiffness) · gastrocnemius fascial flap augmentation · biological / PRP augmentation · percutaneous technique

Nonop rehab — start **within 48 h** · strict adherence · supervised PT · functional bracing → low re-rupture esp. **> 40 yr, women**

Post-op immobilisation — **20° plantarflexion for 4–6 wk** (↓ skin tension, protects repair) · **functional bracing beats casting** (↓ adhesions, ↓ complications)

Open vs percutaneous / minimally invasive repair – wound complications & sural nerve

Outcome	Open	Percutaneous / MIS
Wound complications / infection	**Higher** (superficial infection 6.0%)	**Lower** (0.4%)
Overall complications	**Higher**	Lower
Local scarring	More	**Less**
Sural nerve injury	Nerve seen + protected (0%)	**Higher** (3.4%) – blind needle passage (Orthobullets, OB) · **no difference** (other OBs) – read the options
Re-rupture, PF strength, calf circumference	Same	Same
Return to sport / running, outcome scores	Same	Same
Return to work	–	**More rapid** (Orthobullets)
1-year function	Same	Same

Open technique — incision **just medial to tendon** (sural) · incise **paratenon** · heavy **non-absorbable** suture

Repair techniques – eponyms (TO)

Eponym	Technique
Ma and Griffith	**Percutaneous repair of acute rupture**
Lynn	Repair + **fanned-out plantaris** over it (injury **< 10 days**)
Lindholm	**Two flaps** from proximal tendon + gastrocnemius aponeurosis **turned down 180°** to distal tendon
Maffulli	**Peroneus brevis transfer** (loop across defect) → **NOT an acute repair technique** (TO answer)
Teuffer	**PB loop** when end-to-end impossible
White and Kraynick	Teuffer modification: **PB through calcaneal tuberosity** + plantaris through ends
Turco and Spinella	PB through **midcoronal slit** in distal stump
Bosworth	**Gastrocnemius-soleus turndown** graft
Abraham and Pankovich	**V-Y tendinous flap**
Wapner	**FHL tendon transfer**

Wound complications – most common complication & risk factors

Most common complication of surgical repair — **skin healing problems / wound complications** (**5–10%**, poor posterior blood supply)
✗ Wrong options: infection (common, not commonest) · neurological injury (rare after open) · re-rupture (1–2%) · stiffness · loss of motion (nonop)

– – Active space – –

Risk factors	**tobacco (most common)** · **steroid use** · **female sex** · **diabetes** · **open technique** ✗ Wrong options: ↑ BMI · immediate surgery · male · age > 40
≥ 1 of diabetes / tobacco / steroid	**> 40%** complications vs 6% without

Infected repair – debridement

Deep infection + re-rupture (5 cm)	**debridement of necrotic + infected tendon, no attempt at reconstruction → culture-specific IV antibiotics 6 wk** → later soft-tissue cover + reconstruction ✗ Wrong options: primary repair · V-Y · turndown · FHL transfer (not with active infection)

Chronic rupture – V-Y advancement, turndown & FHL transfer by gap size

Gap (after scar debridement)	Treatment	Source
~1–2 cm / < 2 cm	**End-to-end repair**	OB (most)
< 2–3 cm	Direct repair	OB review
< 3 cm (+ < 3 months)	**Direct repair or V-Y alone**	Orthobullets, OB, TO
2–5 cm	**V-Y advancement (gastroc-soleus) ± FHL** · V-Y fills 2–5 cm	OB
> 3 cm	**FHL transfer ± V-Y** (muscle weak, scarred)	Orthobullets, OB
> 5 cm	**FHL transfer** ± V-Y / turndown · ± allograft	OB
> 10 cm	**Allograft** (not suitable for local transfer)	OB · (> 5 cm in another OB)

4 cm chronic defect	**V-Y ± FHL augmentation** (one OB) · **open FHL transfer** (another OB) – read the options
Gap is measured after scar debridement	1.5 cm + 1.0 cm scar = **2.5 cm → V-Y lengthening + end-to-end** ✗ Wrong options: insertional tear · 0.5 + 1.0 = 1.5 cm → end-to-end alone · 1.5 cm no scar · 3.0 + 2.5 = 5.5 cm → combined procedures
V-Y advancement	**V apex at musculotendinous junction**, limbs diverge to exit tendon · cut **superficial tendinous portion only**, muscle intact · avoids sacrificing a normal muscle–tendon unit
Gastrocnemius turndown	**slip of central third of gastrocnemius tendon** bridges gap · augment with **FHL** (proximity + vascularity)
FHL transfer technique	excise degenerate ends · **release FHL at knot of Henry** · tenodese distal FHL stump to FDL · reroute **through calcaneus** → FHL + triceps surae both act on calcaneus · open / extended approach ✗ Wrong options: percutaneous · limited open

Calcaneal attachment **posterior = greater force** (longer lever arm, advocated) · anterior = more plantarflexion range · transtendinous = transosseous outcomes

Why FHL is preferred
Axis of contractile force replicates Achilles
Fires in phase with gastroc-soleus
Location / harvest avoids NV bundle
Expendable · **stronger** than FDL / PB · **biomechanically strong** bridge
Close + well-vascularised · preserves foot muscle balance

FHL anatomy origin **interosseous membrane + distal 2/3 fibula** · **lateral to FDL at ankle** · **dorsal to FDL at knot of Henry** · inserts **base of 1st distal phalanx** · **tibial nerve**

Option	Tendon	Why wrong
1st distal phalanx, tibial nerve	**FHL** ✓	–
1st proximal phalanx	FHB	Wrong insertion
1st distal phalanx (dorsum), deep peroneal	EHL	Anterior
5th MT base, superficial peroneal	**Peroneus brevis**	**Eversion weakness**, doesn't replicate pull, historical, worse longevity
Medial cuneiform + 1st MT base, deep peroneal	Tibialis anterior	**Out of phase** (anterior compartment)
Medial to FDL, inserts navicular	**Tibialis posterior**	**Chopart weakness, ↓ inversion, pes planus**
Plantar to FHL at knot of Henry	**FDL**	Not adjacent, less power (described 1991; FHL 1993)
Lateral femoral condyle origin	**Plantaris**	Not robust (**absent in 10%**), must be supplemented
–	Peroneus longus, EDL	No (PB acceptable, not PL)
–	**Synthetic / carbon fibre**	**Foreign body reaction**, longevity, no proven benefit
–	Xenograft, hamstring autograft	Poor outcomes / not used

After FHL transfer **only significant change: ↓ hallucal phalangeal pressure** · AOFAS MTP-IP **96.4** · no transfer metatarsalgia · no ↑ 1st MT head pressure
✗ Wrong options: ↑ short flexor recruitment

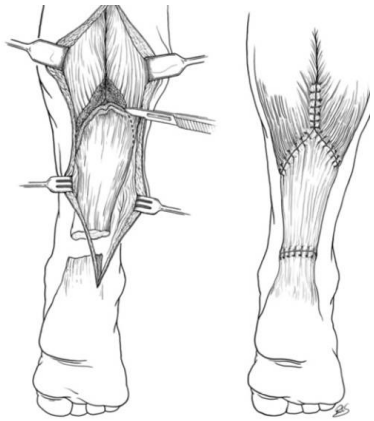
Hallux after FHL transfer **residual hallux PF / IP flexion weakness** (Orthobullets, OB review) · **no functional hallux deficit** (OB study) – read the options

Strength near-normal maximal strength, **↓ endurance** · torque deficit ~35% · FHL hypertrophy · main benefit = pain relief

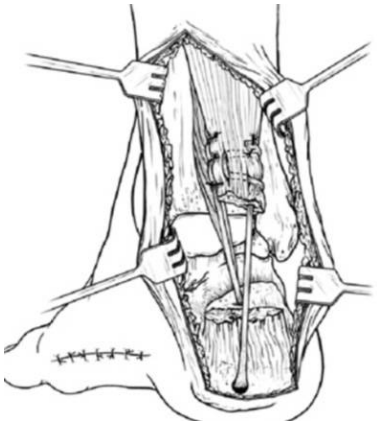
FHL for tendinosis / insertional when **> 50% of tendon involved**

✗ Chronic traps **equinus casting** (retracted ends won't appose) · **primary repair in max PF + stretching** (equinus contracture, re-rupture) · end-to-end repair (contracted gastroc-soleus, gap too big) · percutaneous / limited open

– – Active space – –



V-Y advancement of gastrocnemius-soleus aponeurosis



FHL transfer to the calcaneus

Open insertional rupture on Haglund / insertional tendinopathy

Open insertional rupture, prior steroid injections, Haglund on X-ray	**Haglund excision + gastrocnemius recession + Achilles repair + FHL transfer** ✗ Wrong options: open primary repair alone (distal disease) · functional rehab (never studied for insertional / tendinopathy) · bone-block allograft (> 10 cm) · percutaneous
Chronic tendonitis + Haglund	debridement of osseous spur + repair

6 Complications

Complication	Fact
Re-rupture	**Higher nonop (~10–40% vs 2%)** classically · **no difference** with functional rehab (Level 1) → treat with **surgical repair**
Wound healing	**5–10%** · **smoking = most common** risk factor · female, steroids, open technique
Deep infection	Debride necrotic tendon + **6 wk culture-specific antibiotics**
Sural nerve injury	**Higher with percutaneous** approach
DVT	**7–10%**, **no op vs nonop difference**
FHL harvest	Hallux weakness, harvest morbidity

X vs Y

	Acute midsubstance	Chronic (> 4–6 wk)	Sleeve avulsion	Tennis leg
Site	4–6 cm above insertion	Retracted, scarred stumps	**Insertion** (ossified tendon pulled away)	**Medial gastrocnemius head**
Exam	+ Thompson, gap	+ Thompson, ↑ resting DF, can't toe-walk	Weak squeeze response	Medial calf pain, ecchymosis
Treatment	Nonop rehab or repair	**By gap: end-to-end / V-Y / FHL**	Reconstruction	**Conservative**

How it's asked

"Incorrect statement for TA rupture?"	→ **antithrombolytics have a role** (TO)
"Greatest advantage of surgery + early ROM vs cast 6 wk?"	→ **lower rate of re-rupture**
"Cast 6 wk + PT — applies to conservative treatment?"	→ **higher risk for re-rupture**
"Nonop with early functional rehab vs surgery — expected?"	→ **no significant difference in functional outcomes** ✗ Wrong options: ↓ DVT · ↓ re-rupture · earlier RTS · ↑ complications
"Reported with surgery vs functional rehab?"	→ **increased complication rates**
"Shown to decrease re-rupture after repair?"	→ **early functional rehabilitation protocol**
"Reason to recommend minimally invasive repair vs functional rehab?"	→ **quicker return to work**
"Operative vs nonop outcomes?"	→ **higher nerve injury, quicker return to sport, quicker return to work**
"Not a technique for acute TA repair?"	→ **Maffulli** (TO)
"Nerve at risk in TA repair?"	→ **sural nerve** (TO)
"Percutaneous vs open — clearly reduces?"	→ **wound infection**
"Benefit of limited-open technique vs open?"	→ **decreased local scarring**
"Sural nerve on axial MRI?"	→ **D** (lateral, beside small saphenous vein)
"Open vs percutaneous — open results in?"	→ **increased rate of postoperative wound complications**
"Disadvantage of open vs MIS?"	→ **higher complication rate**
"Why percutaneous over open?"	→ **lower incidence of wound complications**
"Increases wound complications after repair?"	→ **tobacco use**

– – Active space – –

"Most common complication of surgical repair?"	→ **skin healing problems**
"Most common complication to warn of before open repair?"	→ **wound complication**
"Chronic rupture, 5 cm gap after debridement?"	→ **gastrocnemius turndown augmented with FHL transfer**
"Sub-acute rupture with loss of resting tension?"	→ **FHL transfer + sliding V-Y advancement of gastroc-soleus aponeurosis**
"Tendon used to augment V-Y — insertion + nerve?"	→ **1st digit distal phalanx, tibial nerve**
"6 mo neglected, 4 cm defect at surgery?"	→ **open repair with FHL transfer**
"Open insertional rupture, steroid injections, Haglund?"	→ **Haglund excision, gastrocnemius recession, Achilles repair, FHL transfer**
"Best tendon transfer for chronic rupture?"	→ **flexor hallucis longus**
"6 mo injury, 4 cm defect on MRI — reconstruction?"	→ **V-Y advancement with or without FHL augmentation**
"Single-incision FHL transfer — expect?"	→ **significantly decreased hallucal phalangeal pressure**
"Chronic rupture, + Matles — other test + best procedure?"	→ **Thompson test, V-Y advancement with tendon transfer**
"6 wk old, 3 cm defect — procedure likely needed?"	→ **V-Y advancement of the Achilles**
"Intra-op finding supporting V-Y + end-to-end?"	→ **1.5 cm defect + 1.0 cm interposed scar**
"Anatomy of best local autograft after 3.5 cm scar resection?"	→ **lateral to FDL at ankle, dorsal to FDL in midfoot**
"2-yr neglected rupture, 3 cm retraction?"	→ **gastrocnemius-soleus V-Y advancement ± FHL transfer**
"Fever + drainage 4 mo post-repair, 5 cm re-rupture?"	→ **debridement of necrotic + infected tendon, no reconstruction**
"Old cast-treated rupture, ossified tendon pulled off insertion?"	→ **chronic Achilles avulsion (sleeve) rupture**
"Medial gastrocnemius head rupture — expected first-line outcome?"	→ **excellent with conservative care, return within 3 months**
"Bilateral swellings over the Achilles?"	→ **answer key not provided by the source** (module: xanthomas) (TO)
"Heel pain walking upstairs, weak Thompson — investigation of choice?"	→ **USG** (TO)
"Movement tested against resistance in TA tear?"	→ **answer key not provided by the source** (TO)
"False about TA blood supply?"	→ **most abundant zone is the mid-section** (it is the insertion) (TO)
"TA mainly supplied from which surface?"	→ **anterior surface** (TO)

ACHILLES TENDONITIS

– Foot & Ankle · 3 / 48 –

1 Basics

Three conditions

Condition	Who	Site	Key feature
Insertional Achilles tendonitis	**Middle-aged + elderly, tight heel cord**	**At insertion**	Spur, **intratendinous calcification**, progressive bony enlargement
Retrocalcaneal bursitis + Haglund deformity	**Young**	Bursa **between anterior Achilles + posterior calcaneus · posterosuperior calcaneal tuberosity**	Pain **anterior to + 2–3 cm proximal to insertion**
Noninsertional (midportion) tendinopathy	**Active, middle-aged men**	**2–6 cm proximal to insertion**	Thickening, fusiform swelling

Noninsertional thought **more common** · insertional **less common** (OB)

Pathophysiology

Insertional repetitive trauma → inflammation → **cartilaginous then bony metaplasia**

Noninsertional **abnormal vascularity / poor blood supply 2–6 cm proximal** + **repetitive micro-tears**

Risk factors overuse · **dorsiflexor / plantarflexor imbalance** · poor blood supply · genetic · **fluoroquinolones** · inflammatory arthropathy

Inflammation? **degeneration without inflammation** (tendinosis; asymptomatic degeneration from microtrauma + ageing) (OB) · "inflammation and thickening" (another OB) – read the options

Histology

Insertional **disorganised collagen + mucoid degeneration**, **few inflammatory cells**

Tendinosis **anaerobic degeneration** in poorly vascularised segment

3 Presentation

Insertional **posterior heel pain**, swelling, burning, stiffness · **shoe pressure pain** (aversion to closed-heel shoes) · **midline tenderness at insertion** · bony prominence

Retrocalcaneal bursitis **fullness + tenderness medial and lateral to tendon** · **pain with dorsiflexion**

Noninsertional pain, swelling, warmth, **worse with activity**, difficulty running · **thickening + tenderness 2–6 cm proximal** · pain through whole ROM

– – Active space – –

Painful arc sign	**tender intratendinous nodule moves with active ROM** = **tendinosis** (vs **paratenonitis** – stays still)
Dorsiflexion	most have **limited passive dorsiflexion**
Silfverskiöld test	DF improves with **knee flexion → gastrocnemius contracture** · no improvement → **Achilles contracture**

4 Diagnosis & Classification

Key radiology findings

Study	Finding
Lateral foot X-ray	**Bone spur + intratendinous calcification (enthesophyte)** (insertional) · **Haglund deformity** = enlarged posterosuperior tuberosity
MRI	**Intrasubstance intermediate signal** (disorganised tissue) · **thickened tendon** · % degeneration guides surgery · rarely needed for Haglund
Ultrasound	Degeneration · **colour / power Doppler ↑ intratendinous blood flow** (neovascularity) · chronic rupture = **hypoechoic gap between ends**
MRI diffuse thickening, no focal area	→ **tendon transfer** indication



Insertional calcification and spur at the Achilles insertion



Haglund deformity – enlarged posterosuperior calcaneal tuberosity

Noninsertional classification

Type	Feature
Achilles tendinosis	**Tendon thickening**; anaerobic degeneration
Achilles peritendonitis	**Inflammation of tendon sheath**
Paratenonitis	**Inflammation of paratenon**

Brisement for **peritendinitis**, not tendinosis

5 Management

Achilles tendinopathy

- First line → **eccentric strengthening, NSAIDs, heel lift, shoe change**
  (avoid steroid injection – rupture)
- Failed nonop, < 50% degenerated → **debridement ± Haglund resection**
  - gastrocnemius contracture → **+ gastrocnemius recession**
- > 50% degenerated / diffuse / > 55 yr → **debridement + FHL transfer**
  - Achilles contracture (no Silfverskiöld change) → **+ Achilles lengthening**

Operative criteria

Finding	Add to debridement
Haglund deformity	**Calcaneal exostectomy / calcaneoplasty** (prevents recurrence)
< 50% degeneration	**Side-to-side repair** / suture anchor; no transfer
> 50% degeneration (or debrided)	**FHL (or FDL, PB) transfer / augmentation**
DF improves with knee flexion	**Gastrocnemius recession**
DF does not improve with knee flexion	**Achilles lengthening**
> 55 yr, **diffuse MRI thickening**, **subacute rupture** on tendinopathy, heavier patients	**Tendon transfer**
Chronic symptoms **> 6 months** without improvement	Surgery considered

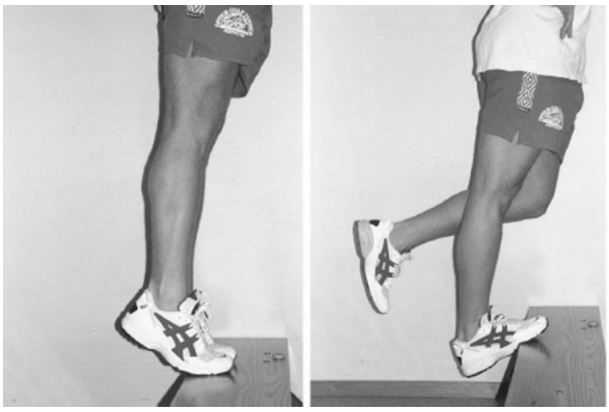
Noninsertional tendinopathy – eccentric loading & PRP

First line **eccentric strengthening** (cornerstone, **heavy-load eccentric gastrocsoleus**) + rest, NSAIDs, heel lift, shoes, orthoses for overpronation, PT **3–6 months** · nonop **65–90% successful**

Not for chronic tendinopathy stair climbing · rope jumping · concentric · isometric

Eccentric = heel drop stand on step, heels off, **slowly lower below step** · **force remains constant as the muscle lengthens** · **closed-chain** · enhances dorsiflexion · most helpful in **later phases** of rehab / after diminished activity

– – Active space – –



Heel drop — eccentric loading on a step

Exercise (✗ Wrong options: traps)	Definition / example
Eccentric ✓	**Force constant while muscle lengthens** · heel drop
Concentric	**Muscle shortens**, tension ∝ external load
Isometric	**Constant length + tension**, no movement · **standing calf-raise hold**, pushing a wall
Isotonic	**Constant force through ROM**
Isokinetic	**Constant velocity (angular) through varied resistance** · Cybex
Plyometric	**Jump training** · may aggravate
Single-leg stance	**Proprioception** → ankle instability
Heel walk	**Dorsiflexor** strengthening

Why eccentric works (theory)	absence of concentric phase disrupts lengthening / shortening cycle → shear in tendon + injury to collagen (exact mechanism unknown)
Doppler neovascularity, intact tendon, 4 months	**eccentric closed-chain exercises** ✗ Wrong options: **steroid** (rupture) · CT chest / abdomen / pelvis · **PRP + delayed arthroscopic debridement** (level IV, not before eccentrics) · resection + FHL + gastrocsoleus turndown (chronic rupture)
PRP	**added to eccentrics may improve outcomes in non-responders** (OB, noninsertional) · listed as nonop option (Orthobullets) · **no evidence** for PRP in tendinosis (OB, insertional) — read the options
Orthotic + eccentrics	**no better than eccentrics alone**
Adjuncts studied	orthotics, PRP, **ESWT**, **topical glyceryl trinitrate**, **polidocanol**, blood, hyaluronic acid, pulsed light, NSAIDs, steroids · GTN, **prolotherapy**, **aprotinin** = evolving, little evidence
Steroid injection	**avoid → Achilles rupture** (Orthobullets, OB) · one OB lists corticosteroid injection among nonop options — read the options
Severe	cast or removable boot
Diagnosis from first-line eccentrics	**Achilles tendinopathy** ✗ Wrong options: **Baxter nerve** (stretching, PT, NSAIDs) · **calcaneal stress fracture** (NWB) · **ATFL tear** (PT + proprioception) · **syndesmotic sprain** (NWB cast / CAM boot)

Insertional tendinopathy – debridement, Haglund resection & FHL transfer

– – Active space – –

Nonop	**heel sleeves + pads (mainstay)** · **small heel lift** · eccentric PT + gastrocsoleus stretching · **locked-ankle AFO 6–9 months** if others fail · **no steroid** · Haglund: ice, external padding · high nonop failure rates
Surgery (< 50% removed)	**retrocalcaneal bursa excision + debridement + calcaneal prominence resection** · tendon detachment → calcaneoplasty → reattachment
Surgery (> 50% removed)	**suture anchor repair + FHL (FDL, PB) augmentation + primary closure** ✗ Wrong options: **side-to-side** (for < 50%) · **tibialis posterior** transfer · wound VAC / staged closure · closure alone
Incisions	**midline, lateral or medial J-shaped**
Haglund + calcifications, failed nonop, loss of function	**debridement + calcaneal exostectomy + possible FHL transfer** ✗ Wrong options: debridement alone · Arizona brace · steroid · ankle arthrodesis
Haglund, normal DF (10°), 30% degeneration	**debridement + exostectomy alone**

Scenario (OB)	Correct plan
Normal heel, DF 10°, **60%**	Debridement + **FHL** (no exostectomy)
Normal heel, DF −10° → 10° with knee flexion, 60%	Debridement + FHL + **gastrocnemius recession**
Haglund, DF 10°, 60%	Debridement + FHL + **exostectomy**
Haglund, DF 10°, **30%**	**Debridement + exostectomy alone** ✓
Haglund, DF −10° not improved with knee flexion, 30%	Debridement + **Achilles lengthening** + exostectomy

Haglund + gastrocnemius contracture, 40% debrided	**calcaneoplasty / Haglund excision + gastrocnemius recession** ("I and III") ✗ Wrong options: FHL (< 50%)
FHL transfer (overweight)	**↑ ankle plantar flexion strength** ✗ Wrong options: ↑ AOFAS · ↑ SF-36 · **↓ great-toe plantar flexion** (not seen) · no difference
Recovery (older, enthesopathy, debridement + ostectomy)	**12 months (a year or more)** · young / pure tendon pathology faster

– – Active space – –

Chronic noninsertional disease – debridement with FHL transfer

Operation	Indication	Success
Percutaneous longitudinal tenotomy	**Mild–moderate**	–
Tendon stripping (suture strips anterior Achilles adhesions)	Adhesions	–
Dorsal wedge calcaneal osteotomy + debridement	**< 50%** involvement	–
Open excision + tubularisation	**Moderate–severe**	**70–100%**
Tendon transfer (FHL, FDL, PB)	**> 50% degeneration**, **> 55 yr**, **diffuse MRI thickening**, **subacute rupture**	–

Fusiform swelling 12 months, > 50% width — **debridement + FHL transfer**
✗ Wrong options: side-to-side (inadequate strength) · steroid · brisement · walking cast

Subacute rupture on tendinosis, 4 cm gap — **debridement + augmentation with tendon inserting on plantar great-toe distal phalanx (FHL)**
✗ Wrong options: primary suture-anchor repair (fails) · plantar 1st MT head · FDL plantar lesser-toe distal phalanges, not MT heads / dorsal

Large-gap options — **V-Y myotendinous flap**, free tendon graft, synthetic graft, local transfer (**PB, FDL, FHL**) · small tear → debridement + primary repair

FHL transfer outcome (Den Hartog) — ↑ ankle–hindfoot scores, mostly good / excellent, **no hallux functional deficit**

Tendon	Insertion
FHL	**Plantar aspect of great-toe distal phalanx**
FDL	**Plantar distal phalanges of lesser toes**

X vs Y

	Insertional	Noninsertional
Site	**At insertion**	**2–6 cm proximal**
Age	Middle-aged + elderly	Active middle-aged
X-ray	**Spur, calcification, Haglund**	Often normal
Nonop	**Heel sleeves / pads**, heel lift, AFO	**Eccentric loading** ± PRP
Surgery	**Debridement + exostectomy** ± FHL	Tenotomy / excision + tubularisation ± FHL

	Tendinosis	Paratenonitis
Painful arc	**Nodule moves** with ROM	Fixed
Treatment	Eccentrics, debridement	**Brisement**

How it's asked

"Exercise to add after a period of reduced activity?" → **eccentric**

"Most effective PT for chronic noninsertional tendinopathy?"	→ **eccentric strengthening**
"Later-phase strengthening in Achilles tendinitis?"	→ **eccentric**
"Exercise brochure for midportion tendinosis?"	→ **Figures B and D — eccentric exercises most likely to relieve pain + return him to previous activity level** ✗ Wrong options: surgery first
"Describe the most effective PT technique?"	→ **force remains constant as the muscle lengthens**
"Most effective exercise for posterior heel pain?"	→ **heel drop**
"Doppler ↑ blood flow, intact tendon — next step?"	→ **eccentric closed-chain exercises**
"First-line eccentric strengthening — diagnosis?"	→ **Achilles tendinopathy**
"PRP in noninsertional tendinopathy — true?"	→ **PRP + eccentrics may improve outcomes in non-responders**
"Haglund + calcification, failed nonop?"	→ **debridement + calcaneal exostectomy + possible FHL transfer**
"55% debrided — after suture-anchor repair?"	→ **FHL augmentation + primary closure**
"Recovery after debridement + ostectomy in 65-yo?"	→ **12 months**
"FHL transfer vs no transfer?"	→ **increased ankle plantar flexion strength**
"When is debridement + exostectomy alone right?"	→ **Haglund, DF 10° knee extended, 30% degeneration**
"40% debrided, Haglund, DF ↑ with knee flexion?"	→ **calcaneoplasty + gastrocnemius recession (I and III)**
"Fusiform swelling, > 50% degenerated?"	→ **debridement + FHL tendon transfer**
"4 cm gap on chronic tendinosis — best function?"	→ **tendon inserting on plantar great-toe distal phalanx (FHL)**

– – Active space – –

ACQUIRED SPASTIC EQUINOVARUS DEFORMITY

– Foot & Ankle · 4 / 48 –

1 Basics

Etiology

Causes	**cerebrovascular accident (CVA)** · **traumatic brain injury (TBI)** → **upper motor neuron** injury
Commonest physical finding after CVA	**spastic equinovarus** deformity of foot and ankle ✗ Wrong options: flaccid · calcaneovalgus · equinovalgus
Associated	joint contractures · **knee hyperextension in stance**

Pathophysiology

Component	Overactive muscle
Equinus	**Gastrocnemius–soleus complex**
Varus	**Tibialis anterior** (relative overactivity) · lesser: **FHL, FDL, tibialis posterior**

3 Presentation

Symptoms	deformity · difficulty with gait
Exam	**spastic equinovarus** · **↑ tone** · **hyperreflexia**



Spastic equinovarus foot after stroke

4 Diagnosis & Classification

Diagnosis	**clinical**: spastic equinovarus foot + prior CVA / TBI

Key radiology findings

Radiographs	**AP, lateral, oblique** of foot and ankle

5 Management

Spastic equinovarus after CVA / TBI

- During neurologic recovery → **PT, botulinum / phenol, AFO**
- Fixed contracture after recovery, not braceable → **surgery**
  - Equinus → **Achilles lengthening (TAL)**
  - Varus → **SPLATT**
  - Recurrence despite soft-tissue surgery → **osteotomies / fusions**

Nonoperative (first line)

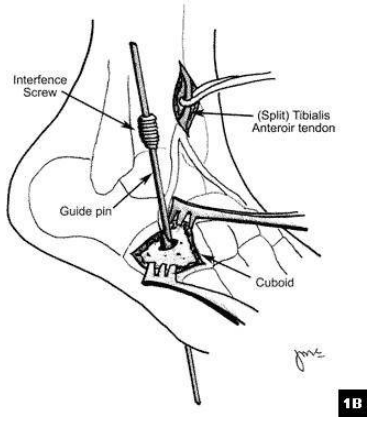
Therapy	stretching + strengthening · maintain joint ROM · splinting
Injections	**phenol blocks** · **botulinum toxin**
AFO	use while in **bed or wheelchair**
Timing	**neurologic recovery 6–18 months after CVA** · **years after TBI** → operate only after recovery period

Operative criteria

Indication for surgery
Fixed contractures persist after neurologic recovery and **not braceable**
Functional deficits
Skin problems secondary to deformity / braces
Recurrence despite proper soft-tissue procedures → **osteotomies + fusions**

Post-stroke equinovarus – SPLATT with TAL

Equinus	**Achilles tendon lengthening (TAL)** (open / percutaneous) or **gastrocnemius recession**
Varus	**split anterior tibialis tendon transfer (SPLATT)** ✗ Wrong options: medializing calcaneal osteotomy + FHL transfer (that's flatfoot)
SPLATT technique	tibialis anterior split → **lateral half to cuboid** through drill hole, sutured · ± lengthening / dorsal transfer of **PTT**



SPLATT – lateral half of tibialis anterior to cuboid

– – Active space – –

Bridle procedure — **posterior tibialis tendon** through **interosseous membrane** to dorsum → dual anastomosis to **tibialis anterior** + rerouted **peroneus longus** (in front of lateral malleolus)

6 Complications

Hindfoot valgus — overcorrection from **inadvertent PTT lengthening**

Prognosis after CVA

Outcome	Rate
Normal ambulation	**25%**
Some level of ambulation	**75%**
Neurologic recovery	**6–18 months** (CVA) · years (TBI)

X vs Y

	SPLATT	Bridle
Tendon moved	**Lateral half of tibialis anterior**	**Tibialis posterior**
Route / insertion	**Cuboid** (drill hole)	Through **interosseous membrane** → dorsum, anastomosed to TA + rerouted PL
Corrects	**Varus** (spastic)	Foot drop / varus

How it's asked

"Foot deformity after stroke?"	→ **spastic equinovarus**
"Post-stroke, correctable contracture, brace skin problems – add to TAL?"	→ **SPLATT** ✗ Wrong options: medializing calcaneal osteotomy + FHL transfer
"Varus after CVA – prime mover?"	→ **tibialis anterior** (relative overactivity)
"Equinus after CVA – cause?"	→ **gastrocnemius–soleus overactivity**
"Surgery indicated when?"	→ **fixed contracture after neurologic recovery, not braceable**
"Neurologic recovery after CVA takes?"	→ **6–18 months**
"Hindfoot valgus after SPLATT?"	→ **inadvertent PTT lengthening**

– – Active space – –

ANKLE ARTHRITIS

– Foot & Ankle · 5 / 48 –

1 Basics

Epidemiology & etiology

Incidence	**30 / 100,000 / yr** · prevalence **~1%** · less common than hip / knee OA
Post-traumatic	**commonest, 75–80%** · mostly **ankle fractures**; rest = **chronic (lateral) ligamentous laxity**
Primary OA	**< 10%**
Other causes	**RA / inflammatory**, osteonecrosis, neuropathic, septic, gout, haemophilia
Risk factors	prior ankle fracture · chronic ligamentous laxity · inflammatory arthropathy
Pathoanatomy	nonanatomic fracture healing → altered contact forces + load bearing → cartilage loss on **talar body + plafond** → narrowing, subchondral sclerosis, **eburnation**

Anatomy

Joint	**ginglymus** (tibia, talus, fibula) · talar dome **biconcave with central sulcus**
ROM	**dorsiflexion 20°** · **plantarflexion 50°**

3 Presentation

Symptoms	pain with weight-bearing · stiffness · locking / catching
Exam	variable effusion · pain + ↓ ROM vs contralateral · angular deformity (trauma) · crepitus
Localising sign (OB)	pain through tibiotalar arc · **painless subtalar inversion / eversion** → isolated tibiotalar disease (→ no need to fuse subtalar)

4 Diagnosis & Classification

Key radiology findings

Modality	Use / finding
Weight-bearing AP, lateral, mortise	Diagnosis: **joint space loss**, subchondral sclerosis + cysts, eburnation, angular deformity
CT	**Surgical planning** (arthrodesis + arthroplasty); quantify healing of a fusion nonunion
MRI	Focal cartilage disease · **↑ sensitivity in early disease** · cartilage injury, subchondral marrow oedema, ligament tears
Labelled WBC scan	Infection workup in fusion nonunion
Scanograms / contralateral films	Malalignment planning before revision

– – Active space – –

Takakura classification (mortise view, varus-type OA)

Stage	Radiograph	Image
I	Osteophytes + early sclerosis, **no narrowing**	
II	**Narrowing of medial joint space**, no subchondral contact	
IIIA	Obliteration **at medial malleolus**, subchondral bone contact	
IIIB	Obliteration **over roof of talar dome**, subchondral contact	
IV	Obliteration with **complete tibiotalar contact**	

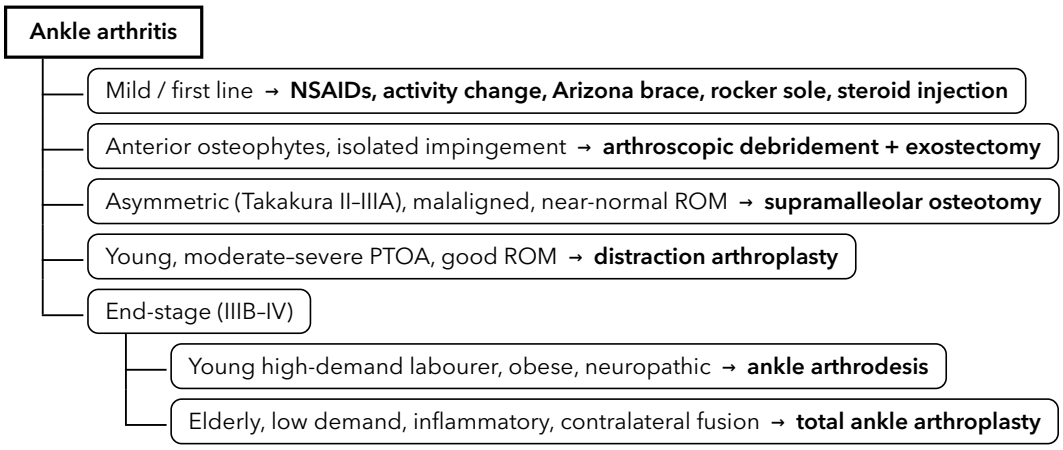
Stage	Treatment (OB)
I	Conservative / symptomatic
II, IIIA	**Supramalleolar osteotomy** (joint preserving)
IIIB	SMO **poor results / higher failure** → TAA or arthrodesis
IV	**Arthrodesis** or **TAA** (whole joint)

Talar osteonecrosis (differential)

Late talar head osteonecrosis (rare)	**talonavicular + subtalar** pain (selective injections) · X-ray / CT = advanced degenerative change of TN + subtalar joints
MRI	**↓ T1, ↑ T2 with dark serpiginous line**
✗ Look-alikes	**PVNS** (erosions + hemosiderin) · **chronic osteomyelitis** (fluid collections, bony destruction) · talar neck / head fracture (no fracture line)

5 Management

– – Active space – –



Nonoperative management – bracing, rocker sole & injection

First line — activity modification, **bracing to immobilise ankle**, NSAIDs · **intra-articular corticosteroid = mainstay injection** · PRP / hyaluronic acid **no consensus** (PRP not proven)

Untried nonop (OB) — **steroid injection + cushioned rocker-bottom shoe** (± cane)
✗ Wrong options: opiates + limited WB 6 wk · cast NWB 12 wk · triple / pan-talar arthrodesis

Arizona (gauntlet) brace — **immobilises ankle + hindfoot** → best brace for tibiotalar OA at any age · PT may worsen pain

Orthosis / shoe	Indication
Single rocker sole	**End-stage tibiotalar (or subtalar) arthritis with limited motion**, or after ankle / subtalar fusion · ↑ toe-off propulsion, ↓ heel-strike pressure, ↓ ankle motion needed
Double rocker sole	**Midfoot prominences (Charcot)**, midfoot OA
Severe angle rocker	**Toe-tip ulcers**
Negative heel rocker	**Fixed ankle dorsiflexion** deformity
CROW boot	Charcot / midfoot
Lateral hindfoot post	**Cavovarus** foot
Hinged AFO	**Foot drop** – does **not** immobilise ankle
Rocker sole (wrong option)	Fixed planovalgus

Arthroscopic debridement & distraction arthroplasty

Debridement + anterior tibial / dorsal talar exostectomy — **isolated anterior impingement** from anterior osteophytes · joint preserving · open option (plan future approaches) · **not for advanced arthritis**
✗ Wrong options: won't correct alignment

– – Active space – –

Distraction arthroplasty — **young**, moderate–severe **post-traumatic** OA · offloading → biologic repair · **needs good pre-op ROM** · static / hinged ex-fix or spatial frame ± microfracture, chondroplasty, orthobiologics · outcomes controversial

Supramalleolar osteotomy – indications (varus, asymmetric arthritis)

Indications — **near-normal ROM** · **minimal talar tilt** · **varus heel alignment** · **medially focused (asymmetric) arthritis = Takakura II / IIIA** · tibiotalar malalignment, eccentric wear · young active patient wanting high activity

Biggest change in ankle forces — **tibial osteotomy**
✗ Wrong options: deltoid / lateral ligament reconstruction · calcaneal osteotomy · PL → PB transfer (adjuncts only)

Deformity	Osteotomy
Varus	**Medial opening wedge** (or lateral closing wedge)
Valgus	**Medial closing wedge** (or lateral opening wedge) ± **oblique fibular lengthening osteotomy**
Dome	Less translation / length change; technically hard
Multiplanar / large	**Gradual correction with spatial frame**

Fibular osteotomy — outcomes **same** whether or not needed

Medial opening wedge best for — **symptomatic medial narrowing + varus alignment**
✗ Wrong options: stage IV valgus → fusion · stage I → conservative · lateral narrowing + neutral alignment → look at knee / hip · IIIB → poor results

Preoperative talar tilt	Five-year survival rate
4–10°	**85%**
> 10°	**65%** (higher failure; also Takakura IIIB)

Contraindications — **< 50% preserved tibiotalar surface** · **age > 70** · hindfoot instability not fixable by ligament reconstruction · significant vascular disease · **inflammatory arthritis** · **neuroarthropathy** ·
✗ Wrong options: **age < 50 is NOT** a contraindication

Complications — **reoperation for hardware removal commonest**, then nonunion

Operative criteria – arthrodesis vs TAA

	Ankle arthrodesis	Total ankle arthroplasty
Indications	Moderate–severe **PTOA or inflammatory**, refractory · **young, high-demand labourers** · malalignment (+ osteotomy) · comorbidities	PTOA or **inflammatory (RA)** · **elderly**, low demand · ? concomitant TN / subtalar fusion · **contralateral ankle fusion** · wants motion (hiking, uneven ground)
Relative contraindications	**Vascular compromise (ABI 0.4, claudication)** – poor wound + fusion healing	**Obesity** (✗ Wrong options: not absolute, BMI 35) · **young labourers** · severe deformity
Absolute contraindications	–	**Active infection** · inadequate bone stock / **severe osteoporosis** · **talus osteonecrosis** · **Charcot / neuropathic joint** · **peripheral neuropathy**
Other contraindications (OB)	–	Uncorrectable deformity · ankle instability · neurologic muscle imbalance · severe malalignment

Arthrodesis vs total ankle arthroplasty – gait, adjacent joints

Not TAA contraindications
post-traumatic arthritis (commonest indication) · inflammatory arthritis / RA · age > 50 · reconstructible ligament damage · **correctable coronal deformity (up to 20–30°)** via osteotomy / soft-tissue balancing (TO: **> 10–15° = contraindication** – read the options) · **preserved subtalar motion = favourable**

Peripheral neuropathy
no protective sensation → microtrauma, instability, wounds, infection, **early loosening**

Arthrodesis
gold standard for end-stage OA · reliable pain relief + ADLs · **union ~90%** (Orthobullets) · **93–94% with cancellous screws**, 74% overall (OB)

Arthrodesis position
neutral dorsiflexion, **5–10° external rotation** (10° OB / TO), **~5° valgus** (0–5° OB), **talus ~5 mm posterior**

Young labourer (28 yo, BMI 31, construction)
ankle fusion
✗ Wrong options: low-profile TAA · bipolar allograft · interposition · arthroscopic debridement (too advanced)

Healthy 66-yo hiker, subtalar pain-free
TAA
✗ Wrong options: TTC nail (spare subtalar) · tibiotalar fusion → younger · distraction → younger, moderate

– – Active space – –

Large OCD + 5° varus, 41 yo joint / cartilage preservation first

Function (TAA vs arthrodesis)	Fact
Gait	TAA patients: **superior gait mechanics compared to AA post-operatively** – **↑ stride length, cadence, stride / gait velocity** → arthrodesis = **↓ stride length** (✗ Wrong options: hip abduction · leg length)
Slopes / stairs	TAA **higher scores walking uphill, upstairs, downstairs**
Uneven surfaces at 12 mo	**Both improve** vs pre-op
Gait function at 12 mo	Arthrodesis **improves** it (✗ Wrong options: "loss of gait function")
Bilateral arthrodesis	Bad for gait → TAA for the second ankle

TAA outcomes up to **90% good–excellent at 5–10 yr** (Orthobullets) · **~90% survivorship at 10 yr** (OB) · **> 80% survivorship at 5 yr if < 50 yo** (Orthobullets) · survivorship ↓ if **< 54 yo** (OB) · **70–98% at 3–6 yr, 80–95% at 8–12 yr** (OB, Easley) · **5-yr up to 60%, reoperation up to 30%** (another OB) – read the options · long-term data pending

Arthrodesis long-term **ipsilateral midfoot / hindfoot degeneration**
✗ Wrong options: revision surgery · hardware failure · walking aids · knee / hip DJD

Subtalar arthrosis after arthrodesis **50% at 10 yr** (Orthobullets) · **> 50% within 10 yr** (OB) · **higher after AA than TAA**
✗ Wrong options: "rare after well-done fusion"

Arthrodesis & arthroplasty technique

Arthrodesis approaches anterior · **transfibular (8–10 cm distal fibula resected)** · mini-open (burr) · arthroscopic

Arthrodesis fixation transarticular compression screws · bridge plate · ex-fix · **hindfoot retrograde nail (TTC)**

TAA approach anterior, **between tibialis anterior + EHL** · protect **SPN, DPN, anterior tibial artery**

TAA design fixed vs mobile bearing · adjuncts: **TAL / gastrocnemius recession**, foot osteotomies, ligament reconstruction

Failed fusion or talar AVN – revision arthrodesis

Workup exclude **infection + metabolic** causes · plain films · CT (bony healing) · **labelled WBC scan** · scanograms if malaligned

Nonunion, subtalar pain-free **revision ankle arthrodesis + bone graft** with **stable compression fixation** (transfibular approach + compression screws) · **spare subtalar** (young)

✗ Nonunion traps external / internal bone stimulator · **dynamisation → instability** · PRP · bone grafting alone · hardware removal + **IM rod / hindfoot nail (TTC)** when subtalar is normal

Revision union **≥ 85%**

6 Complications

Procedure	Complication
Arthrodesis	**Subtalar arthrosis 50% at 10 yr · nonunion 10%**
Nonunion risk factors	**Smoking**, adjacent joint fusion, **failed previous arthrodesis**, **AVN**
TAA	**Intra-op medial malleolus fracture** (prophylactic pinning) · PJI · aseptic loosening · **periprosthetic cysts** · wound + deep infection · osteolysis · periprosthetic fracture
Periprosthetic cysts	Infection or aseptic loosening · small asymptomatic → **observe** · large / progressive / painful → **debridement + bone graft** or **revision TAA**
SMO	Hardware removal (commonest reoperation) → nonunion

TAA complication – syndesmotic nonunion

Pain 10 mo after TAA, stable ligaments, syndesmotic screws

syndesmotic nonunion → ↑ tibial component loosening
✗ Wrong options: poly wear loosening · lateral ligament failure · "normal at 10 mo" · wrong procedure



Syndesmotic nonunion after total ankle arthroplasty

X vs Y

	Arthrodesis	TAA
Patient	**Young labourer**, obese, neuropathic, vasculopath	**Elderly**, RA, low demand
Gait	↓ stride length	**↑ stride length, cadence, velocity**, uphill
Adjacent joints	**Subtalar 50% at 10 yr**, midfoot / hindfoot DJD	Preserved
Talar osteonecrosis	Option	**Contraindicated**

	Supramalleolar osteotomy	Fusion / TAA
Takakura	**II–IIIA** (asymmetric, medial)	**IIIB–IV** (whole joint)
Talar tilt	Minimal (> 10° → 65% survival)	Any
Ages	Young OK (< 50 not CI); **> 70 CI**	–

– – Active space – –

How it's asked

"Single rocker sole best for?"	→ **end-stage tibiotalar arthritis with limited motion**
"Double rocker sole for?"	→ **midfoot prominences (Charcot)**
"Untreated tibiotalar OA — next step?"	→ **steroid injection + cushioned rocker-bottom shoe**
"Nonop tibiotalar OA, which brace?"	→ **Arizona brace** ✗ Wrong options: lateral hindfoot post · hinged AFO · CROW
"Young active woman, varus asymmetric OA — return to sport?"	→ **supramalleolar osteotomy**
"Medial opening wedge SMO most appropriate?"	→ **medial narrowing + varus alignment**
"Greatest change in ankle forces?"	→ **tibial osteotomy**
"Best indication for SMO?"	→ **varus heel alignment** ✗ Wrong options: 30% surface · age > 70 · inflammatory · neuroarthropathy
"True about surgical options in a labourer?"	→ **SMO 5-yr survival 65% if talar tilt > 10°**
"TAA contraindication?"	→ **neuropathic joint disease**
"TAA contraindication (BMI 35, 12° correctable, PTOA)?"	→ **peripheral neuropathy**
"Long-term after ankle fusion?"	→ **ipsilateral midfoot / hindfoot degenerative changes**
"Arthrodesis vs TAA gait?"	→ **decreased stride length**
"28-yo labourer, BMI 31, PTOA?"	→ **ankle fusion**
"66-yo hiker, subtalar spared?"	→ **total ankle arthroplasty**
"Best ankle fusion candidate?"	→ **45-yo construction worker, end-stage OA**
"TAA vs AA true (RA patient)?"	→ **TAA superior gait mechanics**
"TAA vs AA true?"	→ **TAA greater scores walking uphill**
"Pain 10 mo after TAA with syndesmotic screws?"	→ **syndesmotic nonunion**
"Ankle fusion nonunion at 12 mo, subtalar normal?"	→ **revision ankle arthrodesis + bone grafting**
"Distracted fusion, loose distal fixation, 34 yo?"	→ **revision fusion with stable compression + bone graft**
"TN + subtalar pain, T1 dark / T2 bright serpiginous line?"	→ **late osteonecrosis of talar head**

– – Active space – –

ANKLE ARTHRODESIS

– Foot & Ankle · 6 / 48 –

1 Basics

Indications & causes

Definition	fusion of the **tibiotalar joint**; commonest for **end-stage ankle arthritis**; open or arthroscopic
Commonest cause	**post-traumatic arthritis** · **> 60%** of end-stage tibiotalar arthritis (OB)
Other indications	post-infection · chronic instability · **AVN of the talus** · inflammatory arthropathy · primary OA · **neuropathic arthropathy** · tumour resection · **salvage of failed ORIF** · **salvage of failed TAA**
Talar AVN	**trauma = commonest cause (75%)**, mainly talar neck / body fractures · early → core decompression, non-vascularised or vascularised bone graft · late (structural change) → **arthrodesis**
Hindfoot OA causes	calcaneus / talus fractures · RA · OA · end-stage PTTD · tarsal coalition · **Charcot-Marie-Tooth (cavus)**
Talar neck fracture	**varus malunion ~30%** → abnormal ankle loading → arthritis

Effect of hindfoot fusion on motion (talonavicular)

Joint fused	Effect on motion
Talonavicular	**Eliminates almost all hindfoot motion – > 90% ↓** (greatest cumulative effect)
Subtalar (talocalcaneal)	**↓ TN motion 74%** · **↓ CC motion 44%**
Calcaneocuboid	**↓ TN motion 33%** (OB) · **67%** (another OB) – read the options · ↓ subtalar 8% (little effect)
Naviculocuneiform	Little effect on hindfoot
Cuboid–5th MT	Limited effect
Tibiotalar	**Little effect on hindfoot ROM**

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Decides
Weight-bearing X-rays ankle + foot	End-stage tibiotalar arthritis · coronal deformity (varus) · triple malunion (residual subtalar valgus, forefoot abduction)
CT	**Hindfoot (subtalar) degeneration** → spared = **isolated tibiotalar** · involved = **TTC**
MRI	**Talar AVN** – broad-based → TTC
Aspiration	Prior septic ankle – **no growth, WBC 9,800 = no active infection** → arthrodesis (✗ Wrong options: repeat I&D)
Post-op X-ray	**Tibial stress fracture at proximal interlocking screw** of hindfoot nail

Clinical localisation pain with **passive DF / PF** = tibiotalar · pain with **inversion / eversion** + sinus tarsi pain = **subtalar** · **Coleman block** rigid hindfoot = fixed

5 Management

End-stage ankle arthritis, failed bracing / NSAIDs / injections

- Subtalar spared → **TIBIOTALAR ARTHRODESIS (screws)**
- + Subtalar arthritis / broad talar AVN / bone loss → **TTC ARTHRODESIS**
  (lateral transfibular · retrograde IM nail · femoral head allograft if bone loss)
- Infection / draining hardware → REMOVE HARDWARE + I&D → **STAGED ARTHRODESIS**
- Isolated subtalar arthritis, ankle painless → **SUBTALAR ARTHRODESIS**

Operative criteria – arthrodesis vs TAA

Favours arthrodesis	Favours TAA
Young, high-demand / labourer (construction worker, 36–45 yo)	**Elderly, less active**
Peripheral neuropathy (↑ TAA failure) · insensate foot	Preserved sensation
Severe PVD	–
Charcot / neuropathic arthropathy	–
Osteonecrosis of talus (TAA contraindicated)	Good talar bone stock
Active or prior infection (↑ periprosthetic infection)	No infection history
Severe deformity · soft-tissue compromise	Correctable deformity
Gold standard; most reliable pain relief	Keeps ROM → better cadence, ↓ **adjacent joint degeneration**; reasonable 10-yr survivorship

Tibiotalar fusion — **highest success rate** for post-ORIF tibiotalar arthritis
✗ Wrong options: osteochondral allograft · interpositional soft tissue · arthroscopic debridement + microfracture · TTC

Supramalleolar osteotomy — only for **coronal deformity > 10°** with **mild / moderate** arthritis
✗ Wrong options: end-stage

Optimal fusion position

Plane	Position
Sagittal	**Neutral (0°) dorsiflexion / plantarflexion – plantigrade**
External rotation	**5–10° ER** (Orthobullets, OB) · **10° ER** (OB, TO) · **ER ~5°** (TO explanation) – read the options
Coronal	**5° hindfoot valgus** (Orthobullets, TO) · **0–5° valgus** (OB) · **5–10° valgus** (another OB) – read the options
Translation	**5 mm posterior talar translation** under the plafond



Tibiotalar arthrodesis with crossed screws

Why each — neutral flexion ↓ knee extension in gait + **MCL laxity** · slight valgus prevents **lateral column overload** · posterior talus ↓ **lever arm of foot**

Avoid (TO) — **extension** → excess pressure + **intractable heel pain** (slight flexion tolerated) · **any varus** · **anterior talar translation** → **"vaulting" gait + knee pain** (TO)

Wrong positions — **plantarflexion** → equinus gait, poor swing-phase clearance ("never the answer"; goal = plantigrade) · **too much DF or valgus (20°)** → altered gait, ↑ knee stress · **internal rotation** → abnormal foot progression angle · 0° ER · varus → poor outcome, adjacent joint degeneration

Hindfoot arthrodesis — aim **5° hindfoot valgus**

– – Active space – –

Indications – tibiotalar vs tibiotalocalcaneal vs subtalar fusion

Scenario	Answer	✗ Traps
Post-traumatic ankle OA, 36-yo labourer, **CT: hindfoot spared**	**Tibiotalar arthrodesis with screws**	Arthroscopic debridement + steroid (adjunct only) · TAA · TTC IM / extramedullary
Varus end-stage ankle OA, **mild subtalar changes**, diabetic, no neuropathy	**Isolated tibiotalar fusion + slight posterior talar translation**	> 5° PF · IR · combined ST fusion · triple
End-stage ankle OA + **peripheral neuropathy**, no subtalar pain	**Tibiotalar arthrodesis**	TAA (↑ failure) · arthroscopy · subtalar fusion
Prior septic arthritis + talar AVN, aspirate sterile	**Ankle arthrodesis**	Repeat I&D · **TAA (contraindicated in talar AVN)** · SMOT · triple (fixed adult flatfoot)
Talar neck nonunion + **AVN + talar head collapse**	**TTC arthrodesis with femoral head allograft**	Steroid injection · TAA · subtalar fusion alone · triple
Broad-based talar AVN + tibiotalar + subtalar arthritis	**TTC via lateral transfibular approach**	Anterior approach (no subtalar access) · ankle fusion alone · TAA · **talectomy without arthrodesis**
Pes cavus, sinus tarsi pain, painful inversion / eversion, rigid on Coleman block, ankle painless	**Subtalar arthrodesis**	Broström–Gould (instability) · hindfoot scope + os trigonum excision (posterior impingement) · Chrisman–Snook (advanced instability) · **triple (TN + CC + ST arthritis)**
Isolated hindfoot arthritis from **chronic pes planus**	**Subtalar arthrodesis**	–

Techniques & fixation

Arthroscopic arthrodesis	**only if minimal deformity**
Open	**transfibular approach when deformity present** · fixation = **screws** · **plate + screws** · **external fixation**
Staged approach	**clear infection before definitive internal hardware**
TTC fixation options	Steinmann pins · plates + screws · ex-fix · **retrograde IM nail**
Retrograde IM nail	**load-sharing**; **↑ bending stiffness + rotational stability** vs plate-and-screw · useful with distal tibial / talar bone loss or when screw fixation suboptimal



Tibiotalocalcaneal fusion with retrograde hindfoot nail

Nail entry point (TO)	crossing of **bimalleolar (medial malleolus) bisector** (coronal) **+ sagittal line from tip of 2nd toe to centre of heel** (TO)
Lateral transfibular approach	dual joint preparation (**tibiotalar + subtalar**) + **local autograft** (talar osteonecrosis)
TTC indications	end-stage **ankle + subtalar arthritis** · **Charcot neuroarthropathy** · **hindfoot bone loss** (failed TAA, failed arthrodesis) · **osteonecrosis / global AVN of talus** · severe acute trauma · RA · flail ankle

Contraindication – peripheral vascular disease

Retrograde nail TTC	Items
Absolute contraindications	**Active infection** · **severe (profound) peripheral vascular disease** · **severe tibial malalignment**
Relative contraindication	**Normal subtalar joint** – may be sacrificed in non-compliant patients + **diabetic Charcot** for stability
Indications (✗ Wrong options: as contraindication)	**Osteonecrosis of talus** · **bone loss from failed TAA / arthrodesis** · **severe peripheral neuropathy with Charcot**

Salvage of infection/failed TAA – staged arthrodesis

Infected TAA + varus instability + talar bone loss	stage 1: **explant + antibiotic cement spacer + IV antibiotics** → stage 2: **ankle + hindfoot (TTC) arthrodesis with structural allograft**
✗ Infected TAA traps	revision TAA ± structural allograft (allograft remodels incompletely, fails under shear) · ankle arthrodesis in situ / with allograft (**talar bone loss → poor talar purchase**)
Failed TAA salvage	**conversion to fusion** or **transtibial amputation**
Draining wound over old hardware + post-traumatic OA	**removal of hardware, I&D, staged arthrodesis** ✗ Wrong options: keep hardware · simultaneous arthrodesis · simultaneous arthroplasty
Infected post-traumatic OA salvage	arthrodesis or amputation · aggressive debridement + antibiotics before fusion

– – Active space – –

Footwear after fusion – rocker sole

Best gait aid after ankle fusion — **rocker sole (heel-to-toe / rocker-bottom)** → ↓ heel-strike pressure, ↑ toe-off propulsion, "roll off" in late stance, dissipates force across fusion

✗ Option	Actual use
SACH heel	Mimics shock absorption of heel strike; less beneficial than rocker
Extended steel shank	Stiffens sole; worse unless combined with rocker
Total contact insert	Cushion / offload, correct flexible deformity
Lateral sole flare	Varus deformity / lateral ankle instability
Morton's extension (± carbon fibre)	Forefoot pain
Medial heel wedge	Not for well-aligned fusion
AFO, Arizona, double-upright drop-lock	Immobilise an already fused ankle
Pantalar arthrodesis	Unnecessary if hindfoot + transverse tarsal motion painless

Malunited triple arthrodesis – osteotomy

Valgus–supination triple malunion — residual subtalar valgus → **lateral subfibular impingement** + forefoot abduction / midfoot supination via TN joint

Treatment — **medial displacement calcaneal osteotomy + transverse tarsal (derotational) osteotomy**
✗ Wrong options: arthroscopy + lateral ligament reconstruction · tendon transfer + lateral column lengthening + heel cord lengthening (PTTD flatfoot) · ankle arthrodesis with IM nail · TAA + deltoid reefing (ankle normal, patient young)

6 Complications

Complications – adjacent arthritis, nonunion, stress fracture & nerve injury

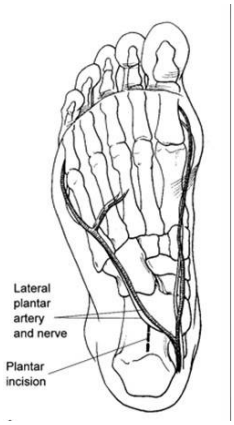
Complication	Key facts
Nonunion	**Commonest complication** (OB) · **10%** (Orthobullets) · **0–47%** across foot and ankle fusions (OB) · **smokers 2.7× risk**
Persistent nonunion after revision	**Neuropathy = greatest risk factor** (✗ Wrong options: diabetes · prior femoral nonunion · post-traumatic arthritis · initial fixation method)
Subtalar (adjacent hindfoot) arthritis	**Commonest long-term complication once united**; nonphysiologic subtalar motion · **common complication (TO)**
Lateral plantar nerve	Hindfoot nail insertion at heel → **numb plantar-lateral foot + 4th / 5th toes** (✗ Wrong options: medial plantar · sural · SPN · DPN)
Superficial peroneal nerve	**Transfibular approach**
Tibial stress fracture	**Unique to IM nail – proximal interlocking screw = stress riser**

Subtalar arthritis (TO) — **common complication** after ankle arthrodesis (TO)

Most common complications (Orthobullets) — **subtalar arthritis + nonunion**

Not ↑ after fusion — talar AVN (rare) · ipsilateral knee arthritis · contralateral ankle arthritis

Adjacent joints — hindfoot, midfoot, forefoot **compensate with ↑ motion** → accelerated arthrosis · TAA may protect



Lateral plantar nerve near the plantar incision for hindfoot nail

Nail stress fracture prevention — **remove proximal screws after union** · **short rod avoiding diaphysis**
✗ Wrong options: late rotatory deformity (controlled by distal transfixion screws) · talar osteonecrosis (minimal dissection) · hardware failure (any construct)



Tibial stress fracture at the proximal interlocking screw

Failed TTC (RA, after failed TAA), unstable pseudoarthrosis, no infection — **revision TTC arthrodesis**
✗ Wrong options: conversion to TAA (subtalar arthrosis, poor talar stock) · supplemental screws alone · **talar resection** (keep talus for vascularity + local graft; femoral head allograft if insufficient) · total contact casting (neuropathic ulcers)

– – Active space – –

X vs Y

	Tibiotalar	TTC	Subtalar	Triple
Joints	Ankle	Ankle + subtalar	Subtalar	TN + CC + ST
Indication	Ankle OA, **hindfoot spared**	**Ankle + subtalar** OA, **global talar AVN**, Charcot, bone loss	Isolated subtalar OA (cavus, chronic planus)	**Fixed adult flatfoot** / TN + CC + ST arthritis
Fixation	**Screws**, plate, ex-fix	**Retrograde IM nail**, plate, ex-fix	Screws	–
Approach	Anterior / transfibular	**Lateral transfibular**	–	–

	Primary fusion failure	Long-term after union
Event	**Nonunion**	**Subtalar arthritis**
Risk factor	**Smoking 2.7×**, **neuropathy** (revision)	Nonphysiologic subtalar motion

How it's asked

"Recommended ankle fusion position?"	→ **neutral DF / PF, 10° ER, 5° valgus** (TO)
"Ankle arthrodesis – ALL true EXCEPT?"	→ **slight anterior translation of talus** (TO)
"Entry point for retrograde TTC nail?"	→ **bimalleolar bisector × line from 2nd toe tip to heel centre** (TO)
"Optimal fusion position?"	→ **0 degrees dorsiflexion / plantarflexion, 0–5 degree hindfoot valgus, 5–10 degree external rotation**
"Optimal fusion position – talus?"	→ **plantigrade, 10° ER, 5° valgus, posterior positioning of talus**
"Varus post-traumatic ankle OA, mild subtalar change – procedure?"	→ **isolated tibiotalar fusion with slight posterior translation of talus**
"Post-ORIF tibiotalar arthritis – highest success?"	→ **tibiotalar fusion**
"Cavus, sinus tarsi pain, rigid hindfoot, ankle painless?"	→ **subtalar arthrodesis**
"Talar neck nonunion with AVN and talar head collapse?"	→ **TTC arthrodesis with femoral head allograft**
"Young construction worker, post-traumatic ankle OA, hindfoot normal on CT?"	→ **tibiotalar arthrodesis with screws**
"Prior septic ankle, sterile aspirate, talar AVN?"	→ **ankle arthrodesis**
"Broad-based talar AVN after subtalar dislocation?"	→ **TTC arthrodesis using lateral transfibular approach**
"Diabetic with neuropathy, end-stage ankle OA?"	→ **tibiotalar arthrodesis**
"Single joint fusion → greatest loss of hindfoot motion?"	→ **talonavicular**

"Fusion causing > 90% ↓ hindfoot motion?"	→ talonavicular joint
"Numb lateral sole + 4th / 5th toes after hindfoot nail?"	→ lateral plantar nerve
"RA, failed TTC after failed TAA, nonunion — next?"	→ revision TTC arthrodesis
"Complication unique to hindfoot IM nail?"	→ tibial stress fracture
"Greatest risk for persistent nonunion after revision?"	→ neuropathy
"United ankle fusion — long-term risk?"	→ subtalar arthritis
"Best footwear to restore gait after ankle fusion?"	→ rocker sole
"Pain-free fusion, hard walking, hindfoot motion painless?"	→ heel-to-toe rocker sole
"Altered gait after successful fusion — device?"	→ rocker-bottom sole
"Infected TAA with talar bone loss — second stage?"	→ ankle and hindfoot arthrodesis with structural allograft
"Draining post-traumatic ankle with hardware — plan?"	→ removal of hardware, I&D, staged arthrodesis
"Absolute contraindication to TTC nail?"	→ severe peripheral vascular disease
"Triple arthrodesis for coalition at 13, now lateral impingement?"	→ calcaneal osteotomy + transverse tarsal osteotomy

– – Active space – –

ANKLE ARTHROSCOPY

– Foot & Ankle · 7 / 48 –

1 Basics

Indications

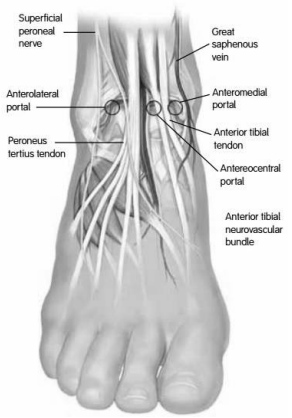
Indications	**osteochondral lesions of talus** (microfracture of OCD) · post-traumatic synovitis debridement · **ATFL / AITFL anterolateral impingement** · **anterior tibiotalar spur resection (anterior bony impingement)** · **os trigonum excision** · loose bodies · cartilage debridement with ankle fusion

Set-up

Position	**supine**, leg over padded bolster · tourniquet + exsanguinate
Distension	**external traction device** distracts tibiotalar joint (↓ cartilage injury, ↑ visualisation) · **20 ml normal saline** injected via marked AM portal site (TO)
Entry	**nick and spread** (skin nick, mosquito spreads subcutaneous tissue) → ↓ neurovascular injury
Pre-tourniquet marking (TO)	dorsalis pedis, DPN, tibialis anterior, great saphenous vein, **peroneus tertius**, malleoli, joint line
Workhorse portals (TO)	**anterolateral, anteromedial, posterolateral**

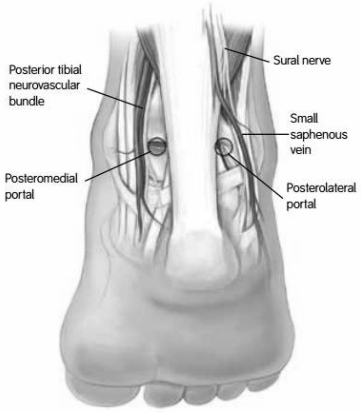
Portals – location & structures at risk

Portal	Location	Role	At risk
Anteromedial (AM)	**Medial to tibialis anterior**, lateral to medial malleolus, at joint line · between TA + saphenous vein	**Primary viewing portal, established first** · 2.7 mm 30° scope (TO)	**Saphenous nerve + great saphenous vein** (mean safe distance **7.4 mm / 9 mm**) · **tibialis anterior tendon**
Anterolateral (AL)	**Just lateral to peroneus tertius** (**~4 mm lateral**, OB) + SPN, medial to lateral malleolus · made under vision with 25G needle	Viewing / **working portal** (TO) · lateral gutter	**SPN – dorsal intermediate cutaneous branch** (**commonest**)
Anterocentral	**Medial to EDL, lateral to EHL** · through extensor tendon if needed (TO)	Rarely used	**Dorsalis pedis artery + DPN** (anterior tibial NV bundle)
Medial midline	**Between EHL and tibialis anterior**	–	**Deep + superficial peroneal nn.** (well lateral to saphenous)
Posterolateral (PL)	**Just lateral to Achilles**, medial to peroneals · **2 cm proximal to LM tip** (Orthobullets) · **at level of line from LM tip parallel to sole** (TO) · prone / semi-decubitus	Posterior viewing (os trigonum) · **vision portal in hindfoot arthroscopy**, 4.0 mm 30° scope (TO)	**Sural nerve + short (small) saphenous vein**
Posteromedial (PM)	**Just medial to Achilles**	Posterior working (os trigonum)	**Posterior tibial artery** (+ tibial nerve, OB)
Trans-Achilles	–	**Abandoned** (TO)	Achilles injury / rupture



Anterior ankle arthroscopy portals and structures at risk

– – Active space – –



Posterior portals – posterolateral (sural, small saphenous) and posteromedial (posterior tibial bundle)

Anterior portals – superficial peroneal nerve injury

SPN	**most commonly injured nerve** in ankle arthroscopy · **anterolateral portal** · variable anatomy · branch of **common peroneal**, runs in **lateral compartment → motor to peroneus longus + brevis** · crosses fibula (lateral → anterior compartment) **~11 cm (3–15 cm) proximal to LM tip** (OB) · fascial exit **10–12 cm** (OB) · **12–15 cm** (Orthobullets) – read the options · terminal **medial dorsal cutaneous (great toe)** + **intermediate dorsal cutaneous (dorsum of foot)**
Protecting SPN	**palpate**, **transilluminate** (shine scope light to see nerve under skin), blunt dissection, trace before incision · **4th toe flexion** used to protect nerve when making AL portal (TO; not 4th toe extension)
Most commonly injured branch	**dorsal intermediate cutaneous branch** (TO + Orthobullets) ✗ Wrong options: dorsal medial cutaneous branch
Dorsal foot numbness after scope	SPN via **anterolateral portal**
✗ SPN distractors	between EHL + EDL = **DPN** (EDB/EHB motor, 1st web) · terminates as **lateral dorsal cutaneous n.** = **sural** · motor to abductor hallucis + FHB = **medial plantar** · terminal branch of femoral = **saphenous**



Transillumination to identify the superficial peroneal nerve

Anteromedial & posterior portals – saphenous & sural nerves

Saphenous nerve	travels along distal medial tibia + ankle · **most at risk with anteromedial portal** ✗ Wrong options: medial midline (well lateral to it)
Sural nerve	**posterolateral portal** (os trigonum resection) · permanent numbness risk

Posterior ankle impingement (os trigonum, Stieda process) (TO)

PAIS — mechanical pinching in **terminal plantar flexion** · chronic > acute · ballet (**en pointe**), football, soccer, downhill running · **acute plantar hyperflexion** or **chronic microtrauma** (TO)

Most common cause (TO) — **os trigonum**
✗ Wrong options: Stieda process · sloping posterior tibial lip · capsular calcification

Osseous causes (TO)	Soft-tissue causes (TO)
Os trigonum · **Stieda process** · downward-sloping posterior tibial lip · posterior tibial osteophyte · prominent posterior calcaneal process · OCL, loose bodies, chondromatosis, subtalar coalition	Thickened posterior capsule · post-traumatic scar · **post-traumatic capsular calcification** · loose bodies · **FHL tenosynovitis** (commonest tendon involved) · abnormal muscles

Os trigonum — **non-fusion of posterolateral talar secondary ossification centre** · lies **lateral to FHL groove** · appears **7–13 yr** · forms **11–13 yr boys / 8–10 yr girls** · fuses within **1 yr** (→ trigonal / Stieda process) · often bilateral · pain from abutment or **synchondrosis disruption** (TO)

Os trigonum prevalence — **7–14%** remain separate (TO) · **7%** don't fuse (TO) · **present in 10–25%** (OB) — read the options

Stieda process (TO) — **elongated posterolateral talar process → causes posterior ankle impingement**
✗ Wrong options: ossification at MCL insertion (Pellegrini-Stieda) · unfused ossicle (os trigonum) · elderly head injury

Imaging (TO) — lateral X-ray: os trigonum / Stieda (may be normal) · **MRI best: bone marrow oedema** in talus, calcaneus or os trigonum

DDx (TO) — Achilles tendinosis / tear, **FHL tenosynovitis**, posterior talar process fracture, Haglund's, OCL, retrocalcaneal bursitis

Treatment — nonop (NSAID, activity modification, PT) → **excision open lateral or posterior arthroscopy** via posterolateral portal (**sural at risk**) · stress fracture ± FHL tenosynovitis / entrapment



Os trigonum on lateral radiograph

- - Active space - -

6 Complications

Complication	Fact
Neurological (overall)	**Commonest reported complication** (OB, Orthobullets, TO) · **neuropraxia** commonest · nerve injury **3.5–10%** · overall complications **6–20%**
SPN	**Commonest nerve** · AL portal · **commonest complication of anterior arthroscopy** (TO + OB)
Cartilage injury	Up to **30%** (TO) · **anterolateral (zone 3) + central (zone 5) talus** · ↓ by distraction + correct portals · called **commonest complication of ankle arthroscopy** in one TO key
Synovial cutaneous fistula	"Somewhat common" · prevent by **immobilisation** until portals heal
Vascular	Rare · anterocentral portal · **anterior tibial artery aneurysm** (case reports)
Tendon / ligament	**EDC, EHL rupture** · ATFL + deltoid injured by accessory portals
Posterior arthroscopy	**Tibial + sural** = commonest nerve injuries · **FHL injury** · Achilles tightness
Other	DVT, infection (**very infrequent**), CRPS

Commonest complication — source conflict

SPN / neurological injury (TO17136845, OB) · **cartilage injury** (TO17136847 explanation) — read the options; if SPN is an option it is the keyed answer

Painful portal neuroma

Burning at lateral portal, Tinel → medial foot / ankle, injection helped 2 wk

SPN medial branch entrapment → neuroplasty (nerve intact) or **excision + burial** (cut / thinned)
✗ Wrong options: sural neuroplasty (posterolateral portal) · DPN · repeat scope for OCD · anaesthetic patch

X vs Y

	Os trigonum	Stieda process
What	**Separate ossicle** (unfused ossification centre)	**Elongated posterolateral talar process** (fused)
Impingement	**Most common cause** of PAIS	Also causes PAIS

	Anterior arthroscopy	Posterior arthroscopy
Position	Supine	Prone / semi-decubitus
Commonest nerve	**SPN**	**Tibial + sural**
Other	Cartilage, fistula	FHL injury, Achilles

How it's asked

"Most common complication of anterior ankle arthroscopy?"	→ **superficial peroneal nerve injury** (TO) ✗ Wrong options: cartilage · EDC · fistula
"True regarding ankle arthroscopy?"	→ **4th toe flexion is used to protect the nerve (CPN/SPN) while making the anterolateral portal** (TO) ✗ Wrong options: 4th toe extension · dorsal medial cutaneous branch · "commonest complication = CPN neurapraxia"
"Most common complication with standard lateral portal?"	→ **injury to the superficial peroneal nerve** ✗ Wrong options: tibial · DPA · RSD · deep infection
"Portal putting SPN at greatest risk?"	→ **anterolateral portal**
"Dorsal foot numbness after scope — portal?"	→ **anterolateral**
"Tendon closest to AL portal?"	→ **peroneus tertius**
"Transillumination protects?"	→ **superficial peroneal nerve**
"Most commonly injured nerve — characteristic?"	→ **motor branches to peroneus longus and brevis** (SPN)
"Lateral portal neuroma, Tinel to medial foot?"	→ **neuroplasty or excision + burial of medial branch of SPN**
"Saphenous nerve most at risk with?"	→ **anteromedial portal**
"Portal 2 (anteromedial) — at risk?"	→ **saphenous nerve**
"Os trigonum resection — portal riskiest for sural?"	→ **posterolateral portal**
"Most common cause of posterior ankle impingement?"	→ **os trigonum** (TO)
"True about Stieda's process?"	→ **causes posterior ankle impingement** (TO)
"Anterocentral portal at risk?"	→ **dorsalis pedis artery (+ DPN)**
"Posteromedial portal at risk?"	→ **posterior tibial artery**

– – Active space – –

ANKLE LIGAMENTS

– Foot & Ankle · 8 / 48 –

1 Basics

Osteology & biomechanics

Ankle joint	tibial plafond + medial malleolus + lateral malleolus + talus · main motion **plantar / dorsiflexion** · secondary inversion / eversion, rotation
Distal tibiofibular joint	distal fibula in **incisura fibularis** (concave distal lateral tibia) · fibula rotates in incisura during gait · **mortise widens from plantar → dorsiflexion** · **syndesmosis screws limit external rotation**
Joint reaction force	**5× body weight** walking on level ground

Ligament overview

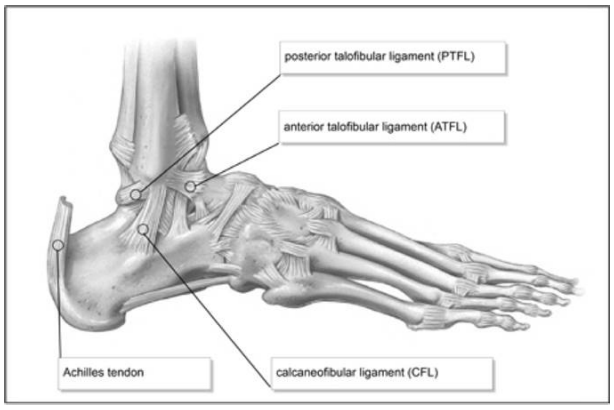
Group	Ligaments
Medial	**Deltoid · calcaneonavicular (spring)**
Lateral	**Syndesmosis** (AITFL, PITFL, TTFL, IOL, ITL) · **ATFL** · **PTFL** · **CFL** · **LTCL**

Lateral ligaments

	ATFL	CFL	PTFL
Function	**Primary restraint to inversion in plantar flexion** · resists **anterolateral talar translation**	**Primary restraint to inversion in neutral / dorsiflexion** · restrains subtalar inversion → limits talar tilt	Supplementary when ATFL + CFL intact · max strain in **dorsiflexion** · limits **posterior talar displacement + ER**
Strength	**Weakest** lateral	–	**Strongest** lateral
Origin	Anteroinferior fibula, **10 mm proximal to tip**	Anterior border fibula, **9 mm proximal to tip**	Posterior border fibula
Insertion	**Talar neck**, just distal to cartilage (**18 mm distal to joint line**)	Calcaneus **13 mm distal to subtalar joint**, **deep to peroneal sheaths**	**Posterolateral tubercle of talus**
Orientation	**45–90°** to tibial axis (Orthobullets) · **perpendicular in neutral → parallel in plantar flexion** (OB)	**Parallel to tibia in neutral**	**Perpendicular** to tibial axis
Injured by	**Inversion in plantar flexion** → commonest, **low ankle sprain**	**Inversion with foot in neutral**	**Complete talar dislocation** / severe sprains with other two torn
Exam	**Anterior drawer in 20° plantar flexion** (Orthobullets; **10° PF** = greatest translation, OB) vs other side · **> 8 mm forward shift** on lateral X-ray = ATFL tear	**Inversion (supination) test** in slight dorsiflexion · **talar tilt** (ATFL + CFL)	No specific test

PTFL if ATFL + CFL incompetent — short fibres restrict IR + ER, talar tilt, dorsiflexion · long fibres only ER, talar tilt, dorsiflexion

LTCL — **lateral process of talus → lateral calcaneus** · **anterior + medial to CFL** · stabilises talocalcaneal joint · injured **with ATFL** · no specific test



Lateral ankle ligaments – ATFL, CFL, PTFL

Syndesmosis

Function — **integrity of ankle mortise**

Components — **AITFL** · **PITFL** (deep part = **inferior transverse ligament**) · **TTFL** · **interosseous ligament (IOL)**

Clinical — **high ankle sprain / syndesmosis injury**, ankle fracture · AITFL + PITFL injured by **eversion + external rotation**

Tests — **external rotation test** · **squeeze test**

Deltoid ligament

Function — **primary restraint to valgus talar tilt** · both layers resist hindfoot eversion · stabilises against **plantar flexion, ER, pronation**

	Superficial	Deep
Crosses	**Ankle + subtalar**	**Ankle only**
Origin	**Anterior colliculus**	**Inferior + posterior medial malleolus**
Insertion	Navicular, talar neck, **sustentaculum tali**, posteromedial talar tubercle	Medial + posteromedial talus
Key	**Tibiocalcaneal (sustentacular) = strongest superficial** · **deep posterior tibiotalar = strongest overall** (TO), resists calcaneal eversion	**Primary medial stabiliser** · prevents **lateral displacement + ER of talus**
Test	**Eversion test** in neutral	**External rotation stress test** (+ syndesmosis)

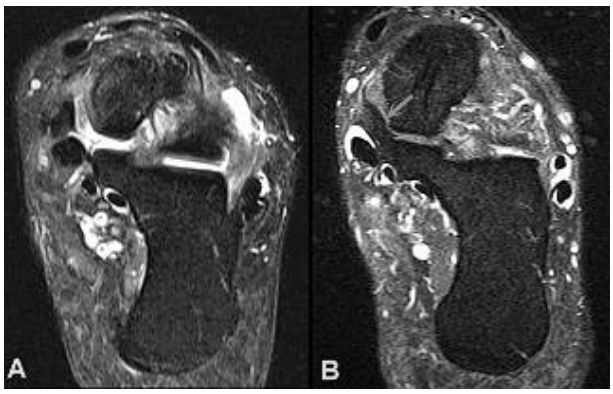
Injury — **pronation (eversion)** → forced ER + abduction · ± medial malleolus fracture · ankle fracture, high ankle sprain

Spring ligament anatomy & flatfoot

Spring (calcaneonavicular) ligament — connects **calcaneus and navicular** · **sustentaculum tali → inferior navicular**
✗ Wrong options: talus–navicular · calcaneus–cuboid · talus–calcaneus

– – Active space – –

Components	**superomedial** + **inferior** calcaneonavicular · **superomedial = largest + most important stabiliser of medial longitudinal arch** ✗ Wrong options: inferior
Superomedial attachments	**anteromedial sustentaculum tali → broadly along medial navicular** ✗ Wrong options: to medial cuneiform · from talus
Function	**static** stabiliser of **medial longitudinal arch + talar head** · arch = dynamic **PTT** + static ligaments / capsule + tarsal interlock · closely related to **deltoid**
Untreated tear	**flatfoot deformity** ✗ Wrong options: syndesmotic widening · TMT widening · hallux valgus · talar tilt
PTTD	spring ligament **most likely attenuated in type II flatfoot** · significant disruption → **reconstruction**
Acute spring tear	forceful landing on flat foot · **tibialis posterior usually normal**



Spring ligament MRI – disrupted (A) vs intact (B)

2 Mechanism

Inversion injury – ATFL

Mechanism	Injury
Inversion in ~20° plantar flexion	**ATFL** (commonest)
Inversion in neutral	**CFL**
Severe inversion (others torn)	PTFL
Eversion + external rotation	**AITFL / PITFL (syndesmosis)**
Eversion	**Anterior colliculus fracture** (deltoid avulsion)

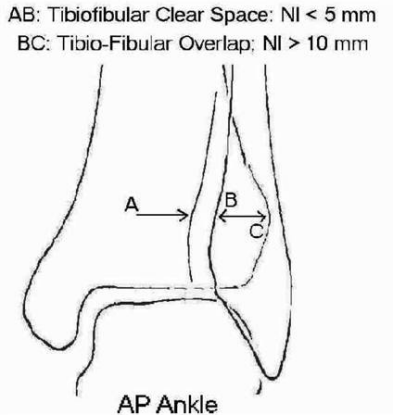
Eversion injury – anterior colliculus

Eversion	**anterior colliculus fracture** = bony **deltoid** avulsion from medial malleolus
✗ Wrong options (all inversion-associated)	**osteochondral talar dome lesions** · **peroneal tendon splits** · **anterolateral soft-tissue (meniscoid) impingement** · **SPN stretch** (paraesthesia) · ATFL + CFL tears

4 Diagnosis & Classification

Key radiology findings

Study	Finding
AP + mortise	**Tibiofibular clear space < 5 mm** · **tibiofibular overlap > 10 mm (AP)** (Orthobullets) · clear space **< 6 mm**, overlap **> 6 mm AP / > 1 mm mortise** (OB) – read the options
Weight-bearing mortise	**Most accurate radiograph** for syndesmosis
CT	**Most accurate** syndesmosis assessment; no validated normals → **compare uninjured side**
Anterior drawer stress X-ray	**> 8 mm** = ATFL tear · stress views more accurate in **chronic** injury
Talar tilt X-ray	Plafond–talar dome angle **< 5° normal** · combined **ATFL + CFL**
Ankle arthrogram	**CFL rupture → dye leaks into peroneal tendon sheath**
US	ATFL – **more accurate than radiographs**
MRI	ATFL, CFL, spring ligament tear · PTFL rarely indicated
Mortise / gravity stress	**Medial clear space widening** → deltoid injury
Arthroscopy	Confirms ATFL tear



Tibiofibular clear space < 5 mm and overlap > 10 mm

Syndesmotic ligament avulsion fragments (AITFL)

Tillaux fracture	**AITFL (anterior tibiofibular ligament)** avulses **anterolateral distal tibial epiphysis** · transitional fracture, **anterolateral physis closes last** ✗ Wrong options: ATFL · CFL · EDB · EHB
Fibular avulsion into syndesmotic space (CT)	**AITFL** insertion → **syndesmosis injury** · repair fragment ± **syndesmotic screw / suture button** in most ✗ Wrong options: excise fragment (incompetent syndesmosis) · Broström / allograft (ATFL / CFL) · deltoid (medial) · SPR (lateral fibular wall avulsion, peroneal subluxation)

– – Active space – –

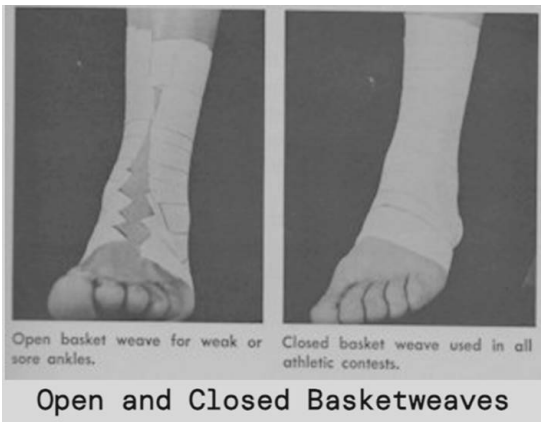
5 Management

Partial tear – physical therapy

Isolated partial ATFL tear (grade I)	stable · **RICE, early protected WB → physical therapy** (peroneal strengthening + proprioception) · brace for sport
✗ ATFL partial-tear traps	**cast + NWB 2–4 / 6–8 wk** (WB recommended) · **early open repair** (only for fracture / intra-articular lesion) · **early arthroscopic repair** (scope only for AITFL impingement / posteromedial impingement)

Acute ankle sprain – taping (TO)

Taping	Use (TO)
Closed basket weave	**Acute ankle injury, maximum support** ✓ (TO) · anchors (below calf + behind MT heads) → **stirrups medial → lateral** + **horseshoes behind heel** woven → **C-strips** → **heel locks**
Open basket weave	Acute sprain with **massive swelling** (TO) · **no overlap in front** → allows swelling + movement · heel locks with **elastic** tape · compression, ↓ inflammation
Low dye	**Zinc oxide** tape for **midfoot, plantar fascia, heel pain** (TO) · **not for ankle sprain**
Medial patellar	Knee (not ankle)



Open vs closed basket weave taping

X vs Y

	ATFL	CFL
Restrains inversion in	**Plantar flexion**	**Neutral / dorsiflexion**
Test	**Anterior drawer** (20° PF)	**Inversion test**, talar tilt
Orientation in neutral	Perpendicular (OB)	Parallel to tibia

	Superficial deltoid	Deep deltoid
Joints crossed	Ankle + subtalar	Ankle only
Test	Eversion	External rotation stress

– – Active space – –

	Superomedial CN	Inferior CN
Size	**Largest**	Smaller
Arch support	**Most important**	Less

How it's asked

"Inversion with foot plantar flexed 20° – ligament?"	→ **anterior talofibular**
"Eversion mechanism associated with?"	→ **anterior colliculus fracture** ✗ Wrong options: peroneal splits · OCL · SPN palsy · anterolateral impingement
"Isolated partial ATFL tear – treatment?"	→ **physical therapy** ✗ Wrong options: cast NWB · open / arthroscopic repair
"Tillaux fragment – attached structure?"	→ **anterior tibiofibular ligament**
"Fibular avulsion into syndesmosis on CT – structure?"	→ **anterior inferior tibiofibular ligament (AITFL)**
"Spring ligament connects?"	→ **calcaneus and navicular**
"Spring ligament complex – true?"	→ **superomedial ligament is the most important stabilizer of the longitudinal arch**
"Untreated spring ligament tear on MRI leads to?"	→ **flatfoot deformity**
"Taping method in runner with acute ankle injury?"	→ **closed basket weave taping** (TO) ✗ Wrong options: open (massive swelling) · low dye (plantar fascia)
"Most accurate radiograph for syndesmosis?"	→ **weight-bearing mortise view**

– – Active space – –

ANKLE SPRAIN

– Foot & Ankle · 9 / 48 –

1 Basics

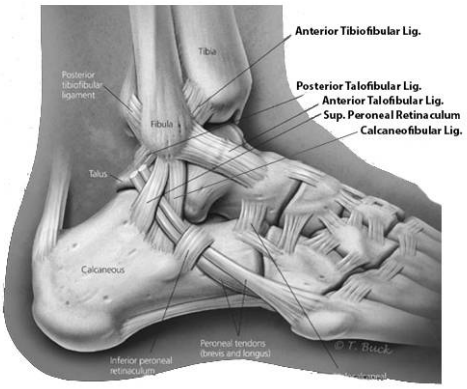
Epidemiology

Commonest reason	**missed athletic participation**
Dancers	ankle sprain = **commonest injury** (Orthobullets) · inversion sprain = **commonest acute skeletal injury in dancers** (OB)
Low vs high	**low (ATFL ± CFL) > 90%** · **high (syndesmosis) 1–10%**
Risk factors	patient: **limited dorsiflexion, ↓ proprioception, poor balance** · environment: **indoor-court sports (basketball, volleyball)** highest

Anatomy & examination – ATFL vs CFL

Ligament	Attachments	Tight in / injured by	Test	Frequency · strength
ATFL	**Anterior fibula → lateral talar neck**	**Plantar flexion + inversion** (parallel to foot axis in PF)	**Anterior drawer in plantar flexion**	**Commonest injured · weakest**
CFL	**Fibular tip → lateral calcaneal tubercle**	**Dorsiflexion + inversion**	**Drawer laxity in dorsiflexion · talar tilt (varus stress)**	**2nd commonest**
PTFL	**Distal fibula → lateral tubercle of posterior talus**	Rarely	Not tested by anterior drawer	**Least injured · strongest + deepest of lateral complex** (OB, TO)

Strength order	**ATFL (weakest) < CFL < PTFL (strongest)**
CFL crosses 2 joints	contributes to **subtalar + ankle** stability → subtalar instability hard to tell from posterior ankle instability ✗ Wrong options: ATFL



Lateral ankle ligaments

Deltoid ligament (TO)

Layer	Bands	Crosses	Limits
Superficial (4)	**Tibiospring** + **tibionavicular** (major) · superficial posterior tibiotalar + tibiocalcaneal (additional)	**Ankle + subtalar**	**Talar abduction**
Deep (2)	**Deep posterior tibiotalar (major)** · anterior deep tibiotalar	**Ankle only**	**External rotation**

Strongest tibiotalar / deltoid part (TO) — **deep posterior tibiotalar ligament** (TO)
✗ Wrong options: anterior deep tibiotalar · "posterior tibiotalar" · talonavicular

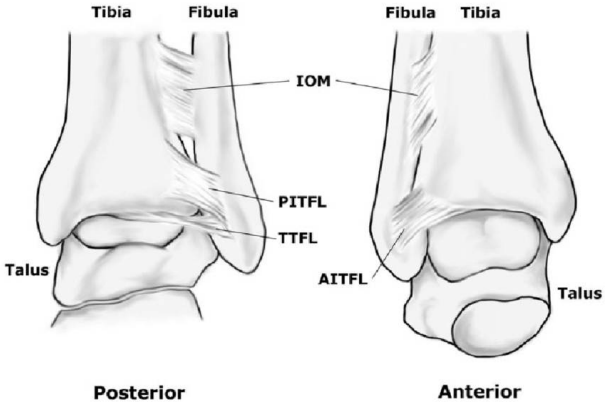
Pronation of talus — superficial + deep **equally** limit it (OB)

Function — fan-shaped; resists **valgus + rotatory** forces · isolated deltoid injury **very rare**

Syndesmosis anatomy (TO)

Ligament	Course	Restrains
AITFL	**Chaput tubercle (anterolateral tibia) → Wagstaffe tubercle (anterior fibula)**	**Fibular external rotation**
IOL	Distal continuation of **interosseous membrane**	**Lateral fibular translation**
PITFL	**Volkmann tubercle (posterior tibia) → posterolateral fibula**	**Posterior fibular translation · strongest component**
Inferior transverse ligament	Distal to PITFL	**Posterior talar displacement**

PITFL > AITFL — **PITFL stronger** (TO) (true statement in "false EXCEPT")



Syndesmosis – AITFL, PITFL, transverse ligament, IOM

Peroneal weakness in dancers

Relative peroneal weakness → inversion ankle sprain · restore full peroneal strength to prevent recurrence
✗ Wrong options: fibular fracture · cuboid subluxation · Achilles rupture · midfoot sprain

– – Active space – –

2 Mechanism

Low sprain	**inversion on a plantarflexed foot** · recurrent sprains → functional instability
Sequence in inversion (TO)	**anterolateral capsule → ATFL → CFL → PTFL** (TO) · ATFL = first or only ligament; total rupture adds CFL + PTFL (TO) ✗ Wrong options: ATFL first then capsule
Eversion (TO)	**deltoid** injury
Syndesmosis mechanism	**hyperdorsiflexion** (TO) · **external rotation**, pain on forced **dorsiflexion + ER** (TO) · **pronation–ER > pronation–abduction > infrequently supination–ER** (Weber C / B) (TO) – read the options ✗ Wrong options: eversion + PF · inversion · hyperplantar flexion

3 Presentation

Symptoms	pain on WB, swelling, ecchymosis · recurrent instability, **catching / popping** after recurrent sprains · "giving way"
Anterior drawer	talus translates anterior to tibia · seated, knee flexed over table edge, stabilise leg, pull heel forward · **10° PF** = greatest translation (OB; 20° PF in another OB stem) · **8–10 mm** = positive · **PF → ATFL**, **DF → CFL** · reflects ATFL ± CFL · limited use acutely
Talar tilt	tibia stabilised, ankle neutral, invert · **> 15° vs contralateral = ATFL + CFL** · **isolated ATFL cannot give + talar tilt** → CFL must be torn ✗ Wrong options: PTFL · AITFL / PITFL
Syndesmosis tests (TO)	**squeeze** (compress tibia + fibula mid-calf → syndesmotic pain) · **fibular translation** (fibula forward/back) · **Cotton** (talus side-to-side, neutral) · **ER stress** (ER + DF, hip + knee 90°) · gross swelling **usually absent**; tenderness at anterior syndesmosis ± deltoid
Cotton / hook test intra-op (TO)	translation considered positive asked (0–1 / 1–2 / 2–3 / 3–4 mm) – **answer key not provided by the source** (TO)



Talar tilt (varus stress) view – lateral ligament laxity

Associated injuries – talar bone bruise & deltoid

– – Active space – –

Associated finding (acute lateral sprain MRI)	Rate
Talar bone bruise	**Commonest**
Talar + medial malleolar bruises	8.1%
Peroneus brevis injury	7.5%
Lateral malleolus tip avulsion	2.7%

Other associations (Orthobullets) — **OCD talus**, **peroneal tendon injuries**, **subtle cavovarus**, **anterior / anterolateral synovitis (commonest intra-articular)**, anterior impingement, deltoid injury, **CRPS**, fractures (**5th MT base, anterior process calcaneus, lateral / posterior process talus**)

Medial pain, ER hurts medially, squeeze −, ER stress X-ray normal — **deltoid ligament complex**
✗ Wrong options: inferior transverse ligament · AITFL (squeeze/X-ray normal) · ATFL (usually recovers nonop) · spring ligament (would give flatfoot)

Deltoid injury — nonop first; **reconstruct if instability persists**

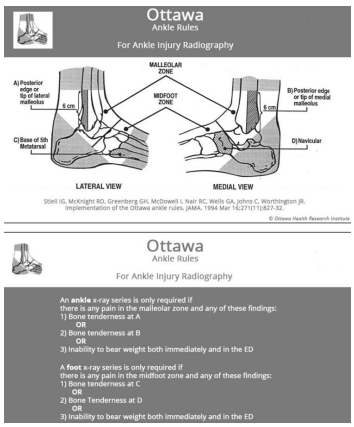
4 Diagnosis & Classification

Diagnostic criteria – Ottawa ankle rules

X-ray series	Pain zone	Any one of
Ankle	Malleolar zone	**Bone tenderness posterior edge / tip of lateral malleolus (distal 6 cm)** · **same for medial malleolus** · **inability to bear weight immediately + in ED**
Foot	Midfoot zone	**Bone tenderness 5th MT base** · **navicular** · inability to bear weight

Sensitivity — **96–99%** for ruling out fracture

Not Ottawa criteria — **swelling**, **ecchymosis** (medial or lateral), **antalgic gait / walking 10 ft** (only inability to bear weight counts), pain with hindfoot motion, **1st MT base tenderness** (peroneus longus), **deltoid / soft-tissue tenderness**



Ottawa ankle rules

– – Active space – –

Key radiology findings

Study	Use / finding
WB AP, lateral, mortise	Lateral: **anterior talar translation = ATFL**
ER stress view	Syndesmosis: asymmetric mortise widening · **MCS > 4 mm** · **TF clear space 6 mm**
Varus stress (talar tilt) view	**CFL** · measures talar tilt
WB stress X-ray (TO)	Suggests instability in obscure syndesmotic cases; **guides management**
MRI	If pain **> 6–8 wk** · **peroneal tendons, OCD, syndesmosis** · not needed acutely (✗ Wrong options: intra-articular contrast)
MRI syndesmosis (TO)	**Gold standard** · **lambda sign = > 2 mm diastasis on coronal** (sensitive + specific) (TO) · also **hyperintense signal contiguous from mortise into tibiofibular joint** (OB)
Peroneal tendon pathology	**MRI** (✗ Wrong options: CT)

Syndesmotic radiographic parameters (TO)

Parameter	Where	Normal	Injury
TF clear space	**1 cm above plafond**; medial fibula → lateral posterior tibial prominence (incisura)	**< 6 mm** (AP + mortise) · **< 5 mm** (Orthobullets ligament note)	**↑ > 6 mm**
TF overlap	1 cm above plafond	**> 6 mm on AP** · **> 10 mm AP** (Orthobullets ligament note)	**↓**
Medial clear space	Mortise	**< 4 mm**	↑ = lateral talar shift
Superior joint space	Mortise	Within 2 mm medially of lateral width	–
Dime (ball) sign	Fibular tip recess → lateral talar process	**Unbroken curve**	**Broken**
Shenton's line of ankle	Mortise	Intact	Broken

Best view for syndesmosis (TO) — **AP** (mortise better for joint congruity) · **"TF overlap < 6 mm on mortise = unstable" is FALSE** (TO)

Medial clear space — **> 4 mm** abnormal (Orthobullets; TO normal < 4 mm) · **> 5 mm = deltoid tear → unstable syndesmosis** (TO) – read the options



TF clear space measured 1 cm above the plafond

Low ankle sprain grading

Grade	Ligament	Ecchymosis / swelling	Pain on WB	Treatment
I	**No tear**	Minimal	Normal	**Functional rehab**
II	**Stretch without tear**	Moderate	Mild	**Functional rehab**
III	**Complete tear (ATFL + CFL)**	Severe	Severe	**Functional rehab** (± 10 d cast NWB, ± semirigid brace)

Syndesmotic injury classification (TO)

By time	By stability
Acute < 6 wk	**Stable** – AITFL ± IOL tear, **deltoid intact** → nonop
Subacute > 6 wk	**Unstable** – **AITFL + IOL + deltoid** (MCS > 5 mm) → surgery
Chronic > 6 months	–

5 Management

Acute low ankle sprain (grade I–III)

- Ottawa + → **X-ray**
- All grades → RICE → **early functional rehab (brace, proprioception, peroneals)**
  (± ~1 wk WB boot; grade III ± 10 d cast)
- Pain / swelling > 6–8 wk → **MRI (OCD, peroneals, syndesmosis)**

Chronic lateral instability

- Untreated / under-treated → **proprioception + peroneal strengthening first**
- Failed nonop, normal alignment → **modified Broström-Gould ± arthroscopy**
- + cavovarus, Coleman corrects → **1st MT dorsiflexion osteotomy + reconstruction**
- + fixed hindfoot varus → **lateralising calcaneal osteotomy + reconstruction**

– – Active space – –

Operative criteria

Situation	Treatment
Grade I–III, persistent pain + instability despite extensive nonop	**Anatomic reconstruction (Broström-Gould)** or tendon transfer / tenodesis
Grade I–III with bony avulsion	Reconstruction
Open ankle dislocation, gross lateral ligament disruption	**Acute repair** (OB)
Recurrent sprains + **impingement lesion** (AITFL, posteromedial)	**Arthroscopic debridement**
Syndesmosis with tibiofibular diastasis	Fixation
Peroneal tear > 50–60%, irreparable, severe tendinopathy	**Peroneus brevis tenodesis**

Acute management – functional rehabilitation & proprioception

Protocol
RICE + elastic wrap → short (**≈ 1 wk / 7–10 d**) WB immobilisation (boot, Aircast, walking cast) if needed → **early functional rehab**: ROM → strengthening → **proprioception** (tilt-board) → sport-specific · **functional brace** (controls inversion + eversion) during strengthening + prophylactically in high-risk activity

Quickest return to activity
early functional rehabilitation · supervised PT helps early, no long-term difference · chronic instability **cannot be predicted** by history, exam or X-ray

Recurrence reduction
functional bracing + early proprioceptive training (✗ Wrong options: NWB cast · WB boot · splint · **neuromuscular training alone** (higher recurrence)) · pathway to retrain = **proprioception** (✗ Wrong options: tactile · pressure · pain · motor)

Proprioception roles (TO)
EXCEPT stem: promotes fracture healing / maintains joint stability / prevents reinjury / improves neuromuscular control + balance — **answer key not provided by the source** (TO)

	Functional management	Cast immobilisation
Return to sport	**Faster, more likely long-term**	**Longer time to RTS**
ROM	**Greater**	Less
Swelling	**Less** (short term)	More
Laxity on stress X-ray	**Less**	**More**
Satisfaction	**Higher**	Lower

FALSE about functional treatment
"**greater risk of ankle laxity**" (it has **less**)

Grade III (ATFL + CFL torn on MRI), no syndesmosis / avulsion
RICE, elastic wrap, then therapy
✗ Wrong options: 6 wk cast · suture button · Broström-Gould · tenodesis

Acute ATFL tear on MRI, laxity PF + DF
early functional rehab, proprioception
✗ Wrong options: Broström · Broström-Gould · Watson-Jones (weaved PB) · Evans (non-weaved PB)

Basketball rebound, ATFL on MRI	**early WB, functional bracing, peroneal strengthening** ✗ Wrong options: sinus tarsi steroid + taping · suture-tape repair · peroneal autograft reconstruction · boot 4–6 wk (deconditioning + recurrence)
After 7–10 d splint, + talar tilt + DF drawer	**early functional rehabilitation** ✗ Wrong options: CT / MRI acutely · 2–4 wk more NWB cast · rigid walking boot; semirigid brace OK for grade III
Lateral sprain, 2 mm drawer, squeeze −, X-ray normal	**functional bracing + PT** ✗ Wrong options: short leg cast 4 wk · MRI · Broström-Gould
Chronic untreated sprain (5 mo), stable drawer	**proprioception + peroneal strengthening** ✗ Wrong options: NSAIDs alone · observation · boot · surgery before full nonop
PT success	**> 80%** (OB)

Chronic instability – Broström-Gould, reconstruction & rehabilitation

Chronic instability	**> 6 wk** · repeated episodes "giving way"
Modified Broström-Gould	**anatomic shortening + reinsertion (imbrication) of ATFL + CFL** · reinforced with **inferior extensor retinaculum + distal fibular periosteum** · **90% good–excellent** · commonest procedure; preferred in athletes with normal alignment ✗ Wrong options: Evans · allograft fibula → 5th MT · added osteotomy if aligned
Gould modification facts	**no biomechanical difference in ankle stability** ✗ Wrong options: superior extensor retinaculum · adds PTFL · ↑ nerve injury · ↓ ROM at 6 wk
Primary repair works even when chronic	ligaments stay attached to capsule · **graft / internal brace = for failed primary repair** · no technique proven superior
Suture-tape augmentation	↑ tensile strength, no better clinical outcome · **over-tightening → ↓ dorsiflexion** ✗ Wrong options: inversion · eversion · PF · pronation
Tenodesis (Watson-Jones, Chrisman-Snook, Colville, Evans)	**nonanatomic** tendon transfer · **subtalar stiffness** · alters biomechanics → discouraged · **correct any malalignment** first
Arthroscopy with Broström	**high incidence of intra-articular pathology** (OCD talus, anterior impingement, synovitis) → **ankle arthroscopy** is the add-on ✗ Wrong options: subtalar fusion · peroneal debridement if no eversion pain · 1st ray / Dwyer osteotomy if aligned
Instability + anterior impingement	**arthroscopic debridement + ATFL + CFL reinsertion with inferior extensor retinaculum + fibular periosteum** ✗ Wrong options: PTFL · superior extensor retinaculum

Post-op mobilisation	**Early (< 3 wk)**	**Delayed (> 3 wk)**
Functional scores	**Better**	Worse
Ankle laxity (drawer, tilt)	↑	Less
Wound complications	↑	Less
Return to sport	Same	Same

– – Active space – –

Failure factors – cavovarus / hindfoot alignment & its correction

Commonest reason for failure — **hindfoot malalignment (native hindfoot varus)** · normal = **5–7° valgus**, axis through medial calcaneus · varus loads lateral ligaments + peroneals

Predispose to failure — **hindfoot varus + 1st ray plantarflexion + cavovarus** = **all of the above**
✗ Wrong options: planovalgus (strains medial side) · low demand · anatomic reconstruction · pre-op acute traumatic instability (all ↓ failure)

(✗ Wrong options: **Traps** – baseline laxity (Beighton)) · **excessively pronated** subtalar (varus foot **supinates**) · **fixed dorsiflexion** of 1st ray (it's **plantarflexion**) · planovalgus

Subtle cavovarus — **mild heel varus + plantarflexed 1st ray** → lateral column overload, **stress fractures**, **peroneal tendon + lateral ligament injury** · forefoot-driven (plantarflexed 1st MT) or hindfoot-driven (tarsal coalition)

Orthotic for cavovarus — **full-length semirigid**, **lateral hindfoot (+ forefoot) wedge / posting**, **1st MT head recess**, **lowered medial arch**
✗ Wrong options: UCBL · medial hindfoot posting ± rigid arch (planovalgus) · lace-up brace alone

Coleman block	Driver	Add to ligament reconstruction
Corrects (flexible hindfoot)	**Forefoot (plantarflexed 1st ray)**	**1st MT dorsiflexion osteotomy** (± **plantar fascia release**)
Negative (fixed hindfoot varus)	Hindfoot	**Lateralising (valgus) calcaneal osteotomy (Dwyer)**

Cavovarus + recurrent sprains, Coleman corrects, failed orthotics — **1st MT osteotomy + lateral ligament reconstruction (peroneus brevis graft)**
✗ Wrong options: sinus tarsi steroid · ligament repair alone · triple arthrodesis

Lowest recurrence, flexible cavovarus — **reconstruction + plantar fascia release + 1st MT dorsiflexion osteotomy**
✗ Wrong options: reconstruction alone · syndesmosis repair · FDL transfer + MDCO (planovalgus)

Cavovarus, Coleman negative, failed nonop — **lateralising calcaneal osteotomy** with Broström
✗ Wrong options: syndesmotic fixation · post-op PT / orthosis / IER reinforcement (less important)

X-ray that predisposes to recurrent sprains — **hindfoot varus**
✗ Wrong options: pes planus · syndesmotic widening (result not cause) · talar OCD (result) · tibiotalar osteophytes

Associated peroneal tendon pathology

Persistent lateral pain after Broström, drawer –, normal labs + X-ray — **peroneus brevis (longitudinal split) tear** · linked to **SPR laxity**
✗ Wrong options: infection · repair failure · PL tear · arthritis (takes > 3 mo)

Pain with resisted eversion after sprain — **MRI**
✗ Wrong options: immobilise 4 wk · more PT · steroid · CT

PB tear	Treatment
Small / ~30% split, repairable	**Debridement, repair + tubularisation**
> 50–60%, irreparable, severe tendinopathy (e.g. only **35% intact**)	**Peroneus brevis tenodesis** (✗ Wrong options: nothing · groove deepening · SPR imbrication alone)
+ instability	Reconstruct SPR, flatten fibular ridge / **deepen groove**

After SPR release + repair — **avoid resisted eversion** early → else peroneal subluxation
✗ Wrong options: inversion, PF, pronation, supination

High ankle (syndesmotic) sprain – mechanism, tests & fixation (TO)

Stable (TO) — **nonop**: RICE + **NWB immobilisation 2 wk** → brace, ROM, strength → neuromuscular + proprioception

Unstable (TO) — **surgery**: **syndesmotic screw or suture button ± AITFL repair** · **deltoid + AITFL tear = indication for surgery** (TO)

Predicting IOM tear / instability in bimalleolar fracture (TO) — **intraoperative stress testing** (TO)
✗ Wrong options: fibular fracture level · Lauge-Hansen class · medial clear space (widens with deltoid + fibula without IOM tear)

Syndesmotic fixation (TO)	
Indicated	**Proximal fibular fracture not fixed + unstabilisable medial injury** · **disruption > 5 cm above plafond**
Not needed	**< 3 cm above plafond** · medial malleolus fixed / deltoid repaired · no medial injury (✗ Wrong options: "fix all")
Controversial	**3–5 cm**, deltoid not repaired
Mechanics altered	**> 4.5 cm** with medial lesion

Screw position (TO) — **2–3 cm above plafond**, **parallel to joint**, **angled 30° anteriorly** (perpendicular to tibiofibular joint) · **2–5 cm** (Orthobullets syndesmosis note) – read the options
✗ Wrong options: 3 cm + 30° posterior · 10–15° cephalad / caudad

Screw error (TO)	Result
Too proximal	**Deforms fibula → mortise widens**
Not parallel	**Fibula shifts proximally**
Not perpendicular	**Fibula stays laterally displaced**

Screw type (TO) — AO: **fully threaded, neutralisation** · others: lag more secure · **max dorsiflexion during fixation** traditional, but may induce **ER moment → malreduction**

Return to play

Injury	Return
Grade I / II	**1–2 wk**
Grade III	**3–4 wk**
High ankle, immobilised	**5–6 wk**
High ankle, screw fixation	**Season**

Prevention — **semirigid orthosis** (cavovarus: lateral hindfoot/forefoot wedge + 1st MT recess) · **peroneal (evertor) strengthening** · proprioception · season-long programme

– – Active space – –

6 Complications

Persistent symptoms — **up to 30%** (Orthobullets) · **up to 40%** residual symptoms (OB) – read the options · pain at 1 yr **5–33%** · pain falls fast in first 2 wk · ↑ risk of sprain both ankles

Commonest cause of chronic pain — **missed injury**: fractures (**anterior process calcaneus, lateral / posterior process talus, 5th MT**), **OCD**, **peroneal tendons**, **syndesmosis**, **tarsal coalition**, impingement, **hindfoot varus**

Impingement — **anteroinferior tibiofibular ligament** + posteromedial lesions → arthroscopic debridement

Complications – SPN neurapraxia & CRPS

SPN stretch neurapraxia — 4 wk after grade III sprain: **burning, exquisite light-touch tenderness anterolateral ankle / dorsolateral foot**, worse with **PF + inversion**, ATFL torn · SPN injury **1.6–4.4%** · **> 80% of grade III sprains have EMG evidence of tibial or CPN injury** (OB)
✗ Wrong options: DPN · deltoid · sural · saphenous

Neuropathic drug — **gabapentin / pregabalin**: **↓ hyperexcitability of voltage-dependent presynaptic calcium channels** · TCA (amitriptyline) also

✗ Wrong option: drug mechanism	Drug
Intracellular **Na channel** block	Lidocaine, bupivacaine
Mu agonist	Opioids
Direct GABA	Diazepam, baclofen
COX-2 selective (not COX-1)	Anti-inflammatory

Nerve complication of lateral ligament reconstruction — **SPN (commonest) → paraesthesia proximal dorsum of foot**

Nerve	Sensory / motor sign
SPN	**Dorsum of foot, anterolateral ankle / distal leg** ✓
DPN	**1st dorsal web space** ± TA / EHL weakness
Sural	**Posterolateral ankle + lateral foot**
Tibial	**Can't plantarflex**
Saphenous	**Anteromedial leg / medial hindfoot**

CRPS — listed association of ankle sprain

X vs Y

	ATFL	CFL
Attachments	Anterior fibula → **talar neck**	**Fibular tip → calcaneal tubercle**
Taut in	**PF + inversion**	**DF + inversion**
Drawer laxity	**Plantar flexion**	**Dorsiflexion**
Talar tilt	Not alone	**Required for + tilt**
Frequency	**1st**	**2nd**

	Low ankle sprain	High ankle sprain
Ligaments	ATFL ± CFL	**AITFL, IOL, PITFL, ITL**
Share	**> 90%**	**1–10%**
Mechanism	**Inversion + PF**	**Hyperdorsiflexion / ER** (TO)
Test	Drawer, talar tilt	**Squeeze, ER stress, Cotton, fibular translation**
RTP	1–4 wk	**5–6 wk / season**

	Cavovarus orthotic	Planovalgus orthotic
Posting	**Lateral** hindfoot / forefoot wedge	**Medial** hindfoot post
Arch	**Lowered** + **1st MT recess**	Rigid arch support, **UCBL**, Arizona

	Broström-Gould	Tenodesis (Evans, Watson-Jones, Chrisman-Snook)
Type	**Anatomic** (ATFL + CFL + IER)	**Nonanatomic** (peroneus brevis)
Problem	Over-tight → **↓ DF**	**Subtalar stiffness**

How it's asked

"Inversion laxity in dorsiflexion, none in PF – ligament?"	**→ calcaneofibular ligament (CFL)**
"Anterior drawer in plantarflexion tests?"	**→ ATFL**
"Talus translates 10 mm in 10° PF – injured?"	**→ ATFL**
"Talar tilt 20° more than other side?"	**→ anterior talofibular and calcaneofibular ligaments**
"Positive talar tilt X-ray – which ligament must be torn?"	**→ CFL**
"Anterior drawer – true statement?"	**→ CFL best tested in dorsiflexion**
"Drawer laxity in DF, stable in PF – attachments?"	**→ fibular tip → lateral calcaneal tubercle**
"Recurrent sprains, drawer lax in DF only – structure?"	**→ CFL**
"Strongest tibiotalar ligament?"	**→ deep posterior tibiotalar** (TO)
"Sequence of ligament failure in inversion?"	**→ anterolateral capsule → ATFL → CFL → PTFL** (TO)

– – Active space – –

"Usual mechanism of syndesmotic injury?"	→ **hyperdorsiflexion** (TO)
"Syndesmosis — false EXCEPT?"	→ **PITFL stronger than AITFL** (TO)
"Syndesmosis X-ray — FALSE?"	→ **TF overlap < 6 mm on mortise = unstable** (TO)
"Most reliable predictor of IOM tear in bimalleolar fracture?"	→ **intraoperative stress testing** (TO)
"Syndesmotic screw position?"	→ **2 cm above plafond, parallel to joint, 30° anterior** (TO)
"Cotton test positive translation?"	→ answer key not provided by the source (TO)
"Role of proprioception EXCEPT?"	→ answer key not provided by the source (TO)
"Pathway to rehabilitate to cut recurrence?"	→ **proprioception**
"Nonop modality shown to minimise recurrence?"	→ **functional bracing + early proprioceptive training**
"Functional vs immobilisation — FALSE?"	→ **functional ↑ laxity** (it's less)
"Cast + NWB vs functional rehab?"	→ **longer time to return to sport**
"Grade III sprain on MRI, no syndesmosis — treatment?"	→ **RICE, elastic wrap, therapy**
"Acute sprain, + drawer, squeeze − — next?"	→ **early functional rehabilitation**
"Untreated sprain 5 months, stable drawer?"	→ **proprioception + peroneal strengthening**
"Ottawa — finding that warrants X-ray?"	→ **distal fibula / lateral malleolus tip tenderness**
"Chronic instability in athlete, normal alignment — best surgery?"	→ **modified Broström**
"Gould modification — true?"	→ **no biomechanical difference in stability**
"Add-on to Broström-Gould for commonly associated condition?"	→ **ankle arthroscopy**
"Post Broström-Gould rehab — true?"	→ **mobilisation < 3 wk ↑ functional scores**
"Over-tightened suture-tape Broström limits?"	→ **dorsiflexion**
"Commonest nerve complication of ATFL reconstruction?"	→ **SPN — proximal dorsum of foot paraesthesia**
"Factor predisposing to reconstruction failure?"	→ **cavovarus foot**
"Most likely cause of failure in high-arched varus heel?"	→ **native hindfoot alignment**
"Cavovarus, Coleman corrects, recurrent sprains?"	→ **1st MT osteotomy + lateral ligament reconstruction**
"Cavovarus, Coleman negative — what cuts recurrence most?"	→ **lateralising calcaneal osteotomy**

"Lowest chance of recurrence, flexible cavovarus?"	→ **reconstruction + plantar fascia release + 1st MT dorsiflexion osteotomy**
"Orthotic for subtle cavovarus runner?"	→ **lateral hindfoot posting + 1st MT head recess**
"Semirigid orthosis, lowered arch, lateral wedge — deformity?"	→ **subtle cavovarus foot**
"Pain + swelling 8 months after Broström, drawer −?"	→ **peroneus brevis tear**
"Eversion pain 7 wk after sprain despite PT — next?"	→ **MRI**
"PB 35% intact at Broström?"	→ **peroneus brevis tenodesis**
"After PB tubularisation + SPR repair, avoid?"	→ **resisted eversion**
"Burning dorsolateral pain 4 wk after grade III sprain?"	→ **SPN stretch neurapraxia**
"Drug mechanism for SPN neuritis?"	→ **↓ voltage-dependent presynaptic Ca channel hyperexcitability (gabapentin)**
"Commonest acute MRI finding with lateral sprain?"	→ **talar bone bruise**
"Medial pain, ER pain, squeeze −, needs reconstruction?"	→ **deltoid ligament complex**
"Peroneal weakness in dancers causes?"	→ **ankle sprain**

– – Active space – –

– – Active space – –

ANTERIOR TIBIALIS TENDON RUPTURE

– Foot & Ankle · 10 / 48 –

1 Basics

Anatomy

Tibialis anterior	**primary ankle dorsiflexor (80%)** · inserts **medial cuneiform + base of 1st MT** · also inverts foot with dorsiflexion
Secondary dorsiflexors	**EHL + EDL** → mask the rupture → **delayed diagnosis common**
Site	rupture **at level of ankle joint**, proximal stump retracts (~3 cm)

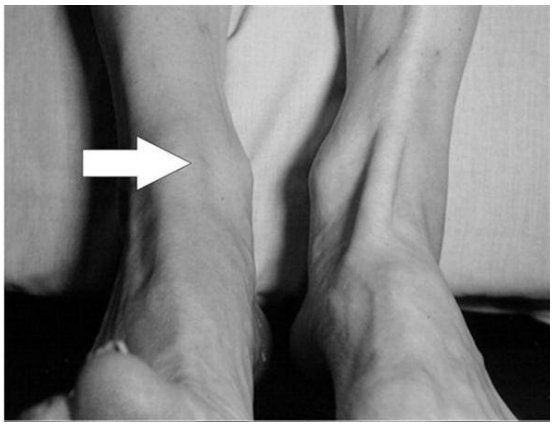
Epidemiology & risk factors

Attritional (atraumatic)	**older patients**, men **> 45 yr** · eccentric load against **plantarflexed ankle** (missed step) · degenerative
Traumatic	**laceration** or closed rupture · strong eccentric contraction in **young**
Risk factors	**older age** · **diabetes** · **fluoroquinolones** · **local steroid injection** · **inflammatory arthritis** · gout

3 Presentation

Diagnosis – foot drop & attritional rupture

Acute	**"pop"** → painful anterior ankle swelling · weak dorsiflexion
Chronic	**painless** · difficulty clearing foot · minimal swelling
Classic triad	**pseudotumour (painless mass) at anteromedial ankle** = retracted stump · **loss of TA contour** (not palpable on resisted DF) · **weak DF + hyperextension of all toes** (EHL / EDL over-firing)
Eversion on dorsiflexion	unopposed EHL + EDL (normal foot **inverts** with DF via TA)
Gait	**steppage gait** (↑ hip flexion in swing) · **foot slap** after heel strike · toe dragging



Loss of tibialis anterior contour after rupture (arrow)

4 Diagnosis & Classification

Key radiology findings

Study	Finding / role
X-ray (3 views foot + ankle)	Exclude osseous injury · usually unremarkable
CT	**Not indicated**
MRI	**Complete vs partial tear** · sagittal **"empty sheath"** · does not decide need for interposition graft



Sagittal MRI — empty tibialis anterior sheath

– – Active space – –

Differential diagnosis

Condition	Distinguishing feature
Attritional TA rupture	**EHL + EDL normal**, painless anterior mass, steppage gait, full passive ROM
L4 radiculopathy / lumbar stenosis	**Tendon palpable, no ankle mass**, dermatomal sensory loss (medial leg), weakness of several groups incl. quadriceps, +ve lumbar MRI
Common peroneal nerve palsy	**EHL + EDL also weak**, sensory loss, compression history
Deep peroneal nerve palsy	Toe extensors weak (not in isolated TA rupture)
Acute TA laceration	Clear sharp injury, **abrupt** onset (not progressive over months)
Anterior ankle impingement	**No** DF weakness, steppage gait or mass

5 Management

Tibialis anterior rupture

- Low-demand / elderly → **AFO**
- Partial rupture → **CAST**
- Acute < 6 wk, active → **DIRECT REPAIR**
  (attempt up to 3 months · gastroc recession if < 5° DF knee extended)
- Chronic > 6 wk → **RECONSTRUCTION**
  - gap → **INTERPOSITION GRAFT (plantaris, hamstring) / EHL transfer**

Operative criteria

Option	Indication
AFO	**Low-demand** patient
Casting	**Partial rupture**
Direct (primary) repair	**Acute < 6 wk**, active high-demand; may be attempted **up to 3 months**
Gastrocnemius recession (before tensioning)	**< 5° passive DF with knee extended**
Reconstruction	**Chronic > 6 wk**
PTT transfer	**Permanent foot drop** only

Surgery even late functional DF + gait improved in **18/19** (early 3 d–6 wk and late 7 wk–5 yr repairs) · traumatic ruptures do better with surgery regardless of timing · atraumatic presenting late → bracing acceptable

Treatment – repair or reconstruction & expected outcome

– – Active space – –

Direct repair	open: through laceration · closed: longitudinal incision over defect · hemostat under **extensor retinaculum** to retrieve proximal stump · **non-absorbable Krackow / Bunnell / Kessler** · oversew frayed ends · avulsion → **suture anchors / bone tunnels**
Sliding graft	**half-width of TA turned down** · secured to medial cuneiform / dorsal navicular distal to retinaculum
Free graft	**autograft (hamstring, plantaris)** or allograft · also Achilles, EDL, EHB interposition
EHL transfer / tenodesis	**distal EHL → EHB** · **proximal EHL stump** used as graft to TA insertion · TA tensioned then sutured to proximal EHL
Augment with	**extensor tendons (EHL, EDL)** ✗ Wrong options: **FHL** (out of phase, too short; FHL for chronic Achilles)
Expected outcome	**restores active DF with some residual weakness + improved activity level** · ↑ AOFAS, high satisfaction · peak torque **72–84%** of other side
Commonest long-term residual	**decreased dorsiflexion strength** (+ ↓ inversion strength) ✗ Wrong options: persistent foot drop / AFO (chronic missed or nonop) · numbness (missed superficial nerve laceration) · persistent pain

(✗ Wrong options: **Wrong treatments** – **VY advancement / turndown flap** (Achilles rupture with gap)) · **SPR repair + groove deepening** (peroneal instability) · AFO in active patient (permanent gait disturbance) · chest CT + skeletal survey + oncology biopsy of the "mass" · MRI of proximal tibiofibular joint for ganglion + peroneal EMG · PTT transfer

6 Complications

List	failure of repair / reconstruction · **weakness of dorsiflexion** · adhesions · neuroma
Prognosis	good with treatment

X vs Y

	TA rupture	**Peroneal nerve palsy**	**L4 radiculopathy**
EHL / EDL	**Normal**	**Weak**	Variable
Anterior mass	**Yes (pseudotumour)**	No	No
TA tendon palpable	**No**	Yes	**Yes**
Sensation	Normal	Sensory abnormality	Dermatomal

	Acute (< 6 wk)	**Chronic (> 6 wk)**
Treatment	**Primary repair ± gastroc recession**	**Reconstruction** – interposition graft, sliding graft, EHL transfer

How it's asked

"Elderly soccer injury, video of left anteromedial mass + toe hyperextension – next?"	→ **left tendon repair and possible graft augmentation** (check the side)

– – Active space – –

"67-yo diabetic, steroid injections, painless anterior mass, EHL / EDL normal?"	**→ attritional rupture of the tibialis anterior tendon**
"Chronic rupture 7 months, 15° DF knee extended — next?"	**→ surgical reconstruction with plantaris tendon interposition augmentation**
"Surgery after acute laceration will most likely?"	**→ restore active dorsiflexion with some residual weakness and improve activity level**
"Most common long-term residual after repair?"	**→ decreased dorsiflexion strength**
"70-yo active, acute rupture — associated procedure?"	**→ primary repair with possible gastrocnemius recession**
"When add gastroc recession?"	**→ < 5° passive DF with knee extended**
"Graft never used for TA?"	**→ FHL**

BLOOD SUPPLY TO THE FOOT

– Foot & Ankle · 11 / 48 –

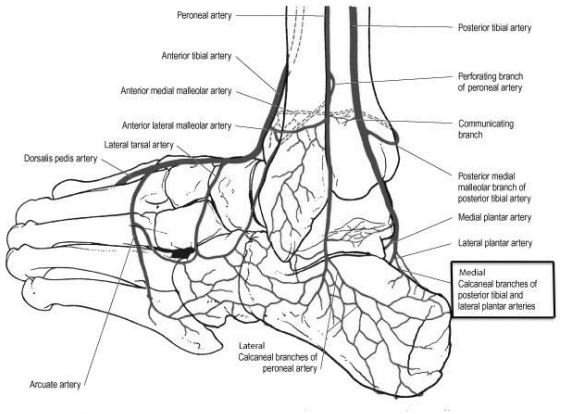
1 Basics

Source arteries

Three source arteries **peroneal (fibular) · posterior tibial (PTA) · anterior tibial (ATA)**

Artery	Origin	Course in leg	Ankle branches	Foot branches
Peroneal	**PTA, ~2.5 cm from its origin**	Pierces interosseous membrane **~5 cm above lateral malleolus** (perforating branch) → joins **anterior lateral malleolar a.** → anterior to syndesmosis → **lateral tarsal a.**	**Posterior lateral malleolar** · communicating	**Lateral calcaneal branch** (terminal)
Posterior tibial	**Larger** terminal branch of popliteal; proximal part = **tibioperoneal trunk**	Between superficial + deep posterior compartment → medial, **palpable behind medial malleolus**	**Posterior medial malleolar** · communicating · **artery of tarsal canal**	**Bifurcates beneath sustentaculum** → medial + lateral plantar
Anterior tibial	**Smaller** terminal branch of popliteal	**Anterior to interosseous membrane**, supplies anterior compartment · **superficial at ankle midway between malleoli**	**Anterior medial + anterior lateral malleolar**	**Dorsalis pedis**

Peroneal supplies (leg) soleus, tibialis posterior, FHL, peroneals



Arterial supply of the foot and ankle

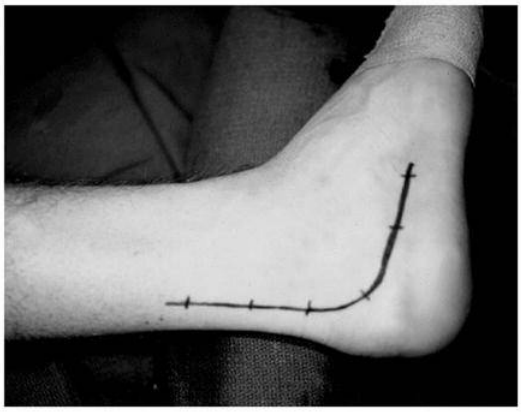
Hindfoot & talus

Artery of tarsal canal branch of **PTA** · **dominant blood supply to talar body**

Artery of sinus tarsi branch of **perforating peroneal a.** · with tarsal canal a. supplies **talus**
✗ Wrong options: heel pad

– – Active space – –

Lateral calcaneal branch of peroneal	**terminal** branch · perfuses **lateral flap of extensile lateral approach to calcaneus** · one of three flap arteries with **lateral malleolar + lateral tarsal** · **minor** contribution to heel pad



Extensile lateral calcaneal incision – flap on lateral calcaneal artery

Heel pad blood supply

Heel pad	**medial calcaneal branch of the posterior tibial artery** = major supply ✗ Wrong options: lateral calcaneal a. (minor) · medial plantar a. · artery of sinus tarsi
Medial calcaneal branch	**first branch of lateral plantar a.** (Orthobullets) · "branch of posterior tibial artery" (OB) – same vessel, read the options
Heel pad avulsion	high-energy (industrial machine) · fibrous honeycomb septa + specialised skin → **irreplaceable** · risk **heel pad necrosis** → poor prognosis, salvage often fails · replantation with **ipsilateral saphenous vein graft** described

Plantar branches (PTA)

Lateral plantar a. branches (in order)	**Medial plantar a.**
1st **medial calcaneal** · 2nd **abductor digiti minimi** · 3rd **digital branch to 5th toe** · terminal **plantar branch**	Given off **under flexor retinaculum** · supplies **medial side of foot** · terminal → anastomoses with **1st dorsal metatarsal a. (DPA)** · superficial digital branches → plantar metatarsal aa. of **first three** IM spaces

Dorsal supply (ATA)

Dorsalis pedis	continuation of ATA · palpable **just lateral to EHL** · branches **arcuate, lateral tarsal, medial tarsal** · ends at **1st IM space → 1st dorsal metatarsal a.** + **deep plantar arch**
Plantar arch	**lateral plantar a. + dorsalis pedis** · plantar digital + plantar metatarsal aa.
Arcuate artery	dorsal midfoot, **deep to extensor tendons** · dorsal metatarsal aa. to **2nd, 3rd, 4th** IM spaces

– – Active space – –



Dorsalis pedis and arcuate artery

Veins

Dorsal venous arch drains into **greater + lesser saphenous** systems

X vs Y

	Medial calcaneal branch (PTA)	Lateral calcaneal branch (peroneal)
Supplies	**Heel pad (major)**	**Extensile lateral calcaneal flap** · heel pad (minor)
Injury context	Heel pad avulsion	Lateral wound necrosis after calcaneus ORIF

	Artery of tarsal canal	Artery of sinus tarsi
Parent	**PTA**	**Perforating peroneal**
Supplies	**Talar body (dominant)**	Talus

How it's asked

"Heel pad avulsion — baseline blood supply of the heel pad?"	→ **medial calcaneal branch of the posterior tibial artery** ✗ Wrong options: lateral calcaneal branch of peroneal (minor) · medial plantar · sinus tarsi a.
"Extensile lateral calcaneal flap perfused by?"	→ **lateral calcaneal branch of peroneal artery**
"Dominant blood supply to talar body?"	→ **artery of tarsal canal (PTA)**

– – Active space – –

BUNIONETTE DEFORMITY

– Foot & Ankle · 12 / 48 –

1 Basics

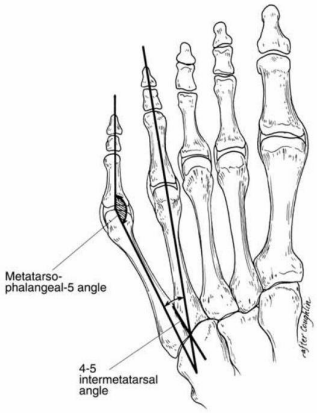
Epidemiology

Tailor's bunion	**lateral prominence of 5th MT head** · adolescents + adults · **women 2–4×** · often **bilateral**
Extrinsic cause	**forefoot compression (tight shoes)** = commonest · abnormal lateral loading
Intrinsic causes	congenital (**splayfoot**, **brachymetatarsia**) · inflammatory arthropathy · residual surgical malalignment
Associated	**varus 5th MTP** · **pes planus** · forefoot widening

Pathoanatomy & normal values

Parameter	Normal	Bunionette
4–5 IMA	**6.5–8°** (Orthobullets, OB) · **6–8°** (OB)	↑
Lateral deviation angle	**0–7°**	↑
5th MT head width	**< 13 mm**	↑ (> 13 mm)
5th MT shaft	Straight	**Lateral bowing**

Lesion bony prominence **± bursitis** over lateral 5th MT head



4–5 intermetatarsal angle

3 Presentation

Symptoms	lateral bunion pain · **plantar callus** · **worse with constrictive shoes** · cosmetic: **medial deviation of 5th toe**, prominent 5th MT head · effect on work
Exam	**plantar or lateral hyperkeratosis** · widened forefoot · erythema, swollen 5th bunion · **painless passive 5th MTP ROM** usually · check shoe wear

4 Diagnosis & Classification

– – Active space – –

Diagnosis **clinical**: lateral 5th MT head prominence ± pain + callus

Key radiology findings

Radiographs **weight-bearing dorsoplantar, lateral, oblique · ↑ 4–5 IMA · ↑ lateral deviation angle · ↑ MT head width**

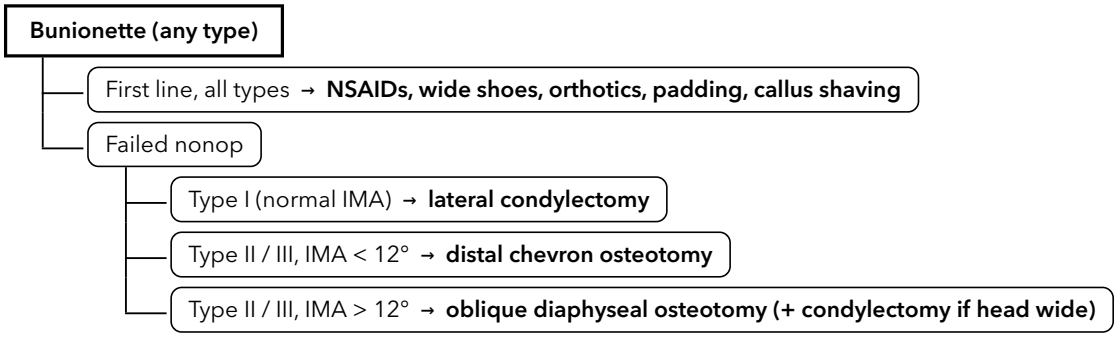
CT rarely needed · associated **trauma or malignancy**

Coughlin classification

Type	Deformity	Surgery
I 	**Enlarged 5th MT head / lateral exostosis** (head > 13 mm)	**Lateral condylectomy**
II 	**Congenital lateral bow of 5th MT** (distal half) · **normal 4–5 IMA**	**Distal chevron osteotomy** (IMA < 12°)
III 	**↑ 4–5 IMA · most common**	Distal osteotomy if < 12° · **oblique diaphyseal osteotomy if > 12°**
IV (OB)	**Combination** of the above	–

Type definitions differ one OB uses **1 = enlarged lateral condyle · 2 = 5th MT varus · 3 = combined** – read the stem

5 Management



– – Active space – –

Nonoperative first-line care

First line for all types	**NSAIDs + shoe wear modification** · orthotics (**semi-rigid inserts**, **custom orthotics** – OB) · **keratosis padding** · **callus shaving** · wide-based shoes · stretch existing shoes · **75–90% success** ✗ Wrong options: condylectomy · osteotomy · MT head resection as first step
Asymptomatic / only shoe rubbing	nonop ✗ Wrong options: lateral condylectomy · condylectomy + distal osteotomy · diaphyseal osteotomy
✗ Wrong option: 1st TMT arthrodesis ± Akin	that's **hallux valgus**, not bunionette

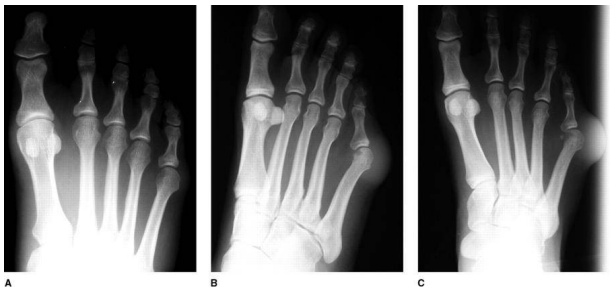
Operative criteria

Finding	Procedure
Type I, normal 4–5 IMA	**Lateral condylectomy (lateral eminence resection)**
Type II (lateral bow, normal IMA)	**Distal chevron osteotomy**
Type II / III, **4–5 IMA < 12°** (Orthobullets) · **≤ 12°** · **8–12°** · **9–12°** (OB)	**Distal MT osteotomy** (chevron) ± condylectomy + lateral capsule tightening
Type II / III, **4–5 IMA > 12°**	**Oblique diaphyseal rotational osteotomy**
↑ IMA + MT head > 13 mm	**Diaphyseal osteotomy + lateral condylectomy**
Plantar callosity	**Shave plantar 5th MT head**
Symptomatic after distal osteotomy	Proximal osteotomy (OB)
Salvage only	**MT head resection**

Procedure by type – lateral condylectomy, distal chevron, diaphyseal osteotomy (4–5 IMA)

Lateral condylectomy	resect **lateral ⅓** of 5th MT head (Orthobullets) · **lateral ¼–⅓** (OB) · or up to lateral articular margin · + **tighten lateral MTP capsule** · **no WB restrictions** / no prolonged immobilisation ✗ Wrong options: for IMA > 10° · lateral half · lowest recurrence · high metatarsalgia
Condylectomy downsides	**highest recurrence** · over-resection · **MTP subluxation**, persistent pain
MT head osteotomy for type I	larger correction than condylectomy, but WB restrictions
Distal chevron	**commonest distal osteotomy** · **Type II: lateral bow, no keratotic lesion, normal IMA** · **biomechanically strongest** vs other osteotomies · fix with **K-wire or screw** ✗ Wrong options: type I who can't tolerate WB restriction · failed prior condylar resection · type III at 13°
Other distal osteotomies	distal transverse · peg-and-slot · step-cut
Oblique diaphyseal rotational osteotomy	**Type II + III, IMA > 12°** · **screw** fixation · ± lateral condylectomy · **causes 5th MT shortening + metatarsalgia** · **rotational deformity** + implant problems (OB)

Proximal osteotomy	**avoid: poor blood supply** (Orthobullets)
Sliding distal MIS osteotomy (S-DMMO)	**type 3, IMA 10°, prominent condyle** → treats deformity, avoids implant complications ✗ Wrong options: contact casting · lateral eminence resection (type 1) · open diaphyseal osteotomy · MT head resection
Large IMA (radiograph)	**MT osteotomy + 5th MT head ostectomy** ✗ Wrong options: head excision (flail joint + transfer lesion) · plantar condylectomy (↓ plantar pressure only) · Silastic implant
Rheumatoid, type 1, IMA 7°, congruent MTP, nodule under head	**simple exostectomy + soft-tissue mass excision** ✗ Wrong options: MT head excision (no metatarsalgia, normal joint) · osteotomy



Coughlin bunionette types I–III



Oblique diaphyseal osteotomy of the 5th metatarsal

6 Complications

Complication	Procedure
Recurrence	**Condylectomy alone** – commonest / highest
Transfer metatarsalgia	**Isolated MT head resection**
MTP instability / flail joint	**MT head resection** (salvage only, poor outcomes)
5th MT shortening + metatarsalgia	**Diaphyseal osteotomy**
Rotational deformity	Diaphyseal osteotomy
Claw toe	–

– – Active space – –

X vs Y

	Type I	Type II	Type III
Problem	Wide head / exostosis	**Lateral bow**	**↑ 4–5 IMA**
4–5 IMA	Normal	**Normal**	**↑**
Surgery	**Condylectomy**	**Distal chevron**	**Diaphyseal (> 12°)** / distal (< 12°)

	Lateral condylectomy	Distal chevron	Oblique diaphyseal
Indication	Type I	IMA < 12°	**IMA > 12°**
WB after	**Unrestricted**	Restricted	Restricted
Problem	**Recurrence**	–	**Shortening**

How it's asked

"Oblique diaphyseal rotational osteotomy – which radiographs?"	→ **type II and type III (B and C)** with IMA > 12°
"Planned distal chevron – which radiograph?"	→ **type II, lateral bow (B)**
"Diaphyseal 5th MT osteotomy is optimal for?"	→ **widened 4–5 IMA + lateral prominence**
"Distal chevron most indicated for?"	→ **laterally bowed 5th MT, normal IMA, no keratotic lesion**
"Distal chevron – which figure?"	→ **type 2 bunionette (Figure C)**
"Enlarged 5th MT head, failed nonop?"	→ **lateral condylar resection**
"Lateral condylectomy – true?"	→ **no weight-bearing restrictions postoperatively**
"Mid-diaphyseal osteotomy planned – pathology?"	→ **type III with 4–5 IMA of 14°**
"Large 4–5 IMA on radiograph – optimal surgery?"	→ **metatarsal osteotomy + 5th MT head ostectomy**
"IMA 13°, head 16 mm, congruent MTP?"	→ **diaphyseal osteotomy + 5th MT condylectomy**
"IMA 10°, prominent condyle, failed pads?"	→ **sliding distal metatarsal MIS osteotomy**
"RA, nodule under 5th MT head, IMA 7°?"	→ **simple exostectomy + soft-tissue mass excision**
"Painless bump, rubs on shoes – initial?"	→ **NSAIDs + shoe wear modification**
"First visit, IMA 9°, callus?"	→ **shoewear modification, custom orthotics, NSAIDs**
"Commonest complication of condylectomy?"	→ **recurrence**
"MT head resection causes?"	→ **transfer metatarsalgia, MTP instability**

CLAW TOE

– Foot & Ankle · 13 / 48 –

1 Basics

Epidemiology

Distribution	typically **multiple toes** · often **bilateral**

Pathophysiology – EDL vs intrinsic imbalance, neurological causes

Primary pathology	**MTP hyperextension** → chronic MTP hyperextension → **unopposed DIP + PIP flexion by FDL** · analogous to **intrinsic minus hand**
MTP joint	**EDL overpowering relatively weak intrinsics** · hyperextended MTP makes intrinsics less efficient plantarflexors → deformity progresses ✗ Wrong options: intrinsics overpowering EDL · plantar plate overpowering EDB · lumbricals overpowering intrinsics · EDL overpowering EDB
IP joints	**stronger flexors overpower weaker intrinsics** (intrinsics act as IP extensors)
Plantar plate	becomes **insufficient / attenuated** → **base of proximal phalanx translates dorsally**
Interossei + lumbricals	**move dorsally** → flexion moment arm shifts to wrong side of MTP centre of rotation
Most common cause	**synovitis** (Orthobullets, OB) · "occurs most commonly in **diabetes, alcoholism, neuromuscular** conditions" (another OB) – read the options
Other causes	trauma · **delayed / missed compartment syndrome** (deep posterior leg compartment or foot) · **cavus** · **inflammatory arthropathy** (soft-tissue attenuation → MTP instability)
Neurological causes	**Charcot-Marie-Tooth**, **Friedreich ataxia**, myelodysplasia, **cerebral palsy**, **multiple sclerosis**, stroke, **diabetic neuropathy** · **clawing of all 4 lesser toes → neurological abnormality**

3 Presentation

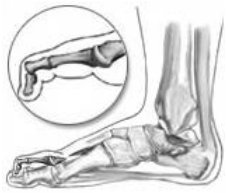
Symptoms	pain at **unstable MTP joint** · **metatarsalgia**
Plantar	**depressed metatarsal head with callus** + tenderness
Dorsal / tip	flexed IP joints with dorsal callosities · **tip-of-toe ulceration** (neuropathic / diabetic)

4 Diagnosis & Classification

Diagnosis	clinical: **MTP hyperextension + PIP flexion + DIP flexion** of a lesser toe

– – Active space – –

Lesser toe deformities

Deformity	MTP	PIP	DIP
Claw toe 	**Hyperextension**	**Flexion**	**Flexion**
Hammer toe 	Slight extension (Orthobullets) · neutral (OB)	**Flexion**	**Extension**
Mallet toe 	Normal / neutral	Normal / neutral	**Flexion**
Curly toe	**Flexed**	Flexed	Flexed
Crossover toe	**Laterally deviated**	Flexed	Neutral

5 Management

Claw toe

- First line → **taping + shoe modification**
- Flexible, no contracture → **Girdlestone-Taylor: EDB tenotomy, EDL lengthening, FDL flexor-to-extensor transfer**
  (diabetic, tip-of-toe ulcer, no infection → isolated percutaneous FDL tenotomy)
- Fixed contracture → **Girdlestone + MTP capsulectomy + proximal phalanx head/neck resection**
- All four lesser toes → **Girdlestone + Weil (distal MT shortening) osteotomy**

Claw toe (TO)

- Flexible → **flexor tendon transfer**
- Fixed → **MTP soft-tissue release (dorsal + medial/lateral capsule, extensor lengthening)**
  - Reduced → **realigned toe**
  - Not reduced → **Weil osteotomy + fixed hammer-toe repair**

Nonoperative — **first line** · plantar padding (**metatarsal / crest pads**, orthotics) to offload plantarly subluxed MT heads · **high toe box** · **sling** holding proximal phalanx parallel to ground

Operative criteria

Indication	Procedure
Painful flexible deformity without contracture · shoe-wear ulcers · otherwise healthy	**Girdlestone(-Taylor)**: EDB tenotomy + EDL lengthening + **FDL flexor-to-extensor transfer**
Fixed contracture	Girdlestone + **MTP capsulectomy** + **proximal phalanx head + neck resection** (tendon transfer + bony resection)
Claw toe of all four lesser toes	Girdlestone + **Weil lesser MT osteotomy**
Flexible deformity, diabetic, tip-of-toe ulcer, no infection	**Isolated percutaneous FDL tenotomy**
Rigid diabetic claw toe with ulcer	Arthrodesis
Rigid mallet toe (rigid DIP flexion)	Bony: **DIP fusion or resection**
Hammer toe (PIP flexion, DIP extension)	**FDL-to-EDL transfer**

Flexible deformity – flexor tenotomy or flexor-to-extensor transfer

Flexible claw toe (TO) **flexor-to-extensor tendon transfer with extensor lengthening** · flexor split into **two slings**, transferred dorsally as for hammer toe · extensor lengthening may or may not be needed (TO)
✗ Wrong options: extensor lengthening + PIP arthroplasty · flexor tenotomy + MTP pinning · distal metatarsal osteotomy

Girdlestone-Taylor FDL to extensor surface → long flexor acts like **intrinsics**: **active MTP plantarflexion + IP extension** · **only for flexible deformity**
✗ Wrong options: PIP arthrodesis · MTP capsulotomy + PP resection arthroplasty (fixed only) · isolated Weil

Percutaneous FDL tenotomy diabetic + **flexible** claw toe + **tip ulcer**, no osteomyelitis (bone not probed) → ↓ tip pressure; minimal exposure (infection, poor healing)

Percutaneous flexor tenotomy outcome	Value
Ulcer healing	**98% at 4 weeks**
Reoperation	**~8%** (✗ Wrong options: 50%)
Ulcer recurrence	**0–18%** (✗ Wrong options: 30%)
Transfer ulceration	Reported complication (✗ Wrong options: "no risk")

✗ Diabetic tip-ulcer traps amputation through 2nd MTP (only with infection / osteomyelitis) · repeat ulcer debridement (doesn't offload) · **DIP fusion** (rigid / arthritic) · **Weil** (all four lesser toes, non-neuropathic; floating toe + transfer metatarsalgia)

Plantar plate repair

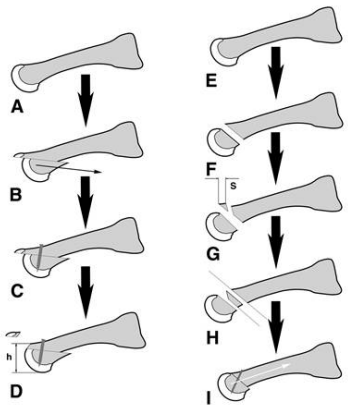
Plantar plate repair **improves the extension deformity at the MTP joint** → toe plantigrade
✗ Wrong options: MTP flexion (claw has MTP hyperextension) · PIP / DIP (addressed indirectly)

Recurrence **persistent plantar plate dysfunction**

– – Active space – –

Weil osteotomy – floating toe complication

Technique	**oblique distal MT shortening osteotomy** · translates head **proximal + plantar** · rigid internal fixation · head moved proximal to callus → axial decompression · **average shortening 4 mm**
Indications	metatarsalgia (2nd MT, high heels) · MTP subluxation · claw toe of all four lesser toes
Floating toe	**most common complication of Weil** = **dorsiflexion contracture at the MTP joint**
Mechanism	plantar depression of head + altered centre of rotation → **interossei move dorsal to MTP axis → converted from plantarflexors to dorsiflexors** (intrinsics act as MTP extensors) ✗ Wrong options: articular damage · disruption of dorsal capsular artery · lengthening (it shortens) · ligament instability
Plantar translation in distal MT osteotomy (TO)	**floating toe**: intrinsic force shifts volar → dorsal + weak toe flexors → **extension at MTP + IP joints** (TO) ✗ Wrong options: claw · hammer · mallet
Cause of 2nd toe "floating" post-op	**technique of surgical osteotomy** ✗ Wrong options: immobilisation · CVA · extensor tethering · infection
Avoid	osteotomy **parallel to plantar surface of foot** or **resect a wafer of bone** so head isn't depressed
Rare after Weil	**osteonecrosis / MTP arthritis** (constant **plantar artery** to MT head + capsule) · nonunion · significant transfer lesions
Weil vs Helal (shaft) osteotomy	Weil: higher satisfaction, less recurrent metatarsalgia, fewer transfer lesions



Weil osteotomy – plantar depression of the head (B–D) vs parallel cut / wafer resection (E–I)

6 Complications

Complication	Cause
Floating toe (most common after Weil)	**Intrinsics migrate dorsal to joint → act as MTP extensors**
Recurrence	**Persistent plantar plate dysfunction**
Transfer ulceration	After percutaneous flexor tenotomy
Transfer metatarsalgia	Weil (not the commonest)

X vs Y

	Isolated FDL tenotomy	Girdlestone-Taylor transfer	Resection / arthrodesis
Deformity	**Flexible**	**Flexible**	**Fixed / rigid**
Patient	**Diabetic, tip ulcer, no infection**	Otherwise healthy	Any
Exposure	Percutaneous	Open	Open

How it's asked

"Claw toe – muscular imbalance from which relatively strong structure?"	→ **EDL overpowering relatively weak intrinsics**
"Multiple sclerosis – associated toe deformity?"	→ **extended MTP, flexed PIP + DIP** ✗ Wrong options: curly · mallet · hammer · crossover
"Plantar translation of MT head in distal MT osteotomy causes?"	→ **floating toe** (TO)
"2nd toe floats after Weil – cause?"	→ **technique of surgical osteotomy**
"Mechanism of commonest Weil complication?"	→ **conversion of interossei from plantarflexors to dorsiflexors**
"Most common complication of Weil osteotomy?"	→ **dorsiflexion contracture at the MTP joint** ✗ Wrong options: osteonecrosis · nonunion · transfer lesions · MTP arthritis
"Flexible claw toe – best treatment?"	→ **flexor-to-extensor tendon transfer with extensor lengthening** (TO)
"Diabetic, flexible PIP + MTP, failed nonop – surgery?"	→ **Girdlestone-Taylor flexor-to-extensor tendon transfer**
"Who benefits most from percutaneous flexor tenotomy?"	→ **diabetic (A1c 8.5), flexible PIP + DIP flexion, ulcer at tip of toe**
"Diabetic tip ulcer, flexible 2nd claw toe, bone not probed – treatment?"	→ **percutaneous flexor digitorum longus tenotomy**
"Percutaneous flexor tenotomy – true?"	→ **ulcer healing rates 98% at 4 weeks**
"Claw toe with lateral deviation of 2nd PP – direct result of plantar plate repair?"	→ **improves the extension deformity at the MTP joint**
"Claw toes of all four lesser toes?"	→ **Girdlestone + Weil osteotomy** · think **neurological cause**

– – Active space – –

DEEP PERONEAL NERVE ENTRAPMENT

– Foot & Ankle · 14 / 48 –

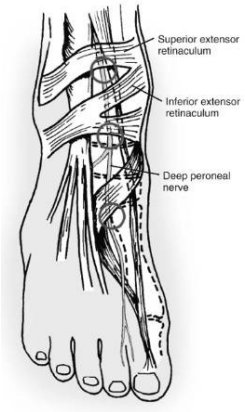
1 Basics

Synonym	**anterior tarsal tunnel syndrome** · rare compression neuropathy of the **deep peroneal nerve (DPN)**
Commonest site	fibro-osseous tunnel under the **inferior extensor retinaculum**
Zone of compression	**1 cm proximal to ankle joint → talonavicular joint**
Risk factors	**high heels**, compressive shoe wear (tight laces, ski boots), previous fracture

Anterior tarsal tunnel anatomy

Border	Structure
Roof (superficial)	**Inferior extensor retinaculum**
Floor (deep)	**Talonavicular joint capsule**
Lateral	Lateral malleolus
Medial	Medial malleolus

Contents	**EDL, EHL, tibialis anterior, peroneus tertius, DPN, dorsalis pedis artery + vein** ✗ Wrong options: superficial peroneal nerve · posterior tibial artery · intermediate dorsal cutaneous nerve · medial calcaneal nerve
At risk at release	**dorsalis pedis artery** (vascular bundle runs with the DPN)
DPN in tunnel	divides into **lateral (mixed motor + sensory → EDB)** and **medial (sensory only → 1st web space)** branches · nerve lies **lateral to EHL**
EHB	crosses superficially over the inferior extensor retinaculum → contributor to DPN compression (OB)



Anterior tarsal tunnel – DPN compression sites under the extensor retinacula

2 Mechanism

Intrinsic	**dorsal osteophytes over tibiotalar or talonavicular joints** (commonest) · pes cavus, post-fracture deformity (**navicular nonunion**) · ganglion · tumour · hypertrophic EHL / EDL / TA belly · peripheral oedema
Extrinsic	**tight laces / ski boots** · **high heels** (plantar flexion) · trauma incl. recurrent ankle instability
Stretch position	**ankle plantar flexion + inversion**

3 Presentation

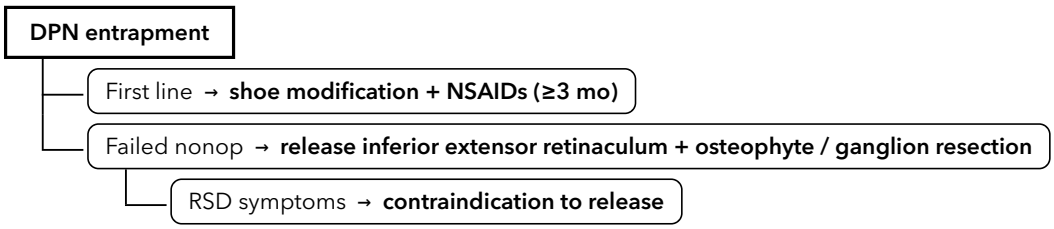
Symptoms	**vague dorsal foot pain** · paraesthesia **lateral hallux, medial 2nd toe, 1st web space**
Motor	**EDB weakness / atrophy**
Sensory	↓ two-point discrimination
Provocative	**Tinel over DPN radiating to 1st web space** · worse with **plantar flexion + inversion** · relief with **lidocaine DPN block**

4 Diagnosis & Classification

Key radiology findings

Study	Use
Lateral foot + ankle X-ray	**Dorsal osteophytes**, fracture sequelae
CT	Bony anatomy of canal
MRI	**Mass lesions** (ganglion, tumour)

5 Management



Nonoperative	**padded shoe tongue**, **alternative lacing** (skip the tender area), full-length rocker-sole steel shank, **night splint** (ankle drifts into plantar flexion), PT if instability, injection, diuretic for chronic oedema · **≥ 3 months** (OB)
Surgery	**S-shaped dorsal incision** ankle → base of 1st/2nd MT · start distal, find nerve, release **both branches** proximally · resect osteophytes, debulk muscle bellies · no compressive shoes after
Outcome	**80% good–excellent** · recovery prolonged; **persistent symptoms** = main complication

– – Active space – –

Foot drop – tendon transfer (TO)

Complete CPN palsy (anterior + lateral groups) (TO)	motor of choice = **tibialis posterior** to dorsum of foot → removes unopposed inversion + restores dorsiflexion · ± **FDL with tib post** (double transfer to EDL + EHL) · or **anterior transposition of lateral gastrocnemius + peroneal neurotization** (TO)
Isolated DPN palsy (peronei spared) (TO)	**rerouting the peroneus longus** (TO) ✗ Wrong options: tib post transfer (removes inversion balance against working evertors) · FDL + tib post · gastrocnemius transposition · foot drop splint

X vs Y

	Deep peroneal nerve	Superficial peroneal nerve
Tunnel	**Anterior tarsal tunnel** (inferior extensor retinaculum)	Not in tunnel
Sensation	**1st web space**	Dorsum of foot
Motor in foot	**EDB** (+ **EHB**, Orthobullets nerves note)	–

How it's asked

"Dorsal foot pain, medial 2nd-toe tingling, worse PF + inversion – structure in same tunnel to protect?"	→ **dorsalis pedis artery** ✗ Wrong options: SPN · PTA
"Commonest intrinsic cause?"	→ **dorsal talonavicular / tibiotalar osteophytes**
"Muscle weak or atrophied?"	→ **EDB**
"Isolated deep peroneal palsy, foot drop – ideal procedure?"	→ **reroute peroneus longus** (TO)
"Complete CPN palsy – tendon transfer motor?"	→ **tibialis posterior** (TO)

DIABETIC CHARCOT NEUROPATHY

– Foot & Ankle · 15 / 48 –

1 Basics

Epidemiology

Definition	chronic, progressive, **non-infectious destructive arthropathy** from **loss of protective sensation** (pain + proprioception) · first described by **Charcot in tabes dorsalis, 1868** (TO)
Incidence	**0.1–1.4%** of diabetics (Orthobullets) · acute Charcot **0.15–2.5%** (OB) · **1% of diabetics after 12 yr of disease** (TO) · **7.5%** of diabetics **with neuropathy** – read the options
Age	**type 1 DM → 5th decade** (20–25 yr after diagnosis) · **type 2 DM → 6th decade** (5–10 yr after diagnosis)
Bilateral	**9–35%**
Site	**foot and ankle commonest** · also shoulder, elbow · knee → ligamentous instability + bone loss
Commonest cause	**diabetic neuropathy** (now) · **tabes dorsalis** = former commonest lower-limb cause, now **2nd** (TO)

Pathophysiology, classification & upper-limb Charcot

Neurotraumatic theory	insensate joint + **repetitive microtrauma** → no protective response · overuse injury (TO)
Neurovascular theory	**autonomic dysfunction → ↑ blood flow (AV shunting, hyperaemia)** → bone resorption, periarticular osteopenia, demineralisation (TO)
Combined (TO)	both neurotraumatic + neurovascular · often initiated by trauma, frequently unrecalled · **osteoclast overactivity**, vascular shunting · twisting / minor injury may trigger (OB)

Cause	Typical joints (TO)
Diabetes (commonest)	**Tarsus + tarsometatarsal (commonest)** · MTP · **ankle > knee + spine**
Tabes dorsalis (2nd commonest)	**Knee (commonest)** · hip · ankle · lumbar spine
Leprosy (Hansen's)	IP joints (hand) · MTP (feet) · lower limbs
Syringomyelia	**Shoulder (commonest)** · elbow · wrist · cervical spine
Myelomeningocele	Ankle · intertarsal

Upper-limb Charcot (TO)	**syringomyelia = commonest cause** (rare overall) (TO) ✗ Wrong options: Hansen's · diabetes · tabes
Other causes	alcoholism / alcoholic neuropathy · leprosy · myelomeningocele / spina bifida · tertiary syphilis · syringomyelia · post-traumatic sensory deficit · **stroke with residual sensory deficit** · HMSN / **Charcot-Marie-Tooth** · congenital insensitivity to pain · amyloidosis · MS · **intra-articular steroids in weight-bearing joints** · RA · methotrexate toxicity · idiopathic
NOT a cause	**Reiter's syndrome**

– – Active space – –

Genes & lab markers

Marker	Role
CGRP (TO)	Normally **antagonises RANKL synthesis** at nerve terminal · **non-functioning in neuropathic foot** → RANKL unchecked · maintains **joint capsule integrity** → loss = destruction + dislocation
RANKL (TO)	Drives **osteoclastogenesis** → excess bone turnover
OPG (TO)	**Decoy receptor** for RANKL · RANKL–OPG balance **disrupted**
IL-1, TNF-α	↑ **NF-κB** + **RANK / RANKL / OPG** pathway
✗ Wrong option: transforming growth factor-beta (TGF-β)	**Not** a key marker (TO)
ESR, CRP, WBC	↑ **in both** Charcot + infection → don't discriminate
Wound healing	**Absolute lymphocyte count > 1500/mm³** · **albumin > 3.0 g/dL**
HbA1c > 8.0%	Poor control → **poor surgical outcomes**
Histology	**Synovial hypertrophy** · **detritic synovitis** (cartilage + bone fragments in synovium)

Bisphosphonates (pamidronate) theoretical role, ↑ osteoclast activity (TO)

3 Presentation

Clinical features

Charcot's triad (TO) **gross joint swelling** · **exaggerated movements (instability)** · **painless** presentation (TO)

Pain **painless classically**; absence of pain diagnostic (TO) · usually **some pain**, less than expected (TO) · **50% painful / 50% painless** (Orthobullets) — read the options

Single joint first · swollen, deformed, subluxed, unstable ("floppy") · **crepitus** from loose bodies (TO)

Acute **swollen, warm, erythematous**; **3.3 °C warmer** (Orthobullets) · **3–7° warmer** (OB) · mimics infection · **intact skin + loss of protective sensation** ≈ pathognomonic

Chronic structurally deformed · bony prominences · **rocker-bottom deformity** · **medial arch collapse** · ligamentous instability

Skin rule (TO) **skin intact → think Charcot** · **skin broken → most likely infection** (TO)

Undiagnosed patient with Charcot foot **evaluate for diabetes mellitus**

4 Diagnosis & Classification

Key radiology findings

Modality	Finding / fact
X-ray (AP + lateral foot, ankle series)	**Early** — mimics OA · **Late** — joint space obliteration, **fragmentation of both articular surfaces → subluxation / dislocation**, bone "chunks" in fibrous tissue, soft-tissue oedema, distension, HO
X-ray (TO)	Marked destruction, **periarticular erosions**, JSN, **osteophytes**, **subchondral sclerosis**, **loose bodies** — simulates OA, **absence of pain diagnostic**
Technetium bone scan	**Positive in both** neuropathic joint + osteomyelitis (high turnover, non-specific)
Indium WBC scan	**Cold (negative) in neuropathic joint** · **hot in osteomyelitis**
Sulfur colloid marrow / indium-111	Added to Tc for specificity
MRI (with + without contrast)	**Most sensitive for soft tissue infection + osteomyelitis** · **best for abscess vs soft-tissue swelling** · **cannot reliably separate Charcot from infection** (T2 / STIR oedema in both)
Ultrasound	Fluid collections; less specific
Biopsy	**Guides antibiotics** in osteomyelitis / abscess



Charcot ankle — destruction and fragmentation of the hindfoot (TO)

Diagnosis – elevation test, neuropathy & Charcot vs infection imaging

Elevation test	**erythema ↓ / resolves with elevation** (15–30 min, pillows, overnight) = **Charcot** · **unchanged in infection** (cellulitis, osteomyelitis, abscess) = **key differentiator**
Dependent rubor	redness only when leg dependent → Charcot
Semmes-Weinstein monofilament (5.07)	**confirms neuropathy** · **most sensitive test** · sensitivity **40–95%** · neuropathy has greatest effect on diabetic foot pathology ✗ Wrong options: Thompson · Cotton · syndesmosis squeeze · Babinski
Favours Charcot over infection	no ulcer / intact skin · **normal WBC + normal serum glucose** · no fever / sepsis · fracture without ulceration
Low-grade fever + normal CRP + erythema settles	still **neuropathic arthropathy** ✗ Wrong options: navicular stress fracture · osteomyelitis · embolic ischaemia · RA

– – Active space – –

Not useful as next step inflammatory markers (↑ in both) · MRI (low specificity) · Tc scan alone

Diagnostic criteria – Charcot vs infection

Feature	Charcot	Infection / osteomyelitis
Elevation	**Erythema decreases**	**Unchanged**
Skin	**Intact**	**Broken / ulcer**
WBC, glucose	Normal	**Leukocytosis, ↑ glucose**, fever
ESR / CRP	↑	↑
Tc bone scan	**+**	**+**
Indium WBC	**Cold**	**Hot**
MRI	Oedema	Oedema; **abscess**

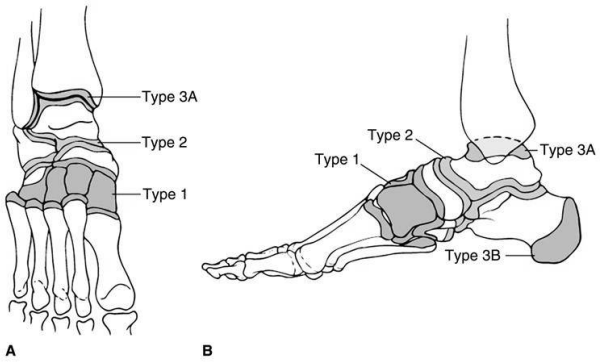
✗ Mimic	Distinguishing point
Cellulitis	Erythema persists on elevation · no radiographic destruction
CRPS	Aberrant inflammation, **no** Charcot X-ray changes
DVT	Swelling + **posterior calf tenderness**; erythema (if any) persists
Occult fracture	Typically **spinal cord injury** patients
Gout	**Normal uric acid** · no rapid bony destruction
Charcot-Marie-Tooth, Lisfranc injury, Freiberg	Different history / X-ray

Brodsky classification (anatomical)

Type	Joints	Feature	Frequency
1 (midfoot)	**TMT + naviculocuneiform**	**Fixed rocker-bottom with valgus** · **commonest**	**60%**
2 (hindfoot)	**Subtalar, TN, CC**	**Unstable**, immobilise **up to 2 yr** · 2nd commonest (TO)	**10%** (Orthobullets) · **30–35%** (TO) – read the options
3A	**Tibiotalar (ankle)**	Late varus / valgus → **malleolar ulcer + osteomyelitis**; significant deformity + disability	**20%**
3B	**Calcaneal tuberosity fracture**	Distal foot changes / **proximal migration of tuberosity**	< 10%
4	Combination	–	< 10%
5	**Forefoot only**	–	< 10%

Brodsky trap (TO) **"Type 3A = forefoot" is FALSE** → 3A = **ankle** · forefoot = type 5 (TO)

– – Active space – –



Brodsky classification – types 1, 2, 3A, 3B

Eichenholtz classification (staging)

Stage	Name	Clinical	X-ray	Treatment (TO)
0	–	Oedema, erythema, **loss of protective sensation**, instability	**Negative** (bone scan + in all stages)	**Frequent follow-up + serial X-rays**
1	**Fragmentation** (dissolution, acute)	**Warm, oedema, ↑ ligamentous laxity**	**Osteopenia, periarticular fragmentation, subluxation / dislocation**	**TCC**
2	**Coalescence** (subacute, reparative)	**↓ oedema + warmth**	**Coalescence**, absorption of fine debris, early fusion, sclerosis	**TCC → CROW**
3	**Reconstruction** (Orthobullets) · **consolidation** (TO) – chronic	**No inflammation**, stable, often deformed	**Consolidation + remodelling**, arthrosis, osteophytes	Plantigrade → **custom inlay shoes / CROW** · non-plantigrade / recurrent ulcer → **debridement ± arthrodesis**

Time to consolidation (TO) **forefoot 6 mo · midfoot 12 mo · hindfoot 18 mo** (TO)

5 Management

Charcot foot – goal: stable plantigrade foot in shoewear

- Acute (Eichenholtz 0–2), skin intact → **TOTAL CONTACT CAST**
  - then → CROW → **depth shoes + double rocker**
- Braceable, equinus + focal prominence / ulcer → **EXOSTECTOMY + TAL**
- Unbraceable / unstable deformity → **REDUCTION + ARTHRODESIS**
  (ring fixator if ulcer / poor bone · TTC nail for ankle + subtalar)
- Failed surgery / recurrent infection → **AMPUTATION**

– – Active space – –

Operative criteria

Procedure	Indication
TCC (nonop)	**First line** – acute Charcot **without skin breakdown** · **75% success**
Exostectomy + TAL	**"Braceable" foot** with **equinus** + **focal bony prominence** causing skin breakdown / ulcer failing TCC
Deformity correction, arthrodesis ± osteotomy	**Severe deformity not "braceable"** · unstable deformity · **persistent plantar ulcer** · **unstable Charcot ankle** · failed orthotics
External (ring) fixation	**Poor bone quality** or **compromised soft tissues / ulcer**
Amputation	**Failed previous surgery (unstable arthrodesis)** · **recurrent infection / osteomyelitis** · troublesome ulcers
Avoid surgery in acute stage	Not enough evidence for stage 1 surgery; avoid during inflammation (some authors propose early fixation)

Operative indications (Orthobullets)	recurrent infection · deformity · severe skin breakdown
Joint replacement	**relatively contraindicated** (TO)

Acute stage – total contact casting

First line / gold standard	**total contact cast (TCC)** + offloading for **acute (Eichenholtz 1–2) Charcot without ulcer**
Duration	cast changed **every 2–4 wk for 2–4 months** (Orthobullets) · **up to 4 months** (OB) · **weekly cast changes until erythema + swelling decrease** (OB) · **2–4 months** (TO) – read the options
Weight bearing	**NWB or limited WB = gold standard** until swelling resolves + consolidation on X-ray (✓ "TCC, no weight bearing, weekly cast changes") · **WB TCC effective** in early stages, no harm (De Souza, OB) · NWB **not conclusively necessary** (OB) – read the options
Alternative	pneumatic bracing
Medications	bisphosphonates · neuropathic pain drugs · antidepressants · topical anaesthetics
Stage 1 on cast, still warm at 2 months	**continue TCC** ✗ Wrong options: gout treatment (uric acid normal) · PT (aggravates stage 1) · ORIF of navicular · biopsy + ID referral

Goal in fragmentation **prevent further collapse + deformity**, avoid high-contact-stress ulceration

– – Active space – –

✗ Wrong first step	Why
IV antibiotics / admission	No infection
I&D / emergent debridement	No ulcer, **negative indium scan**, erythema settles
MRI, inflammatory markers, Tc scan alone	Don't change management (low specificity)
Midfoot / TN + TMT / subtalar / triple arthrodesis, ORIF, ring fixator	Only after failed nonop, gross deformity or ulceration
Plantar exostectomy	Not in acute phase
BKA / transtibial amputation	Salvage only
Shoe modification, custom orthotics (Jones bar, carbon shank, AFO leaf spring, plastizote insole)	Not for acute stage

Footwear & bracing after casting or partial amputation

CROW (TO) **Charcot restraint orthotic walker** · used **in Charcot joint** (TO)
✗ Wrong options: subtalar arthritis · ankle arthritis · gout of great toe

CROW design custom **bivalved total-contact AFO** · full foot enclosure · **rigid rocker sole** · custom insole · rigid polypropylene anterior + posterior shell · **DF stop**, heel lift · rocker sole counters **"nutcracker effect"** · Velcro (TO)

CROW use **after TCC** (oedema reduced first) · controls oedema · enables ambulation · **prevents deformity** during coalescence · **removable → hygiene, wound care, comfort**

CROW drawbacks **high cost** · bulky → poor compliance · **knee + lumbar pain** (TO)

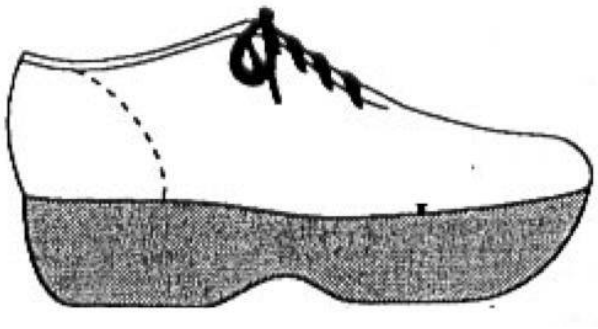


Charcot restraint orthotic walker (CROW)

Sequence **TCC → CROW → custom shoe / depth-inlay shoe with accommodative orthoses**

Eichenholtz 3 shoe **double rocker sole** → ↓ ulcer risk at **plantar apex / midfoot prominence**

– – Active space – –



Double rocker sole

Rocker sole	Use
Double rocker	**Plantar midfoot prominence (Charcot rocker-bottom)**
Severe-angle rocker	Pressure **distal to ball of foot** (toe-tip ulcers)
Mild rocker	Mild **MT head relief**, propulsion
Negative heel rocker	**Fixed dorsiflexion**, forefoot pressure → shifts load to midfoot + heel
Heel-to-toe rocker	**After ankle / subtalar fusion** · fixed lesser toe deformities

Partial 1st ray amputation (TA intact, 10° DF)

steel shank (flat 1-inch steel strip) → **extends foot lever, prevents deformity at toe break** ± rocker sole
✗ Wrong options: snug shoe (volume fluctuation) · custom shoe (unnecessary expense) · leather sole (no catching with 10° DF + intact TA) · lateral wedge

Chronic deformity & ulceration – exostectomy & reconstructive arthrodesis

Ulcer cause

malpositioned tarsal bones after joint collapse

Recurrent midfoot ulcer, failed TCC, stable deformity, equinus

exostectomy + Achilles tendon lengthening + protective brace · limb salvage **up to 90%**
✗ Wrong options: ex-fix · BKA (viable tissue) · observation · exostectomy alone (equinus needs lengthening)

Silfverskiöld test

positive → tight gastrocnemius (gastroc recession) vs whole Achilles complex (TAL)

Talar head ulcer (Chopart dislocation), failed 2 TCC cycles

reduction + arthrodesis of Chopart joint with ring fixator (no hardware near ulcer, soft tissue monitored)

✗ Option for talar head ulcer	Why wrong
CROW / repeat offloading	Failed already → contraindicated
Achilles lengthening alone	Doesn't reduce talar head
In-situ TTC with IM nail	TTC suits **Brodsky 2–3**, but **in-situ** fusion contraindicated — must **reduce**
Midfoot osteotomy + Lisfranc plates	For **Brodsky 1**; **plates contraindicated** with poor bone + ulcer

Charcot ankle + subtalar, failed bracing, no wounds, active

TTC fusion with retrograde nail (small incisions)
✗ Wrong options: anterior plate tibiotalar fusion (wound complications + misses subtalar) · BKA · TCC (for ulcers) · CAM orthosis

Fixation	internal: screws, pins, plates, **tibiocalcaneal nail** (internal may need **staging**) · **external fixation** (poor bone / soft tissue; usually **single stage**)
Post-op	**minimal WB for 3 months**

Amputation level (Chopart)

Forefoot gangrene with necrosis up to navicular (smoker, hypertensive) (TO)	**Chopart amputation** (TO) ✗ Wrong options: Pirogoff · Syme · Lisfranc
Chopart advantages (TO)	**conserves heel support** · technically easier · **shoe + AFO** instead of knee-high prosthesis · little shortening · **heel skin = weight-bearing end** · easier muscle balancing (TO)
Equinus (TO)	common cause of foot amputation failure · Chopart → **tibialis anterior transfer proximally + tendo-Achilles lengthening** · Syme → only TA lengthening, slight equinus ruins ambulation (TO)
Goal	partial / limited amputation if vascularity allows · BKA for severe deformity with soft-tissue involvement or failed reconstruction

6 Complications

Complication	Fact
Arthrodesis complications	**Up to 70%** – hardware failure, infection, malposition, **recurrent ulceration**, fracture
Overall complication rate	**50–56%** (Sämann)
Recurrence (secondary episodes)	**Type 1 DM 41.6% vs type 2 18.3%**
Amputation rate	**3–9%** – depends on avoiding ulceration (TO)
Late complications	Chronic ulceration → infection → **osteomyelitis**
Ankle (Brodsky 3A)	**Malleolar ulcer + osteomyelitis**

X vs Y

	Technetium bone scan	Indium WBC scan	MRI
Charcot	**Hot**	**Cold**	Oedema
Osteomyelitis	**Hot**	**Hot**	Oedema / abscess
Key fact	Sensitive, non-specific	**Differentiates**	**Most sensitive**, can't separate

	TCC	CROW
Stage	**Acute, first line**	**After TCC**
Removable	No	**Yes** (hygiene, wounds)
Downside	Cast changes	**Cost**, bulk

– – Active space – –

	Exostectomy + TAL	Reconstructive arthrodesis
Foot	**Braceable**, stable, focal prominence	**Unbraceable / unstable**
Fixation	–	Ring fixator (ulcer) · TTC nail (ankle)

How it's asked

"Diabetic, painless ankle swelling, prior ray amputation, X-ray destruction?"	→ **Charcot arthropathy** (TO)
"Charcot foot in undiagnosed patient – also evaluate?"	→ **diabetes mellitus**
"Floppy ankle, erythema resolves with elevation – exam?"	→ **Semmes-Weinstein monofilament testing**
"Charcot knee – possible causes EXCEPT?"	→ **Reiter's syndrome**
"Midfoot erythema diminishes with 15-min elevation?"	→ **neuropathic arthropathy**
"Type 1 diabetic, crepitus after twist, hyperaemia ↓ with elevation?"	→ **Charcot arthropathy**
"Diagnosis + associated finding?"	→ **Charcot arthropathy; erythema decreases after elevation**
"Most sensitive imaging for soft tissue infection?"	→ **MRI with and without contrast**
"Redness gone after 30 min elevation?"	→ **Charcot arthropathy**
"Infection vs neuropathic – true?"	→ **indium WBC scan negative (cold) for neuropathic joints**
"Erythema resolves on pillows, type 1 DM?"	→ **Charcot arthropathy**
"Charcot foot, 6-mo gait disturbance – initial management?"	→ **total contact casting**
"Swelling settles overnight, fragmentation on X-ray?"	→ **total contact cast and non-weight bearing**
"Normal indium WBC scan – next?"	→ **total contact casting**
"Red foot, ESR 40, normal WBC + glucose, fracture on MRI?"	→ **TCC, no weight bearing, weekly cast changes until erythema and swelling decrease**
"Stage 1, still warm after 2 months in cast?"	→ **continued total contact cast immobilization**
"Diabetic + prior stroke, red foot settles with elevation?"	→ **total contact casting**
"HbA1c 8.5%, dependent rubor – initial?"	→ **total contact casting**
"Recurrent midfoot ulcer, failed TCC, equinus?"	→ **exostectomy & Achilles tendon lengthening with protective brace**
"Talar head ulcer after 2 TCC cycles?"	→ **reduction and arthrodesis of the Chopart joint using a ring fixator**
"Charcot ankle + subtalar, failed bracing, active golfer?"	→ **tibiotalar and subtalar fusion with retrograde TTC nail**

"CROW orthosis used in?"	→ **Charcot joint** (TO)
"Eichenholtz 3 after TCC – shoe to prevent breakdown?"	→ **double rocker sole** (Figure D)
"Partial 1st ray amputation – shoe?"	→ **steel shank to extend the foot lever**
"Forefoot gangrene to navicular – level?"	→ **Chopart** (TO)
"Brodsky – NOT true?"	→ **Type 3A represents forefoot changes** (TO)
"Commonest cause of upper-limb Charcot?"	→ **syringomyelia** (TO)
"Key Charcot markers EXCEPT?"	→ **transforming growth factor-beta (TGF-β)** (TO)

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DIABETIC FOOT ULCERS

– Foot & Ankle · 16 / 48 –

1 Basics

Epidemiology

Incidence	**~12% of diabetics** have foot ulcers · **commonest medical complication** making diabetics seek treatment
Amputation	ulcers cause **~85% of lower-extremity amputations** · diabetes = **15–40× ↑ amputation risk** · **diabetic foot ulceration = most likely predictor of eventual amputation**
PVD	**60–70%** of patients with diabetes > 10 yr · **> 60% of diabetic ulcers** have ↓ blood flow
Diabetic foot infection	cause of amputation in **25–50%** · osteomyelitis ↑ amputation risk **up to 4×**
Charcot	**7.5% of neuropathic diabetics** · neuropathy in 30–50% of diabetics, neuropathic fracture only 1–2.5%

Pathophysiology

Neuropathy	**largest effect** · **sensory → loss of protective sensation = primary risk factor** for ulcer · **autonomic → dry skin** (no glandular function) · motor → intrinsic dysfunction → dynamic toe deformity
Angiopathy	**lesser effect** than neuropathy
Mechanical	**equinus → forefoot (MT head) overload** · ↓ MTPJ motion → ↑ IP pressure on hallux · **peroneus longus overpull → plantarflexed 1st MT** → 1st MT head ulcer
Three pillars of treatment	**adequate blood supply** · **treat infection** · **pressure relief**

Risk factors for ulceration & amputation (neuropathy, smoking)

Most likely to ulcerate	**cannot feel 5.07 Semmes-Weinstein monofilament** – vascular + nutrition tests predict **healing**, not ulcer formation ✗ Wrong options: TcpO2 30 · ABI < 0.6 · albumin 3.1 · autonomic dry skin · Silfverskiöld result
Concurrent variables in amputees	**ulceration 84%** · neuropathy 64% · infection 59% · gangrene 55% · ischaemia 46% · + ↑ HbA1c, ↓ oxygen tension, ↓ ABI
Least likely to predict amputation	**ABI 1.07** (normal) ✗ Wrong options: ulcer · monofilament loss · infection · HbA1c 10.7
Only consistent modifiable risk factors	**smoking + glycaemic control** (Rossboth)
Single best programme to cut amputations	**smoking cessation with counselling + pharmacotherapy** ✗ Wrong options: annual plantar-pressure screening + custom footwear · mandatory TCC for all · routine arterial US · GLP-1 weight-loss clinic
Infection risk factors	ulcer **> 30 days** · recurrent ulcer · trauma · **probe to bone** · PAD

Microbiology

Usually polymicrobial — **aerobic gram-positive cocci (S. aureus) commonest · gram-negatives ↑ in chronic wounds / recent antibiotics · obligate anaerobes with ischaemia / gangrene**

Gold standard — **deep cultures + bone biopsy** guide management

3 Presentation

Symptoms — often **painless**

Exam — **depth (probe for bone)** · infection (cellulitis, pus, gangrene) · **Silfverskiöld** · **DP + PT pulses** · hair on toes (absent = poor flow)

Signs of infection blunted — ↓ WBC function, PAD, poor metabolic control, neuropathy · commonest signs = **↑ exudate + worsening glycaemic control** · ↑ WBC / ESR / CRP / fever **absent in 50%**

Ulcer location	Driver	Treatment
Plantar MT heads (1st / 2nd)	**Equinus**	**TCC + TAL / gastrocnemius recession**
1st MT head	Equinus **+ PL overpull**	**Gastrocsoleus recession + PL → PB transfer**
Plantar hallux IP joint	**Stiff / rigid MTPJ**	**Keller resection arthroplasty**
Toe tip (apical)	**Flexible claw / hammer toe**	**Percutaneous flexor tenotomy**
Plantar midfoot	**Charcot rocker-bottom**	**Exostectomy + TAL**
Heel	Bed-bound pressure · calcaneal osteomyelitis	Offloading / padding boots · **partial calcanectomy + TAL**
Medial malleolus	**Venous stasis** (rarely diabetic)	–
Dry toe gangrene	**Arterial insufficiency**	Toe disarticulation

Silfverskiöld test

Technique — **forefoot inverted, hindfoot subtalar neutral** (locks transverse tarsal joints) · compare passive DF **knee flexed vs extended**

Finding	Contracture	Procedure
DF improves with knee flexion	**Isolated gastrocnemius**	**Gastrocnemius recession (Strayer)**
DF equal with knee flexed + extended	**Gastrocsoleus / Achilles**	**Achilles tendon lengthening (Hoke)**

– – Active space – –



Silfverskiöld test – ankle dorsiflexion with knee extended vs flexed

4 Diagnosis & Classification

Diagnostic criteria

Condition	Criterion
Neuropathy (TO)	**10 g (5.07 Semmes-Weinstein) monofilament** not perceived **all 3 times** (TO)
Osteomyelitis	**Bone biopsy with cultures + pathology = gold standard** · **exposed bone → presume osteomyelitis**
Infected ulcer	↑ exudate · ↑ WBC / ESR / CRP · worsening glucose · erythema **persisting despite elevation**
Healing potential	**TcpO2 = gold standard** (see labs)

Monofilament technique (TO)

10 g monofilament · eyes closed · **90° to skin**, pressed **until it buckles 1 cm** · patient says "yes" · **repeat 3 times** (TO)
✗ Wrong options: 1 g · 5 g · 1 g polyfilament

Monofilament accuracy

sensitivity **41–93% / 57–93%**, specificity **68–100% / 75–100%** · **3-site test: plantar great toe, 3rd MT, 5th MT** (Feng)

Genes & lab markers – healing predictors (vascular studies & nutritional labs)

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Marker	Favours healing	Poor healing / risk
TcpO2 (transcutaneous oxygen)	**> 30 mmHg** (or **40**, by source) – **gold standard** for healing potential	**< 30–40** · **< 20** · **15**, 24, 25 mmHg = poor
ABI	**> 0.45** (Orthobullets, most OB) · **> 0.60** to heal an ulcer (one OB) – read the options	**< 0.45** · **falsely ↑ by arterial calcification** (↓ compliance)
Toe pressure	**> 40 mmHg** (OB) · **> 45 amputation, > 60 ulcer** (Orthobullets)	**< 30 mmHg** (amputation wound risk)
Albumin	**> 3.0 g/dL**	**< 3.0** (most) · **< 3.5** (one OB) – read the options
Total lymphocyte count	**> 1,500 /mm³**	**< 1,500**
Haemoglobin (amputation)	**> 11.0 g/dL**	< 11
Total protein (amputation)	**> 6.2 g/dL**	< 6.2
HbA1c	–	**> 8.0**
Doppler / exam	**Triphasic waveforms**, hair on toes	Absent pulses, no hair
Systolic BP > 140	**Not predictive**	–

Next test with absent pulses ABI + TcpO2 (noninvasive vascular) · with **palpable pulses → Silfverskiöld**

Pre-TMA labs – which is the risk? **albumin 2.9 g/dL**
✗ Wrong options: Hb 12.5 · total protein 6.9 · TLC 1,700 · toe pressure 55

Key radiology findings

Study	Use / sign
X-ray (AP, lateral, oblique foot + ankle)	Demineralisation · **periarticular erosions** · **cortical erosions = late, nonspecific** (confused with Charcot) · abnormal film LR+ 2.3
MRI	**Best for abscess vs soft-tissue swelling** · **most sensitive for osteomyelitis** (↑ T2 marrow signal) · LR+ 3.8 / LR– 0.14, accuracy 89% · **hard to separate infection vs Charcot** · less accurate in **ischaemic** ulcers
Bone scan (Tc-99m, Ga-67, In-111)	Separates **soft-tissue infection vs osteomyelitis vs Charcot** · **↑ indium-labelled WBC = osteomyelitis** · **leukocyte scintigraphy** best for infection vs Charcot
Charcot X-ray	**Midfoot destruction, fragmentation, osteopenia, subluxation** · rocker-bottom

Midfoot plantar ulcer, metformin + gabapentin – X-ray? **Charcot midfoot**
✗ Wrong options: Lisfranc · hallux valgus · CMT cavus · flatfoot without fragmentation

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Osteomyelitis – bone biopsy diagnosis

Finding	Value
Probe to bone	**67%** osteomyelitis (Orthobullets) · **~70%** / nearly ⅔ (OB) · **most specific clinical finding** · sens **66%**, spec **85%**, PPV 89%, NPV 56% · LR+ 6.4
Exposed bone	Highly predictive – presume osteomyelitis
Ulcer > 2 cm²	LR+ 7.2
ESR > 70 mm/h	LR+ **11**
Bone biopsy (culture + histology)	**Most accurate / most specific / gold standard**
Wound / swab culture	**Positive at all stages → not predictive** of osteomyelitis
CBC	Unhelpful (immunocompromise, vascular insufficiency)
CRP / ESR	Helpful, **not diagnostic**

No single history / exam finding excludes osteomyelitis

✗ Biopsy traps HbA1c > 15% · ESR + CRP · probe-to-bone · cortical erosions · CBC · technetium scan · WB radiographs

Wagner classification & monofilament screening (TO)

Grade	Orthobullets	TO table	Treatment
0	Skin intact, bony deformity – "**foot at risk**"	No open lesion; may have deformity or cellulitis	**Shoe modification + serial exams**
1	**Superficial** ulcer	Superficial, **partial or full thickness** (not exposing ligament, deep fascia, bone)	**Office debridement + TCC**
2	**Deeper, full-thickness** extension	To **ligament, tendon, joint capsule, deep fascia**; **no abscess / osteomyelitis**	**Operative debridement + TCC**
3	**Deep abscess or osteomyelitis**	+ **joint sepsis**	**Operative debridement + TCC** (I&D + antibiotics before casting)
4	**Partial forefoot gangrene**	Gangrene of portion of forefoot **or heel**	**Local vs larger amputation**
5	**Extensive gangrene**	Entire foot	**Amputation**

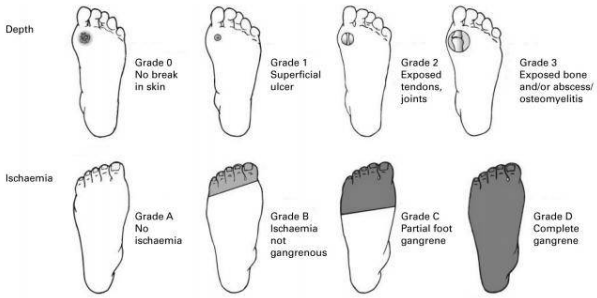
Exposed tendons, no bone / abscess (TO) **Wagner grade 2** (TO)
✗ Wrong options: 1 · 3 · 4

TO wording conflict TO text: ulcers "breaking through superficial fascia… with **exposed bone or tendon** = grade 2" · TO table + Orthobullets: **abscess / osteomyelitis = grade 3** – read the options

Brodsky depth-ischaemia classification

Depth	Definition	Treatment
0	At-risk foot, no ulcer	Education, accommodative footwear, regular exams (**shoe modification**)
1	Superficial, not infected	**Offload – TCC**, walking brace, special footwear
2	**Deep, exposing tendons / joints**	Surgical debridement, wound care, offload, culture-specific antibiotics
3	**Extensive ulcer or abscess** (exposed bone, osteomyelitis)	Debridement / partial amputation, offload, culture-specific antibiotics

Ischaemia	Definition	Treatment
A	Not ischaemic	–
B	Ischaemia, no gangrene	Noninvasive vascular testing → **angioplasty / bypass**
C	**Partial forefoot gangrene**	Vascular reconstruction + **partial foot amputation**
D	**Complete gangrene**	Full vascular work-up + **major amputation**



Brodsky depth-ischaemia classification

Mycetoma (Madura foot) (TO)

Definition (TO) — **chronic granulomatous infection of skin + subcutaneous tissue** · fungi (**eumycetoma**) or bacteria (**actinomycetoma**) · **bacteria ~60%** (TO)

Commonest organism (TO) — **Actinomadura madurae** (TO)
✗ Wrong options: Pseudallescheria boydii · A. pelletieri · Madurella mycetomatis

Actinomycetes (bacteria) (TO)	Eumycetes (fungi) (TO)
Actinomadura madurae · Actinomadura pelletieri · Streptomyces somaliensis · Nocardia	Pseudallescheria boydii · Madurella mycetomatis

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Madura foot – swollen deformed foot with sinuses

5 Management

Diabetic foot ulcer

- Infected / abscess / osteomyelitis → I&D + bone culture → **culture-specific abx**
  - infection cleared → **TCC ± TAL**
- Clean plantar ulcer, perfused → **wound care + TCC**
  - forefoot + equinus → **add TAL / gastrocnemius recession**
- Ischaemic → vascular studies → **revascularise**
- Gangrene → **amputation (level by PT pulse + labs)**

Forefoot ulcer with limited dorsiflexion

- Pulses absent → **ABI / TcpO2 first**
- DF improves with knee flexion → **gastrocnemius recession (Strayer)**
- No improvement with knee flexion → **Achilles tendon lengthening (Hoke)**

Operative criteria

Procedure	Indication
I&D + antibiotics before casting	**Wagner ≥ 3** · deep full-thickness ulcer failing TCC
Flexor tendon tenotomy	**Flexible toe deformity + toe-tip ulcer** (high healing **if no osteomyelitis**)
TAL / gastrocnemius recession	**Plantar forefoot ulcer + equinus** (DF ≤ 5°)
Ostectomy ± TAL	**Bony prominence** causing internal pressure (TAL if tight Achilles)
Partial calcanectomy ± TAL	**Large heel ulcer + calcaneal osteomyelitis**
Keller resection arthroplasty	**Plantar hallux IPJ ulcer + stiff MTPJ, failed TCC**
Deformity correction + TAL	**Nonplantigrade, rigid Charcot deformity** with recurrent ulcers
Syme amputation	**Forefoot gangrene + palpable posterior tibial pulse**

Total contact casting, offloading & NPWT

Wound care	**first line** · moist environment · absorb exudate · barrier · offload
TCC	**gold standard for mechanical relief of plantar ulcers** · snug below-knee · protects **insensate** limb, keeps patient ambulatory · lower infection, admission, lost income
TCC contraindications	**absolute: infection** · relative: marginal arterial supply, poor compliance with cast care, cast claustrophobia
TCC duration	changed every **3–4 wk**, prevents daily wound care · total **several months (up to 4 months)** · 83% heal, 17% complications (skin irritation) ✗ Wrong options: ≤ 4 weeks
Recurrence after TCC	typically **3–4 wk after removal** = **greatest risk within 1 month** (newly healed tissue weak to vertical + shear stress) · most within first 6 months
Sequence	**TCC (non-removable) → CROW (removable) → custom shoe / orthotic**
Clean 3-month plantar ulcer, no bone, normal labs	**total contact casting** ✗ Wrong options: TCC + empiric IV antibiotics · debridement + antibiotics · CROW
Infected heel ulcer – NOT indicated	**total contact casting** · indicated: surgical debridement, tighter glycaemic control, imaging (MRI), culture-directed antibiotics
✗ TCC traps	replace weekly with active infection · cast ≤ 4 wk · avoid in insensate feet · custom insoles equally effective (orthoses **prevent**, TCC **treats**) · orthotics > TCC (TCC heals faster, compliance)
Non-removable > removable	non-removable casts heal better than removable casts / dressings · **pneumatic walker** = alternative, better wound surveillance, but **removable → compliance**; no evidence of superiority
NPWT	**↓ ulcer size + time to healing** · for perfused, uninfected, offloaded ulcers failing standard care / deep after debridement · granulation, ↓ oedema, removes exudate, ↑ local flow ✗ Wrong options: custom shoes · electrophysical therapy · hyperbaric oxygen (mixed) · topical PDGF
Without offloading	debridement or NPWT alone **won't** heal · VAC does **not** reduce recurrence

Achilles / gastrocnemius lengthening for forefoot ulcers (Silfverskiöld)

Rationale	equinus → cyclic forefoot overload → MT head ulcers · percutaneous TAL **↓ forefoot peak pressure ~25%**
Mueller RCT (TCC ± TAL, DF ≤ 5°)	healing **100% vs 88%** (no difference) · recurrence **↓ 75% at 7 mo, ↓ 52% at 2 yr** · **81% vs 38%** recurrence at 2 yr
Lengthening effect	**↓ recurrence**, not faster healing of active ulcers vs TCC alone · Colen **25% vs 2%** recurrence (94% RRR) – avoid over-lengthening
TAL + TCC vs TCC alone	**lower forefoot ulcer recurrence at 2 years** (✗ Wrong options: higher / same · ↑ BKA) · **↑ heel peak pressure → more heel-pad breakdown** (✗ Wrong options: lower)
Static calf stretching	**no effect** on ROM or plantar pressure

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Gastrocnemius recession (Strayer) benefits most — **chronic plantar forefoot ulcer**
✗ Wrong options: peroneal tendinitis · Freiberg · retrocalcaneal bursitis · calcaneal osteomyelitis

Scenario	Answer	✗ Traps
Forefoot ulcer, no bone, no infection, **palpable pulses** – what next?	**Silfverskiöld test**	MRI, ABI, TcpO2, HbA1c
Wagner 1, **failed TCC 8 mo**, gastrocnemius contracture	**Strayer procedure**	Integra + STSG, STSG, more TCC, Weil osteotomy
Bilateral **5th MT head** ulcers 2 yr, minimal DF	**Gastrocnemius release + wound care**	Plastazote cut-out, HBO + NWB, healing shoe, **TMA (salvage)**
1st MT head grade II, failed orthosis + TCC, no deformity	**Gastrocsoleus recession + PL → PB transfer**	Plastazote, 1st MT head resection, ray / TMA
DF 10° short, **no change with knee flexion**, failed 3 mo TCC	**Achilles lengthening**	3 more months TCC, debridement, NPWT, MT head excision
MT head ulcer – heals + ↓ recurrence	**Achilles lengthening procedure**	Keep HbA1c 9.2, TcpO2 24, **narrow toe box**, flexor tenotomy
Progressed to toe gangrene, no Silfverskiöld change – early step?	**Achilles tendon lengthening**	FDL tenotomy, **isolated gastrocnemius recession**, Keller, pneumatic walker first
Healed recurrent forefoot ulcer, rigid equinus	**Ostectomy + Achilles tendon lengthening**	Ostectomy + gastrocnemius recession, revascularisation, TCC in neutral

Midfoot / forefoot gastrocsoleus recession — 10/11 midfoot + 19/20 forefoot healed · add **PL lengthening (1st MT head)** or **tib post lengthening (5th MT head)** (Laborde)

Osteomyelitis – treatment

Highest success — **IV antibiotics based on percutaneous bone biopsy culture sensitivity** · only factor affecting remission · remission only **64% at 1 yr**
✗ Wrong options: empiric IV vancomycin · oral levofloxacin · ulcer-swab sensitivities · elevation + NWB

Deep full-thickness ulcer, failed 2 mo TCC, erythema persists despite elevation — **intra-operative debridement + irrigation with culture** → then antibiotics + casting
✗ Wrong options: continue TCC · 6 wk empiric IV · office wound culture (nonspecific) · skin graft

Surgery — **remove infected + necrotic bone** · pathogens **polymicrobial or S. aureus**

Greatest predictor of failure → amputation — **presence of osteomyelitis** (e.g. sesamoid) · nonop sesamoid osteomyelitis → **15/18 needed surgery**
✗ Wrong options: first ulcer (first ulcers heal better) · HbA1c 7.2 · TcpO2 48 · TAL

Surgical offloading – exostectomy, deformity correction, phalangeal resection, partial calcanectomy

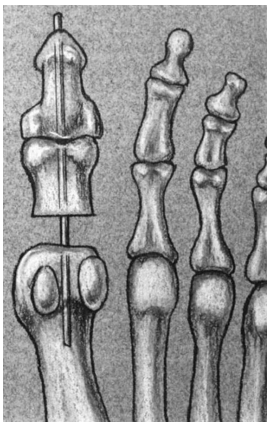
Wagner 3 heel ulcer + calcaneal osteomyelitis, failed debridement + TCC — **partial calcanectomy** (+ bone biopsy, debridement, antibiotics) · preserves limb length, ↓ morbidity vs Syme / BKA
✗ Wrong options: ankle disarticulation · BKA · fasciocutaneous / free flap (only for uninfected ulcers)

Partial calcanectomy technique	remove posterior process; cut **1 cm posterior to subtalar + CC joints** · **Achilles released, retracts** · custom-moulded orthosis · 10/12 healed



Partial calcanectomy — line of resection

Charcot plantar midfoot ulcer, recurrent 2 yr, no osteomyelitis	**midfoot exostectomy + Achilles tendon lengthening** (+ debridement, TCC) ✗ Wrong options: CROW · VAC · in-situ midfoot / subtalar fusion (none ↓ recurrence)
Nonplantigrade rigid Charcot, ulcer under talar head	**surgical correction of deformity + TAL + therapeutic footwear** · recurrence likely unless corrected ✗ Wrong options: depth-inlay shoes + orthosis · CROW · TAL + TCC + CROW · Syme
Charcot surgery goal	**stable, plantigrade, ulcer-free foot** · ulcer + deep infection → **15% major amputation**
Hallux IP plantar ulcer, rigid MTPJ, failed 3× TCC	**Keller: resect base of proximal phalanx** → restores MTPJ motion, ↓ IP pressure · K-wire keeps plantigrade toe, **5 mm MTP gap**, prevents cock-up ✗ Wrong options: CROW · distal phalanx base · proximal phalanx head · MT head



Keller resection arthroplasty with K-wire

Toe-tip ulcers – flexor tenotomy

Indication	**apical toe ulcer + passively correctable claw / hammer toe** · dynamic **flexor stabilisation** (toes curl at heel lift) linked to apical ulcers · prophylactic with callus
Outcome	healing **> 95%** · complications **< 15%** · incision wound complications **< 5%** · **office / outpatient**, percutaneous needle

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Known complication	**similar ulcer on adjacent toe (transfer lesion) 8%** → adjacent tenotomy
✗ Tenotomy traps	> 50% wound complications · **MT head ulcers** (persist → gastrocnemius recession) · osteomyelitis irrelevant (failures mostly had osteomyelitis) · needs inpatient stay

Footwear & orthoses (therapeutic shoe bill, rocker sole, depth-inlay, Plastazote)

Shoe modification	**prevention** (Brodsky 0 / Wagner 0) · deep / wide shoes, custom insoles, rocker soles · wide toe box + rocker → better healing, ↓ recurrence
Best forefoot offloading shoe	**rocker sole** ✗ Wrong options: polyethylene foam insole · open-toe sandals · custom in-depth shoes · hard post-op shoe
Healed first Wagner II ulcer + hammertoe, insensate	moderate risk → **depth-inlay shoes + custom accommodative foot orthosis** ✗ Wrong options: nothing until second ulcer · Oxford rubber-sole shoes · DF MT osteotomy · TAL
Plastazote	**cross-linked polyethylene** · **surface area distribution → ↓ peak pressure** · **heating + moulding ↓ properties** (stiffer) ✗ Wrong options: ↑ stiffness · no cross-linking · ↑ peak pressure

Therapeutic Shoe Bill (Medicare, per calendar year)	
Option A	**1 pair custom-moulded shoes** (+ its inserts) **+ 2 extra pairs of inserts**
Option B	**1 pair extra-depth shoes** (≥ 3/16 inch) **+ 3 pairs of inserts**
Modifications covered	Rocker soles, MT bars, **wedges**, flared heels, **Velcro closures**, **inserts for missing toes**
Not covered	**Two pairs of custom-moulded shoes**

Amputation level selection

Forefoot gangrene + palpable PT pulse, not septic	**Syme amputation** → highest function ✗ Wrong options: TMA (too near infected tissue, only to clear infection first) · BKA · AKA · debridement + antibiotics
Syme	**ankle disarticulation, malleoli removed, heel pad anchored** · heel pad fed by **PT artery** branches · needs **patent PT artery + stable heel pad** · success **85%** with palpable PT pulse vs 1/4 without · **90%** with triphasic / normal PT vs **57%**
Sepsis resolved, albumin 1.9, ABI 0.70	**local wound care, parenteral antibiotics, metabolic support, re-evaluate in 1 week** · albumin **≥ 2.5 g/dL** → good Syme healing ✗ Wrong options: bypass · Syme now · guillotine / closed transtibial
Most likely to fail + need revision	**transmetatarsal amputation** in uncontrolled diabetic (**HbA1c > 8**) with forefoot ulcer · **HbA1c = main predictor of TMA success**, then need for debridement ✗ Wrong options: long / short transtibial · transfemoral · Syme
Surgical site infection	complicated diabetics **7.25×** vs non-diabetics

6 Complications

– – Active space – –

Complication	Fact
Ulcer recurrence after TCC	**3–4 wk** after cast removal
Heel-pad breakdown	After TAL (↑ heel pressure)
Transfer lesion	Adjacent toe after flexor tenotomy (8%)
TMA failure	HbA1c > 8, persistent forefoot ulcer
Amputation	Infection 25–50% · osteomyelitis ↑ 4× · Charcot ulcer + infection 15%
Partial calcanectomy	Needs custom orthosis to cushion heel

X vs Y

	Osteomyelitis / infection	Charcot arthropathy
Elevation	Erythema **persists**	Erythema **abates**
Inflammatory markers	↑	Usually normal
MRI	Excellent for osteomyelitis	Hard to distinguish
Best separator	**Leukocyte (indium WBC) scintigraphy**	–
Definitive	**Bone biopsy**	Bone biopsy if doubt

	Gastrocnemius recession (Strayer)	Achilles lengthening (Hoke)
Silfverskiöld	**DF ↑ with knee flexion**	**No change**
Contracture	Isolated gastrocnemius	Gastrocsoleus complex

	TCC	CROW / pneumatic walker	Custom orthoses / shoes
Role	**Treat** active ulcer (gold standard)	After TCC / alternative	**Prevent** ulcer
Removable	No	**Yes** (compliance ↓)	Yes
Recurrence	↓ only with TAL	doesn't ↓	Moderate-risk prevention

How it's asked

"Diabetic most likely to develop a foot ulcer?"	**→ unable to feel the 5.07 Semmes-Weinstein monofilament on the plantar foot**
"Highest risk of developing the wound shown?"	**→ absent plantar sensation with 5.07-gauge monofilament testing**
"Least likely to predict amputation?"	**→ ankle-brachial index of 1.07**
"One programme to reduce amputations?"	**→ smoking cessation with counselling and pharmacotherapy**

– – Active space – –

"Recommended neuropathy screen for touch?"	→ **10 gm monofilament** (TO)
"Exposed tendons, no bone or abscess — Wagner grade?"	→ **grade 2** (TO)
"Commonest organism of Madura foot?"	→ **Actinomadura madurae** (TO)
"Good healing predictor EXCEPT?"	→ **TcpO2 of 15 mm Hg**
"NOT predictive of poor healing?"	→ **systolic blood pressure > 140 mmHg**
"Pre-TMA lab that raises wound-healing risk?"	→ **albumin of 2.9 g/dL**
"Septic patient stabilised, albumin 1.9 — best next?"	→ **local wound care, parenteral antibiotics, metabolic support, reevaluation in 1 week**
"Clean 3-month plantar ulcer, normal labs — treatment?"	→ **total contact casting**
"Midfoot ulcer treated in TCC — radiograph?"	→ **Figure C, Charcot midfoot**
"Infected heel ulcer — NOT indicated?"	→ **total contact casting**
"True of TCC?"	→ **ulcers at greatest risk of recurring within 1 month of total contact cast discontinuation**
"Offloaded, uninfected ulcer not shrinking — proven adjunct?"	→ **negative pressure wound therapy**
"Silfverskiöld positive — which condition benefits most from Strayer?"	→ **chronic plantar forefoot ulcer**
"Forefoot ulcer, palpable pulses — next information?"	→ **results of Silfverskiöld test**
"Failed TCC, gastrocnemius contracture — next?"	→ **Strayer procedure (gastrocnemius lengthening)**
"5th MT head ulcers for 2 years, minimal DF — best?"	→ **a gastrocnemius release and supportive wound care**
"Recurrent 1st MT head ulcer after orthosis + TCC?"	→ **gastrocsoleus recession and peroneus longus to brevis tendon transfer**
"TCC + percutaneous TAL vs TCC alone?"	→ **lower rate of neuropathic plantar forefoot ulcer recurrence at two years**
"Failed TCC, DF short of neutral not improving with knee flexion?"	→ **Achilles lengthening**
"Heals ulcer + ↓ recurrence (MT head ulcer)?"	→ **performing an Achilles lengthening procedure**
"Could have prevented progression to gangrene?"	→ **Achilles tendon lengthening**
"Healed recurrent forefoot ulcer, Silfverskiöld negative — keep plantigrade?"	→ **ostectomy and achilles tendon lengthening**
"2nd MT head osteomyelitis — highest success?"	→ **IV antibiotics based on percutaneous bone biopsy culture sensitivity**

"Strong osteomyelitis indicator EXCEPT?"	→ positive wound culture
"Deep ulcer failing TCC, erythema despite elevation — next?"	→ intra-operative debridement and irrigation with culture
"Metatarsal head visible on probing — best test for bone involvement?"	→ bone biopsy
"Most specific for osteomyelitis?"	→ bone biopsy with cultures and pathology
"Sesamoid osteomyelitis treated nonop — predictor of failure?"	→ presence of osteomyelitis in his sesamoids
"Wagner 3 heel ulcer + calcaneal osteomyelitis — surgery?"	→ partial calcanectomy
"Charcot midfoot ulcer, no osteomyelitis — prevent recurrence?"	→ midfoot exostectomy and Achilles tendon lengthening
"Hallux IP ulcer, stiff MTPJ, failed TCC?"	→ resect the base of the proximal phalanx
"Rigid nonplantigrade Charcot, healed talar-head ulcer — prevent recurrence?"	→ surgical correction of the deformity, Achilles tendon lengthening, and therapeutic footwear
"Forefoot gangrene, palpable PT pulse — definitive, best function?"	→ Syme amputation
"Level most likely to fail, HbA1c 8.5 smoker?"	→ transmetatarsal amputation
"Not covered yearly by the Therapeutic Shoe Bill?"	→ two pairs of custom-molded shoes
"Shoe modification to cut forefoot pressure?"	→ rocker sole shoes
"Healed first Wagner II ulcer + hammertoe — next?"	→ depth-inlay shoes with a custom accommodative foot orthosis
"Why Plastazote?"	→ surface area distribution
"True of flexor tenotomy for toe ulcers?"	→ patients can develop a similar ulceration to an adjacent toe after tenotomy
"Which ulcer suits percutaneous flexor tenotomy?"	→ Figure A, apical toe ulcer

– – Active space – –

FHL TENDONITIS & INJURIES

– Foot & Ankle · 17 / 48 –

1 Basics

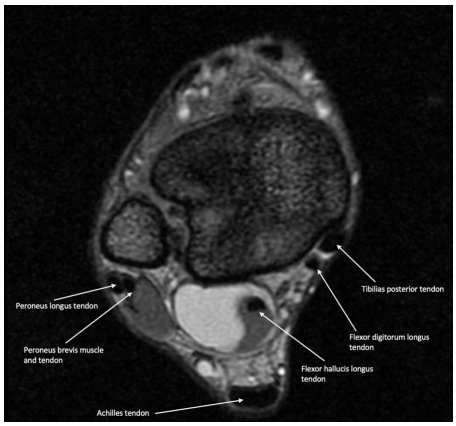
Epidemiology

Site	**posterior ankle** (behind talus, fibro-osseous tunnel) · great toe
Risk factor	**excessive / maximal plantarflexion → ballet dancers en pointe / demi-pointe**, **gymnasts**
Dancers	FHL tendinitis = **common cause of posteromedial (medial-sided) ankle pain** · dancers **endure symptoms longer** before seeking care than non-dancers
Associated conditions	**posterior ankle impingement (PAIS)** · **os trigonum** (posterolateral tubercle)

Diagnosis in dancers & FHL anatomy

Origin	**inferior 2/3 of posterior surface of fibula + lower part of interosseous membrane**
Course	between **posteromedial + posterolateral tubercles of talus** in a **fibro-osseous tunnel → beneath sustentaculum tali →** crosses **dorsal to FDL at knot of Henry** (FHL "higher", FDL "down") **→** stays dorsal to FDL + NV bundle · multiple FHL–FDL connections
Insertion	plantar base of **distal phalanx of great toe**
Action	primary: **plantarflexion of hallux IP + MTP** · secondary: ankle plantarflexion
Os trigonum link	FHL runs **directly medial to lateral tubercle of posterior talar process →** coexistent tendinopathy ✗ Wrong options: peroneus brevis / longus · FDL · tibialis posterior (not as close)
Os trigonum	unfused **secondary ossification centre of lateral tubercle of posterior talar process** · or united **stress fracture** of that tubercle

Origin (as in the options)	Muscle
Inferior 2/3 posterior fibula + lower interosseous membrane	**FHL** ✓
Anterior fibula + adjacent interosseous membrane	EHL (wrong option)
Cuboid + lateral cuneiform	FHB (wrong option – doesn't cross ankle)
Fibular head + upper 1/2–2/3 lateral fibula + septa	Peroneus longus (wrong option)
Inferior 2/3 lateral fibula + septa	Peroneus brevis (wrong option – subfibular pain with eversion + dorsiflexion)



Labelled axial T2 MRI – fluid in the FHL sheath behind the talus

2 Mechanism

Mechanism	activities with **maximal plantarflexion → impingement of FHL** along posterior ankle → tendinitis → even **rupture**
Pathoanatomy	posterior to talus, **within fibro-osseous tunnel** (posteromedial ankle) · or secondary to **os trigonum**
Chronic	**nodule formation → triggering** of great toe
PAIS	**scarring + tendinopathy of FHL** · Achilles in vicinity but **remains normal**
FHL laceration	direct acute trauma · **acute laceration = most common form of injury**

3 Presentation

Pain	**posteromedial ankle** · worse **en pointe / demi-pointe** ✗ Wrong options: posterolateral = os trigonum
Hallux	**triggering**, **locking**, **clicking / catching**, **snapping behind medial malleolus** with great-toe motion
Key sign	**painful crepitus of tendon with passive motion of great toe** (passive hallux motion moves inflamed FHL at ankle)
Other signs	pain with **resisted flexion of hallux IP** (resisted great-toe plantarflexion) · **↓ passive extension of great toe** · posteromedial tenderness, fullness, crepitus · **no tenderness at 1st MT head**
Forced ankle plantarflexion	painful in FHL tendinitis (Orthobullets, OB PAIS stem) · **no pain** on forced passive PF in a classic FHL stem; forced PF pain = **posterior impingement** (OB) – read the options
Tib post working	**heel inverts on toe rise** → excludes PTT
Laceration	**loss of active IP joint flexion**

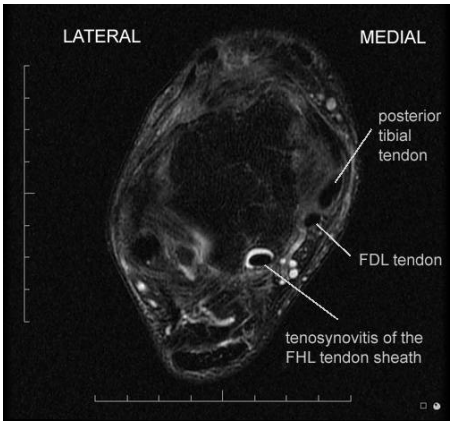
Exam finding	Diagnosis
Painful crepitus with passive great-toe motion	**FHL tendinitis** ✓
Pain on percussion of posterior tibial nerve	Tarsal tunnel (wrong)
Posterior ankle pain with forced passive plantarflexion	Posterior impingement (wrong)
Medial pain with resisted inversion	Posterior tibial tendinitis (wrong)
Foot drop, weak dorsiflexion	Tibialis anterior rupture (wrong)

– – Active space – –

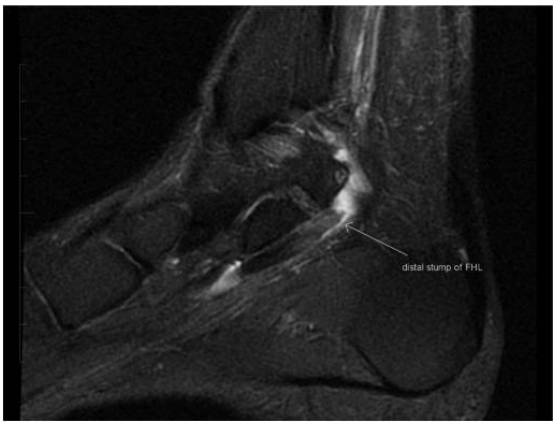
4 Diagnosis & Classification

Key radiology findings

Study	Finding
MRI (axial T2)	**Fluid around FHL tendon at level of ankle joint** = **tenosynovitis of FHL sheath** as it wraps posteromedial ankle · intrasubstance signal
MRI – laceration	**Retracted tendon ends** · fluid in FHL sheath
X-ray / bone scan / MRI in ballet student	**Os trigonum** → look for associated **FHL** pathology



Axial MRI – tenosynovitis of the FHL sheath, medial to the talus



Sagittal MRI – FHL laceration with retracted ends

5 Management

FHL tendinitis

- First line → **rest, ice, NSAIDs, activity modification, arch support, PT**
  (no steroid injection – posterior tibial nerve close)
- Fails nonop (e.g. 6 mo) → **release FHL from fibro-osseous tunnel at ankle**
  (arthroscopic or open posteromedial; tenosynovectomy ± repair)
- Acute laceration of FHL + FHB → **acute surgical repair**

Operative criteria

– – Active space – –

Indication	Procedure
Recalcitrant symptoms · athletes with symptoms despite rest + nonop	**Release of FHL from fibro-osseous tunnel**, tenosynovectomy, ± tendon repair
Lacerations of both FHL and FHB	**Acute surgical repair**

Treatment – FHL decompression

Next step after failed 6 mo nonop — **decompression of FHL at the level of the ankle**
✗ Wrong options: at metatarsal (ankle level commoner than distal tarsal tunnel in midfoot) · peroneal decompression (lateral pain) · 1st MT osteotomy (hallux valgus) · gastrocnemius recession (isolated gastroc equinus)

Steroid injection — **contraindicated** (posterior tibial nerve close by)

Compression — contracture of **medial retinaculum** over FHL → posteromedial pain

Outcomes — **> 90% good / excellent in dancers** · good in dancers + non-dancers · professional dancers do better than amateurs (Hamilton)

X vs Y

	FHL tendinitis	Os trigonum / posterior impingement
Pain	**Posteromedial**	**Posterolateral**
Hallux motion	**Painful, clicking, triggering, crepitus**	Not painful
Forced passive PF	± (see conflict above)	**Reproduces posterior pain**
History	Insidious, en pointe	After ankle sprain / PF injury (os trigonum fracture)

Posterior / medial ankle structure	Why wrong in a dancer's PAIS stem
FHL	✓ impinged in fibro-osseous tunnel
Tibialis posterior	Heel inverts on toe rise = functioning
Achilles	In vicinity, remains normal
EHL, tibialis anterior	Opposite (dorsal) side
Tibial nerve	Tarsal tunnel (Tinel)

How it's asked

"Dancer, posteromedial pain in demi-pointe, fluid around FHL on MRI – exam finding?" — → **painful crepitus with passive great-toe motion**
✗ Wrong options: tibial nerve percussion · forced passive PF · resisted inversion · foot drop

"Ballet dancer, snapping behind medial malleolus + great-toe triggering – impinged structure?" — → **flexor hallucis longus tendon**
✗ Wrong options: PTT · tibial nerve

– – Active space – –

"Ballet student, os trigonum on X-ray / bone scan / MRI – tendon most often involved?"	**→ flexor hallucis longus**
"En pointe, posteromedial tenderness, no pain on forced PF, pain with hallux motion?"	**→ FHL tendinitis** ✗ Wrong options: painful os trigonum · knot of Henry stricture · FDL tendinitis
"En pointe ballerina, posteromedial crepitus, heel inverts on toe rise – structure?"	**→ flexor hallucis longus tendon** ✗ Wrong options: tib post · Achilles
"Origin of the implicated muscle (fluid in FHL sheath)?"	**→ inferior 2/3 posterior surface of fibula + lower interosseous membrane**
"Dancer, failed 6 months of rest / NSAIDs / arch support – next step?"	**→ decompression of FHL at the level of the ankle**
"Loss of active hallux IP flexion after a cut?"	**→ FHL laceration → acute repair**

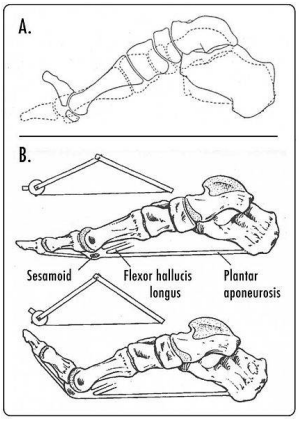
FOOT ANATOMY AND BIOMECHANICS

– Foot & Ankle · 18 / 48 –

1 Basics

Plantar fascia

Origin	**medial calcaneal tuberosity**
Insertion	**plantar plate + bases of all 5 proximal phalanges** · lateral band → **5th MT base**
Function	**windlass mechanism**: toes dorsiflex at toe-off → fascia tightens → **↑ arch height, foot shortens** · **2nd most important medial arch support**
Load transfer	**plantar aponeurosis = primary structure of force transfer hindfoot → forefoot in stance** · tension peaks in **late stance**, up to **96% body weight** (OB) ✗ Wrong options: Achilles · Lisfranc ligament · tibialis posterior · tibialis anterior



Windlass mechanism – toe dorsiflexion tightens the plantar fascia and raises the arch

Hindfoot – subtalar joint

Facets	**posterior = largest** · **middle = medial, on sustentaculum** · **anterior = continuous with talonavicular joint**
Motion	**inversion / eversion** only · **no role in plantar / dorsiflexion**

Transverse tarsal (Chopart) joint

Components	**talonavicular + calcaneocuboid**
Talonavicular support	**spring ligament** = **superomedial + inferior calcaneonavicular ligaments**
Calcaneocuboid	**saddle-shaped** · plantar support = **superficial + deep inferior calcaneocuboid ligaments** · superior support = **lateral limb of bifurcate ligament**

– – Active space – –

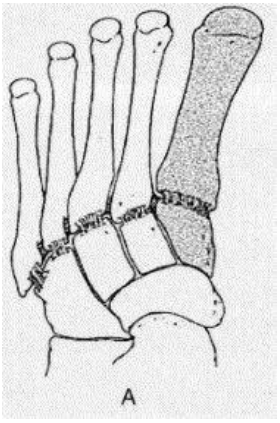
Midfoot – Lisfranc (TMT) joint

Articulations 1st–3rd metatarsocuneiform + 4th–5th metatarsocuboid · naviculocuneiform + intercuneiform joints = dense ligaments, **little motion**

Column	Bones	Mobility / role
Medial	**1st MT, medial cuneiform, navicular**	**Carries most load in standing**
Middle	**2nd + 3rd MT, middle + lateral cuneiforms**	**Least mobile** → rigidity for push-off
Lateral	**4th + 5th MT, cuboid**	**Most mobile** → uneven ground

TMT ligament layer	Strength
Interosseous (contains **Lisfranc ligament**: **plantar medial cuneiform → base of 2nd MT**)	**Strongest**
Plantar	Next strongest
Dorsal	**Weakest**

Transverse Roman arch dorsal surface wider than plantar · **2nd MT base = keystone**



Medial column of the foot

Forefoot

1st MT **shortest + widest** · takes **50% of weight during gait** (Orthobullets)

2nd MT **longest** · **most stress during gait → commonest MT stress fracture**

Which MT bears most weight walking? **1st MT 50% in gait** (Orthobullets) · **standing = 1st MT, walking = 2nd MT** (TO) – read the options

Transverse metatarsal ligament **holds hallucal sesamoids in place** as 1st MT head drifts medially → **false image of sesamoid subluxation** (head moves, not sesamoids)

Adductor hallucis conjoined tendon lateral aspect of **lateral sesamoid** + lateral base of proximal phalanx

2nd MTP plantar plate dense phalangeosesamoidal complex · must become **lax before dorsal translation** of proximal phalanx · once attenuated, main deforming force = **EDL** · **weakest at origin (metatarsal neck)**

Seymour fracture (hallux distal phalanx) remove nail + interposed physeal tissue, repair nail bed, splint in extension, **antibiotics**

Intrinsic muscle function

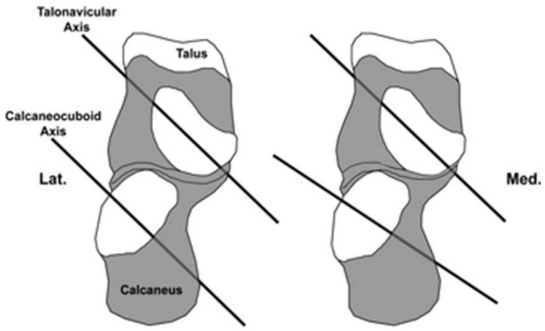
Foot intrinsics — **flex MTP joints + extend IP joints**
✗ Wrong options: extend MTP + flex IP · flex both · extend both

2 Mechanism

Subtalar & transverse tarsal joint axes in gait

Phase	Subtalar / calcaneus	Chopart axes	Foot
Heel strike	**Eversion**	**Parallel → unlocked**	**Supple** → shock absorption, accommodates ground
Push-off / toe-off	**Inversion**	**Divergent → locked**	**Rigid lever arm** for propulsion

✗ Gait traps — eversion **unlocks** (✗ Wrong options: locks) · calcaneus **everted** at heel strike (✗ Wrong options: inverted throughout stance) · **tibialis anterior contracts eccentrically** in stance (✗ Wrong options: concentrically) · push-off foot is **rigid** via inversion (✗ Wrong options: flexible via eversion)



Subtalar eversion – parallel Chopart axes; inversion – divergent axes

Windlass mechanism & push-off

Flexible → rigid foot at toe-off — (1) **tightening of plantar fascia** · (2) **progressive external rotation of lower extremity** (pelvis → ankle → subtalar) · (3) **subtalar inversion locks transverse tarsal joint**

Does NOT occur at push-off — **loosening of the spring ligament**
✗ Wrong options: subtalar inversion · Chopart locking · external rotation · plantar fascia tightening (all do occur)

Flatfoot causes (OB) — triceps surae insufficiency, obesity, PTTD, laxity of **spring ligament** or other ligaments

– – Active space – –

X vs Y

	Heel strike	Push-off
Subtalar	Eversion	**Inversion**
Transverse tarsal	**Unlocked** (parallel axes)	**Locked**
Limb rotation	–	**External rotation**
Plantar fascia	–	**Tight** (windlass)

	1st MT	2nd MT
Size	**Shortest, widest**	**Longest**
Load	**Standing / 50% in gait** (Orthobullets)	**Walking** (TO) · most stress
Classic	Sesamoids	**Stress fracture**, keystone, Lisfranc ligament

How it's asked

"Primary site of force transfer hindfoot → forefoot in stance?"	→ **plantar aponeurosis**
"NOT at normal push-off?"	→ **loosening of spring ligament**
"Subtalar–transverse tarsal relationship in gait?"	→ **Chopart axes parallel at heel strike**
"Eversion of subtalar joint does what to Chopart?"	→ **unlocks** (flexible)
"Tibialis anterior in stance?"	→ **eccentric contraction**
"Intrinsic muscles act on toes by?"	→ **flexing MTP + extending IP joints**
"Metatarsal bearing most weight during walking?"	→ **2nd** (TO)
"Lisfranc ligament runs?"	→ **plantar medial cuneiform → base of 2nd MT**
"Strongest TMT ligament layer?"	→ **interosseous**
"Least mobile column?"	→ **middle**

FOOT MUSCLE FORCES & DEFORMITIES

– Foot & Ankle · 19 / 48 –

1 Basics

Simple deformities – strong vs weak muscle

Deformity	Strong (deforming)	Weak
Equinus	**Gastrocnemius–soleus**	Dorsiflexors
Cavus	**Plantar fascia, intrinsics**	Dorsiflexors
Varus	**Tibialis posterior, tibialis anterior**	**Peroneus brevis**
Supination	**Tibialis anterior**	**Peroneus longus**
Flatfoot	**Peroneus brevis**	**Tibialis posterior**

Complex deformities

Deformity	Strong	Weak
Equinovarus + supination	Gastroc–soleus, tib post, tib anterior	**PB + PL**
Equinovalgus	Gastroc–soleus, **peroneals**	**Tib post, tib anterior**
Calcaneovalgus	Dorsiflexors / evertors (**L4, L5**)	Plantarflexors / invertors (**S1, S2**)

Muscle antagonists & tendon transfers in cavovarus (PL→PB, EHL transfer)

Muscle	Antagonist / force couple	Fact
Tibialis posterior	**Peroneus brevis = primary antagonist**	Inverts, secondarily plantarflexes
Tibialis anterior	**Peroneus longus**	TA secondarily inverts → only **partial** antagonist of TP · **TA eccentrically contracts after heel strike**
Peroneus longus	Tibialis anterior	**Plantarflexes the first ray**
Peroneus brevis	Tibialis posterior	**Primary evertor**; does **not** contribute to cavus

Cavus driver **weak tibialis anterior + strong peroneus longus → plantarflexed first metatarsal**

Nerve	Motor supply
Tibial	**Tibialis posterior**, long toe flexors (deep + superficial posterior compartments)
Deep peroneal	Anterior compartment: **TA, EDL, EHL, peroneus tertius**
Superficial peroneal	Lateral compartment: **PL, PB**
Sural, saphenous	**No motor** (sensory only)

– – Active space – –

DPN / SPN / L5 root injury	anterior + lateral compartment weakness → **unopposed tib post + long toe flexors** → hindfoot + forefoot **varus**
Tendon transfer requirements	adequate **strength + excursion** · insertion close to substituted tendon, routed in **direct line** · **intact nerve + blood supply** · **functional (supple) joints** · donor must be **innervated** ✗ Wrong options: TA / peroneals in CPN palsy

3 Presentation

Foot drop

Definition	inability to dorsiflex ankle ± toes · **commonest cause = peroneal nerve palsy**
Causes	central (brain, cord, roots) · peripheral (sciatic, peroneal) · **traumatic (knee dislocation, laceration, blunt)** · compressive (mass, deformity correction) · systemic (**diabetic polyneuropathy**, mononeuropathy) · iatrogenic (casting, positioning, surgery) · mechanical (muscle debridement, tumour excision) · **anterior compartment syndrome** (devitalised muscle) · **CMT** · polio

Nerve	Motor loss	Sensory loss
DPN	**Ankle / toe dorsiflexion**	**First dorsal web space**
SPN	**Ankle eversion**	**Lateral leg + dorsal foot**

Gait	**foot slaps floor at heel strike** = tibialis anterior weakness · **toes extend in swing** = extensors substituting for weak TA
Knee-level injury	lateral leg sensory loss + ↓ eversion + ↓ dorsiflexion

Equinovarus after stroke

Commonest ankle–foot deformity after stroke	**equinovarus**
Equinus	overactive **gastrocnemius–soleus** · **Varus** — overactive **tibialis anterior** (+ tib post, FHL, FDL)
Other features	clonus, intrinsic-minus foot, supple claw toes, compensatory **knee hyperextension**, premature triceps surae firing

4 Diagnosis & Classification

Silfverskiöld test & gastrocnemius recession indications

Test	ankle dorsiflexion with **knee extended**, then **knee flexed** (hindfoot locked)
Why it works	**gastrocnemius crosses the knee** (origin posterior femur) → relaxes with knee flexion · **soleus does not**



Silfverskiöld test: dorsiflexion with knee extended then flexed

Diagnostic criteria

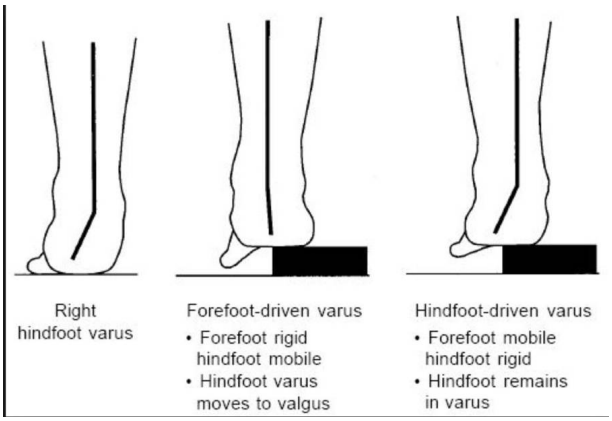
Silfverskiöld result	Diagnosis	Surgery
↑ **DF with knee flexed** (e.g. −10° ext → +10° flex; −5° → 10°; 0° → 15°; 3° → 16°)	**Isolated gastrocnemius contracture** (positive test)	**Gastrocnemius recession (Strayer)**
Same DF knee flexed + extended, can't pass neutral in either	**Achilles (gastroc-soleus) contracture**	**Achilles lengthening** (Hoke triple-cut, percutaneous, open) or gastroc + soleus recession
DF past neutral in both positions	**Normal**	–
✗ Wrong option: DF better with knee extended	Opposite of positive test	–

Thresholds	gastroc contracture if **DF < 10° with knee extended**, improving with knee flexion (OB) · **difference > 10° = significant** (OB) · **shy of neutral → past neutral** with knee flexion (OB) – read the options
Not a Silfverskiöld finding	calf squeeze ± plantarflexion = **Thompson test** (Achilles rupture) · resting ankle position irrelevant
Other exams (wrong options)	**Ober** = IT band · **Dial** = knee posterolateral corner · **Thessaly** = meniscus · **Thomas** = hip flexion contracture · **Coleman** = hindfoot flexibility
Gastroc contracture causes	↑ **forefoot strike** → ↑ plantar tension → **plantar fasciitis**, **arch collapse**, **Achilles tendinopathy**, **metatarsalgia** (plantar callus under MT heads), **neuromas**
Lateral head of gastrocnemius	muscle tested by the Silfverskiöld test

Coleman block test – forefoot-driven cavovarus

Method	block under **lateral border (lateral column / hindfoot)**, first ray allowed to drop off
Heel corrects to neutral / slight valgus	**flexible subtalar / hindfoot** → varus is **forefoot-driven** = **plantarflexed first ray** (valgus forefoot) ✗ Wrong options: dorsiflexed first ray · varus forefoot · rigid flatfoot
Heel stays in varus	**hindfoot-driven (rigid hindfoot)** → needs calcaneal osteotomy

– – Active space – –



Coleman block test: forefoot-driven vs hindfoot-driven varus

Subtle cavus associations	**lateral column overload**, recurrent ankle sprains / functional ankle instability, **5th MT stress fracture**
5th MT stress fracture (esp. chronic / recurrent)	look for **hindfoot varus deformity** ✗ Wrong options: lateral ligament instability · talocalcaneal coalition (causes flatfoot, protects 5th MT) · PB rupture · anterior process fracture

5 Management

Equinus

- Silfverskiöld positive (gastroc only) → **gastrocnemius recession (Strayer)**
- Fixed in both knee positions → **Achilles lengthening (TAL)**

Cavovarus

- Supple, Coleman corrects → dorsiflexion 1st MT osteotomy (± PL → **PB)**
- Coleman does not correct → **+ lateralising calcaneal (Dwyer) osteotomy**
- Rigid / degenerative → **triple arthrodesis**

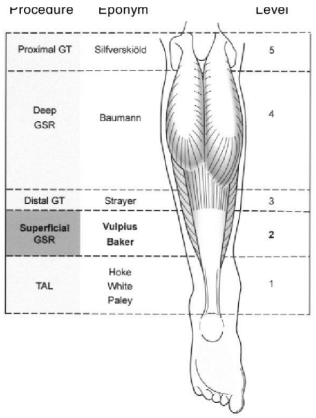
Chronic flexible foot drop

- < 6–12 mo, nerve may recover → **AFO + observation + therapy**
- Acute laceration → **nerve repair, graft or transfer**
- No recovery, supple joints → **tib post transfer to dorsum (lateral cuneiform)**
  - Equinus contracture → **+ gastroc / Achilles lengthening**

Gastrocnemius recession vs Achilles lengthening

Strayer — **fascial release of gastrocnemius** (distal GT) for recession + gentle lengthening · after **failed nonop**
✗ Wrong options: failed PT alone is not an indication (try orthotics / injections)

Level	Procedure	Eponym
5	Proximal gastrocnemius tenotomy	Silfverskiöld
4	Deep gastroc–soleus recession	Baumann
3	**Distal gastrocnemius tenotomy**	**Strayer**
2	Superficial gastroc–soleus recession	Vulpius, Baker
1	**TAL**	**Hoke (triple cut)**, White, Paley



Levels of gastrocnemius–soleus lengthening

Sural nerve — at risk with gastrocnemius intramuscular aponeurotic recession

Plantar fasciitis + positive Silfverskiöld — **gastrocnemius recession**
✗ Wrong options: unchanged test · 3 mo failed PT · flexible planovalgus · too-many-toes

Cavovarus + positive Silfverskiöld (0° → 15°) — **gastrocnemius recession**
✗ Wrong options: heel cord / percutaneous / open TAL · gastroc + soleus recession (those for combined contracture)

Runner, callus under 1st MT head, −5° → 10° — **Strayer**
✗ Wrong options: FHL → Achilles (chronic Achilles rupture / degeneration) · Hoke TAL · 1st MT extension osteotomy (cavus with flexible hindfoot) · posterior capsule release

Metatarsalgia 2nd–4th, normal X-ray, 3° → 16° — **gastrocnemius recession**
✗ Wrong options: Weil osteotomies · PTT debridement + MDCO · triple-cut TAL · MTP steroid

Stage IIA AAFD after MDCO + FDL → navicular — gastroc recession if **can't DF above neutral with knee extended but can past neutral with knee flexed**

Isolated Achilles lengthening — **can't DF past neutral with knee extended or flexed**
✗ Wrong options: CP gastroc-only → gastroc lengthening · runner anterior impingement → arthroscopic debridement · foot drop → tendon transfer · normal Silfverskiöld

– – Active space – –

Cavovarus correction

Flexible forefoot-driven (Coleman corrects)	**dorsiflexion first metatarsal osteotomy** → hindfoot resumes slight valgus, **avoids calcaneal osteotomy** ✗ Wrong options: triple / subtalar arthrodesis (degenerative only) · PB lengthening · medialising calcaneal osteotomy (flexible flatfoot)
Coleman corrects + Silfverskiöld unchanged (−5° both)	**first ray dorsiflexion osteotomy + Achilles lengthening** ✗ Wrong options: gastroc recession · calcaneal osteotomy · triple arthrodesis (deformity supple)
Rearfoot orthotic	won't correct a forefoot cause · may still need **lateralising calcaneal osteotomy**
Subtle cavus, lateral column overload, mild instability, no joint-line pain	**PT + cavus orthotic: lateral forefoot posting + first metatarsal head recessing** (another OB: **lateral hindfoot posting** + 1st MT head recess) ✗ Wrong options: MRI (no joint-line pain) · boot (stress fracture) · NSAIDs + lace-up (ignores lateral column) · arch support + **medial posting – contraindicated** in cavus
Decreases plantarflexion force on 1st MT head	**peroneus longus → peroneus brevis transfer** (also improves eversion when PB weak / torn)

Transfer	Indication
PL → PB	**↓ plantarflexed 1st MT** (cavovarus), weak PB / chronic PB tear
Tib post → dorsum (lateral cuneiform)	**Weak dorsiflexion / foot drop**
SPLATT	**Equinovarus**, excessive **forefoot inversion in swing**, spastic TA
FDL → tib post / navicular	**PTT dysfunction** (flexible flatfoot)
FDL → EDL	**Flexible hammer / claw toes**
EHL → 1st MT neck (Jones)	**Hallux claw / hyperextended 1st MTP**
FHL → Achilles	Chronic Achilles rupture / degeneration
TA → talus dorsum	**Midfoot amputation** with firing TA (prevents equinus)

Jones procedure	**hallux IP fusion + EHL transfer to 1st MT neck** for **hyperextended 1st MTP** in cavus ✗ Wrong options: FHL, EHB, EDL, TA (wrong length / direction)
Polio, flexible cavus, claw toes, equinus, foot slap	soft tissue only: **plantar fascia release + Achilles lengthening + EHL → 1st MT neck + tib post → dorsum + FDL → EDL** ✗ Wrong options: calcaneal / metatarsal osteotomies (flexible) · triple arthrodesis (young)

Foot drop – tibialis posterior transfer to dorsum

Nonop	**AFO** + observation + therapy (stretch, keep joints supple) · AFO downsides: ↓ ankle ROM, discomfort
Wait	if no return in **6–12 mo** → transfer · **no dorsiflexion 1 yr after nerve repair → assume no recovery**
Operation	**tibialis posterior (tibial nerve) → through interosseous membrane → lateral cuneiform / dorsum of foot** (or circumtibial) ± **gastroc / Achilles lengthening** · converts deforming invertor into **strongest available dorsiflexor** · ↓ plantarflexion + inversion torque

– – Active space – –

CPN laceration, best single-tendon transfer (MRI label B)	**tibialis posterior** ✗ Wrong options: A = TA · E = peroneals (non-functional) · C = FDL (adjunct for toe extension only) · D = Achilles (transfer not described; lengthen it instead)



Axial T1 MRI ankle: B = tibialis posterior, C = FDL, A = tibialis anterior, E = peroneals, D = Achilles

Combined PTT + FHL + FDL transfer (Movahedi)	foot drop correction excellent **60%**, toe extension excellent **47%**
Equinovarus 13 mo after MVC, flexible	**tib post → lateral midfoot + Achilles lengthening** ✗ Wrong options: ankle / subtalar / pantalar arthrodesis (fixed deformity or end-stage arthritis only) · SPLATT (TA not working)
CMT, no TA / PB, 5° equinus, TP + FDL 5/5	**gastrocnemius lengthening + tib post → dorsum** ✗ Wrong options: PB → dorsum · FDL → dorsum · ankle fusion · triple (supple)
Knee dislocation, CPN transection, no recovery at 1 yr	**tib post → dorsum** ✗ Wrong options: sural cable graft (less reliable) · PL → TA · EHL → TA (peroneal-innervated) · ankle fusion in young
Anterior compartment syndrome, DF 0/5, eversion 4/5, Silfverskiöld unchanged at 20° DF	**tib post through interosseous membrane to dorsum** ✗ Wrong options: PL → PB (eversion only) · Achilles lengthening + AFO (no fixed equinus) · **FHL → navicular — inadequate excursion + strength** · TA → talus (TA dead)

Spastic hemiplegia – bracing & SPLATT

Timing after stroke	**AFO + PT for ≥ 6 mo** (Orthobullets, Miller) · **18–24 mo** before reconstruction (Younger & Hansen) — read the options
Early (3 mo after stroke)	**AFO + physical therapy** (± phenol / Botox, stretching, ROM) ✗ Wrong options: Strayer · SPLATT · FHL transfer + toe flexor release · tib post fractional lengthening (all reasonable later)
Surgical options (after recovery plateau)	**SPLATT** · **FHL → dorsum of foot + release FDL / FDB at each toe base** · gastrocnemius / Achilles lengthening
Dynamic varus, TA active in swing + stance, weak evertors, clonus, equinus, 2 yr	**percutaneous Achilles lengthening + split TA transfer to lateral cuneiform** ✗ Wrong options: TAL alone · **out-of-phase tib post transfer** or **Bridle** (TA still active, risk **planovalgus**) · toe flexor tenotomy
TAL in spasticity	restores DF + **↓ clonus** (stretch response) · if DF still inadequate → **tib post lengthening**

– – Active space – –

Flatfoot – FDL transfer & orthotic posting

Supple pes planoabductovalgus, can't single-leg heel raise	bony surgery + **FDL tendon transfer** (similar dynamic function to PTT) · alternatives FHL, PB ✗ Wrong options: EHL · PL repair · **spring ligament release – worsens flatfoot**; repair / reconstruction is OK · lateral ligament reconstruction
Flexible planovalgus + lateral hindfoot (impingement) pain	**medial hindfoot posting / wedge** = simplest offload ✗ Wrong options: lateral hindfoot posting ↑ lateral pressure · rigid / semi-rigid / accommodative orthotic alone (foot still pronates)

X vs Y

	Gastrocnemius contracture	Achilles contracture
Silfverskiöld	**DF improves with knee flexion**	**No change**
Treatment	**Gastroc recession (Strayer)**	**TAL (Hoke, percutaneous, open)**

	Tib post transfer to dorsum	SPLATT
Problem	**Flaccid foot drop** (CPN palsy, CMT, compartment syndrome)	**Spastic** equinovarus (stroke, TA overactive)
TA	**Dead**	**Active / overactive**
Insertion	Lateral cuneiform / dorsum via interosseous membrane	Lateral cuneiform

	Forefoot-driven cavovarus	Hindfoot-driven cavovarus
Coleman	**Heel corrects**	Heel stays varus
Surgery	**Dorsiflexion 1st MT osteotomy**	+ **calcaneal osteotomy**

How it's asked

"Test in figure (knee extended vs flexed)?"	**→ Silfverskiöld – gastrocnemius tightness vs Achilles contracture**
"Indication for gastroc recession in plantar fasciitis?"	**→ 10 degrees less than neutral with knee in extension versus 10 degrees beyond neutral with knee flexed**
"Cavovarus, DF 0° → 15° with knee flexion – Achilles management?"	**→ gastrocnemius recession**
"Runner, 1st MT callus, −5° → 10°?"	**→ Strayer procedure**
"Test for structure at white arrow (lateral gastrocnemius)?"	**→ Silfverskiöld test**
"Gastroc recession after stage IIA flatfoot correction when?"	**→ can't DF above neutral with knee extended, can with knee flexed**
"Metatarsalgia, 3° → 16°, failed orthotics?"	**→ gastrocnemius recession**
"Who benefits from isolated Achilles lengthening?"	**→ can't DF past neutral with knee extended or flexed**

"Coleman corrects, Silfverskiöld unchanged — procedures?"	→ first ray dorsiflexion osteotomy + Achilles lengthening
"Coleman block correction proves varus due to?"	→ plantar flexed first ray
"Heel corrects to valgus on Coleman — procedure?"	→ dorsiflexion first metatarsal osteotomy
"Associated pathology with 5th MT stress fracture?"	→ hindfoot varus deformity
"Subtle cavus, lateral midfoot pain, recurrent sprains?"	→ PT + orthotic with lateral forefoot posting + first metatarsal head recessing
"CPN laceration, best single tendon (MRI label)?"	→ B — tibialis posterior
"Transfer for anterior + lateral compartment deficit — true?"	→ antagonist is peroneus brevis
"Flexible post-traumatic equinovarus, no DF / eversion?"	→ tib post transfer to lateral midfoot + Achilles lengthening
"CMT foot drop, supple, 5° equinus?"	→ gastrocnemius lengthening + tib post to dorsum
"No DF 1 yr after CPN repair?"	→ tibialis posterior transfer to dorsum of foot
"Nerve to the transferred muscle in foot drop?"	→ tibial nerve
"Foot drop after compartment syndrome — restore DF?"	→ posterior tibialis transfer through interosseous membrane to dorsum
"Polio, foot slap, claw toes, flexible cavus?"	→ plantar fascia release, TAL, EHL → 1st MT neck, tib post → dorsum, FDL → EDL
"Hyperextended 1st MTP after cavus reconstruction — transfer?"	→ extensor hallucis longus (Jones)
"Transfer that ↓ plantarflexion force on 1st MT head?"	→ peroneus longus to peroneus brevis
"Peroneus brevis is primary antagonist of?"	→ posterior tibialis
"Stroke 3 months ago, spastic TA varus?"	→ AFO with physical therapy
"Spastic hemiplegia 2 yr, TA active swing + stance, weak evertors?"	→ percutaneous TAL + split TA transfer to lateral cuneiform
"Soft-tissue procedure with bony flatfoot surgery?"	→ flexor digitorum longus transfer
"Simplest shoe change for lateral hindfoot pain in planovalgus?"	→ medial hindfoot posting

– – Active space – –

FOOT PUNCTURE WOUNDS

– Foot & Ankle · 20 / 48 –

1 Basics

Epidemiology

Setting	common **workplace** injury (construction sites) · stepping on a **nail** or stick through sock / sole
Commonest object	**nails** (TO) · also needles, glass, wood, plastic, sharp metal (TO) ✗ Wrong options: needles commonest
Infection	**~10%** · **osteomyelitis 1–2%** (penetration to cartilage / periosteum) (TO)
Complications (TO)	soft-tissue infection · deep abscess · **pyogenic granuloma** · **epidermal inclusion cyst** · osteomyelitis

Risk factors for infection (TO)

Wound (TO)	presentation **> 6 h** · location · depth · contamination of object · **retained foreign body**
Host (TO)	elderly · immunocompromised · PVD · **diabetes = most significant risk factor** → often osteomyelitis

Microbiology – S. aureus vs Pseudomonas (rubber-soled shoes)

Infection	Commonest organism	Features asked
Soft tissue (cellulitis, abscess)	**Staphylococcus aureus** (Staph + Strep species)	**Coagulase positive, haemolytic gram-positive cocci** · commonest in diabetics + non-diabetics · S. aureus + group A strep in children
Osteomyelitis (non-diabetic)	**Pseudomonas aeruginosa**	**Gram-negative rod, single polar flagellum** (motile) · **secretes blue-green pigment in vitro** · grape-like odour · biofilm, exotoxin, moist environments
Through **rubber sole of shoe**	**Pseudomonas** (TO)	Grows in **moist inner layers of a used athletic shoe** (not new / very old) → inoculated with rubber
Diabetic osteomyelitis	S. aureus more common · **polymicrobial**	Diabetics → more osteomyelitis, more organisms
Bodies of water (TO)	Pseudomonas, **Aeromonas hydrophila**, **Vibrio**, **Mycobacterium marinum**	–
Toothpick (TO)	**Eikenella** (oropharyngeal flora)	–

Pseudomonas ✗ traps no flagellum · spore-forming · ferments lactose · clear colonies (colonies **blue-green** on nutrient agar)

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✗ Distractor organism	Description	Classic association
Staph epidermidis	Coagulase-negative, non-haemolytic GPC	**Periprosthetic joint infection**
Salmonella	GN rod, **multiple (peritrichous) flagella**	**Osteomyelitis in sickle cell**
Clostridium tetani	Spore-forming GP rod, multiple flagella, anaerobe	**Tetanospasmin** blocks glycine + GABA release
Pasteurella multocida	–	**Animal bites**
Eikenella corrodens	–	**Human bites** (OB) · toothpick (TO)
Vibrio	–	Marine environments

3 Presentation

Timing often **weeks to a month** after injury · limp · fever, drainage

Exam swollen, tender foot · obvious wound ± tract · **well-demarcated erythema** · ± lymphadenopathy



Plantar nail puncture wound with surrounding erythema

Relapse after initial improvement suggests **osteomyelitis or retained foreign body**

Child, nail 1 wk ago, heel pain, tender calcaneal tuberosity **calcaneal osteomyelitis** (Pseudomonas)

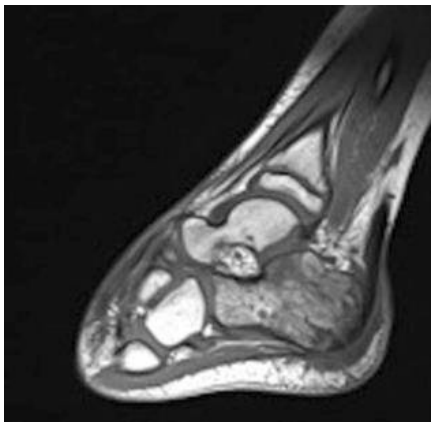
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4 Diagnosis & Classification

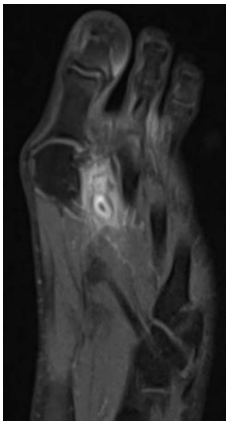
Key radiology findings

Study	Use
X-ray (AP + lateral)	**Normal early** · bone destruction later · **exclude radiopaque foreign body**, fracture · multiple views for **all puncture wounds except knife wounds** (TO)
Ultrasound	**Most accurate for wood / toothpick** in plantar arch (abrupt density change) · **non-radiopaque organic FB** + abscess · detects wooden FB **≥ 5 mm** (TO)
MRI	**Rule out osteomyelitis** (1–2%) · obtain **before operative I&D** · osteomyelitis: **low T1 / high T2 marrow** · abscess + retained FB on post-contrast T1
Bone scan	not indicated when imaging shows no osteomyelitis

Missed foreign bodies **~40%** missed on initial examination (TO)

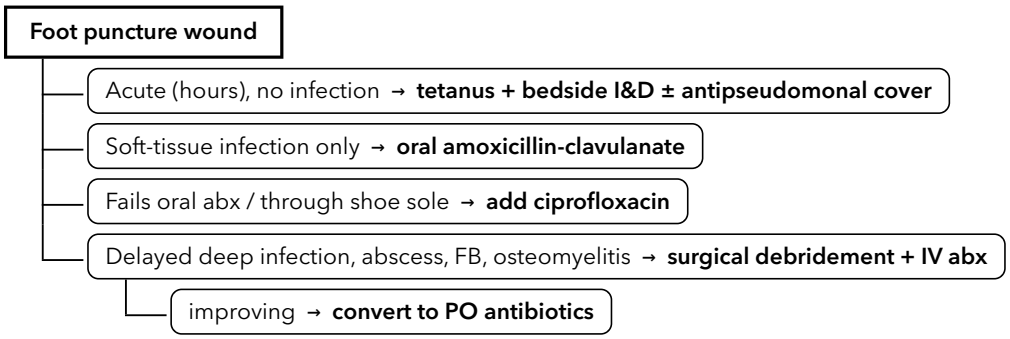


T1 MRI – calcaneal osteomyelitis after nail puncture in a child



Post-contrast MRI – forefoot abscess around a retained rubber foreign body

5 Management



Prophylaxis & tetanus (TO)

Prophylactic antibiotics — **controversial · not advised for all** (TO) · no standard regimen; acute injury should cover **Pseudomonas** (Orthobullets)

Indications (TO) — **highly contaminated** · presentation **> 6 h** · **at-risk host (diabetes, immunosuppression)**

Agents (TO) — 1st-generation oral cephalosporin · antistaphylococcal penicillin · fluoroquinolone · macrolide

Through the sole of a shoe — **antibiotic prophylaxis against Pseudomonas**: oral **fluoroquinolone (ciprofloxacin)** ✓ (TO)

Tetanus booster — all puncture wounds with **outdated or unknown** immune status

Previous tetanus toxoid doses (TO)	Clean, minor: TT vaccine	Clean, minor: HTIG	All other wounds: TT vaccine	All other wounds: HTIG
< 3 or unknown	**Yes**	No	**Yes**	**Yes**
> 3	Yes if last dose **> 10 yr**	No	Yes if last dose **> 5 yr**	No

TO "all FALSE EXCEPT" key — **antipseudomonal prophylaxis through shoe sole** · keyed false: osteomyelitis = S. aureus, needles commonest, **TT for clean minor wound if status unknown** (the TO's own table says Yes) — read the options

Microbiology & antibiotic choice

Situation	Antibiotic
Soft-tissue only (S. aureus, β-haemolytic strep)	**Amoxicillin-clavulanate** = first line
Pseudomonas, adult	**Ciprofloxacin 750 mg PO q12h** · levofloxacin 750 mg PO q24h — **fluoroquinolones = only oral class active vs Pseudomonas**
Pseudomonas, **child**	**Ceftazidime or cefepime** (no fluoroquinolones in children)
Fluoroquinolone contraindicated	Parenteral **ceftazidime**, piperacillin, ticarcillin-clavulanate
Post-debridement	**IV antipseudomonal** → PO once clinically improved

✗ Antibiotic traps — doubling amoxicillin dose · **TMP-SMX** (MRSA; poor Pseudomonas cover) · **rifampin** (TB) · **vancomycin** (gram-positive, MRSA) · **cephalexin** (superficial gram-positive skin) · **ampicillin-sulbactam** (human bite, polymicrobial)

– – Active space – –

Deep infection – surgical debridement & IV antibiotics

Indications	**late / delayed presentation with deep infection ± osteomyelitis** · **foreign body removal** · **no improvement with oral antibiotics**
Technique	**tract + soft-tissue debridement** · deep culture · **bony curettage** (if osteomyelitis) · **pack with wick** → heal by secondary intention
Established infection 24–36 h after plantar puncture	**admit for parenteral antibiotics** (Eidelman)
Rubber-soled shoe, months of pain + swelling	retained **rubber foreign body → abscess → surgical wound exploration, débridement + IV (broad-spectrum) antibiotics** ✗ Wrong options: bone scan · IV antibiotics alone · reinstituting oral antibiotics · expectant · percutaneous / CT-guided drainage (too small to retrieve FB) · débridement + hydrotherapy
Calcaneal osteomyelitis	**IV antibiotics + debridement**

X vs Y

	Soft-tissue infection	Osteomyelitis
Rate	~10%	1–2%
Organism	**S. aureus** (coag +, haemolytic GPC)	**Pseudomonas** (GN rod, polar flagellum, pigment)
Oral drug	Amoxicillin-clavulanate	**Ciprofloxacin** (child: ceftazidime)
Surgery	If abscess / FB / no response	Debridement + bony curettage + IV abx

How it's asked

"Commonest organism, soft-tissue infection after puncture?"	→ **Staphylococcus aureus**
"Characteristic of that organism?"	→ **coagulase positive, haemolytic gram-positive cocci**
"Commonest organism, osteomyelitis after puncture (non-diabetic)?"	→ **Pseudomonas**
"True of the osteomyelitis organism?"	→ **secretes a pigment in vitro** ✗ Wrong options: no flagellum · spores · lactose fermenter
"Nail through shoe, failed amoxicillin-clavulanate — add?"	→ **ciprofloxacin**
"Child, calcaneal osteomyelitis after nail — drug?"	→ **may be treated with ceftazidime** ✗ Wrong options: TMP-SMX · cephalexin · amox-clav
"Puncture wounds — only true statement?"	→ **antipseudomonal prophylaxis if through shoe sole** (TO)
"Commonest object causing puncture wounds?"	→ **nails** (TO)
"Most significant host risk factor?"	→ **diabetes mellitus** (TO)

"Rubber-soled shoe, 2–3 months pain, abscess on MRI — next?"	→ **surgical wound exploration, debridement + IV antibiotics**
"Imaging to locate a toothpick in the plantar arch?"	→ **ultrasound**
"Imaging to rule out osteomyelitis?"	→ **MRI**

– – Active space – –

– – Active space – –

FREIBERG'S DISEASE

– Foot & Ankle · 21 / 48 –

1 Basics

Epidemiology & diagnosis (2nd metatarsal head, female predominance)

Definition — **infarction (infraction) + fracture of the metatarsal head** · **osteochondrosis** · = **AVN / idiopathic osteonecrosis** of MT head · rare

Sex — **female > male**: **~4:1** (Orthobullets, OB) · **5:1** (OB) – read the options

Age — **adolescents > adults**: **13–18 yr** (Orthobullets, OB) · **11–17 yr** (OB) · **female adolescent athletes**
✗ Wrong options: 30s–40s

MT head involved	Rate
2nd (most common), **dorsal** aspect	**68%**
3rd (second commonest)	**27%**
4th	**3%**
4th + 5th (lateral rays)	**Rare**
1st	**Rarely / not typically** – concurrent 1st involvement uncommon

Pathophysiology

Cause — disrupted blood supply from **repetitive microtrauma / stress overloading** or **idiopathic osteonecrosis → collapse of 2nd MT head** · multifactorial: vascular compromise, **genetic predisposition**, altered biomechanics

Risk factors — **long 2nd metatarsal** · **gastrocnemius contracture** (OB)

Natural history — **dorsal articular collapse first**, plantar surface preserved until late → **degenerative arthritis** of 2nd MTP in adulthood · early disease may **resolve spontaneously with MT head remodelling**

3 Presentation

Symptoms — forefoot pain, swelling, stiffness at **2nd MT head / base of 2nd toe**, **worse with WB** · plantar or dorsal forefoot pain · difficulty with push-off · insidious, no trauma

Exam — peri-articular swelling · **↓ (passive) ROM of 2nd MTP** · **pain on passive motion of 2nd toe** · pain worse with **distraction (early)** and **compaction (late)**

Labs — **ESR + CRP normal** → osteomyelitis unlikely

4 Diagnosis & Classification

Key radiology findings

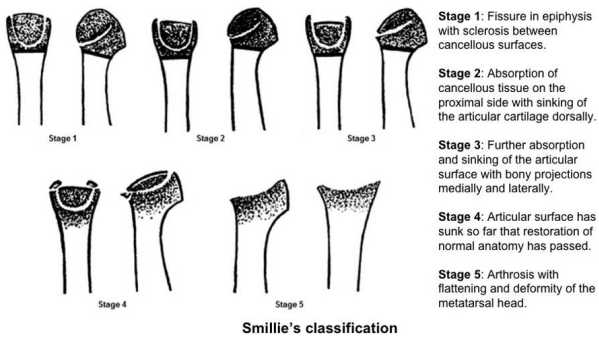
Study	Finding
X-ray AP, lateral, oblique foot	**Early: usually normal → subchondral sclerosis → flattening of MT head** (later, not an early finding) → **joint destruction** (late)
X-ray site	Defect in **upper (dorsal) half of articular surface**
MRI	**Early disease (stage 1)**: subchondral fracture, **patchy oedema** in MT head · confirms **plantar cartilage sparing**



Freiberg disease – flattening of the 2nd metatarsal head

Smillie classification

Stage	Finding	Treatment if nonop fails
1	**Subchondral fracture visible only on MRI**	Nonop
2	**Dorsal collapse of articular surface** on X-ray	**Debridement + MT osteotomy / DCWO**
3	**Collapse of dorsal MT head, plantar articular portion intact** (± medial + lateral bony spurs)	**Debridement + MT osteotomy / DCWO**
4	**Collapse of entire MT head**, joint-space narrowing	**DuVries partial head resection / interposition arthroplasty**
5	**Severe arthritis, joint-space obliteration**	**DuVries / interposition arthroplasty**



Smillie stages 1–5

– – Active space – –

Differential diagnosis\

✗ Wrong options: distractors

✗ Option	Why not
Morton neuroma	**Normal X-ray**; pain at **(3rd) web space**, not MT head
Osteomyelitis	**Normal ESR / CRP**
Mucous cyst	**DIP joints of hand**, end-stage arthritis
Post-traumatic arthritis	Isolated progressive 2nd MTP OA without trauma unlikely at 17
Synovial sarcoma	Young adults; **commonest sarcoma of foot** but commonest **around knee**; **soft-tissue calcification**
Benign bone tumour / normal X-ray	X-ray abnormal

5 Management

Freiberg disease

- All stages first → **NSAIDs, activity change, boot / cast, MT pad**
- Failed nonop, early–mid (Smillie I–III, plantar cartilage spared) → **DCWO ± debridement**
- Late (IV–V, no plantar cartilage) → **DuVries partial head resection / interposition**
  (avoid MTP fusion · polyvinyl implant)

Nonoperative care – metatarsal pad

Indication	**early stage** · first line for **all stages**
Sequence	**short-leg walking cast or boot 4–6 wk** (severe / orthotics fail) → **stiff-soled shoe with MT bars or pads** (after casting)
Metatarsal pad	**offloads 2nd MT head** → relieves pain ✗ Wrong options: steroid · arthrotomy · shortening osteotomy · neck dorsiflexion osteotomy (before nonop fails)
12-yo, normal ESR / CRP	**osteochondrosis → boot immobilisation + observation** ✗ Wrong options: osteomyelitis imaging · AVN → osteotomy / resection · tumour · "normal X-ray, reassure"
Steroid injection	**↑ intra-articular pressure → may worsen osteonecrosis** · may help **adults with secondary DJD** · short-term relief only, doesn't stop collapse
Early surgery	not to prevent progression; no aggressive initial osteotomy

Surgery – debridement, dorsal closing-wedge osteotomy & contraindications

Indications	**progressive pain, joint destruction, deformity** after failed nonop
Stage II after 2 months failed nonop (16-yo)	**joint debridement + metatarsal osteotomy** ✗ Wrong options: interposition · core decompression · MTP arthrodesis · more immobilisation
MRI plantar cartilage spared, failed 6 wk nonop	**dorsal closing-wedge (dorsiflexion) osteotomy of 2nd MT head** ✗ Wrong options: 6–8 wk more nonop · steroid · interposition (late) · **debridement + cancellous grafting** (second-line, less predictable)

DCWO goals — **shortens → offloads MT head** · **resects collapsed dorsal bone + cartilage** · **rotates intact plantar cartilage into contact with proximal phalanx** · lesser-toe MTP approach



Dorsal closing-wedge osteotomy of the 2nd metatarsal

DCWO	Finding
Contraindication	**End-stage OA with severe joint collapse** (no plantar congruity) · end-stage destruction without plantar sparing → poor outcome
Indication ✓	**AVN of 2nd MT head** (= Freiberg) · **degenerative changes with plantar articular congruity** · **dorsal collapse with moderate arthritis**
Not a contraindication	**Age** (65-yo)

MTP arthrotomy + loose-body removal — very rarely; after extensive nonop fails · ± **drilling of MT head**, **subchondral bone grafting**, **interposition arthroplasty** · effective in young + adults with DJD

Interposition tendon — **EDL** (Orthobullets) · **EDB** (OB, Çevik – ↑ pain, ROM, function in late disease) – read the options

DuVries arthroplasty (partial MT head resection) — **severe stage 4–5**, plantar cartilage insufficient · ± capsular interposition

Full MT head resection — **avoid → overloads adjacent MT heads**

Polyvinyl (implant) arthroplasty — **not approved for lesser toes**
✗ Wrong options: even for diffuse collapse + arthritis

Joint-preserving vs sacrificing — **joint-preserving → higher satisfaction** (✗ Wrong options: MTP fusion superior) · **lesser-toe fusion → unacceptably high nonunion** (✗ Wrong options: reliable union)

6 Complications

Degenerative joint disease — **2nd MTP arthritis in adulthood** (natural history)

– – Active space – –

X vs Y

	Early–mid (I–III)	Late (IV–V)
Plantar cartilage	**Intact**	Lost
X-ray	Normal / sclerosis / dorsal collapse	Whole-head collapse, arthritis
Surgery	**DCWO ± debridement**	**DuVries / interposition**

How it's asked

"15-yo girl, 2nd MT head flattening – true?"	**→ greater prevalence in females**
"Subchondral sclerosis + flattening of 2nd MT head – true?"	**→ more common in females**
"12-yo, normal ESR / CRP – pathology + next step?"	**→ osteochondrosis → immobilisation in boot + observation**
"Field-hockey player, failed walking cast – diagnosis?"	**→ Freiberg's disease**
"17-yo male, pain on passive 2nd-toe motion – true?"	**→ second MTP joint most commonly affected**
"Plantar pain at 2nd MT head, 3 months – management?"	**→ metatarsal pad**
"Stage II, failed 2 months nonop?"	**→ joint debridement + metatarsal osteotomy**
"Finding that prevents DCWO?"	**→ end-stage OA with severe joint collapse**
"Plantar cartilage spared, failed nonop?"	**→ dorsal closing-wedge (dorsiflexion) osteotomy**
"X-ray supporting polyvinyl arthroplasty?"	**→ none – not approved for lesser toes**
"Early radiographic finding?"	**→ normal X-ray** ✗ Wrong options: flattening (late)

– – Active space – –

GAIT CYCLE

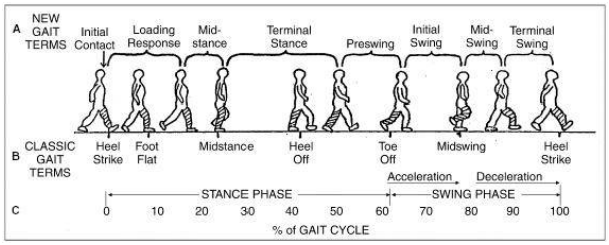
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1 Basics

Definitions

Gait cycle	**heel strike → heel strike** of the same limb
Stance vs swing	**60% / 40%** (Orthobullets, OB) · **62% / 38%** (another OB) – read the options
Stride	consecutive initial contacts of the **same (ipsilateral) foot**
Step	initial contacts of **alternating (contralateral) feet**
Stance phases (5)	initial contact → loading response → mid-stance → terminal stance → pre-swing
Swing phases (3)	initial swing → mid-swing → terminal swing

Phase (Rancho terms)	Classic term	Definition	% cycle
Initial contact	Heel strike	Foot contacts ground	0–2%
Loading response (1st double support)	Foot flat	Initial contact → **elevation of opposite limb**; weight transferred to stance limb	0–10%
Mid-stance (single support)	Midstance	Opposite limb elevation → **both ankles aligned in coronal plane**	10–30%
Terminal stance (single support)	Heel off	**Supporting heel rises** → opposite heel contacts	30–50%
Pre-swing (2nd double support)	Toe off	Opposite initial contact → just before ipsilateral elevation	50–60%
Initial swing	Acceleration	Limb elevation → **maximal knee flexion**	60–73%
Mid-swing	Midswing	Knee flexion → **tibia vertical**	73–87%
Terminal swing	Deceleration	Tibia vertical → just before initial contact	87–100%



Gait phases: Rancho vs classic terms and % of cycle

Prerequisites of gait – Gage (TO)

Five prerequisites (TO)	**stability in stance** · **adequate step length** · **clearance in swing** · **foot prepositioning** (in swing) · **conservation of energy**

– – Active space – –

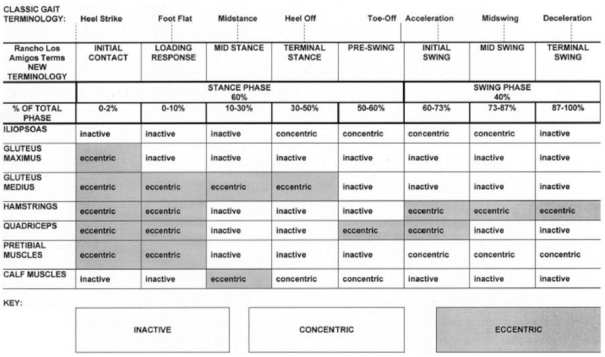
NOT prerequisites (TO)	**foot clearance in stance** (clearance is a swing requirement) · **adequate stride length** (it is **step** length) (TO)

Tibialis anterior – eccentric at heel strike / loading response

TA eccentric	**heel strike (initial contact) + loading response** → lowers foot to ground, controls **plantar-flexion moment**, limb loading + shock absorption
TA concentric	**initial swing + mid-swing** (→ terminal swing) → dorsiflexion for **foot clearance**, prepositions foot for contact
TA inactive	**mid-stance, terminal stance, pre-swing** ✗ Wrong options: eccentric at any of these
Gastrocsoleus	**dormant at heel strike → eccentric at foot flat / mid-stance** (weight moves forward over foot) → **concentric at toe-off (push-off)** → relaxed in swing ✗ Wrong options: GS eccentric at heel strike · GS concentric at heel strike · GS concentric in swing · TA concentric at toe-off
TA weakness (peroneal nerve palsy)	mild–moderate → **foot slap** at heel strike · in swing → **steppage gait** (↑ hip + knee flexion to clear foot)
Primary antagonist of TA	**peroneus longus** (PF + everts) → **superficial peroneal nerve** · TA dorsiflexes + inverts
Other antagonist pair	**peroneus brevis vs tibialis posterior**
Strength	ankle plantarflexors **6×** stronger than dorsiflexors

Muscle activity by phase

Phase	Active muscles
Initial contact	**Hip extensors** stabilise hip · **quadriceps eccentric** · **TA eccentric**
Loading response	**TA eccentric** (controls PF moment) · **quads** stabilise knee vs flexion moment
Mid-stance	**Gluteus medius + calf eccentric**
Terminal stance	**Toe flexors + tibialis posterior** most active
Pre-swing	**Hip flexors** propel advancing limb
Initial swing	**Hip flexors concentric** advance the leg · TA concentric
Mid-swing	**Ankle dorsiflexors** → foot clearance
Terminal swing	**Hamstrings** decelerate thigh



Muscle activity through the gait cycle

Swing mechanics

Swing momentum	generated by **ankle plantar flexors (gastrocsoleus) + hip flexors at terminal stance** ✗ Wrong options: hip flexors + dorsiflexors in early swing · knee flexors in early swing · knee extensors
Knee in swing	flexion in early swing + extension in terminal swing are **passive**
Main concentric contraction in swing	**tibialis anterior**
Activity	**more muscle activity in stance** than swing (stabilise plantigrade foot)
FHL	plantarflexes hallux at **push-off**; **inactive in swing**

Tendon function – tibialis posterior

Origin	**tibia + fibula + interosseous membrane** ✗ Wrong options: FDL = **tibia only** · FHL = **fibula only**
Activity	**concentric in terminal stance + pre-swing** · EMG biphasic: initial contact + midstance and pre-swing ✗ Wrong options: eccentric · initial swing
Function	**most powerful inverter** · dynamic stabiliser of medial arch · **inverts hindfoot + locks Chopart's joint** → gastrocsoleus acts through MT heads (**TP + GS synergistic**)
Insertions	anterior: **navicular tuberosity** + plantar 1st cuneiform · middle: bases **MT 2–4**, 2nd + 3rd cuneiforms, cuboid, FHB · posterior recurrent: **sustentaculum tali**
TP loss	GS pulls **talar head down** → stretches **spring ligament** → medial arch collapses
TP dysfunction EMG	**↑ activity in 2nd half of stance** (delayed / right shift)
Rupture site	distal to medial malleolus: flattened, fibrocartilaginous, **hypovascular** zone

Determinants & centre of gravity

Variable	Value / effect
Pelvic rotation	**4° medially (anteriorly)** on swing side → lengthens limb
Pelvic tilt	Pelvis **drops 4°** on swing side → lowers COG at midstance (Orthobullets) · contralateral drop **~5°** minimal, gluteal-controlled (TO)
Knee flexion at heel strike	**15°** (Orthobullets) · **10°** (TO) – lowers COG, absorbs shock · read the options
Ankle	**Plantar flexion** at heel strike + first part of stance
Knee at midstance	**Extends** as ankle PF + foot supinates → restores leg length
Lateral pelvic shift	Over stance limb (COG over base of support)
COG (standing)	**5 cm anterior to S2**
COG vertical	**Highest at midstance** · **lowest at double-limb support**
COG horizontal	**5 cm** per adult male step

– – Active space – –

Stance & running biomechanics (TO)

Heel at loading (TO)	**force acceptance + shock absorption, NOT force generation** (TO) ✗ Wrong options: "maximum force for further movement generated by heel" = FALSE
Heel strike posture (TO)	rearfoot in slight **inversion 0–5°** · foot supinates · pelvis level, **anterior tilt 10°**, internally rotated · hip **ER 5–10°, flexed 20–30°** · knee **flexed 10°** (TO)
Shock-absorption cascade (TO)	laterally directed GRF → rearfoot **everts** + tibia / femur **IR** + hip adduction + knee flexion (peaks **~45°**) · all **eccentric** (TO)
Anterior pelvic tilt up to 10° (TO)	**acceptable** in early stance (heel strike → foot flat) · excess strains lumbar spine / hamstrings (TO)
Slow vs fast running (TO)	slow = **heel strikes first** · fast = **forefoot strikes first** (TRUE) (TO)
Forefoot strike (TO)	↓ vertical GRF, ↓ knee + hip loads · ↑ **Achilles tendinopathy, plantar fasciopathy, MT stress fractures** (TO)
Tibiotalar pronation (TO)	present **when foot is on ground** (stance) = **dorsiflexion + eversion + abduction** (TRUE) (TO)
Midstance (TO)	ankle → **max dorsiflexion ~20°** · max pronation after max dorsiflexion · peak rearfoot eversion **~10°**, forefoot abduction **~5°** (TO)

3 Presentation

Pathological gait – gluteus maximus paralysis (TO)

Gluteus maximus	**chief thigh extensor** · erect posture · lateral rotation · abduction · **inferior gluteal nerve (L5, S1, S2)** (TO)
Gluteus maximus lurch (TO)	at heel strike **trunk lurches backwards** · line of gravity shifts **posterior to hip** in early stance → compensates weak hip extension (TO)
NOT seen (TO)	**Trendelenburg gait** (gluteus medius) (TO)

Quadriceps weakness

Quadriceps atrophy	affects **stance → midstance** ✗ Wrong options: swing phases (quads largely silent in swing)
Compensation	leans forward at hip → COG **anterior to knee** → knee **hyperextension**
Hamstring / hip-flexor weakness	affects **swing** (limits limb advancement)

5 Management

Gait after TAA vs arthrodesis

Greatest difference (level walking)	**peak plantarflexion power generation** (↑ with TAA)
TAA	**~43% more sagittal ankle ROM** (21.1° vs 14.7°) · ↑ forefoot-tibia + hindfoot-tibia motion · closer to normal but **neither restores normal gait** · differences amplified on **stair ascent**

Arthrodesis compensations	✗ Wrong options: **↑ hip ROM** · ↑ transient force at heel strike · may have **better coronal eversion** · walking speed / cadence difference only modest
PROMs	similar improvement after both

X vs Y

	Step	**Stride**
Measured between	Initial contacts of **alternating** feet	Initial contacts of the **same** foot
Gage prerequisite	**Yes** (adequate step length)	**No**

	Gluteus maximus gait	**Trendelenburg gait**
Muscle	Gluteus maximus (hip extensor)	Gluteus medius (abductor)
Nerve	**Inferior gluteal**	Superior gluteal
Sign	**Trunk lurches backwards** at heel strike	Pelvic drop

	Tibialis anterior	**Gastrocsoleus**
Heel strike	**Eccentric**	**Dormant**
Foot flat / mid-stance	Inactive	**Eccentric**
Toe-off	Inactive	**Concentric**
Swing	**Concentric**	Relaxed

How it's asked

"TA contracts eccentrically during?"	→ **heel strike**
"Phase of eccentric TA contraction?"	→ **loading response** ✗ Wrong options: mid-stance · terminal stance · pre-swing · initial swing
"Correct muscle activity in normal gait?"	→ **TA contracts eccentrically at heel strike**
"Primary antagonist of TA innervated by?"	→ **superficial peroneal nerve** (peroneus longus)
"% of gait cycle in stance?"	→ **60%**
"Swing momentum generated by?"	→ **ankle plantar flexors + hip flexors at terminal stance**
"Quadriceps atrophy affects which phase?"	→ **midstance**
"NOT a prerequisite of gait?"	→ **foot clearance in stance** (TO)
"NOT a prerequisite of gait (Gage)?"	→ **adequate stride length** (TO)
"False statement about gait?"	→ **heel generates maximum force for further movement** (TO)
"Gluteus maximus paralysis — all true EXCEPT?"	→ **Trendelenburg gait** (TO)

– – Active space – –

"Ruptured tendon on axial MRI behind medial malleolus – true?"	**→ originates tibia + fibula, concentric in terminal stance**
"TAA vs ankle arthrodesis – greatest gait difference?"	**→ peak plantarflexion power generation**

– – Active space – –

HALLUX RIGIDUS (MTP JOINT ARTHRITIS)

– Foot & Ankle · 23 / 48 –

1 Basics

Epidemiology

Commonest site of OA in the foot	1st MTP (Orthobullets) · **2nd commonest great-toe pathology after hallux valgus** (OB)
Incidence	**2.5% of > 50-yr-olds** · **F : M = 2 : 1** · peak **5th–6th decade**
Risk factors	**trauma to 1st MTP** (**~80%** of unilateral cases; e.g. old **turf toe**) · high-impact sport
Causes of 1st MTP arthritis (OB)	**trauma**, **hallux valgus**, **elevated 1st metatarsal**

Etiology & associations

Primary cause	**unknown** · acute trauma + repetitive microtrauma predispose
Pathoanatomy	osteophytes + cartilage loss **start dorsally** → whole joint later
Anatomic variants	metatarsus adductus · MT head morphology · **long 1st MT** · hallux valgus / varus
Associated	**sesamoid arthritis** · gout · RA · seronegative arthropathies

Anatomy & biomechanics

1st MT	**shortest and widest**
Medial branch of medial dorsal cutaneous nerve	**overlies 1st MTP** → irritated by **dorsal osteophytes** · plantar-medial hallucal nerve also supplies joint
Load	1st MTP carries **~120% body weight** each step
Running	needs **60° dorsiflexion**

2 Mechanism

Checkrein deformity (TO)

Cause (TO)	**fracture of distal third of tibia → FHL adheres to fracture callus →** bowstring to hallux insertion (TO) ✗ Wrong options: FHL tear · IP joint fracture · **FHL tethering at 1st MTP**
Sign (TO)	**ankle dorsiflexion → great toe flexes** · **ankle plantarflexion → hallux IP extends** (TO)

– – Active space – –

3 Presentation

Pain	1st ray / 1st MTP · **worse at push-off (toe-off / heel-rise)** · swelling · gait abnormality · difficulty with shoes
Inspection	**dorsal prominence** (osteophyte) · shoe irritation + redness · severe → **hyperextension deformity**
Motion	**↓ dorsiflexion** · **pain at terminal dorsiflexion** early → **pain through whole arc** as it progresses
Neuro	**dorsomedial paraesthesia** / ↓ sensation distal medial dorsal foot (**medial dorsal cutaneous nerve**, ± Tinel)
Grind test +	**severe disease, central chondral wear**

Associations & sequelae (transfer metatarsalgia)

Hallux rigidus leads to	**restricted dorsiflexion** → forefoot **internal rotation** → **↓ push-off strength** → **inverted gait** → **transfer metatarsalgia** ✗ Wrong options: **flexor hallucis longus (FHL) rupture** · everted gait · ↑ push-off · Haglund deformity

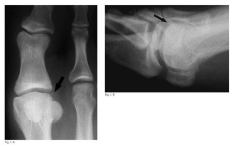
4 Diagnosis & Classification

Key radiology findings

Study	Finding
Weight-bearing AP, lateral, sesamoid, oblique	**Dorsal osteophytes** · **joint space narrowing** · **subchondral sclerosis + cysts**
MRI	Rarely needed · **osteochondral lesion with normal X-ray** · characterises mild OA

Radiographic progression	may **not correlate** with symptoms

Coughlin & Shurnas classification

Grade	Exam	Dorsiflexion	Radiograph	X-ray
0	**Stiffness**, no pain	**40–60°** (10–20% lost)	**Normal**	–
1	**Mild pain at extremes**	**30–40°** (20–50% lost)	**Mild dorsal osteophyte**, normal joint space	
2	**Moderate pain at extremes**, more constant	**10–30°** (50–75% lost)	**Moderate osteophyte**, **< 50% JSN**	
3	Significant stiffness, near-constant pain at extremes, **no mid-range pain**	**≤ 10°** (75–100% lost)	**Severe osteophyte**, **> 50% JSN**	
4	As grade 3 **+ pain at mid-range**	As grade 3	As grade 3	

Scenario → grade (OB) stiffness + ↓ push-off, **no pain** = **0** · dorsal osteophyte, no stiffness, **pain at terminal DF** = **1** · moderate pain at extremes + mild osteophyte + **< 50% JSN** = **2** · JSN + stiffness + pain at end-range = **3** · large osteophyte + stiffness + **mid-range pain** = **3** (OB) · severe JSN + **pain throughout** = **4**

Other staging (OB) "**Stage III** = end-stage, cartilage lost from **phalanx + metatarsal**" · "**Stage 4** = severe dorsal osteophyte + **> 50% JSN**" – read the stem

5 Management

Hallux rigidus

- Grade 0–1 → **NSAIDs, activity mod, injection, Morton's extension**
- Grade 1–2, pain only at terminal DF → **dorsal cheilectomy**
  - < 30–40° DF after cheilectomy / runner → **+ Moberg**
- Grade 3–4, mid-range pain, + grind → **MTP arthrodesis**
- Elderly, low demand → **Keller resection arthroplasty**

Failed 1st MTP arthroplasty

- Silastic synovitis, great-toe pain only → **implant removal + synovectomy**
- Infection → antibiotic spacer → **staged arthrodesis**
- Loosening / bone loss / shortening + lesser metatarsalgia → **removal + structural graft + arthrodesis**

– – Active space – –

Operative criteria

Grade / situation	Treatment
Grade 0 (motion loss, no X-ray change)	Nonop · **plantar release** (OB)
Grade 0–1	**Nonop** (NSAIDs, injections, **Morton's extension**) · **up to 55%** succeed
Grade 1–2 · select grade 3 with pain only at terminal DF	**Dorsal cheilectomy** (± synovectomy, debridement)
Cheilectomy gives **< 30–40° DF** · **runner**	**+ Moberg** dorsal closing wedge osteotomy
Grade 3–4 · **mid-range pain** · **+ grind** · failed cheilectomy · grade 3 with **< 50% MT head cartilage** left	**MTP arthrodesis** (gold standard)
Grade 3–4, wishes to keep motion	Prosthetic / **interpositional arthroplasty**
Elderly, low demand, fast rehab	**Keller resection arthroplasty**
Mild disease + **intra-articular loose bodies**	Dorsal synovectomy + debridement

Nonoperative care – Morton's extension & carbon plate

Morton's extension
stiff insert with rigid extension **under 1st ray, past the 1st MTP → limits 1st MTP dorsiflexion** at push-off · ↓ plantar loading across MTP · **mainstay / first line for hallux rigidus**
✗ Wrong options: hallux valgus · hallux varus · Morton's neuroma · spring ligament · 2nd MTP synovitis · Jones · hammer / mallet toe (high toe box) · midfoot arthritis · Freiberg

Limits
hallux MTP dorsiflexion
✗ Wrong options: midfoot collapse · hindfoot varus · MTP plantarflexion · adduction

Poor surgical candidate, + grind
Morton's carbon fiber extension
✗ Wrong options: **UCBL** + **Arizona brace** (flexible pes planovalgus) · **medial flare** (ankle instability) · **negative heel rocker** (↑ PF in terminal stance)

Other nonop
NSAIDs · avoid excessive great-toe DF · **stiff-sole shoe** / carbon fiber shank · toe-box stretching · rocker-bottom shoes · molded inserts

Injections
corticosteroid + sodium hyaluronate → ↓ pain at 4 + 8 wk in low-grade disease

Not first line (OB)
CAM boot, AFO, hard-soled shoes · **PT aggravates** · extending Morton's to 2nd MT adds nothing

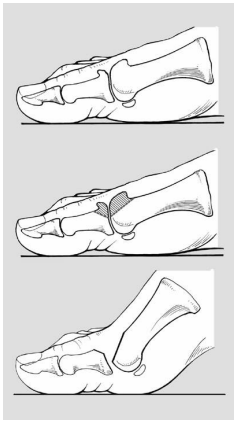


Morton's extension orthosis

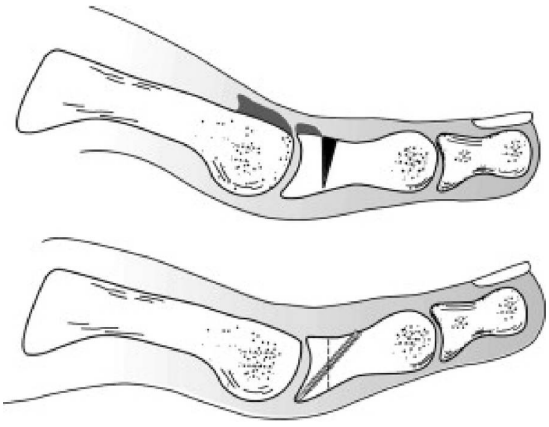
Cheilectomy ± Moberg osteotomy – indications

Ideal candidate
pain at terminal DF only + **shoe-wear irritation** from dorsal prominence · **grade 1–2** · **no mid-range pain**

Contraindications	**mid-range pain** on passive motion · **+ grind test** · **sesamoid arthritis** · comorbidities (poor DM control, PVD)
Technique	dorsal approach · **complete synovectomy** · resect dorsal osteophyte + **up to 30% dorsal MT head** (Orthobullets) · **25%** (OB) · **> 30% → subluxation / instability**
Goal	**70–90% of dorsiflexion** intra-op (Orthobullets) · **70–90° DF** (OB)
Outcomes	grade 1–2: **97% excellent**, **92% pain relief at 10 yr** · higher grades: **25–56% → arthrodesis** · progression needing fusion **7–9% within 10 yr**
Grade 2, 35° arc, failed Morton's	**cheilectomy + joint debridement** ✗ Wrong options: observation · sesamoidectomy · Keller · fusion
Carpenter, terminal-DF pain, mild OA	**cheilectomy** · cheilectomy ≠ Helal / Weil / chevron osteotomy ✗ Wrong options: Lapidus · fusion · resurfacing · chevron
Moberg	**dorsal closing wedge osteotomy of proximal phalanx** → ↑ DF by sacrificing plantarflexion arc · with cheilectomy: **88% satisfaction at 2 yr**
Moberg indications	**runners** (need 60°) · cheilectomy gives **< 30–40° DF** (Orthobullets) · **< 30°** (OB) · failed cheilectomy
Runner, 30% resected, only 20° DF	**Moberg procedure** ✗ Wrong options: close + WBAT · FDL → EDL · Keller (high demand) · fusion
Ballet dancer, stage 2	**cheilectomy, débridement ± Moberg** ✗ Wrong options: **fusion = career-ending** · Keller · uni- / bipolar replacement



Dorsal cheilectomy



Moberg dorsal closing wedge osteotomy

– – Active space – –

MTP arthrodesis – indications & optimal position

Indications	**standard of care / gold standard** for **grade 3–4** · **commonest procedure** · **pain throughout MTP motion** · failed cheilectomy · failed arthroplasty
Best long-term function + pain relief	**arthrodesis** · restores **WB + propulsion of 1st ray** ✗ Wrong options: synovectomy · cheilectomy · hemiarthroplasty · interposition (worse long-term)
Failed cheilectomy at 1 yr	**1st MTP arthrodesis** ✗ Wrong options: Moberg (won't relieve pain)
Grade 4, young active golfer	**great-toe MTP fusion** ✗ Wrong options: Keller (not for young / active) · bipolar replacement · cheilectomy · fascia lata interposition
Stage 3, + grind, 10° DF	**arthrodesis** ✗ Wrong options: cheilectomy · Moberg · Keller (↓ propulsion + transfer metatarsalgia) · resurfacing hemiarthroplasty (not with phalangeal cartilage loss)
Fixation	wires, pins, lag screws, crossed screws, plates · **dorsal plate + compression screw = strongest** · medial→lateral compression screw common

Position	Orthobullets	OB
Valgus (vs MT shaft)	**5°**	**10–15°** · **5–10°** · **15°** – read the options
Dorsiflexion (vs floor)	**15°**	**10–15°** · **10°**
Rotation	–	**Neutral**
Footplate clearance	**4–8 mm** toe tip off plate	**~5 mm**

Footplate test	flat sterile tray, ankle neutral, simulates WB · **5 mm clearance → irrigate and close** ✗ Wrong options: Moberg · superior capsulotomy · add screw · revise plate



Intra-op footplate test – toe tip clears plate

Outcomes	fusion **90–100%** (Orthobullets) · union **~94%** (OB) · **95–96% satisfaction** · return: **hiking 92%, golf 80%, jogging 75%, tennis 75%**
Tennis player, stage 4, failed orthosis	**arthrodesis** ✗ Wrong options: cheilectomy + Moberg (stage 1–2) · interpositional (not for athletes) · hemi / total arthroplasty (subsidence, radiolucency)

Mid-range pain, 10° DF, eroded joint intra-op	**arthrodesis** ✗ Wrong options: cheilectomy · bunionectomy + cheilectomy · resection · interposition

Alternatives – synthetic cartilage implant & Keller resection arthroplasty

Keller	**resect base of proximal phalanx** ± capsular interposition · **elderly, low demand → fastest rehab** · interposition: **90% pain improvement, 4% reoperation at 4.5 yr**
80-yo minimally ambulatory, plantar-flexed toe	**Keller + osteophyte removal** ✗ Wrong options: silicone implant · Moberg · cheilectomy · arthrodesis
Keller contraindications	**rigid hyperextension deformity** · **sesamoid arthritis** · young active (instability)
Keller risks	**cock-up (hyperextension)** · **weak push-off** · **transfer metatarsalgia** (↓ with capsular interposition) · gross MTP instability
Synthetic cartilage interposition	keeps motion · **equivalent pain reduction + function vs arthrodesis** · **longer time to max pain relief** · **no prolonged WB restriction** · **not contraindicated in grade 2** · not if **HV / varus > 20°**
Interpositional arthroplasty (capsule / EHB)	younger **< 60 yr** late-stage, **strongly opposed to fusion** · limited toe function → not for athletes

Prosthetic arthroplasty

Indication	**grade 3–4** wishing to **preserve motion** · indications unclear
Contraindications	**active infection** · **insufficient bone stock** · **sesamoid arthritis**
Hemiarthroplasty	unipolar MT head or phalanx base · **keeps 1st ray length** · **easier conversion to fusion** · hemi = total for function, complications, satisfaction
Generations	1st–2nd **silastic** (silicone) · 3rd **press-fit metal** · 4th **threaded-stem metal**
Silastic	good short-term, **~60% failure** · **silicone wear, synovitis, osteolysis** · **abandoned**
Current implants	**80–90% survival at ~5 yr** · later **loosening + complex revision with bone loss** · **worse than primary fusion**; fusion after failed arthroplasty worse than primary fusion

– – Active space – –

6 Complications

Complication	Fact
Arthritis progression after cheilectomy	**7–9% → fusion within 10 yr**
Instability	**Excessive cheilectomy resection** (> 30%)
Malunion (fusion)	**16%** → transfer metatarsalgia, **hallux IP arthritis**, poor push-off
Nonunion (fusion)	**5–10%** · **< 40% symptomatic** → revision fusion + bone graft
Hallux IP OA	**15%** after MTP fusion · usually asymptomatic · ↑ with **excess valgus**
Cock-up deformity	**Keller** → treat with **1st MTP arthrodesis**
Failed arthroplasty	Silicone / early implants; loosening, osteolysis, technical error
Infection	**< 2%**
Symptomatic hardware	–

Arthrodesis malposition complications

Fused position	Result
Excess dorsiflexion	Pain at **toe tip, IP joint, under 1st MT head** · **sesamoid / plantar pain** · **dorsal shoe irritation**
Excess plantarflexion (too straight)	**↑ pressure at toe tip** · can't roll over → **walks on lateral foot**, **lateral calluses**, lateral sole wear (supinates)
Excess valgus	**Pressure on 2nd toe → soft corn / ulcer** · **IP joint degeneration**
Varus	**Shoe-wear problems** (narrow toe box) · **medial hallux irritation** (✗ Wrong options: sesamoid pain · rollover problem)
Shortened	**Transfer metatarsalgia** (2nd MT) + **varus of 2nd toe** (lost medial support) · IP degeneration

2nd MTP plantar pain + new 2nd-toe deformity after fusion	**fused shortened** ✗ Wrong options: varus · valgus · plantarflexion · dorsiflexion
Lateral foot pain, lateral calluses, stands on outside of foot	**fused in plantarflexion**
Sesamoid / plantar 1st MT head pain, fixed MTP	**first MTPJ arthrodesis in 5 degrees of valgus** (relation to metatarsal shaft), **30 degrees of dorsiflexion** (relation to floor) → ↑ plantar pressure under 1st MT head ✗ Wrong options: cheilectomy 10% or 30% · 10° valgus / 15° DF (ideal) · 5° varus

Dorsiflexion malunion after prior surgery

Dorsal great-toe irritation in closed shoes despite hardware removal	**dorsiflexion malunion** · functional, but shoe problems ✗ Wrong options: neuropathy · **nonunion** (activity pain + swelling, relieved by stiff shoes) · **transfer metatarsalgia** (plantar 2nd MT head pain) · **plantarflexion malunion** (gait difficulty, walks on lateral foot)

Failed implant/hemiarthroplasty – removal & bone-block arthrodesis

Commonest failure modes	**aseptic loosening**, **subsidence**, **infection**
Silastic, 15 yr, synovitis, normal ESR / CRP / WBC, erythema settles on elevation	**implant removal + synovectomy** ✗ Wrong options: steroid injection · orthosis · I&D + IV antibiotics · revision silastic
Silastic long-term complication complicating revision fusion	**osteolysis** · also loss of fixation, pathologic fracture ✗ Wrong options: recurrent deformity · infection (not ↑) · chronic synovitis (less impact) · **floating toe** (Weil osteotomy)
Lucency around phalangeal component, pain since return to labour, normal labs	**mechanical failure** ✗ Wrong options: septic arthritis · periprosthetic fracture · **aseptic loosening from PE debris** (needs prior stable period) · transfer metatarsalgia
Failed arthroplasty + lesser-toe metatarsalgia + callosities, negative aspirate	**implant removal, synovectomy, structural bone graft, arthrodesis** ✗ Wrong options: long-stem revision · silastic conversion · spacer + reimplant (no infection) · removal + synovectomy alone (weak push-off)
Failed silastic, osteolysis, metatarsalgia	**implant removal + structural bone graft + MTP fusion** ✗ Wrong options: Morton's extension · in situ fusion · Keller (worsens metatarsalgia) · titanium total toe
Failed hemiarthroplasty with bone loss	**bone-block arthrodesis** ✗ Wrong options: **in situ arthrodesis** (short 1st ray → transfer metatarsalgia) · Keller · Moberg · silicone
Failed hemiarthroplasty + shortening + cock-up	**arthrodesis with structural bone graft** ✗ Wrong options: tenotomy (treats consequence not cause) · osteotomy · resection · **particulate graft** (minimal bone loss only)
Severe 1st MT shortening after arthroplasty	**distraction bone block arthrodesis** ✗ Wrong options: antibiotic spacer · shortening arthrodesis · **ex-fix lengthening** (brachymetatarsia) · revision arthroplasty

Structural bone graft indication
1st MT shortening not rebalanced by lesser MT osteotomy (usually **> 5 mm**) – commonest after **failed arthroplasty**
Proximal phalanx bone loss – not enough bone for fixation without violating IP joint
Shortening with loss of medial support of 2nd toe → 2nd MTP varus

Graft	**tricortical iliac crest** (allograft commonest, Orthobullets) · autograft or allograft (OB)

– – Active space – –



Bone-block arthrodesis of the 1st MTP

X vs Y

Orthosis	Condition
Morton's extension (carbon fiber)	**Hallux rigidus**
UCBL	Flexible pes planovalgus
Arizona brace (UCBL + lace-up)	Flexible pes planovalgus
Medial flare	Ankle instability
Negative heel rocker	↑ PF in terminal stance
High toe box	Hammer / mallet toe

	Cheilectomy	Moberg	Arthrodesis	Keller
Grade	**1–2**	Adjunct (1–2)	**3–4**	Any, elderly
Pain	**Terminal DF**	Insufficient DF	**Mid-range**, + grind	–
Patient	Active	**Runner, dancer**	Young labourer, athlete	**Low demand**
Pitfall	> 30% → instability	Loses PF	Malposition	**Cock-up**, metatarsalgia

	Weil osteotomy	Helal osteotomy
Site	**Intra-articular**, parallel to sole	**More proximal**
Fixation	**Screw perpendicular** to cut	**None**
Use	Shortening for **MTP dislocation** · floating toe risk	Shortening

How it's asked

"Morton's extension is used for?"	→ **hallux rigidus**
"Morton's extension limits which motion?"	→ **hallux MTP dorsiflexion**
"Poor surgical candidate, + grind – orthotic?"	→ **Morton's carbon fiber extension** ✗ Wrong options: UCBL · Arizona
"Which radiograph benefits from this orthosis?"	→ **hallux rigidus (Figure F)**
"Hallux rigidus associations EXCEPT?"	→ **FHL rupture**

"Hallux rigidus can lead to?"	→ transfer metatarsalgia
"Grade 2, failed Morton's, 35° arc?"	→ cheilectomy + joint debridement
"Terminal-DF pain, mild OA, dorsal osteophyte?"	→ cheilectomy
"Carpenter, shoe irritation, failed carbon shank?"	→ resect dorsal osteophyte + 25% of dorsal MT head
"Pain at extremes, < 50% JSN, cheilectomy planned — grade?"	→ grade 2
"Best cheilectomy candidate?"	→ dorsal osteophyte, no stiffness, pain at terminal DF
"Pain only at terminal DF, failed rigid orthotics?"	→ dorsal cheilectomy
"Persistent pain at extreme DF despite Morton's?"	→ dorsal cheilectomy
"Runner, only 20° DF after cheilectomy?"	→ Moberg procedure
"Ballet dancer, stage 2?"	→ cheilectomy, débridement ± Moberg
"+ grind, 20° painful arc — best long-term?"	→ fusion in neutral rotation, 15° valgus, 10° dorsiflexion
"Pain at 2nd MTP + 2nd-toe deformity after fusion?"	→ fused in shortened position
"Lateral foot pain + calluses after fusion?"	→ fused in plantarflexion
"Sesamoid pain under 1st MT head after surgery?"	→ 5° valgus, 30° dorsiflexion arthrodesis
"Tennis player, stage 4?"	→ arthrodesis
"Mid-range pain, 10° DF, eroded joint?"	→ arthrodesis
"Failed cheilectomy — best pain relief + function?"	→ 1st MTP arthrodesis
"Positive grind, 10° DF, stage III?"	→ 1st MTP arthrodesis
"Grade 4 golfer, wants no further surgery?"	→ great-toe MTP fusion
"Pain through arc, 5° DF, + grind — most reliable?"	→ MTP arthrodesis
"Toe tip clears footplate by 5 mm?"	→ irrigate and close
"Best arthrodesis candidate?"	→ pain throughout MTP motion
"Silastic implant, synovitis, normal labs?"	→ removal of implant + synovectomy
"Failed arthroplasty + lesser metatarsalgia?"	→ implant removal, synovectomy, structural bone grafting, arthrodesis
"Failed hemiarthroplasty with shortening and cock-up?"	→ arthrodesis with structural bone graft
"Failed silastic, metatarsalgia + great toe pain?"	→ implant removal + structural bone grafting + MTP fusion
"Lucency around phalangeal component in labourer?"	→ mechanical failure
"Severe 1st MT shortening after prior arthroplasty?"	→ distraction bone block arthrodesis

– – Active space – –

"Failed hemiarthroplasty with bone loss — most reliable?"	→ **bone-block arthrodesis**
"Silastic complication that complicates revision?"	→ **osteolysis**
"80-yo, low demand — fastest rehab?"	→ **Keller resection arthroplasty + osteophyte removal**
"Synthetic cartilage implant — true?"	→ **equivalent pain reduction vs arthrodesis**
"Dorsal shoe irritation after fusion despite hardware removal?"	→ **dorsiflexion malunion**
"Checkrein deformity of great toe follows?"	→ **fracture of distal third of tibia** (TO)

HALLUX VALGUS

– Foot & Ankle · 24 / 48 –

1 Basics

Epidemiology

Prevalence	**~23%** of 18–65 y · up to **30% of females** · **women up to 15:1** · **most common toe deformity** (TO)
Forms	**adult** vs **juvenile / adolescent** (see 5 Management)
Shoe-wearing societies	almost exclusively with constricting footwear (OB) · still occurs in **barefoot populations** → footwear **contributory, not sole cause** (TO)

Genes & lab markers

Family history	**70%** (Orthobullets) · **60–70%** (TO) – **strongest risk factor**, **autosomal dominant, incomplete penetrance** (TO)
COL5A1 mutation	**Ehlers-Danlos** → hallux valgus → **MTP arthrodesis** (OB)
Connective tissue	**Ehlers-Danlos, Marfan** (ligamentous laxity) (TO)

Risk factors

Type	Factors
Intrinsic (TO)	**Genetic** (strongest) · **rounded (convex) 1st MT head** (flat / chevron-shaped head resists subluxation) · **metatarsus primus varus** (↑ IMA, **primary skeletal driver**) · **1st metatarsal-cuneiform hypermobility** · **pes planus** · **ligamentous laxity** · **long 1st MT (Egyptian foot)**
Intrinsic (Orthobullets)	**↑ DMAA** · 1st TMT instability · **2nd toe deformity / amputation** · RA · CP · metatarsus adductus
Extrinsic	**Narrow toe box + high heel** = most significant environmental factor (TO)
Systemic (TO)	**RA** (synovitis destroys medial capsule + collaterals) · **CP, poliomyelitis** (muscle imbalance) · **EDS, Marfan** (laxity)
Not a risk factor	**Pes cavus** (TO) · flexible cavovarus (OB)

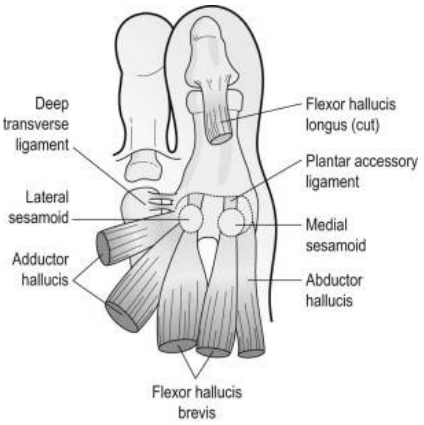
Pes planus	associated with **progression** · elevation + functional lengthening of 1st MT ✗ Wrong options: cross-over toe · tarsal coalition · hammer toe · 1st MT supination
Progression predictors	**DMAA** + **AP talo–1st MT angle** at presentation (OB)

Pathoanatomy & deforming forces (pronation, adductor, abductor hallucis, conjoined tendon)

Cascade	**valgus phalanx → varus 1st MT** → MT head drifts **medially** → sesamoid complex **laterally translated relative to MT head**
Sesamoids	sit in the two heads of **FHB**; attached to proximal phalanx base by sesamoid-phalangeal ligament / plantar plate

– – Active space – –

Sesamoids don't move (TO)	anchored to 2nd MT by **deep transverse intermetatarsal ligament → MT head migrates medially off them** = apparent "sesamoid subluxation" (TO) · Orthobullets: sesamoids displace **lateral** to MT head — read the options
Medial	capsule + **MCL stretched, attenuated** (TO) · early lesion = failure of medial sesamoid ligaments + MCL (OB)
Lateral contracture	lateral capsule + **adductor hallucis** + **lateral head FHB** → lateral **bowstring** (TO)
Adductor hallucis	**primary deforming force** · inserts **fibular (lateral) sesamoid + lateral base proximal phalanx** ✗ Wrong options: abductor hallucis · EHB · FHB (more plantar, contributes less) · FHL (not part of sesamoid complex, not primary)
Abductor hallucis	inserts with **medial FHB tendon** into medial base proximal phalanx · migrates **plantar + lateral → flexion and pronation** ✗ Wrong options: ↑ abduction · extension · pronation alone
Conjoined tendon	**FHB + adductor hallucis** at lateral base proximal phalanx · **do not release → iatrogenic hallux varus** · release adductor (transverse + oblique heads) **adjacent to lateral sesamoid** instead
EHL + FHL	bowstring **lateral to MTP axis** → dynamic lateral deviators, perpetuate deformity (TO) · Orthobullets: FHL + adductor = plantar-plate muscles that become deforming forces
Plantar plate muscles (Orthobullets)	abductor hallucis · FHB · **FHL (deforming)** · **adductor hallucis (deforming)**
Hallux position	**valgus + pronation** · nail faces **laterally** (TO) ✗ Wrong options: valgus + supination · varus
1st MT rotation (TO)	severe HV = **pronation (internal rotation) of the first metatarsal** → sesamoid displacement · **87%** pronated on CT (OB) ✗ Wrong options: supination / external rotation · inversion of 1st ray · pure sagittal dorsiflexion
Triplanar deformity (OB)	valgus + **dorsiflexion** + **pronation** of 1st ray → offloads 1st MT head → load to lesser MTs
Bunion (TO)	**medial eminence** = normally covered medial MT head now uncovered · **not an exostosis** · **adventitious bursa** → painful bursitis · osteophytes later
Windlass mechanism	less effective → **transfer metatarsalgia**
HV interphalangeus (TO)	laterally angled distal articular surface of proximal phalanx · persists after MTP correction unless addressed
2nd toe (TO)	**crossover toe** · **2nd MTP instability** (plantar plate attenuation, dorsal subluxation) · hammer / claw · transfer metatarsalgia · lesser-toe crowding
Associated	hammer toe · callosities · pes planus · Marfan · EDS · metatarsus adductus



Plantar 1st MTP: sesamoids in FHB, adductor hallucis laterally, abductor medially

3 Presentation

History	slowly progressive deformity · pain with ambulation · **difficulty with shoe wear** (medial eminence)
Commonest complaint	**pain over the bunion** (TO) · bursitis, callus over medial eminence
Lateral overload (TO)	**transfer metatarsalgia**, **stress fractures**, **callosities of lateral MTs** · long-standing → **MTP OA** (TO)
Digital nerve compression	symptoms possible
Exam	hallux **valgus + pronated** · 1st MTP ROM, crepitus, **grind** · **1st TMT hypermobility** · callus · sesamoid pain · pes planus, lesser toes, midfoot / hindfoot
Flexible vs fixed (TO)	flexible = toe passively pulled to normal → nonop · **fixed** = cannot → operative (TO)
Ligamentous laxity	Beighton (thumb to forearm, knee hyperextension) + excessive passive PF / DF of 1st TMT → **Lapidus**
Transfer metatarsalgia sign	**plantar focal tenderness and keratosis under the second metatarsal head**, **improves with metatarsal pad** ✗ Wrong options: Mulder click → Morton neuroma · insertional Achilles · fixed equinus with negative Silfverskiöld · fixed hindfoot varus on Coleman block (cavus is not a HV risk)
DDx	gout · **hallux rigidus** (stiff 1st MTP; gout, pseudogout, OA; **mostly bilateral**) (TO) · RA · turf toe · HV interphalangeus

Bunionette (tailor's bunion)

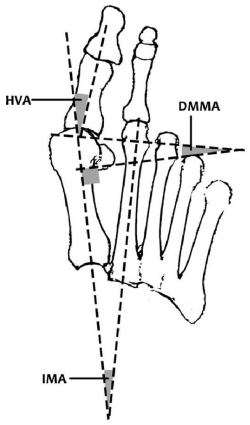
Site (TO)	**lateral side of 5th metatarsal** ✗ Wrong options: medial 1st MT · lateral 1st MTP · medial 5th MTP

4 Diagnosis & Classification

Key radiology findings

Views — **weight-bearing AP, lateral, oblique** · non-weight-bearing films **underestimate** deformity (TO) · **axial sesamoid (skyline) view** – sesamoid position (TO)

Measurement	Definition	Normal	Use
HVA	1st MT axis vs proximal phalanx axis	**< 15°**	MTP deformity; **> 15° diagnostic** (TO)
IMA	1st vs 2nd MT axes	**< 9°** (normal **9°**, TO)	MT deformity (**metatarsus primus varus**) → decides distal vs proximal
DMAA	1st MT axis vs line through base of distal articular cap	**< 15°** (Orthobullets) · **< 10°** (OB) · **< 6°** (TO) – read the options	**MTP congruence** (TO)
HVI	proximal vs distal phalanx	**< 10°**	**> 10° → Akin**
Metatarsus adductus angle	2nd MT vs lesser tarsus axis (4th / 5th MT-cuboid reference)	**< 10°**	Predisposes; Sgarlato, Engel, Kilmartin methods



HVA, IMA and DMAA

Measure of congruence — **Distal Metatarsal Articular Angle (DMAA)** (TO)
✗ Wrong options: talar–1st MT angle · IMA · HVA

Round sign (TO) — rounded lateral MT head on AP = **pronation of the first metatarsal** (TO) · **squared / flat lateral border = normal rotation** (OB) · positive round sign → **recurrence risk** (OB)
✗ Wrong options: subchondral cyst · sesamoid obliteration · AVN

Sesamoid position (TO) — **Hardy & Clapham grades 1–7** or tibial sesamoid position · position 1 = fully covered · 7 = completely lateral · **≥ position 4 significant**

1st MC joint instability (TO) — **joint-line obliquity**, **divergence of 1st / 2nd MT bases** → Lapidus · plantar-flexion **sag** at 1st TMT (OB)

MTP arthritis — narrowing, **dorsal osteophytes** (hallux rigidus component), sclerosis, cysts → arthrodesis (TO)

Diagnostic criteria

Criterion	Threshold
Diagnosis	**HVA > 15°** (TO)
Surgery (symptomatic)	**IMA > 10° + HVA > 20°** (Orthobullets, OB)
Severe (TO)	**HVA > 40° · IMA > 16° · HVI > 10° · DMAA up to 6° acceptable** (TO)
Significant sesamoid subluxation	**≥ position 4** (TO)

Congruent vs incongruent joint

	Congruent	Incongruent (subluxated)
Joint surfaces	**Parallel**, matched; hallux deviates but no subluxation	Proximal phalanx **subluxed laterally** on MT head
Cause	**Bony** (↑ **DMAA**, articular tilt)	**Soft-tissue imbalance** (tight lateral, lax medial)
DMAA	**↑** (congruent HV needs ↑ DMAA)	Usually **low**
Treatment	**Osteotomy correcting DMAA** (biplanar chevron, Akin, double osteotomy)	**Lateral release + medial imbrication** (modified McBride) ± osteotomy
Trap	**Lateral release in congruent joint → creates incongruency → OA** (TO)	Akin **contraindicated**



Congruent 1st MTP joint with ↑ DMAA

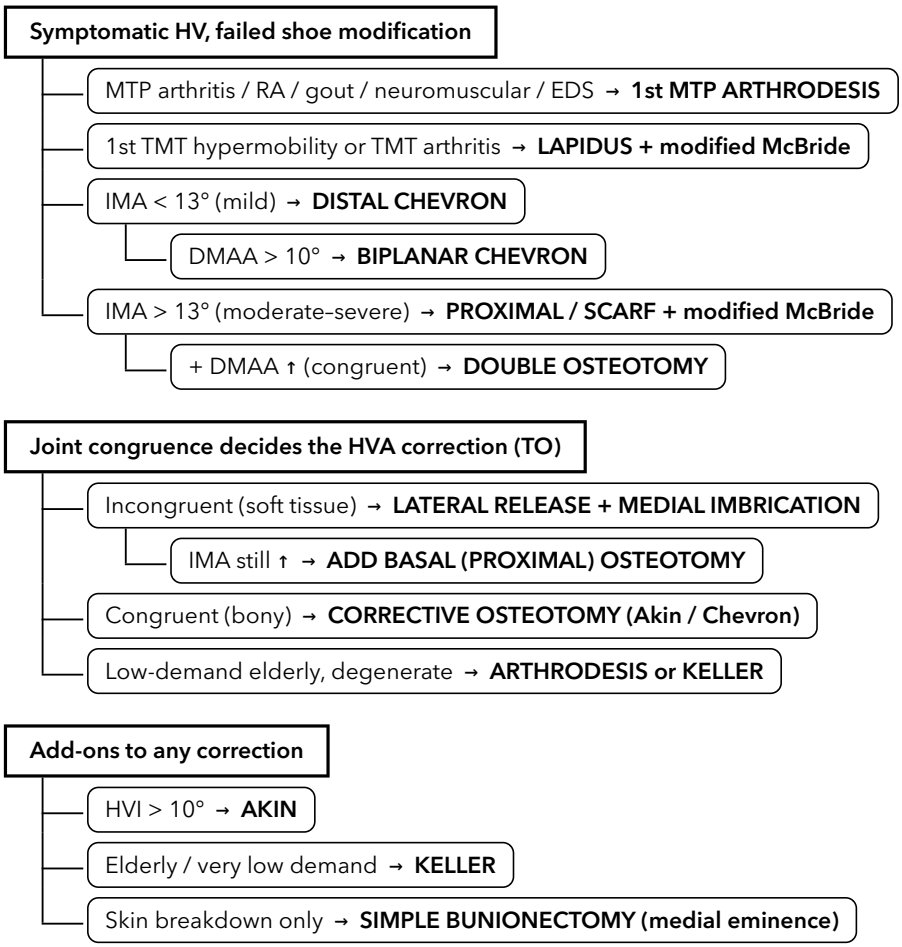
Severity grading – sources disagree

Grade	TO	Orthobullets	OB explanations
Normal	HVA **< 15°** · IMA **< 9°**	HVA **< 15°** · IMA **< 9°**	HVA **< 15°** · IMA **< 9°**
Mild	HVA **15–20°** · IMA **< 13°** (IMA grade **9–11°**)	HVA **< 30°** · IMA **< 13°**	HVA **< 25°** (chevron **< 25–30°**) · IMA **< 13°**
Moderate	HVA **20–40°** · IMA **13–16°** (IMA grade **11–16°**)	HVA **30–40°** · IMA **13–20°** · proximal if HVA **> 25°**, IMA **> 13°**	**IMA > 13°** → proximal
Severe	HVA **> 40°** · IMA **> 16°**	HVA **> 40°** · IMA **> 20°** · double osteotomy HVA **41–50°**, IMA **16–20°**	HVA **> 40–41°** · IMA **> 15–16°** (**> 20°**, Robinson)

Read the options **IMA 13°** = the commonest distal vs proximal divide · HVA cut-offs vary most

- - Active space - -

5 Management



Nonoperative (TO)

Role (TO)	**first-line for all** · definitive for mild symptoms, unfit, early deformity · **does not correct deformity** – relieves symptoms, slows progression
Footwear (TO)	**single most important** · **wide toe box**, adequate depth · **low heel < 2.5 cm (1 inch)** · soft uppers · custom / extra-depth for severe
Pads + spacers (TO)	bunion pads (silicone, felt, gel) · interdigital spacers · toe wedges / spacers worn **2–3 years** (TO)
Orthoses	help with **pes planus or metatarsalgia** (Orthobullets, TO) · **Morton's extension** limits MTP motion if hallux rigidus component (TO) · don't correct HV
Night splints (TO)	temporary relief only · **no evidence** they alter natural history
Anti-inflammatory (TO)	NSAIDs, ice · **steroid into bursa** short-term · repeated → **skin atrophy + capsule weakening**
Physio (TO)	intrinsic strengthening (towel curls, marble picks) · **gastrocnemius-soleus stretching** (tight → ↑ forefoot load)
Surgery indication	persistent symptoms despite shoe modification · **never for cosmesis alone** · choice by HVA, IMA, DMAA, congruency, arthritis, MC stability, age / demand (TO) · **> 150 procedures** described (TO) · > 130 (OB) ✗ Wrong options: asymptomatic contralateral bunion

Operative criteria

Situation	Procedure
Juvenile / adolescent, open physis	**First cuneiform osteotomy**
Hypermobile 1st MT / 1st TMT arthritis	**Lapidus**
Recurrence + 1st TMT pain	**Lapidus**
DJD of 1st MTP, gout, **RA**	**MTP arthrodesis**
Down's, CP, EDS, neuromuscular	**MTP arthrodesis**
Skin breakdown	**Simple bunionectomy** (medial eminence removal)
HVI > 10°	**Akin**
DMAA > 10° (Orthobullets, OB) · **> 15°** (OB)	**Biplanar chevron** / distal medial closing wedge
IMA > 13°	**Proximal / shaft osteotomy**
Severe + DMAA > 15° · juvenile DMAA > 20°	**Double osteotomy**
Elderly / very low demand, severe	**Keller**

Procedure selection by HVA / IMA / DMAA – chevron, scarf / proximal osteotomy, Akin, MTP fusion

Distal & soft tissue	Technique	Indication	Complications
Modified McBride (DSTP)	**Adductor release from lateral sesamoid** / PP · **lateral capsulotomy** · **medial capsular imbrication** · ± transverse intermetatarsal ligament (TO)	**Incongruent** joint · HVA **< 35°**, IMA **< 11°**, HVI **< 15°** (Orthobullets) · HVA **< 25°**, IMA **< 15°** (OB) · 30–50 y · **almost never alone**	Recurrence (high alone), hallux varus
Original McBride	+ **lateral sesamoidectomy** – **abandoned**	Not indicated	**Hallux varus**
Silver	Medial eminence resection / exostectomy only	Mild; skin breakdown	–
Chevron (Austin)	Distal **intra-articular** V-cut, **55–60°** · head **translated laterally 5–10 mm** (TO) · 1 screw / 2-mm absorbable pin · medial approach, L-capsulotomy	**Mild** (IMA **< 13°**), adults **and children**, congruent or incongruent · corrects IMA only **5–6°** (TO) · beats nonop (RCT)	**AVN**, recurrence, dorsal malunion
Biplanar chevron	Extra **medial wedge** (dorsomedial + plantar-medial limbs) → closing wedge ↓ DMAA	**DMAA > 10°**	–
Mitchell	Distal **extra-articular**, more proximal than chevron	As chevron, rare	Recurrence, **malunion**, transfer metatarsalgia
Akin	Proximal phalanx **medial closing wedge** · keep lateral cortex · staple / screw · moves FHL to midline	**HVI > 10°** · congruent + **DMAA < 10°** · secondary if chevron / DSTP under-corrects · adjunct	incongruent joint, DMAA > 15°

– – Active space – –

Proximal, fusion & salvage	Technique	Indication	Complications
Scarf / Ludloff / Mau	**Shaft Z (step-cut)** · plantar-sloping longitudinal cut + chevron limbs · **2 screws** · translate, **rotate (pronation)**, shorten / lengthen · IMA **8–10°** (TO)	HVA **> 25°**, IMA **> 13°** · DMAA normal or ↑ · **first choice for moderate** (OB)	**Dorsal malunion** + transfer metatarsalgia · **troughing** (TO)
Proximal crescentic / broomstick	Proximal MT + mod McBride · crescentic blade, distal fragment rotated laterally · **inherently unstable**	Severe · IMA **> 13°**, HVA **> 25°**	**Hallux varus** · **dorsiflexion malunion**
Proximal opening / closing wedge + DSTP (TO)	Medial opening (graft) or lateral closing	Severe IMA **without MC instability** (TO)	–
Double osteotomy	Proximal + distal (biplanar)	Severe, **DMAA > 15°** · juvenile **DMAA > 20°**	–
Lapidus	1st TMT arthrodesis + DSTP	Hypermobility, TMT arthritis, severe	Nonunion, shortening
1st cuneiform opening wedge	75% depth, parallel to TMT · graft · plate	**Open physis**, laxity, flatfoot, hypermobile ray	Nonunion
1st MTP arthrodesis	Medial / dorsal approach · conical reamers	HVA **> 40°**, IMA **> 20°** · DJD · neuromuscular · callosities under lesser MTs · failed surgery	Malposition → transfer metatarsalgia
Keller	Resect **base of proximal phalanx** + medial eminence	**Abandoned** · elderly, low demand	**Cock-up toe** · floppy weak hallux

Severe HV with high IMA (TO) (✗ Wrong options: **Chevron's osteotomy** (mild–moderate only; translation would exceed shaft width)) · ✓ scarf, 1st MTPJ fusion, Lapidus (TO)

IMA 18° (TO) **proximal (basal) metatarsal osteotomy** (TO)
✗ Wrong options: Akin · isolated Silver · distal chevron

Picture of HV – diagnosis + treatment (TO)

hallux valgus, Akin osteotomy (TO)
✗ Wrong options: hallux varus + MT osteotomy · hallux rigidus + Keller · hammer toe + MTP arthrodesis

Case (HVA / IMA / DMAA / HVI)	Answer	✗ Distractors
23 / 12, passively correctable	**Chevron**	Akin + McBride, Silver, Lapidus
25 / 12, HVI **9**	**Chevron** (+ Akin only if HVI > 10°)	IMA 13–15 → proximal
23 / 11 / 7	**Chevron**	↑ IMA → proximal + distal
20 / 13 / 12	**Biplanar chevron + modified McBride**	DMAA 6 → plain chevron
20 / 10 / 20, congruent, no TMT laxity	**Biplanar chevron**	+ TMT hypermobility → Lapidus
35 / 12 / 9 / 12, congruent	**Distal osteotomy, head displaced laterally + Akin medial closing wedge**	DSTP (joint congruent), head medially, lateral opening wedge
31 / 16, incongruent	**Proximal (Scarf) + modified McBride**	Keller, chevron, Lapidus, McBride alone
40 / 15 / 15	**Scarf**	Akin, DSTP, MTP fusion, Silver
35 / 16 / 10	**Proximal osteotomy + DSTP**	30/9/16 → biplanar distal
IMA 16, no TMT laxity	**Proximal osteotomy + modified McBride**	McBride alone, proximal alone
35 / 16, HVI 13, one foot symptomatic	**Proximal 1st MT osteotomy + Akin, right foot only**	Bilateral surgery
45 / 17	**Crescentic + Akin**	Akin + McBride, Lapidus + Keller
30 / 16 / 16, HVI 8	**Proximal + distal MT medial closing wedge**	Incongruent → no distal DMAA cut
69 y, push-off pain, MTP DJD	**First MTP arthrodesis**	Chevron, proximal, Lapidus

Akin wedge

medial closing: faster healing, better bony contact, earlier hardware removal, no graft
✗ Wrong options: lateral opening wedge

Distal osteotomy limits

corrects IMA, HVA, DMAA · **cannot correct 1st MT pronation** (OB) · don't use chevron if preop IMA **> 15.5°**; aim post-op IMA **< 10°** (OB)

Newer evidence

distal chevron with **long plantar limb** may match proximal osteotomy for moderate–severe (OB)

Metatarsus adductus + HV (OB)

MA **< 20° → rotational scarf osteotomy of the first metatarsal alone** gives **adequate correction without lesser metatarsal procedures** · MA **≥ 20° →** first-ray + lesser MT osteotomies
✗ Wrong options: 2nd / 3rd MT osteotomies mandatory if MA > 15° · first-three-ray TMT arthrodesis · disregard IMA · soft tissue contraindicated

General principles of correction (TO)

Lateral release (TO)

essential **except congruent deformity** · dorsal 1st web space or medial transarticular · **adductor from lateral sesamoid** + lateral capsulotomy + **lateral metatarsosesamoid ligament**

– – Active space – –

Medial eminence resection (TO)	**flush with medial shaft** · **over-resection → hallux varus**
Medial capsulorrhaphy (TO)	plication of attenuated medial capsule
Sesamoid reduction (TO)	sesamoids back under head in **crista** · persistent subluxation = inadequate lateral release or MT correction

Lapidus (first TMT arthrodesis) – hypermobility indications & modified McBride

Indications	**1st TMT hypermobility** (laxity) · **1st tarsometatarsal arthritis** · severe HV (HVA **41–50°**, IMA **16–20°**, OB · HVA **> 40°**, IMA **> 15°**, OB · IMA **> 16°**, TO · IMA **> 20°**, Robinson) · metatarsus primus varus · recurrence with TMT pain · **concomitant pes planus** · adolescent bunion (physis closed)
Not indications	IMA **12°** · HVA **40°** alone · **age > 50** · **flexible cavovarus** · **MTP** (not TMT) hypermobility · MTP arthritis (→ MTP fusion) · spastic CP (→ MTP fusion)
Contraindications (OB)	**open physis** · already **shortened 1st MT**
Typical answers	**HVA 37, IMA 18 + TMT joint hypermobility in young patient** · **HVA 42, IMA 24 + TMT hypermobility** · laxity + hypermobile TMT + flexible planus → **tarsometatarsal joint arthrodesis with modified McBride** · large IMA + TMT laxity → **first tarsometatarsal (TMT) joint arthrodesis** ✗ Wrong options: MTP fusion · **1st + 2nd TMT** (2nd not fused) · distal / proximal osteotomy (don't fix hypermobility)
Technique	2 incisions or extended medial · correct with **adduction, supination, neutral plantar flexion** · screws ± plate · resect cartilage, **keep subchondral plate** (↓ shortening) · Lapidus 1934 (TO) · original = 1st–2nd MT bases + intercuneiform
Better outcome	**multiple screw fixation** · **bone grafting** · **correction of IMA** · failed conservative care · **anatomic plantarflexion of 1st MT** ✗ Wrong options: **dorsiflexion unloading of the first metatarsal** → lesser MT overload
Lapidus alone	corrects valgus + dorsiflexion but **fails to correct pronation** → add **modified McBride** (derotates / supinates, ~10°) = "modified Lapidus"
Modified McBride components	**release of the adductor from the lateral sesamoid** · **lateral** capsulotomy · **medial** capsular imbrication ✗ Wrong options: medial capsulotomy · lateral imbrication · resect base of proximal phalanx (Keller)
Nonunion	**~10%** (Orthobullets, with MTP fusion) · **up to 12%** (OB) · **3–10%** (TO) · **4%** (OB) – read the options · ~33% symptomatic · NWB **6–8 wk** historically (TO)

Juvenile hallux valgus – DMAA & double osteotomy

	Juvenile / adolescent	Adult
Hallmark	**Metatarsus primus varus** (↑ IMA)	Soft-tissue + bony
DMAA	↑, **congruent** joint	Variable
Laterality	**Bilateral, familial**	–
Pain	Not the primary complaint	Primary
HVA	**Smaller** than adults	Larger
Bursa / medial eminence	**Less** prominent	Prominent
MTP joint	**Flexible** (↑ deformity)	not rigid in juvenile
Foot	**Flexible flatfoot**	–
Commonest complication	**Recurrence > 50%**	Recurrence 10–47%

Timing nonop **until physis closes** · surgery for symptoms with **IMA > 10° + HVA > 20°**

Open physis **no proximal MT osteotomy** · severe → **first cuneiform (opening-wedge) osteotomy** · soft tissue alone fails

Closed physis as adult, but congruent ↑ DMAA needs **second (distal) osteotomy → double first metatarsal osteotomy** (HVA **> 25°** moderate / **> 40°** severe + **DMAA > 15°**)

Cases HVA 36, IMA 16, DMAA 21, physis closed → **double first metatarsal osteotomy** (✗ Wrong options: Ludloff + DSTP · cuneiform · MTP fusion · biplanar chevron alone) · ↑ IMA + congruent ↑ DMAA → **proximal and distal first metatarsal osteotomy** (✗ Wrong options: phalanx osteotomy · DSTP) · **HVA 32 + DMAA 25** → double (✗ Wrong options: HVA 32 + DMAA 12 · skeletal immaturity · neuromuscular · TMT hypermobility)

Results (OB) double osteotomy corrects HVA ~22–24°, IMA ~8–9° · **4–5×** chevron correction · no recurrences

Commonest complication (TO) **recurrent hallux valgus**
✗ Wrong options: hallux varus · MTP OA · AVN · physeal arrest

Causes of recurrence (TO) inadequate lateral release · insufficient medial plication · **missed metatarsus adductus** · failure to recognise **congruency** · **TMT hypermobility**

Neuromuscular / connective tissue disorders – MTP arthrodesis

Rule **CP, Down's, EDS (COL5A1), RA, gout, severe DJD**, neuromuscular → **MTP arthrodesis** (high recurrence after osteotomy) · even for mild–moderate deformity

(✗ Wrong options: **Traps** – Akin) · **Evans** (calcaneal lengthening for planovalgus) · modified McBride · **Keller** (elderly, cock-up toe)

CP recurrence after distal osteotomy + McBride cause **cerebral palsy → 1st metatarsophalangeal joint arthrodesis** · CP adolescents: no recurrence, 75% excellent (OB)
✗ Wrong options: revision distal osteotomy · proximal osteotomy · "increased HVA"

– – Active space – –

1st MTP arthrodesis & salvage

Position	Valgus	Dorsiflexion	Rotation	Source
Orthobullets	**5–10°**	**5–10°**	**Neutral**	flat platform simulates WB
OB	**5–10°**	**5–15° relative to floor**	**Neutral**	–
TO	**10–15°**	**10–15° relative to 1st MT**	–	–

Read the options DF reference differs (floor vs 1st MT)

Technique avoid **EHL + dorsomedial cutaneous nerve** · dorsal plate + compression screw / crossed lag screws (TO)

Gold standard (TO) HV + **MTP arthritis (HV with hallux rigidus)** · **failed prior surgery** · neuromuscular (CP, polio) · rheumatoid · very severe in elderly · also subluxed / dislocated MTP, forefoot inflammatory disease (OB)

Outcomes **~90% satisfaction** · eliminates pain, prevents recurrence · **no high heels** (TO) · superior to **metallic hemiarthroplasty** (24% hemi failed, OB)

Keller (TO) shortened, **floppy, weak hallux** · loss of push-off · transfer metatarsalgia · replaced by MTP fusion · very elderly sedentary only

MTP arthroplasty (TO) silicone (**Swanson**), hemi, total · wear, loosening, **osteolysis** (silicone) · only if motion wanted and fusion declined

6 Complications

Complication	Incidence	Cause	Treatment
Recurrence	**10–47%** by procedure (Orthobullets) · **4–11%** after proximal osteotomy (OB) · Scarf **30% at 10 yr** (OB) · juvenile **> 50%**	**Insufficient preop assessment / failure to follow indications**	Revise **if symptomatic** (pain ≠ recurrence)
AVN MT head	**Rare** (Orthobullets) · **up to 20%** (OB) · 7.7% scintigraphy, none clinical (OB)	Distal (chevron) osteotomy	**MTP arthrodesis** ± structural graft
Dorsal malunion + transfer metatarsalgia	**~5%** · up to **82%** with single screw ± K-wire (OB)	Proximal crescentic, Lapidus (shortening)	**Plantarflexion osteotomy** · fusion + graft
Hallux varus	**6%**	Overcorrection	Reverse scarf / chevron / Akin · MTP arthrodesis
Cock-up toe	**Up to 41%** with Keller	**FHL injury** · most severe Keller complication	**Z-lengthen EHL** · revision Keller · MTP fusion
2nd MT transfer metatarsalgia	**Up to 50%** of HV patients	Concomitant; **malpositioned MTP fusion**	**Weil** + extensor + capsular release
Neuropraxia	**~3%**	**Dorsomedial cutaneous nerve**	Improves over months
Nonunion	**~10%** Lapidus + MTP fusion · **~33% symptomatic**	Arthrodesis	Revision + graft (± locked plating)
Dissatisfaction	**10–50%**	**Unmet expectations**, not outcome	–

Recurrence – sesamoid reduction & IMA

Risk factors
↑ preop IMA + **↑ preop HVA** · **rounded lateral 1st MT head** (round sign = pronation) · **residual increased tibial sesamoid displacement** (lateral) · DMAA ↑ missed (> 10°) · **inadequate IMA correction** · **no lateral release / adductor not released** · **no medial capsule closure** · **Akin alone** for moderate–severe · juvenile · noncompliance with WB

Sesamoids
unreduced sesamoids → 10× odds of recurrence (OB) · reduction under head **prevents recurrence of the hallux valgus deformity**
✗ Wrong options: extensor strength · sagittal correction · supination · nonunion

Unrelated to recurrence
resection of the lateral sesamoid → hallux varus, not recurrence (EXCEPT answer) · post-op IMA **15°** → recurrence
✗ Wrong options: varus

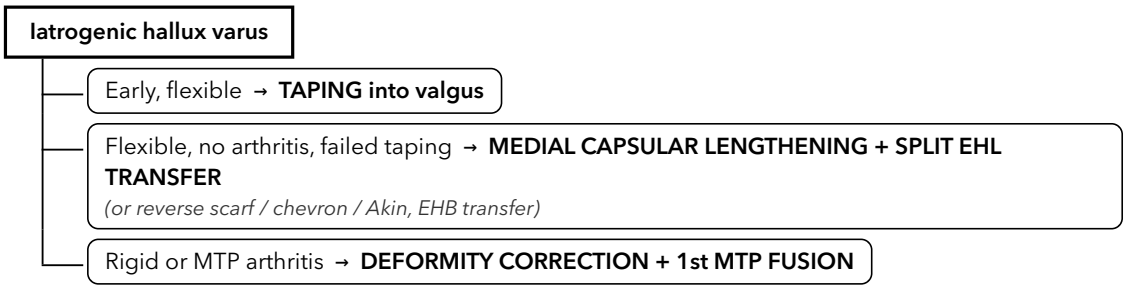
Revision target
adductor hallucis (label D) left unreleased
✗ Wrong options: FHL · abductor hallucis (pronator, not driver) · FHB bellies · remove lateral sesamoid (varus)

– – Active space – –

Bilateral case	right (squared head, tibial sesamoid reduced) recurred vs left (rounded, sesamoid overlapping lateral border) → **increased preoperative 1-2 intermetatarsal angle on the right compared to the left** ✗ Wrong options: head rotation · sesamoid alignment (those favour left recurrence) · medial eminence resection → varus
Persistent after isolated distal osteotomy	**pronation of the first metatarsal** · rotation best corrected by **fusion / proximal osteotomy** (OB) ✗ Wrong options: IMA · HVA · DMAA (distal osteotomy corrects these) · HVI 9° (normal)
Scarf	double-chevron diaphyseal osteotomy · **more stable** than others

Iatrogenic hallux varus – causes (fibular sesamoid excision, over-release, medial eminence)

Causes	**fibular (lateral) sesamoid resection** = **commonest** · **excessive medial capsular reefing** (+ adductor + lateral FHB detached) = **second commonest** · **over-resection of medial eminence** · overcorrection of IMA (< 9° normal) · **excessive lateral release** / **conjoined tendon release** · medial tibial sesamoid subluxation · excessive postoperative bandaging · MTP OA · AVN collapse into varus · malunion of distal osteotomy · original McBride
Mechanism	fibular sesamoidectomy releases lateral FHB slip → **unopposed medial FHB + abductor hallucis** → medial rotation + translation · usually **with supination** ✗ Wrong options: under-corrected pronation
Not a cause	**resection of the tibial sesamoid** (never resected in HV surgery; EXCEPT) · use of chevron / Akin per se · MTP arthritis (rarely varus) · complete medial sesamoidectomy · complete abductor release · **excessive lateral capsulorrhaphy** (it is **medial** capsulorrhaphy)
Medial eminence cut	saw blade placed too **lateral** (cut A) → **over-resection** → **hallux varus** ✗ Wrong options: recurrence · cock-up (FHL / Keller) · transfer metatarsalgia (shortening) · AVN
"Toe too straight"	**complete excision of medial eminence**



Flexible painless varus	**medial capsular lengthening and split extensor hallucis longus tendon transfer** ✗ Wrong options: MTP fusion (salvage) · lateral sesamoid excision (worsens) · lateral closing wedge phalanx osteotomy (misses MTP) · soft-tissue release alone
Varus + MTP arthritis, can't fit shoes	**deformity correction and first metatarsophalangeal (MTP) fusion** · standard fusion **without structural graft** ✗ Wrong options: cheilectomy · Keller · Moberg · Lapidus



Iatrogenic hallux varus

Nerve injury – dorsomedial cutaneous nerve

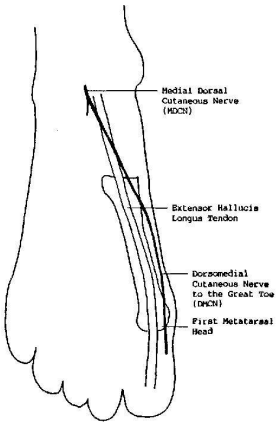
Nerve — **dorsomedial (medial dorsal) cutaneous nerve of the hallux** (DMCN) = **terminal branch of the superficial peroneal nerve** · sensation **dorsal + medial half of hallux** · lies directly over **medial eminence** · commonest nerve injured (~3%)

Injury — **dorsal / dorsomedial approach** · medial capsular imbrication / MT osteotomy → numbness or **neuritic pain medial hallux**, painful neuroma against shoe · prefer **direct medial / mid-medial** incision · transverse vein **superficial + proximal** to nerve (OB)

Course — mostly **neuropraxia → resolves** over months · recalcitrant → **resect DMCN + bury in proximal 1st MT or medial cuneiform** (OB)

✗ Distractor nerve	Territory per explanations
Deep peroneal (medial branch)	**1st web space**, lateral hallux; extensors
Dorsolateral cutaneous	"Branch of deep peroneal" (OB)
Saphenous	Dorsomedial hindfoot · medial ankle · medial midfoot (OB, three wordings)
Medial plantar / plantar hallucal	Plantar medial foot (tibial)
Intermediate dorsal cutaneous	Mid-dorsum, doesn't reach medial hallux
Sural	Lateral foot + ankle
Tibial	Posterior to medial malleolus → medial + lateral plantar

– – Active space – –



Dorsomedial cutaneous nerve over the 1st MT head

Complications – osteonecrosis, dorsiflexion malunion, shortening & transfer metatarsalgia

AVN – classic link	**distal Chevron osteotomy** (✗ Wrong options: proximal osteotomies) · collapse → **osteonecrosis of the metatarsal head and MTP arthrodesis** (✗ Wrong options: PTOA dx · intra-articular osteotomy · varus / recurrence)
AVN – blood supply conflict	**medial capsulotomy = primary insult**; distal osteotomy + lateral release **do not ↑ risk** (Orthobullets, OB) · **laterally based head supply** → chevron + **extensive lateral release** ↑ risk (OB) – read the options
AVN + shortening + lesser MT overload	**metatarsophalangeal joint fusion with an autologous bone block** (restores length) · AVN with bone loss = only complication needing **bone block arthrodesis** (OB) ✗ Wrong options: revision Lapidus · Keller (worsens overload) · Moberg · allograft MT head (experimental)
Dorsiflexion malunion	**first metatarsal proximal crescentic osteotomy** · inherently unstable · **smooth pins / single screw** · **plate** stronger (OB) · presents: **loss of MTP dorsiflexion**, **transfer metatarsalgia** ("marble" under 2nd–3rd MT heads) ✗ Wrong options: AVN (distal) · cock-up · varus · neuroma
Dorsiflexion malunion treatment	**plantar flexion metatarsal osteotomy** · plantarflexion malunion → revision osteotomy ± MTP fusion ✗ Wrong options: capsular release (bony block) · biplanar / double (DMAA) · phalanx osteotomy (HVI)
Transfer metatarsalgia – causes	**1st metatarsal shortening** (aggressive resection in MTP fusion, **modified Lapidus procedure** done incorrectly, chevron) · dorsiflexion malunion · residual pronation · **functional load transfer from altered forefoot biomechanics** after modified Lapidus · MTP fusion in **plantarflexion** (5° PF relative to floor) · **long 2nd MT** · equinus
✗ Transfer metatarsalgia traps	Cotton osteotomy (plantarflexion medial cuneiform for forefoot varus) · Akin · lateral sesamoidectomy (→ varus) · modified McBride (no shortening) · Morton's neuroma (Mulder click, paraesthesia) · **Freiberg** (2nd MT head AVN, visible on X-ray) · 1st MT nonunion · recurrence · pseudoarthrosis · TMT instability · cock-up toe
Prevention	**minimal bone resection** at 1st MT base · avoid shortening · fuse MTP in correct position
First MT	bears **half the forefoot weight** (OB)

Weil osteotomy technique

Indication	2nd MTP metatarsalgia / long 2nd MT (e.g. after MTP fusion) → **second metatarsal osteotomy (Weil) with extensor tendon and dorsal capsular release** ✗ Wrong options: 2nd MTP fusion · Girdlestone-Taylor · **Helal** (shaft, ↑ nonunion) · head resection
Weil	**intra-articular** MT head osteotomy · longitudinal decompression by **shortening**
Saw cut	**parallel with the plantar surface of the foot** → minimises **plantar displacement** of capital fragment · remove wafer / thicker blade also ↓ plantar shift ✗ Wrong options: perpendicular to shaft · parallel to MT inclination · 45° · 10°

Sesamoid excision – FHB preservation

Medial (tibial) sesamoidectomy	preserve / repair **flexor hallucis brevis tendon** → prevents **hallux valgus** ✗ Wrong options: FHL · abductor · adductor · EHB
Complications of medial sesamoidectomy	stiffness · claw toe · **hallux valgus** · (lateral sesamoidectomy → **hallux varus**)

Prognosis

Recovery	return to work **6–12 wk** · improvement up to **1 yr** · patient-reported outcome not tied to degree of correction (OB)

X vs Y

	Recurrence of HV	**Iatrogenic hallux varus**
Sesamoid	**Unreduced** (tibial sesamoid lateral)	**Fibular sesamoid resected**
Lateral release	**Inadequate** (adductor intact)	**Excessive** / conjoined tendon cut
Medial capsule	Not closed / insufficient plication	**Over-tightened**
IMA	**Under-corrected** (↑ preop)	**Over-corrected**
Medial eminence	–	**Over-resected**

	Chevron	**Mitchell**	**Scarf**	**Proximal crescentic**
Level	Distal, **intra-articular**	Distal, **extra-articular**	**Diaphyseal Z**	**Proximal**
IMA correction	**5–6°** (TO)	Low	**8–10°** (TO)	Large
Signature complication	**AVN**	**Malunion**	**Troughing**, dorsal malunion	**Dorsiflexion malunion**, varus

– – Active space – –

	Lapidus	1st MTP arthrodesis	1st cuneiform osteotomy
Key indication	**TMT hypermobility / arthritis**	**MTP arthritis, neuromuscular, RA, EDS**	**Open physis**
Pronation	Not corrected alone → add McBride	–	–
Pitfall	Shortening → transfer metatarsalgia	Malposition / shortening → transfer metatarsalgia	–

	Akin	Biplanar chevron	Modified McBride
Fixes	**HVI > 10°**	**DMAA > 10°**	**Incongruent** joint
Congruent joint	✓ (DMAA < 10°)	✓	No (creates incongruency)

How it's asked

"NOT a risk factor for hallux valgus?"	→ **pes cavus** (TO)
"Rotational deformity of 1st MT in severe HV?"	→ **pronation (internal rotation)** (TO)
"Round sign on DP radiograph?"	→ **pronation of the first metatarsal** (TO)
"Structure released from lateral sesamoid + proximal phalanx?"	→ **adductor hallucis tendon** (TO)
"Bunion, deformity of the hallux?"	→ **valgus and pronation**
"Abductor hallucis action once hallux is in valgus?"	→ **flexion and pronation**
"Conjoined tendon not to release?"	→ **FHB + adductor hallucis**
"Primary deforming tendon in HV?"	→ **adductor hallucis**
"Pes planus associated with progression of?"	→ **hallux valgus**
"HV + metatarsus adductus 18°?"	→ **rotational scarf alone may suffice** (no lesser MT procedures)
"Picture of great-toe deformity – diagnosis + treatment?"	→ **hallux valgus, Akin osteotomy** (TO)
"NOT recommended for severe HV with high IMA?"	→ **Chevron osteotomy** (TO)
"Does NOT indicate severe HV?"	→ **DMAA 5°** (TO)
"IMA 18° – procedure?"	→ **proximal (basal) metatarsal osteotomy** (TO)
"Measurement of MTP congruence?"	→ **DMAA** (TO)
"HVA 40, IMA 15, DMAA 15?"	→ **scarf osteotomy**
"HVA 23, IMA 12, passively correctable?"	→ **distal chevron**
"HVA 31, IMA 16, incongruent?"	→ **proximal (scarf) + modified McBride**

"Biplanar chevron is best for?"	→ **HVA 20, IMA 10, DMAA 20, congruent**
"IMA 16, HVA 35, HVI 13, only right foot painful?"	→ **proximal MT osteotomy + Akin, right foot only**
"Chevron picture – preop values?"	→ **HVA 25, IMA 12, HVI 9**
"Proximal osteotomy + DSTP – preop values?"	→ **HVA 35, IMA 16, DMAA 10**
"HVA 35, IMA 12, DMAA 9, HVI 12, congruent?"	→ **distal osteotomy (head lateral) + medial closing wedge phalanx osteotomy**
"69-year-old, push-off pain, MTP DJD?"	→ **first MTP arthrodesis**
"Proximal + distal medial closing wedge osteotomy for?"	→ **HVA 30, IMA 16, HVI 8, DMAA 16**
"Severe HVA 45, IMA 17, no TMT pain?"	→ **crescentic + Akin osteotomy**
"IMA 16, no TMT laxity – plan?"	→ **proximal osteotomy + modified McBride**
"Chevron picture – preop values (DMAA given)?"	→ **HVA 23, IMA 11, DMAA 7**
"Biplanar chevron + modified McBride – deformity?"	→ **HVA 20, IMA 13, DMAA 12**
"Lapidus – better outcome EXCEPT?"	→ **dorsiflexion unloading of the first metatarsal**
"HV, large IMA, 1st TMT laxity – bony procedure?"	→ **first TMT arthrodesis**
"Indication for Lapidus?"	→ **1st tarsometatarsal arthritis**
"Laxity (Beighton), hypermobile TMT, flexible planus?"	→ **TMT arthrodesis + modified McBride**
"Lapidus most appropriate?"	→ **HVA 37, IMA 18, TMT hypermobility, young patient**
"Component of modified McBride?"	→ **release of adductor from lateral sesamoid**
"Findings needing Lapidus?"	→ **HVA 42, IMA 24, TMT hypermobility**
"Commonest complication after juvenile HV correction?"	→ **recurrent hallux valgus** (TO)
"14-year-old, HVA 36, IMA 16, DMAA 21, physis closed?"	→ **double first metatarsal osteotomy**
"14-year-old, ↑ IMA, congruent joint, ↑ DMAA?"	→ **proximal and distal first metatarsal osteotomy**
"Juvenile vs adult bunion?"	→ **metatarsus primus varus**
"17-year-old, ↑ DMAA, congruent, high IMA, no hypermobility?"	→ **double metatarsal osteotomy**
"Indication for double osteotomy in juvenile HV?"	→ **HVA 32° + DMAA 25°**
"HV in Ehlers-Danlos (COL5A1)?"	→ **MTP arthrodesis**
"HV in cerebral palsy?"	→ **MTP arthrodesis**

– – Active space – –

"CP, recurrence after distal osteotomy — cause + treatment?"	→ cerebral palsy; 1st MTP arthrodesis
"Recurrence risk EXCEPT?"	→ resection of the lateral sesamoid
"Right foot recurred (squared head, sesamoid reduced) — why?"	→ ↑ preop IMA on the right
"Recurrence related to?"	→ residual increased tibial sesamoid displacement
"Recurrent HV — labelled structure to address?"	→ adductor hallucis (D)
"Left abnormal after isolated distal osteotomy?"	→ pronation of the first metatarsal
"Why reduce sesamoids during scarf?"	→ prevents recurrence
"Causes of post-op hallux varus EXCEPT?"	→ resection of the tibial sesamoid
"Flexible painless hallux varus — reconstruction?"	→ medial capsular lengthening + split EHL transfer
"Toe 'too straight' one year after correction — cause?"	→ complete excision of medial eminence
"Medial eminence cut too lateral (A not B)?"	→ hallux varus
"Post-op complication needing bone block arthrodesis?"	→ AVN with bone loss (C)
"Varus + MTP stiffness after chevron + Akin + adjuncts — cause?"	→ resection of fibular sesamoid
"Dorsal + medial hallux numbness after dorsal chevron approach?"	→ dorsomedial cutaneous nerve of the hallux
"Nerve most at risk in bunion surgery?"	→ terminal branch of superficial peroneal nerve
"Paraesthesia after HV correction — nerve?"	→ medial dorsal cutaneous nerve
"Neuritic medial hallux pain after dorsomedial incision — true of the nerve?"	→ terminal branch of SPN
"Persistent medial hallux numbness after distal osteotomy + Akin?"	→ dorsomedial cutaneous nerve
"2nd MT callus after MTP fusion — what should have been added?"	→ Weil osteotomy + extensor tendon + dorsal capsular release
"Dorsal toe pain, malunion after single-screw bunion surgery — procedure?"	→ proximal crescentic osteotomy
"Collapse of MT head 2 years after chevron?"	→ osteonecrosis → MTP arthrodesis
"17-year-old, dorsiflexion malunion after crescentic osteotomy?"	→ plantar flexion metatarsal osteotomy
"AVN + shortening + lesser MT overload after chevron?"	→ MTP fusion with autologous bone block
"Hallux varus + MTP arthritis, can't fit shoes?"	→ deformity correction + 1st MTP fusion

"Aggressive bone resection in MTP fusion causes?"	→ **transfer metatarsalgia**
"2nd–3rd MT pain + keratosis after prior bunion surgery – which procedure?"	→ **modified Lapidus**
"MTP fused in 5° plantarflexion – complication?"	→ **transfer metatarsalgia**
"'Marble' under 2nd–3rd MT heads after crescentic + smooth wires?"	→ **dorsiflexion malunion**
"Callus after modified Lapidus – cause?"	→ **functional load transfer from altered forefoot biomechanics**
"Lesser MT pain 3 years after MTP fusion?"	→ **1st metatarsal shortening**
"Associated finding with HV + plantar callus?"	→ **keratosis under 2nd MT head, improves with metatarsal pad**
"Weil osteotomy saw cut?"	→ **parallel to plantar surface of foot**
"Medial sesamoid excision – preserve what?"	→ **FHB tendon**
"Tailor's bunion is over?"	→ **lateral side of 5th metatarsal** (TO)

– – Active space – –

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HALLUX VARUS

– Foot & Ankle · 25 / 48 –

1 Basics

Definition	**medial deviation of the hallux relative to 1st metatarsal** · most often **iatrogenic overcorrection after bunion surgery**
Incidence	**2–14% after hallux valgus surgery** · **more common in women**
Three components	**medial deviation of hallux** at 1st MTP · **supination of phalanx** · **claw-toe deformity**

Causes – excessive lateral release & over-correction

Cause	Examples
Congenital	**Metatarsal physeal bracket**
Iatrogenic (commonest)	Overcorrection at bunion surgery
Trauma	–
Inflammatory	**RA**, **ankylosing spondylitis**
Neurological	**CMT**, **post-polio**

Iatrogenic pathoanatomy	**excessive lateral soft-tissue / capsular release** · **fibular (lateral) sesamoid excision** · **excessive medial eminence resection** · **overplication of medial capsule** · **overcorrection of IMA** (e.g. **excessively large proximal medial opening-wedge osteotomy**) or of hallux valgus interphalangeus · loss of osseous support · malaligned osteotomy, soft-tissue laxity
Medial release	listed as a contributor (OB7610) · not **excessive medial capsular release** (OB220893 – lateral release + medial overplication cause it) – read the options
Not causes	**tibial sesamoidectomy** · **inadequate** medial eminence resection · **1st TMT hypermobility** · **inadequate IMA correction** · failed pin fixation · choosing soft-tissue procedure over fusion
After fibular sesamoid resection (TO)	**unopposed pull of abductor hallucis → medial dislocation of tibial sesamoid** → medial hallux translation · also **MTP joint hyperextension** + **buttonholing of metatarsal head** (TO) ✗ Wrong options: **adductor hallucis**

3 Presentation

Complaint	**appearance** (toe "too straight" → excessive medial deviation) · **difficulty wearing shoes** · often **asymptomatic**
Pain	suggests **joint arthritis** or trauma · ↓ ROM, instability, weak push-off
Exam	varus angulation · **dorsal contracture of MTP** ± IP contracture · **EHL displaced medially → "bowstring"** · **tibial sesamoid displaced medially** · medial blister · **fixed vs flexible (passively correctable)** decides treatment

4 Diagnosis & Classification

Key radiology findings

View / finding	Detail
WB AP + lateral foot	Standard · + NWB oblique + **sesamoid axial**
Hallux valgus angle	**< 0°** (normal **5–15°**) · e.g. HVA −19°
IMA	**Reduced** (overcorrected; e.g. 6°)
Medial eminence	**Excessively resected**
Sesamoids	**Tibial sesamoid subluxed medially** · **lateral (fibular) sesamoid absent**
Osteotomy	Overcorrected
Joints	**Degenerative change at MTP / IP** → fusion
CT / MRI	Rarely needed · **osteonecrosis of 1st metatarsal**



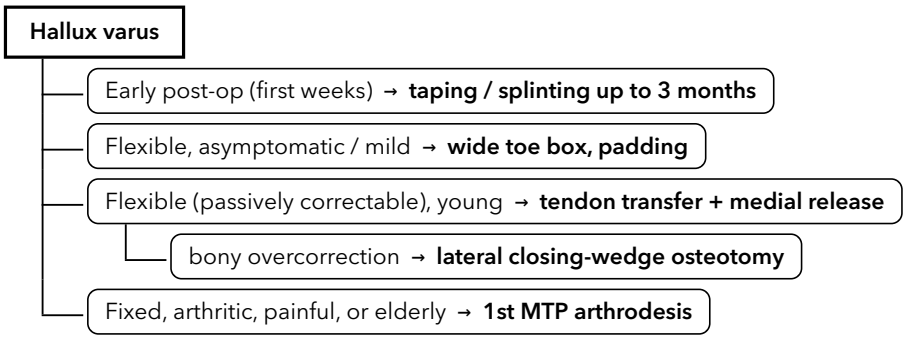
Hallux varus after bunion surgery



Iatrogenic hallux varus after bunion correction

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5 Management



Treatment – reducible vs fixed deformity (tendon transfer vs MTP fusion)

Taping / splinting — **early post-op varus**; may correct if started **within first weeks**; continue **up to 3 months** / until soft tissues heal · not for fixed deformity

Shoe modification — **wider, flexible toe box** + padding · **flexible, longstanding, asymptomatic** · not for chronic fixed + arthritic · established varus → nonop role limited

Lateral closing-wedge osteotomy — **overcorrected proximal / distal MT or proximal phalangeal osteotomy** · ± scar release + lateral ligament repair · not when MTP degenerative

Tendon procedure (flexible deformity)	Detail
Adductor hallucis reattachment + medial release	**Difficult after McBride**
Abductor hallucis transfer → lateral base of proximal phalanx	+ reattach / reef **conjoined tendon** in web space
EHL or EHB transfer + medial release ± IP fusion	Under **transverse intermetatarsal ligament → distal MT neck** (lateral → medial) · **split EHL** + MT osteotomy + medial capsule release for **young, passively correctable**

Operative criteria

1st MTP arthrodesis	Indication
Absolute	**Fixed (not passively correctable) MTP** with significant deformity + non-functioning hallux · **painful MTP arthritis**
Relative	**Medial eminence resected beyond tibial sesamoid sulcus**
Age	**Elderly – 72-yo, moderately reducible, painful → fusion** (soft tissue not indicated for age)

Fusion gives **most predictable pain relief, alignment, activity, shoe fit, no recurrence**

✗ Alternative	Why not
MTP replacement / hemiarthroplasty	Unpredictable; **radiolucency, subsidence**, high failure (hallux rigidus)
Keller resection arthroplasty	Unpredictable pain relief, activity, recurrence
Interposition arthroplasty	**No realignment**
Lateral MT opening-wedge osteotomy	Doesn't correct **arthritic, incongruent MTP**
1st MT–cuneiform (Lapidus) fusion	**Salvage for failed hallux valgus**, not varus
Cheilectomy + medial release / soft tissue alone	Won't correct fixed deformity
Great-toe amputation	Unreasonable in healthy patient

X vs Y

	Flexible (passively correctable)	Fixed / arthritic
Patient	**Young**	**Elderly** or degenerative MTP
Surgery	**Tendon transfer + medial release** ± osteotomy	**1st MTP arthrodesis**
Splint / tape	Useful early	No

How it's asked

"Hallux varus after fibular sesamoid resection – all EXCEPT?"	→ **unopposed pull of adductor hallucis** (it's the abductor) (TO)
"Progressive varus 2 yr after bunion surgery – factor?"	→ **excessive lateral soft-tissue release**
"HVA −19°, IMA 6° after bunionectomy – error?"	→ **excessively large proximal metatarsal medial opening-wedge osteotomy**
"Rigid varus, pain mid-range, 70-yo?"	→ **hallux metatarsophalangeal joint fusion**
"Not completely reducible, failed taping – predictable alignment + pain relief?"	→ **MTP joint arthrodesis**
"72-yo, moderately reducible, painful – most predictable?"	→ **first MTP fusion**
"Fixed varus + DJD after fibular sesamoidectomy?"	→ **first MTP arthrodesis**

– – Active space – –

HAMMER TOE

– Foot & Ankle · 26 / 48 –

1 Basics

Epidemiology

Commonest lesser-toe deformity	hammer toe · **older / middle-aged women** · **2nd toe** usually
Associated	**painful corn on dorsal PIP** · secondary to **hallux valgus** or curly toe · neurogenic, degenerative, metabolic causes

Plantar plate attenuation – MTP deformity

Pathoanatomy	**plantar plate injury** · **overpull of EDL** · **intrinsic imbalance / weakness**
Plantar plate rupture → hammer toe	MT head no longer suspended → **translates plantarly** → **MTP extension** → **EDL overpull** → **PIP flexion** to keep toe plantigrade ✗ Wrong options: mallet toe · TMT arthritis · Freiberg · hallux valgus (not the direct result)
Plantar plate attenuation causes	**MTP joint extension** ✗ Wrong options: DIP flexion · PIP extension · MTP / DIP malrotation
Plantar 2nd MTP pain	"**stepping on a pebble**" · worse in high heels · MRI = degenerative plantar plate tear

3 Presentation

Symptoms	**dorsal PIP pain with shoe wear** · deformity · plantar 2nd MTP pain
Exam	**PIP flexion + DIP extension** (+ slight MTP extension)
Ankle plantarflexion test	**flexible deformity corrects** · **fixed / rigid does not correct**
Push-up test	dorsal pressure under the MT head → **flexible deformity reduces** (removes extrinsic overpull)



Hammer toe – clinical appearance

4 Diagnosis & Classification

Diagnosis **clinical**: rigid or flexible lesser toe with PIP flexion, DIP extension, slight MTP extension

Key radiology findings

Imaging **not required** for diagnosis or treatment

Claw toe vs hammer toe vs mallet toe

Joint	Claw toe	Hammer toe	Mallet toe
			
DIP	Flexion	**Extension**	**Flexion**
PIP	Flexion	**Flexion**	Normal
MTP	**Hyperextension**	**Slight extension**	Normal
Cause	–	**EDL overpull** + intrinsic weakness	**FDL overpull / contracture** or EDL rupture
Callus	–	**Dorsal PIP**	**Toe pulp** (ulcers / calluses)

5 Management

- **Hammer toe**
  - Pain / dorsal PIP corn → **high toe box, foam / silicone sleeves, dorsal PIP padding**
  - Flexible, failed nonop → FDL → **EDL transfer**
  - Rigid, failed nonop → **PIP resection arthroplasty (± arthrodesis)**
  - MTP involved → Girdlestone (FDL → **EDL)**
    - Mild MTP hyperextension → **EDL Z-lengthening / tenotomy**
    - Moderate–severe → **EDL Z-lengthening + dorsal capsular release**

– – Active space – –

Operative criteria

Situation	Procedure
Flexible, failed nonop	**FDL → EDL tendon transfer**
Rigid / fixed, failed nonop	**PIP resection arthroplasty** ± tenotomy / transfers · or **PIP arthrodesis**
MTP involvement	**Girdlestone** (FDL → EDL) – as for claw toe
Mild MTP hyperextension	**EDL Z-lengthening or tenotomy**
Moderate–severe MTP hyperextension	**EDL Z-lengthening + dorsal capsular release**
2nd hammer toe + hallux valgus	**Correct hallux valgus** (arthrodesis if severe) · **amputation** if HV touches 3rd toe in elderly / poor health / declines reconstruction

Surgery indicated for progressive deformity · **fixed contractures** · **dorsal toe ulceration**

Flexible vs rigid deformity – padding & PIP resection arthroplasty

Fixed deformity
precludes success with soft-tissue procedures alone → indication for **PIP resection arthroplasty**
✗ Wrong options: pain with shoes · all lesser toes involved · no MTP deformity · no hallux valgus

Rigid 2nd hammer toe sequence
dorsal PIP padding → shoe modification → PIP resection arthroplasty
✗ Wrong options: **plantar** padding (pain is dorsal) · MTP capsular release · MT head excision

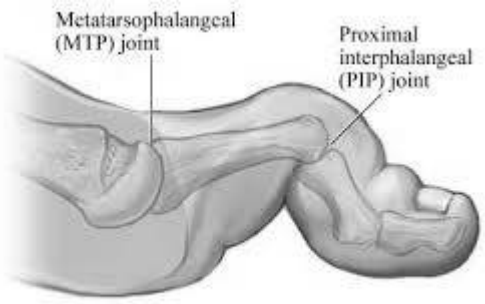
Extensor tenotomy / lengthening
with (not before) resection arthroplasty in severe cases → for **MTP dorsiflexion contracture**
✗ Wrong options: tenotomy preceding arthroplasty

Rigid hammer toe, failed nonop
PIP resection arthroplasty with K-wire fixation
✗ Wrong options: **Weil osteotomy** (irreducible MTP subluxation / transfer metatarsalgia) · **FDL transfer** (flexible) · **EDL lengthening** (claw toe) · **DIP fusion** (rigid mallet toe)

Resection arthroplasty technique
resect **head + neck (distal condyle) of proximal phalanx** (± middle phalanx articular surface) → **fibrous joint** or arthrodesis (**true fusion not necessary**) · **K-wire 2–3 wk** · then **tape PIP in extension 3 wk**

PIP arthrodesis
option for rigid deformity · **high nonunion rate** · **K-wire = most cost-effective implant**

Girdlestone
EDL Z-plasty lengthening · **MTP capsule release** · ± oblique **MT shortening osteotomy** · **FDL → EDL transfer**



Rigid hammer toe – PIP flexion, DIP extension

X vs Y

	Flexible hammer toe	Rigid hammer toe
Ankle PF / push-up test	**Corrects**	**Does not correct**
Surgery	**FDL → EDL transfer**	**PIP resection arthroplasty** / arthrodesis

Procedure	Treats
FDL transfer	Flexible hammer toe
PIP resection arthroplasty	Rigid hammer toe
EDL lengthening	Claw toe / MTP hyperextension
DIP fusion	Rigid mallet toe
Weil osteotomy	Irreducible MTP subluxation · transfer metatarsalgia

How it's asked

"Indication for PIP resection arthroplasty over soft-tissue balancing?"	→ **fixed deformity**
"Rigid 2nd hammer toe — sequence?"	→ **dorsal PIP padding, shoe modification, PIP resection arthroplasty**
"Rigid hammer toe, failed high toe box and sleeves?"	→ **PIP resection arthroplasty with K-wire fixation**
"Hammer toe — plantar plate attenuation causes?"	→ **MTP joint extension**
"Degenerative plantar plate tear — likely to develop?"	→ **hammer toe**
"Commonest lesser-toe deformity?"	→ **hammer toe** (2nd toe)
"Flexible hammer toe, failed nonop?"	→ **FDL → EDL transfer**
"PIP arthrodesis — cheapest implant?"	→ **K-wire** (high nonunion rate)

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HIGH ANKLE SPRAIN & SYNDESMOSIS INJURY

– Foot & Ankle · 27 / 48 –

1 Basics

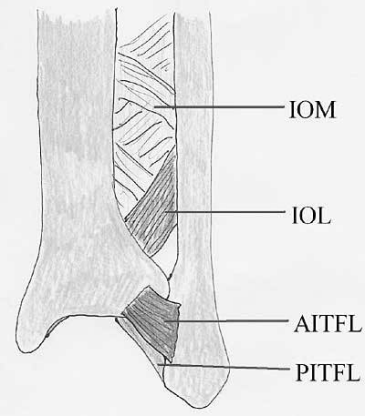
Epidemiology

Incidence **0.5% of ankle sprains** without fracture · **13% of all ankle fractures**

Associated injuries OCD **15–25%** · peroneal tendon injury **up to 25%** · **deltoid ligament** · loose bodies · fractures: **Weber C**, Weber B, 5th MT base, anterior process calcaneus, lateral / posterior process talus

Anatomy – distal tibiofibular syndesmosis

Ligament	Origin → insertion	Key fact
AITFL	**Chaput tubercle** (anterolateral tibia) → **Wagstaffe tubercle** (anterior fibula)	Injured in isolated high ankle sprain
PITFL	**Volkmann tubercle** (posterior tibia) → posterior lateral malleolus	**Strongest component**
Interosseous ligament (IOL)	Distal continuation of IOM	**Main restraint to proximal migration of talus**
Interosseous membrane (IOM)	–	Disrupted → synostosis risk
Inferior transverse ligament (ITL)	–	–



Syndesmosis anatomy – IOM, IOL, AITFL, PITFL

Not syndesmosis **ATFL** (anteroinferior fibula → **talar neck**) · CFL, PTFL (lateral complex) · **bifurcate (Chopart) ligament** (subtalar + midtarsal) · IPR (peroneal tendons)

Fibular incisura morphology **crescentic commonest (61.3%)** · trapezoid · chevron · widow's peak · **flat least (3.1%)**

Biomechanics

Function maintains tibia–fibula integrity · resists **axial, rotational, translational** forces · widens **1 mm** in normal gait

Ankle position	Talus in mortise	Distal fibula
Dorsiflexion	**Wider anterior talus** engages	**External rotation** + proximal, posterolateral translation
Plantarflexion	Narrow posterior talus	**Internal rotation** + distal, anteromedial translation

Inversion / eversion mainly subtalar → little effect on fibular rotation · **supination** (PF + inversion + adduction) → fibular IR

Deltoid **indirectly stabilises** medial mortise · **superficial = restrains eversion** · **deep = prevents excessive ER**

Sectioned (Massri-Pugin, cadaver)	Result
Isolated AITFL or PITFL	**No significant instability**
AITFL + PITFL + IOL	Diastasis **posterior incisura only**
All syndesmotic + deltoid	Diastasis **anterior + posterior** incisura

Instability needs rupture of **≥ 2 ligaments** (Rammelt) · instability ↑ **immensely** once **deltoid** also cut (Zalavras)

2 Mechanism

Commonest **external rotation** of foot (or **foot planted, body internally rotates**) → talus rotates laterally, pushes fibula away from tibia

Consequences ↑ tibial compressive stress · lateral fibular subluxation · mortise incongruence

Eversion injury medial + lateral pain, positive calf squeeze

3 Presentation

Symptoms **anterolateral ankle pain proximal to AITFL** · ± medial tenderness / swelling · **difficulty bearing weight** (lateral sprains usually can WB)

Clue nagging pain with dynamic moves, **relieved by tight circumferential wrap above ankle → AITFL** (wrap holds fibula in incisura)

Syndesmosis tenderness **single best predictor of return to play**

Test	Technique	Positive
Squeeze (Hopkin)	Compress tibia + fibula at **midcalf**	Pain at syndesmosis
External rotation stress	**Knee + hip 90°**, ankle neutral; ER / dorsiflex foot	Pain / widening
Cotton (lateral stress)	**Lateral pull on fibula**	Syndesmosis widening
Fibular translation	AP drawer on fibula, tibia fixed	↑ translation + pain · **most false positives** (Beumer)

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4 Diagnosis & Classification

Diagnosis – examination, stress radiographs & MRI; Maisonneuve

Views	AP, lateral, mortise ankle **+ AP, lateral entire tibia / fibula** (**proximal fibula fracture = Maisonneuve**)
Optional	**external rotation stress view** · **gravity stress view** (deltoid competence) · **contralateral ankle** (widening vs normal variant)
Persisting WB pain 2 wk after "sprain"	**ankle stress radiographs + ipsilateral tibia-fibula radiographs** (bilateral comparison) ✗ Wrong options: boot WBAT · NWB boot without further films · MRI first · tib-fib films alone
Fibular fracture + syndesmotic injury	**ORIF fibula, then assess syndesmosis** (repair may stabilise; most still need fixation)

Key radiology findings

Measurement	Normal	Injury
Tibiofibular overlap	**> 6 mm AP** · **> 1 mm mortise** · **> 10 mm AP** (Orthobullets ligament note)	↓
Tibiofibular clear space	**< 6 mm AP + mortise** · **< 5 mm** (Orthobullets ligament note)	↑
Medial clear space	**≤ 4 mm** (best on mortise)	↑ · good specificity, **very poor sensitivity**
Lateral tibiofibular ratio	APTF ratio avg **0.94**; anterior fibular cortex crosses **middle of tibial shadow**	**Most reliable plain-film measure of reduction**



AP measurements — 1 tibiofibular overlap, 2 tibiofibular clear space, 3 medial clear space

– – Active space – –



Lateral tibiofibular ratio — segment 4 (anterior tibia → anterior fibula) over 5 (anterior fibula → posterior tibia)

AP measurements limited sensitivity, poor reliability · **cannot reliably predict** injury (Krähenbühl)

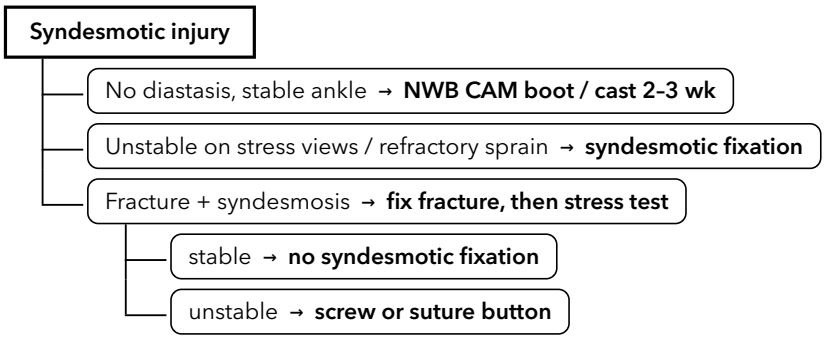
Study	Use
CT	Suspicion with normal X-rays · **more sensitive than X-ray for minor injury** · **post-op check of reduction** · Wagstaffe tubercle (AITFL insertion) on axial
MRI	Suspicion with normal X-rays · **> 90% sensitivity / specificity** (Han) · **~100%** (Krähenbühl) — diagnostic tool of choice
Lambda sign (coronal MRI)	**Hyperintense signal contiguous between mortise and tibiofibular articulation** (OB) · **> 2 mm diastasis on coronal** (TO) · "**highly sensitive + specific**" (Orthobullets) · not an independent predictor, sensitive + specific **with positive exam** (Ryan)
Deltoid discontinuity on MRI	Poor indicator — no deltoid ruptures in Han's 8 syndesmotic injuries



Coronal T2 MRI lambda sign — fluid tracking from mortise into syndesmosis

– – Active space – –

5 Management



Operative criteria

Nonoperative	Operative (screw or suture button)
Sprain without diastasis or ankle instability	**Sprain (no fracture) unstable on stress radiographs**
–	**Sprain refractory to conservative treatment**
–	**Fracture still unstable after fracture fixation**

Nonop delayed WB until pain-free · PT in **brace limiting ER** · recovery **notoriously prolonged + variable**, up to **2× standard sprain**

Intraoperative assessment & reduction

Best intra-op test **external rotation stress test**: **fewest false positives, smallest inter-observer variance**
✗ Wrong options: squeeze · fibular translation · Cotton · anterior drawer

Cotton test previously reported least false positives · produces **greater tibiofibular clear space** widening than ER test in Weber C (Stoffel) · ER stress gives **35% more medial clear space** widening (Matuszewski) – read the options

Dorsiflexion **externally rotates fibula** (wide anterior talus) · aids rotational reduction · **maximal dorsiflexion before screw not required** (Tornetta: PF fixation did not limit DF)

Clamp **lateral malleolar ridge of fibula → centre of AP width of tibia** = most accurate reduction (Gardner)

Goal restore **fibular length, rotation, position in incisura** · **outcome depends on reduction, not fixation type**

Persistent talar tilt after suture button **deltoid** incompetent
✗ Wrong options: interosseous (addressed by button) · bifurcate · CFL · PTFL

Reconstruction order proximal → distal: **IOL + AITFL first → repair deltoid** if medial talar tilt persists (Wake)

AITFL repair restores mortise stability · syndesmotic fixation unnecessary once repaired · reconstruct with **4th EDL autograft** or suture (Nelson)

Fixation – screw vs suture button

Screw technique **two 3.5 or 4.5 mm** screws · **3 or 4 cortices** · **2–5 cm above plafond** (Orthobullets) · **2–3 cm, parallel to joint, angled 30° anterior** (TO) – read the options · NWB **6–12 wk**

No difference stainless steel vs **titanium** · **3 vs 4 cortices** (pullout) · **bioabsorbable** similar · two screws preferable

– – Active space – –

	Syndesmotic screw	Suture button (TightRope)
Malreduction	**Higher – up to 40%** (**39%** vs 15%, Sanders RCT)	**Lower**
Reoperation / implant removal	**30%** – often requires removal	**4%** – **no removal needed**
Function	Same	Same · early results: **earlier return to activity**
Mechanics	Medial + anterior talar translation	Closer to native
Zone 2 fibular screws	↑ malreduction	–

6 Complications

Complication	Fact
Malreduction	Commoner than believed · **up to 40% with screws**
Missed injury	**End-stage ankle arthritis**
Chronic instability	Inadequate reduction → instability + post-traumatic arthritis
Tibiofibular synostosis	**~10% after Weber C** (Orthobullets) · 15 / 230 operative ankle fractures, mostly Weber C (Albers)

Post-injury synostosis / heterotopic ossification

Cause high ankle sprain with **IOM disruption** · prior **tibial stress fracture** (Henry) · Weber C fixation

Timing HO develops within **6–12 months**

Presentation **activity-related leg / anterior shin pain** for 1–2 yr · **loss of ankle dorsiflexion** + tib-fib motion · often **asymptomatic**; most return to sport

Treatment symptomatic first · **surgical excision** (anterolateral approach) only for **persistent pain despite nonop + no uptake ("cold") on bone scan**

Next step, synostosis with ongoing uptake **observation + repeat bone scan → resect when no uptake**

✗ Synostosis traps **indomethacin 25 mg TID 6 wk** + **700 cGy** = HO prophylaxis after acetabular surgery · **60 Gy** (sarcoma / myeloma; wounds won't heal) · chemotherapy (only with biopsy-proven neoplasm) · excise when "hot" · instability or ↓ plantarflexion as indication



Post-traumatic distal tibiofibular synostosis

– – Active space – –

X vs Y

	High ankle sprain	Low (lateral) ankle sprain
Ligaments	**AITFL**, PITFL, IOL, IOM, ITL	**ATFL, CFL, PTFL**
Mechanism	**External rotation**	**Inversion**
Weight bearing	**Difficult**	Usually possible
Test	Squeeze, ER stress	Anterior drawer
Recovery	**≈ 2× longer**	Shorter

	ER stress test	Cotton test
Manoeuvre	ER foot, tibia stabilised	Lateral pull on fibula
Intra-op	**Fewest false positives, least variance** (recent)	Least false positives (older) · more clear-space widening in Weber C

How it's asked

"CT arrow at Wagstaffe tubercle — structure + next step?"	**→ anterior inferior tibiofibular ligament (AITFL); ORIF of the fibula with subsequent syndesmotic assessment**
"Foot planted, body turned, better with a tight wrap above ankle — injured?"	**→ AITFL**
"Sprain, WB pain at 2 wk, ?MCS widening — next?"	**→ ankle stress radiographs and ipsilateral tibia-fibula radiographs**
"Most sensitive + specific imaging finding?"	**→ coronal MRI hyperintense signal contiguous between mortise and tibiofibular articulation** (lambda sign)
"Intra-op test with fewest false positives?"	**→ external rotation stress test**
"Manoeuvre that externally rotates the fibula?"	**→ ankle dorsiflexion**
"Most reliable X-ray measure of syndesmotic reduction?"	**→ lateral tibiofibular ratio (measurements 4/5)**
"Persistent talar tilt after suture button?"	**→ deltoid**
"True of syndesmotic screws?"	**→ higher rate of malreduction with screw fixation versus flexible fixation**
"Syndesmotic synostosis, pain, uptake on bone scan — next?"	**→ observation and repeat bone scan, then surgical resection if no increased uptake**
"Indication + timing for synostosis excision?"	**→ persistent pain despite non-operative treatment and no uptake on bone scan**
"Single best predictor of return to play?"	**→ syndesmosis tenderness**

INTERDIGITAL (MORTON'S) NEUROMA

– Foot & Ankle · 28 / 48 –

1 Basics

Epidemiology

Demographics	**middle-aged women** · **F:M 9:1** (Orthobullets) · peak **6th decade** (OB)
Commonest site	**3rd web space** (3rd + 4th interdigital nerves, between 3rd + 4th MT heads) · **2nd web space = second commonest** ✗ Wrong options: 1st web space · 1st MTPJ (hallux rigidus / valgus) · 5th MT head
Other OB wording	"2nd or 3rd web space" / "2nd and 3rd interdigital nerves" — read the options

Anatomy & pathophysiology

Interdigital nerve	**plantar to the transverse (deep) intermetatarsal ligament** between MT heads ✗ Wrong options: dorsal to it
3rd web space	**confluence of medial + lateral plantar nerves** (lateral branch MPN + medial branch LPN share a common perineurium) ✗ Wrong options: anterior tarsal tunnel (deep peroneal nerve)
Nature	**compressive (entrapment) neuropathy → perineural fibrosis + entrapment** of the interdigital nerve
Mechanism	compression / tension around **transverse intermetatarsal ligament** at toe-off · **repetitive microtrauma** · traction from tethering between fixed medial + mobile lateral MTs
Aggravators	**high heels**, **narrow toe box**, wide feet

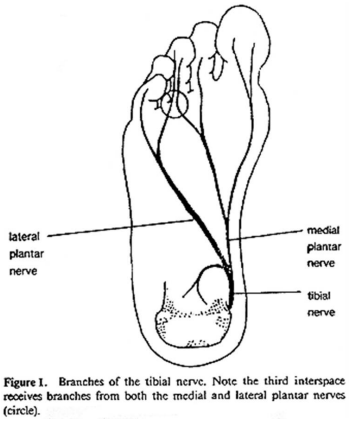


Figure 1. Branches of the tibial nerve. Note the third interspace receives branches from both the medial and lateral plantar nerves (circle).

Third interspace receives branches of both medial and lateral plantar nerves

– – Active space – –

3 Presentation

Diagnosis & location (third web space, Mulder's click)

Pain	**plantar forefoot**, between MT heads · worse with **weight-bearing + narrow toe-box shoes** · relieved by **removing shoes + massaging foot** · "**walking on a marble**"
Paraesthesia	burning / tingling into adjacent toes (3rd + 4th toes for 3rd web space) · plantar aspect of web space
Most consistent sign	**tenderness of involved web space** · **positive web-space compression test** · neuroma may be palpable
Mulder's click	squeeze forefoot (MTs together) + dorsal pressure between MT heads → painful / bursal click
Drawer test at MTPJ	assesses **MTPJ instability** (differential)



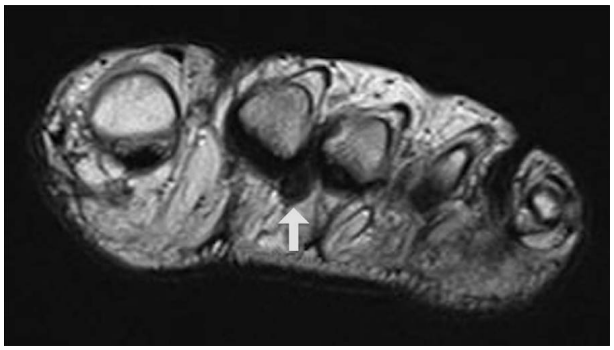
Mulder's click – squeezing the forefoot

4 Diagnosis & Classification

Diagnosis — **clinical** (history + exam); advanced imaging generally not required

Key radiology findings

Study	Use / finding
WB AP / lateral / oblique X-ray	**Usually normal** · may show bony deformity · excludes Freiberg (MT head flattening), MTP arthritis
Ultrasound	**Non-palpable neuroma** with clear clinical picture · **oval / well-circumscribed hypoechoic mass parallel to MTs** · **operator dependent**
MRI	**Not required** (✗ Wrong options: "needed to confirm") · rules out other pathology / stress fracture when exam non-diagnostic



MRI: interdigital neuroma (arrow) between MT heads

Diagnostic criteria

Test	Positive result
Common digital nerve block (diagnostic lidocaine)	**Confirms diagnosis** → numbness of lateral toe surface + **relief of pain**
No relief after well-placed block	Think **MTP synovitis**
Histology	**Perineural fibrosis + nerve entrapment, no fibrinoid necrosis**

Differential – rheumatoid nodule & histology

Lesion	Histology
Morton's neuroma	**Perineural fibrosis**, entrapped interdigital nerve, **no fibrinoid necrosis**
Rheumatoid nodule	Subcutaneous · **central fibrinoid necrosis + palisading fibrous cells**
Neurofibroma	Hypocellular fibroblastic network, **wire-like collagen**, myxoid areas
Schwannoma (benign PNST)	Encapsulated · **Antoni A** (palisading spindle cells) + **Antoni B** (hypocellular mucinous)
Plantar fibromatosis	From **plantar fascia** · dense mature collagen + fibrocytes · not provoked by squeeze

Other differentials **MTP synovitis**, metatarsalgia, **stress fracture**, MTPJ arthritis, MT head osteonecrosis (Freiberg), neoplasm, lumbar radiculopathy
✗ Wrong options: **anterior tarsal tunnel** (deep peroneal nerve) · **tarsal tunnel**

5 Management

Morton's neuroma

- First line → **wide toe-box shoe + firm sole + metatarsal pad**
  (± short-term steroid injection, dorsal)
- Failed nonop → **dorsal neurectomy + DTML release, bury stump**
- Recurrent / stump neuroma → **revision neurectomy (plantar or dorsal) + burial**

– – Active space – –

Nonoperative

First line	**flat shoe with wide toe box + firm sole + metatarsal pad** ✗ Wrong options: MRI · PT (limited role) · serial steroid injections · custom orthotics before shoe change
Metatarsal pad	offloads MT heads by **↑ intermetatarsal space** → ↓ nerve pressure
Results	unpredictable · **~20% complete resolution** (Orthobullets) · **roughly half** eventually need surgery (OB) · "most" need surgery (another OB) – read the options · NSAIDs rarely help
Corticosteroid injection	**short-term relief only**, **no long-term relief** · usually **dorsal** · steroid + LA **not superior** to LA alone · ~50% still request excision · **serial injections → local tissue (fat-pad) atrophy**
Alcohol sclerosing injection	not recommended (no long-term data, risk to surrounding tissue)

(✗ Wrong options: **Wrong orthoses** – **Morton extension** = hallux rigidus) · **longitudinal arch support** = PTT tendinitis · **carbon-fibre insert** = midfoot arthritis · **AFO** = peroneal palsy / foot drop
✗ Wrong options: "orthotics with ↑ arch height first line"

Treatment – injection & surgical excision (dorsal vs plantar)

Failed nonop	**neurectomy** (most effective, most durable) · **dorsal approach most common** ✗ Wrong options: repeat steroid · isolated DTML release (less effective) · distal oblique shortening osteotomy / MTP capsular release + extensor lengthening (these are for **MTP synovitis**)
Percutaneous nerve ablation	not used

Dorsal neurectomy step	Detail
Incision	**3–4 cm** just proximal to web space
Dissection	**Blunt** → protect **superficial peroneal nerve** branches (✗ Wrong options: deep peroneal)
Exposure	Spread MTs → tension + **transect transverse intermetatarsal ligament** (✗ Wrong options: "reduces success")
Resection	Identify nerve proximal + distal to bifurcation → resect **≥ 3 cm proximal to ligament**
Burial	Bury proximal stump in **intrinsic muscles**
Closure	**Repair DTML** → prevents intermetatarsal instability

Dual neuromas (2nd + 3rd web spaces)	**resect 2nd web space + neurolysis (DTML release) of 3rd** → equal relief, protects **digital arteries** to 3rd + 4th toes ✗ Wrong options: resect both via two dorsal incisions (highest arterial risk) · leave 3rd untreated · FDL transfer · single plantar incision

6 Complications

Complications of neurectomy

Complication	Fact
Stump neuroma – inadequate resection	**Commonest** cause of persistent / recurrent pain
Inadequate retraction (traction neuritis)	Plantar branches tether nerve → no retraction (✗ Wrong options: commonest)
Painful plantar scar	**~5%**, with **plantar** incision
Intermetatarsal instability	DTML not repaired
Wrong structure resected	Digital artery / neurovascular bundle or **lumbrical**
Toe ischaemia	Digital artery injury in dual web-space surgery

Recurrent neuroma – inadequate proximal resection & revision

Cause — **inadequate proximal resection** → residual **plantar-directed nerve branches** tether nerve to plantar skin → no retraction out of WB zone of MT head
✗ Wrong options: painful scar · infection · Freiberg · MTP arthritis · tarsal tunnel · MTP synovitis

Presentation — same symptoms as primary + **Tinel's sign** at prior resection site

Resection level — **≥ 3 cm proximal** to DTML (Orthobullets, OB) · plantar branches **absent 4 cm proximal** → cut there (another OB) – read the options · "5 mm proximal" = inadequate

Revision — **revision neurectomy through plantar OR dorsal approach + nerve burial**
✗ Wrong options: steroid + PT · fishmouth web-space incision (poor view, wound problems) · lumbrical transection → MTP extension deformity · dual incisions

Plantar approach in revision — preferred by most after a dorsal primary → avoids scar, **direct view of nerve bifurcation**, easier **proximal** dissection, DTML seen directly

Dorsal approach in revision — advantage = **lower risk of plantar scar tenderness + wound complications** · **earlier weight-bearing** but **recurrence not lower** · needs DTML release · harder proximal dissection

Revision outcomes — function similar · no society recommendation favouring either

– – Active space – –

X vs Y

	Dorsal approach	Plantar approach
Use	**Most common primary**	**Preferred for revision**
Complete excision	Less likely · revision **8–10%**	**More likely** · revision **< 5%**
Sensory loss	**Greater**	Less
Scar	No plantar scar · ✓ fewer wound problems	**Painful plantar scar** (~5%)
Weight-bearing	**Earlier**	Longer NWB
DTML	Must be **released**	Seen directly
Outcomes / satisfaction / ADL limits	Comparable	Comparable

	Morton's neuroma	MTP synovitis
Site	Web space, plantar	MTP joint
Test	Mulder's click, web-space compression	**Drawer test** (instability)
Digital nerve block	**Relief**	**No relief**
Surgery	Neurectomy	Shortening osteotomy (long 2nd MT) · capsular release + extensor lengthening

How it's asked

"Commonest location?"	**→ third web space**
"Cyclist, pain between 3rd + 4th toes, worse in shoes – next step?"	**→ shoes with a wide toe box**
"Tingling into 3rd + 4th toes, Mulder's click – true?"	**→ confluence of medial + lateral plantar nerves**
"Burning between 3rd + 4th toes, forefoot click – initial management image?"	**→ metatarsal pad** (Figure D)
"Plantar vs dorsal approach – plantar has?"	**→ increased likelihood of complete neuroma excision**
"Commonest complication of the most effective treatment?"	**→ neuroma secondary to inadequate resection**
"Failed shoe change, brief relief from steroid – next?"	**→ dorsal neurectomy**
"Walking on a marble – true of treatment?"	**→ corticosteroid injections do not provide long-term relief**
"2nd + 3rd web-space neuromas – safest plan?"	**→ resect 2nd, neurolysis of 3rd by DTML release**
"Excised 5 mm proximal to ligament, Tinel's positive – cause?"	**→ inadequate proximal resection**

"Commonest cause of persistent pain after excision?"	→ **inadequate resection of an interdigital neuroma**
"Recurrence after dorsal surgery — best step?"	→ **revision neurectomy through plantar or dorsal approach with nerve burial**
"Primary advantage of dorsal approach for revision?"	→ **lower risk of plantar scar tenderness + wound complications**
"Polyarthropathy, ANA+, perineural fibrosis without fibrinoid necrosis?"	→ **Morton's neuroma** ✗ Wrong options: rheumatoid nodule

– – Active space – –

– – Active space – –

LEG NERVE ENTRAPMENT SYNDROMES

– Foot & Ankle · 29 / 48 –

1 Basics

Entrapment sites – overview

Nerve	Site of compression	Name / clue	Treatment
Ilioinguinal	Hypertrophied abdominal muscles (intensive training)	Hyperaesthesia · pain ↑ with **hip hyperextension**	Surgical release if nonop fails
Obturator	Well-developed adductors (**skaters**)	Chronic **medial thigh pain** · **nerve conduction studies** establish diagnosis	Supportive
Lateral femoral cutaneous	Tight belts, prolonged hip flexion	**Meralgia paresthetica** (lateral thigh pain)	PT, NSAIDs, posture, release compressive devices
Sciatic	**Ischial tuberosity** · **piriformis**	**Piriformis syndrome**	–
Saphenous	**Adductor (Hunter's, subsartorial) canal**	**Surfer's neuropathy** · prolonged kneeling · pain **inferomedial to knee**	**Knee pads** first line
Common peroneal	**Behind fibula / fibular neck** – ganglion cyst, direct blow, pressure	Foot drop	**Proximal tibiofibular joint fusion** to prevent cyst recurrence
Superficial peroneal	**Fascial exit ~10–12 cm** (OB) · **12–15 cm** (Orthobullets) **proximal to lateral malleolus**	Dorsal foot numbness ↑ with **PF + inversion**	Observation → fascial release
Deep peroneal	**Inferior extensor retinaculum**	**Anterior tarsal tunnel syndrome** · **1st web space**	–
Posterior tibial	**Flexor retinaculum** / tarsal tunnel	**Tarsal tunnel syndrome** · medial + plantar foot	–
1st branch lateral plantar (Baxter)	**Abductor hallucis deep fascia vs medial quadratus plantae**	**Distal tarsal tunnel syndrome** · runners	**Release abductor hallucis fascia**
Medial plantar	**Knot of Henry** (FDL × FHL) · **foot orthotics**	**Jogger's foot**	**Discontinue orthosis**
Sural	**12–15 mm distal to fibular tip** (equinus) · gastrocnemius–Achilles musculotendinous junction (calf)	Lateral dorsum + heel	Surgical release
Interdigital	–	**Morton's neuroma**	–

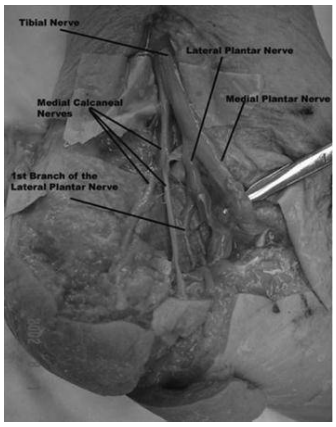
Tarsal tunnel anatomy

– – Active space – –

Tarsal tunnel — fibro-osseous tunnel **directly inferior to medial malleolus** · roof = **flexor retinaculum** · **abductor hallucis = plantar border** · compression by fascial septae, flexor retinaculum, abductor hallucis

Tibial nerve — divides into **medial plantar**, **lateral plantar**, **medial calcaneal** branches · plantar nerves = **terminal sensory branches** → most plantar sensation

Dissection label (medial ankle)	Structure
A	**1st branch of lateral plantar nerve (Baxter) — most commonly entrapped**
B	Medial calcaneal nerve branches
C	Tibial nerve
D	Lateral plantar nerve
E	Medial plantar nerve (jogger's foot)



Medial ankle dissection — tibial nerve, plantar nerves, medial calcaneal branches, 1st branch of LPN

Dorsal sensation — **SPN = dorsum except 1st web space** · **DPN = 1st web space** · **sural = lateral foot**
✗ Wrong options: plantar surface

Saphenous nerve entrapment (TO)

Saphenous nerve (TO) — **purely sensory** · **longest terminal branch of posterior division of femoral nerve** · **L3–L4**

Course (TO) — from below inguinal ligament with femoral artery · **deep to sartorius** throughout thigh · enters **subsartorial / adductor / Hunter's canal** (proximal third of thigh, per TO) — fibrous band **vastus medialis ↔ adductor magnus**

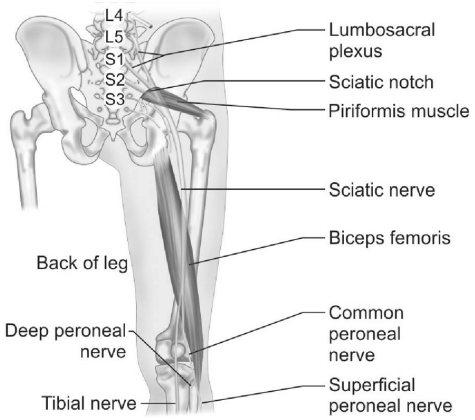
Exit (TO) — joins **saphenous branch of descending genicular artery** · both **pierce roof of the canal just proximal to the adductor hiatus**

Commonest entrapment site (TO) — **adductor canal**
✗ Wrong options: below inguinal ligament · at adductor hiatus · below patella

– – Active space – –

Common peroneal nerve compression (TO)

Why CPN (TO)	**most fixed nerve, minimum tolerance to mobility** · fixed at origin **under biceps femoris** → fixed to **fibular neck** as it winds around
Unconscious, lying in lateral position, bruised lateral knee (TO)	**CPN** injury ✗ Wrong options: tibial · ulnar · trigeminal



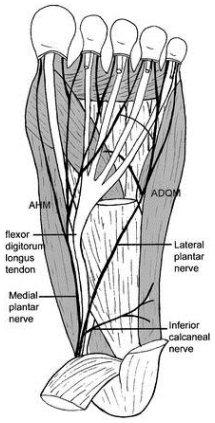
Course of the common peroneal nerve behind biceps femoris and around the fibular neck

Branches	**CPN → DPN + SPN** · SPN in **lateral compartment** → supplies **peroneus longus + brevis** → dorsal foot sensation

3 Presentation

Baxter's nerve (first branch of lateral plantar nerve)

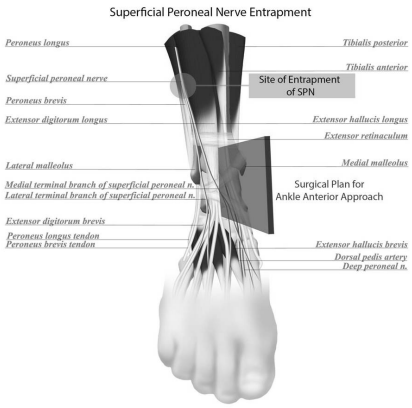
Who	**running athlete** · normally arched or **cavus** foot · responsible for **~⅕ of hindfoot pain**, widely missed
Symptoms	**chronic plantar medial heel pain** (often 9–12 months) · start-up pain like plantar fasciitis **plus neuritic burning pain unrelated to loading** · worse at end of day / prolonged activity
Signs	**tenderness at abductor hallucis origin** · **Tinel's just proximal to abductor hallucis → radiates to lateral foot / 5th MT base** · **no Tinel's over posterior tibial nerve** · chronic → **abductor digiti minimi atrophy**
Innervates	**abductor digiti quinti (minimi)** (OB) · also **FDB + quadratus plantae** (one OB) ✗ Wrong options: abductor hallucis · FDL · FHL · plantar fascia
Course	**deep to abductor hallucis + FDB** · **superficial to quadratus plantae** · **medial to calcaneus** · **just anterior to medial calcaneal tuberosity** / just superior to plantar fascia insertion
Compression point	between **deep fascia of abductor hallucis** and **medial caudal margin of medial head of quadratus plantae** (Orthobullets writes "abductor hallucis longus")
Mechanism	stretching + tethering in abductor hallucis fascial leashes · hypertrophy of small foot muscles · hindfoot microtrauma in runners



Plantar nerves — lateral plantar nerve and inferior calcaneal (Baxter) nerve to ADQM

Superficial peroneal nerve entrapment

Mechanism	**inversion injury** · **fascial defect** (± muscle herniation)
Site	where it **exits deep crural fascia of anterolateral leg**, **~12 cm** (Orthobullets) · **10–12 cm** (OB) proximal to lateral malleolus · lateral → anterior compartment
Symptoms	**numbness + tingling over dorsum of foot** · ↑ with **passive plantarflexion + inversion** · ± eversion weakness
Sridhara signs	↓ dorsal sensation **sparing 1st web space** · **soft-tissue bulge ~10 cm above lateral malleolus** · **Tinel's over bulge** · bulge ↑ with resisted dorsiflexion / WB · tenderness over bulge or terminal branches
Variable course	SPN in **lateral compartment 72%**, anterior 23% (Barrett) · **anterior in almost 50%** (Rosson) → **explore both compartments**



Site of SPN entrapment at its fascial exit in the distal anterolateral leg

- - Active space - -

4 Diagnosis & Classification

Key diagnostic findings

Test	Finding
EMG	Lateral plantar nerve entrapment positive in **88%** of neurogenic subcalcaneal heel pain
MRI	**Abductor digiti minimi atrophy** → Baxter's neuropathy
Nerve conduction studies	Help establish **obturator** entrapment
Electrical studies	Confirm **peroneal compression at fibular neck**

5 Management

Baxter's nerve entrapment

- First → **rest, ice, PT, orthotics, NSAIDs**
- Persistent > 3 months → **release abductor hallucis fascia** *(good–excellent 89%)*

Superficial peroneal nerve entrapment

- Most → **observation**
- Refractory / elite athlete → **fascial release + SPN neurolysis**
  - fascial defect → **release, never close (worsens)**

Operative treatment

Condition	Operation	✗ Traps
Baxter's (distal tarsal tunnel)	**Release of abductor hallucis fascia** → decompresses **lateral plantar nerve** branch	partial plantar fasciotomy / gastrocnemius recession (= recalcitrant **plantar fasciitis**)
SPN	**Fascial release + superficial peroneal neurolysis** (80% satisfied)	**repair herniation + close fascial defect** · 4-compartment fasciotomy · neurectomy · lumbar discectomy
CPN ganglion	Excision · **proximal tibiofibular fusion** prevents recurrence	–
Medial plantar (jogger's foot)	**Stop orthosis**	–
Saphenous	**Knee pads** first line	–
Sural	Surgical release	–

Baxter release complications persistent heel pain needing more surgery · neuroma · superficial infection · RSD

6 Complications

Peroneal nerve palsy – order of recovery

Short leg cast → foot drop	compression of **peroneal nerve at fibular neck**
Last muscle to recover	**extensor hallucis longus** (innervated **most distally**) ✗ Wrong options: EDL · peroneus longus · tibialis anterior · FDL (tibial nerve)

X vs Y

	Baxter's nerve	Plantar fasciitis	Tarsal tunnel	Jogger's foot
Nerve	**1st branch LPN**	–	Posterior tibial	**Medial plantar**
Site	AbH fascia / quadratus plantae	Calcaneal insertion	Flexor retinaculum, below medial malleolus	**Knot of Henry**, orthotic arch
Pain	**Medial plantar heel, burning**, ↑ tapping	**Plantar heel**, first steps, ↑ great-toe dorsiflexion	Medial + plantar foot	Medial arch
Tinel's	**At AbH origin → lateral foot**	–	**Over tibial nerve**	–
Patient	Runner, cavus	–	–	Runner, **hyperpronation, heel valgus**
Surgery	**AbH fascia release**	Partial fasciotomy / gastrocnemius recession	Tarsal tunnel release	Stop orthosis

	SPN	DPN	Sural
Site	Fascial exit **10–12 cm** above lateral malleolus	**Inferior extensor retinaculum**	Anywhere; Achilles MTJ; 12–15 mm distal to fibula tip
Sensory loss	**Dorsum, spares 1st web**	**1st web space**	**Lateral dorsum + heel**
Name	–	**Anterior tarsal tunnel**	–

How it's asked

"Runner, heel pain, Tinel's at abductor hallucis origin radiating laterally?"	**→ compression of the first branch of the lateral plantar nerve (Baxter's nerve)**
"Nerve to abductor digiti quinti compressed under abductor hallucis fascia?"	**→ first branch of the lateral plantar nerve**
"Baxter's nerve — muscle affected?"	**→ abductor digiti minimi**
"Abductor hallucis fascia release relieved heel pain — which nerve?"	**→ lateral plantar nerve**

– – Active space – –

"Anatomy of the Baxter nerve?"	→ **runs deep to the flexor digitorum brevis** ✗ Wrong options: superficial to AbH · deep to quadratus plantae · branch of medial plantar
"Medial ankle dissection — nerve most commonly entrapped?"	→ **A = 1st branch of lateral plantar nerve**
"Inversion injury, dorsal foot numbness ↑ PF + inversion — cause?"	→ **compression of the superficial peroneal nerve by fascial band proximal to lateral malleolus**
"Commonest SPN compression site?"	→ **proximal to lateral malleolus, exiting fascia of anterolateral leg**
"Marathoner, Tinel's over distal lateral fibula bulge — surgery?"	→ **fascial release and superficial peroneal neurolysis**
"Foot drop after short leg cast — last to recover?"	→ **extensor hallucis longus**
"Unconscious in lateral position, bruised lateral knee — nerve?"	→ **CPN** (TO)
"Commonest site of saphenous nerve entrapment?"	→ **adductor canal** (TO)

– – Active space – –

LISFRANC INJURY

– Foot & Ankle · 30 / 48 –

1 Basics

Epidemiology

Definition TMT fracture-dislocation · disruption between **medial cuneiform and base of 2nd metatarsal** · purely ligamentous or fracture-dislocation (treatment implications)

Incidence **0.2% of all fractures** · **1 per 55,000 / yr** · **males > females** · **3rd decade**

Missed **easily missed**, up to **20%** at initial encounter · missed → **progressive planovalgus**, chronic pain

Lisfranc equivalents **contiguous proximal metatarsal fractures** or **tarsal fractures** · may involve multiple TMT joints

Anatomy – Lisfranc & plantar ligaments

Lisfranc joint complex **tarsometatarsal** + **intermetatarsal** + **intertarsal (intercuneiform)** articulations · 5 MTs with 3 cuneiforms + cuboid

Column	Joints	Mobility
Medial	**1st TMT**	Little / none
Middle	**2nd + 3rd TMT**	Little / none
Lateral	**4th + 5th TMT**	**Most mobile** – accommodates uneven ground

✗ Mobility trap (TO) 1st–3rd cuneiform–MT joints have **greater** mobility than 4th–5th (false – they are **stable, rigid lever for push-off**)

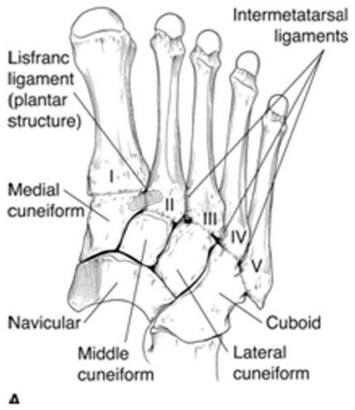
– – Active space – –

Bony stability — **trapezoidal bases of 1st–3rd MTs → transverse "Roman arch"** · **2nd TMT = keystone**, 2nd MT recessed in mortise between **medial and lateral (recessed middle) cuneiforms** · **shallow 2nd TMT mortise = risk factor** (TO)

Ligament	Key facts
Lisfranc ligament	**Plantar interosseous** · **lateral surface of medial cuneiform → medial base of 2nd MT** (near plantar surface) · **largest + strongest** interosseous TMT ligament · **8–10 mm wide, 5–6 mm thick** · **tightens with pronation + abduction** of forefoot · stabilises 1st/2nd TMT + midfoot arch (✗ Wrong options: "dorsal interosseous" (TO trap))
Plantar TMT ligaments	**Plantar ligament medial cuneiform → 2nd + 3rd MTs** + Lisfranc ligament both needed for **transverse instability**
Dorsal TMT ligaments	**Weaker than plantar** (TO ✓) → **displacement usually dorsal**
Intermetatarsal ligaments	**2nd–5th MT bases only** · **no IM ligament between 1st and 2nd MTs** (✗ Wrong options: "1st–2nd IM ligament is strongest" (TO trap))
Lisfranc ligament complex (TO)	**TMT ligaments of 1st + 2nd rays + Lisfranc ligament**
1st MT base ↔ medial cuneiform	**Joint capsule** (not Lisfranc ligament)

Point on X-ray — Lisfranc ligament runs **base of 2nd MT → medial cuneiform**
✗ Wrong options: 1st MT → 2nd MT · 2nd MT → middle cuneiform · medial → middle cuneiform

✗ Ligament distractors — **spring** (calcaneus → navicular, supports talar head) · **plantar fascia** (plantar calcaneus → MT heads) · **AITFL** (tibia → fibula) · bifurcate, long plantar, Chopart, CFL



Lisfranc ligament – plantar, medial cuneiform to 2nd MT base

2 Mechanism

Indirect (commonest) — **rotational force + axial load through hyper-plantarflexed forefoot** → hyperflexion / compression / **abduction** moment → MTs displaced **dorsal + lateral** · usually involves at least **2nd TMT**

Settings — MVA, fall from height, **athletes** (football player fallen on, stirrup in equestrian, soccer pivot, twist under motorcycle, dancer)

Direct — crush (workplace) · **hockey puck to medial foot ≠ Lisfranc** → occult trabecular fractures (below)

3 Presentation

Symptoms	severe midfoot pain · **inability to bear weight** · pain at push-off
Plantar ecchymosis	**medial plantar** bruising = **highly suggestive of Lisfranc** (TO) · also seen in **calcaneal fracture** (TO) ✗ Wrong options: mycetoma · hydatid cyst of talus · vertical talus
Exam	midfoot swelling, TMT tenderness · pain with **pronation + abduction** of forefoot · midfoot squeeze · **piano key test** (hypermobility)
Instability test	grasp MT heads, **dorsal force** while palpating TMTs → **dorsal subluxation = instability** · 1st + 2nd MTs displace medially + laterally → **global instability → surgery** · **plantar ligaments intact → no dorsal subluxation → nonop possible**



Plantar ecchymosis — Lisfranc injury

4 Diagnosis & Classification

Key radiology findings

Sign	View
Medial base 2nd MT ↔ medial middle cuneiform discontinuity	**AP** – diagnostic
Widening 1st–2nd ray interval ± **fleck sign** (avulsion of Lisfranc ligament from 2nd MT base, in **1st IM space**)	**AP** – diagnostic
Dorsal displacement of proximal base of 1st or 2nd MT	**Lateral**
Medial 4th MT base ≠ medial cuboid	**Oblique**
Disrupted medial column line (tangent to medial navicular + medial cuneiform)	**Oblique**

– – Active space – –

Study	Use
AP, lateral, oblique	Initial study of choice
Weight-bearing (bilateral, comparison)	**Non-WB films normal + suspicion → next step** · reveals subtle diastasis · **contralateral** for low-energy injuries
Physician-assisted stress exam / stress fluoroscopy	Dynamic alternative · **EUA reserved for equivocal cases**
CT	Comminution, **pre-op planning**, subtle fractures, fleck sign
MRI	**Confirms purely ligamentous injury** · plantar ligament integrity · doesn't assess **dynamic instability** as well as WB films

Mulcahy alignment check

AP: 2nd MT–middle cuneiform, 3rd MT–lateral cuneiform, 1st MT–medial cuneiform · oblique: middle cuneiform–2nd MT, lateral cuneiform–3rd MT · lateral: talus–2nd MT axis, medial cuneiform–5th MT, dorsal TMT surfaces



Fleck sign in the 1st intermetatarsal space

Diagnosis – weight-bearing/stress radiographs, MRI & plantar ecchymosis

Midfoot pain, plantar ecchymosis, non-WB X-ray normal

standing / weight-bearing AP radiographs (✗ Wrong options: CT · MRI · bone scan (expensive, not warranted) · compartment pressures) · **"of both feet"** when CT shows no fracture (✗ Wrong options: MRI first · RTP stiff-soled shoe · fix · cast 6 wk before stability known)

Football player, plantar flexed foot, nonstanding films normal

weight-bearing AP radiograph
✗ Wrong options: RTP · PT · orthosis · NWB cast

WB radiograph most likely shows

dorsal displacement of 1st MT proximal base on lateral
✗ Wrong options: fleck lateral to 5th MT base (5th MT avulsion) · calcaneal sclerotic line (stress fracture) · proximal medial sesamoid (**turf toe**) · talar head directly over navicular (**medial subtalar dislocation**)

MRI: dorsal capsule torn, plantar ligament intact, WB films normal

stable → protected WB in CAM boot
✗ Wrong options: bridge plate · strict NWB cast 6 wk · stress fluoro EUA · primary arthrodesis

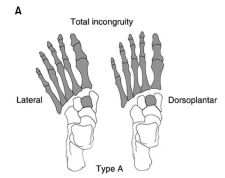
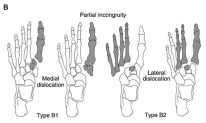
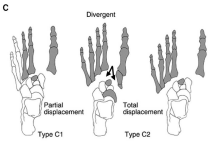
Hockey puck direct blow to medial foot, X-rays normal

MRI → occult fractures **medial malleolus, medial talar neck, medial cuneiform, 1st MT** (trabecular stress fractures + marrow oedema; medial malleolus can displace)
✗ Wrong options: bilateral WB X-rays (mechanism inconsistent with Lisfranc) · WBAT · cast without diagnosis · RTP

Diagnostic criteria – instability

Finding	Meaning
> 2 mm displacement / shift between 1st + 2nd MT on WB film	**Unstable → ORIF / anatomic reduction**
Diastasis 2nd MT base ↔ medial cuneiform	**Pathognomonic** of Lisfranc ligament rupture
Lisfranc + plantar ligament (medial cuneiform → 2nd/3rd MT) torn	**Transverse instability**
Plantar ligament intact on MRI	**Stable** → nonop
Medial gapping on stress view	Injury of that TMT even without frank displacement
> 5 mm diastasis, neglected pure ligamentous (TO)	**Medial column arthrodesis**

Hardcastle & Myerson classification

Type	Pattern	Diagram
A	**Complete homolateral** dislocation (all 5 TMTs, same direction)	
B1	**Partial**, **medial column** dislocation	
B2	**Partial**, **lateral** dislocation	–
C1	**Partial divergent**	
C2	**Complete divergent**	–

Primary arthrodesis may give better function + reduction — **complete** injuries **Type A or C2**

DDx — MT base fracture · MT stress fracture · tarsal fracture

5 Management

Suspected Lisfranc

- Non-WB X-ray normal → **bilateral WB / stress X-ray (MRI if equivocal)**
  - stable, plantar ligament intact → **CAM boot / cast**
- Unstable (> 2 mm) or displaced → **operative**
  - massive swelling → **temporary pinning, definitive at 2–3 wk**
  - soft tissues OK → **ORIF (< 24 h) or primary arthrodesis**

– – Active space – –

Operative choice

- Bony fracture-dislocation, acute → **ORIF rigid 1–3 TMT**
- Purely ligamentous → **primary arthrodesis medial ± middle column**
- Delayed / chronic / arch collapse → **arthrodesis**
- Lateral column (4–5) displaced → **temporary K-wires** *(never fuse 4th–5th TMT)*

Operative criteria

Nonoperative	ORIF	Primary arthrodesis (1st–3rd TMT)
Non-displaced, stable on WB · partial Lisfranc rupture, no displacement	**Any instability (> 2 mm shift)**	**Purely ligamentous** (controversial)
Nonambulatory, serious vascular disease, **severe neuropathy** (insensate diabetic)	**Bony fracture-dislocation**, acute	**Delayed treatment** (e.g. 8 wk, 6 mo)
Medically unfit	Young active (soldier)	**Chronic deformity**, missed injury
Cast 8 wk (Orthobullets) · **NWB cast 6 wk** then gradual return (OB)	–	**Complete A / C2** · severe intra-articular comminution · failed fixation

Nonop of displaced / unstable — significantly **worse** functional + radiographic outcomes · persistent midfoot instability

ORIF indications & fixation technique (medial vs lateral columns)

Goal — **anatomic reduction + stable fixation** = the outcome predictor · **open > closed reduction** (see reduction, **loose osteochondral fragments, soft-tissue interposition** – ligament / bony avulsion may be **incarcerated**)

Standard construct — **rigid fixation 1st–3rd TMT (screws or spanning plates)** + **K-wires for 4th + 5th TMT** (flexible, removed later) · stiff fixation of all 5 TMTs → poor function

Plate vs screw vs wire — **cortical screws > K-wires** (more rigid) · **pin fixation inferior** (ligament needs prolonged fixation) · plates span TMTs when **MT base comminuted** · more screws = more rigid

Fix only what's injured — 1st + 2nd TMT ligamentous injury → **ORIF of 1st and 2nd TMT** (✗ Wrong options: CR + K-wires · 1–3 + K-wire 4–5 (only when all five injured, e.g. **homolateral**)) · no lateral instability → **ORIF medial two columns alone** (✗ Wrong options: add lateral column stabilisation)

Closed manipulation + percutaneous screw — acceptable **only if anatomic alignment achieved closed**

Bridge vs transarticular — acute displaced → open reduction with **spanning-bridge plate or transarticular screws** · acute reducible → CRPP option

Timing — **within 24 h** or wait for swelling (up to **2–3 wk**) · massive swelling → **temporary percutaneous pinning** of medial + lateral columns → delayed ORIF / arthrodesis (↓ wound infection, low-level evidence)

Approach — single or dual **longitudinal** incisions · **1st–2nd web space** · 1st TMT exposed **between EHL and EHB** · **protect deep NV bundle (DPN + dorsalis pedis)**

Sequence	**reduce intercuneiform instability first** → fix 1st–3rd TMT transarticular screws · check intercuneiform + naviculocuneiform joints
Post-op	early midfoot ROM, protected WB · **K-wires out 6–8 wk** · **screws out 3–6 mo** · **full WB 8–10 wk** (Orthobullets) · **protected WB 3–5 mo** + passive midfoot ROM (OB) – read the options · asymptomatic screws across only 1st–3rd TMT may stay · **no vigorous sport 9–12 mo**



Temporary lateral column pinning

ORIF vs primary arthrodesis for ligamentous injuries

Purely ligamentous – preferred treatment (TO)	**medial tarsometatarsal joint arthrodesis** (TO) ✗ Wrong options: NWB cast · anatomical screw fixation · WBAT cast
Why fuse ligamentous injuries	**ligament healing less reliable than bone** → deformity recurs after ORIF · ligamentous injuries do **worse** than bony · nonop → uniformly poor (redislocation)
Ly & Coetzee RCT (purely ligamentous)	primary arthrodesis 1st–3rd rays → **higher AOFAS at 2 yr** · **25% of ORIF** → early arthritis → revised to fusion
Best result, acute purely ligamentous	**open reduction + primary arthrodesis with screw fixation** ✗ Wrong options: CRPP · open + pins · metallic or bioabsorbable screws alone

– – Active space – –

Alternative to ORIF (ligamentous) — **open reduction + arthrodesis of medial and middle columns, reduction + stabilisation of lateral column**
✗ Wrong options: fuse all three columns · CRPP all · closed reduction + cast · boot

Outcome	ORIF	Primary arthrodesis (PA)
Hardware removal	**Higher** (~**75%**, often planned)	**Lower** (~**20%**) ✓
Overall reoperation	**Higher** (driven by HW removal)	**Lower** ✓
Reoperation **excluding HW removal**	Equal (Buda: 163 ORIF vs 54 PA)	Equal
Functional outcome	**Equivalent** overall (level 1, meta-analyses, delayed cases)	**Better for purely ligamentous** (Ly & Coetzee) · **improved return to activity**
Patient-reported outcomes	**No difference** (OB)	**Higher with PA** (prospective, OB) – read the options
Cost	**Higher overall** (OB) · **ORIF less expensive** (another OB) – read the options	–
Infection, malreduction, hardware failure, alignment, satisfaction, NWB time	**Equal**	**Equal**
PTOA	**Higher**	Lower
Adjacent joint stress	–	**Greater concern**

✗ PA traps — "higher implant removal with PA" · "lower PTOA with ORIF" · "higher malreduction with ORIF" · "worse function with PA" · "higher wound infection with PA" · "decreased reduction quality with PA" · "**medial + lateral column arthrodesis better**"
✗ Wrong options: – inferior

Lateral column — **never fuse 4th + 5th TMT → inferior functional outcomes** (Mulier: total arthrodesis worse than medial arthrodesis + lateral ORIF) · use **temporary K-wires** · **medial column fusion superior to combined medial + lateral**

All 5 TMT fused (B2 injury) — **inferior functional outcomes** vs medial ray arthrodesis + temporary lateral K-wires

Homolateral fracture-dislocation (medial + lateral columns) — **arthrodesis of medial column + K-wire fixation of lateral column**
✗ Wrong options: cast · percutaneous screw / ORIF medial column alone · fuse medial + lateral

MVC, medial pain, plantar ecchymosis — **open reduction + arthrodesis of medial two TMT joints**
✗ Wrong options: ex-fix + staged screws · percutaneous wires 1–3 · Morel-Lavallée debridement · screws + ligament reconstruction

Equestrian, stirrup, 1 wk — **arthrodesis of medial TMT joints** · operative reduction mandatory whether fusion or ORIF
✗ Wrong options: aircast NWB · walking cast · Broström · delayed osteotomy

Crush at work — **ligamentous patterns better with primary arthrodesis than ORIF**
✗ Wrong options: workplace = no difference · fusion only as salvage (post-ORIF salvage fusion does worse than primary) · rigid 4th–5th TMT fixation · suture fixation alone for complete dislocations

Suture-button	possible role in **isolated Lisfranc ligament** injury · complete dislocations need rigid fixation
Fusion technique	denude TMT cartilage → **cortical screws or square plate** · cast / splint 6 wk → WB in boot 6–12 wk → **full WB in shoes by 12 wk**

6 Complications

Complication	Fact
Post-traumatic midfoot arthritis	**Commonest complication** · ↑ with delay + ORIF · **up to 80% non-anatomic ORIF** · **54% symptomatic OA at ~10 yr after ORIF** (Orthobullets) · **> 25% even with anatomic reduction** (OB) → **midfoot arthrodesis**
Malunion	Non-anatomic ORIF → **shoe modification** (nonsurgical) or **corrective arthrodesis**
Nonunion	**Smoking** → revision arthrodesis + graft (unless elderly low-demand)
Hardware removal	**~75% ORIF** · **~20% arthrodesis**
Deep infection	**3–4%** · swelling at definitive surgery → I&D ± HW removal
Planovalgus deformity	**Nonop** or non-anatomic ORIF · missed injury → arch collapse + **forefoot abduction**
Dorsal foot numbness	**Dorsal osteophytes impinging SPN** (tight shoes)

Prognosis	military: **~70%** resume occupational running at ~3 yr after ORIF or PA · **missed diagnosis = poor prognosis**

Chronic/missed injuries – midfoot arthritis & arthrodesis

Most common complication (10 yr after boot NWB 6 wk)	**midfoot arthritis** ✗ Wrong options: nonunion · planovalgus · 1st TMT hypermobility ("not associated") · dorsal numbness
Untreated trauma → chronic pain + progressive deformity	**Lisfranc injury → pes planovalgus** ✗ Wrong options: navicular (no planovalgus) · **calcaneus / talar neck → varus** · ATFL (ankle instability)
Chronic PTOA nonop	**shoe inserts / modifications + NSAIDs** · fail → **arthrodesis of affected joints**
Malunion, pain at push-off / stairs	**shoe modification with cushioned heel + rocker sole** (unloads TMT in terminal stance) ✗ Wrong options: neutral-posted rigid orthosis · UCBL · AFO locked 90° · Arizona
2 yr after cast, push-off pain, arch loss	**TMT joint arthrodesis** ✗ Wrong options: Evans lateral column lengthening · TN fusion · Dwyer · triple
6 mo old (ICU head injury), 1st–3rd TMT	**arthrodesis of 1st through 3rd TMT** ✗ Wrong options: orthotics · CR + screws · ORIF · add triple
> 6 mo, instability + arch collapse	**open reduction + primary arthrodesis ± tricortical ipsilateral iliac crest autograft** for bone loss ✗ Wrong options: RTW · PT / orthotics (only if unfit) · cast WBAT (insensate diabetic / unfit) · ORIF (late presenters fuse)
Factors favouring primary arthrodesis	**primarily ligamentous injury + delay in treatment** ✗ Wrong options: significant bony injury → **ORIF**

– – Active space – –

Midfoot arthrodesis best indicated	**chronic ligamentous Lisfranc injury** (ligament won't heal functionally) ✗ Wrong options: acute **nutcracker cuboid** (restore lateral column length) · diabetic **equinus** (stretching / TAL / gastroc recession) · **TN nonunion** (revise that joint) · **ulcers under 2nd–3rd MT heads** (relative plantarflexion / 1st ray dorsiflexion)
Chronic Lisfranc + progressive flatfoot (alcoholic, supple hindfoot)	**midfoot arthrodesis** ✗ Wrong options: PT / AFO (PTTI first line) · MCO + transfers + LCL (flexible flatfoot) · triple (rigid arthritic hindfoot)
Midfoot arthrodesis technique	denude 1st–3rd TMT + bone graft · **reduce with windlass mechanism** · cast 6 wk → boot 6–12 wk → WBAT at 12 wk if healed

X vs Y

	ORIF	**Primary arthrodesis**
Best for	**Bony fracture-dislocation**, acute	**Purely ligamentous**, delayed, chronic, A / C2
Hardware removal	**~75%**	**~20%**
Function	Equivalent overall	Better in ligamentous (Ly & Coetzee)

	Medial + middle columns (1st–3rd TMT)	**Lateral column (4th–5th TMT)**
Motion	Rigid lever for push-off	**Mobile** – uneven ground
Fixation	**Rigid screws / plate** or fuse	**Temporary K-wires**
Fusion	Acceptable	**Avoid – inferior function**

	Lisfranc planovalgus	**PTTI planovalgus**
History	Trauma, plantar ecchymosis	Insidious
Treatment	**Midfoot (TMT) arthrodesis**	PT / AFO → MCO + transfer + LCL → triple

How it's asked

"True about Lisfranc joint anatomy?"	**→ dorsal TMT ligaments are weaker than plantar TMT ligaments** (TO)
"Ligament from medial cuneiform to base of 2nd MT?"	**→ Lisfranc ligament**
"Lisfranc's ligament connects?"	**→ base of the second metatarsal to the medial cuneiform**
"Origin / insertion of the injured ligament on labelled X-ray?"	**→ point D (2nd MT base) to point A (medial cuneiform)**
"Ligaments needed for transverse instability?"	**→ interosseous first cuneiform–second metatarsal + plantar ligament from first cuneiform to second and third metatarsals**
"Plantar ecchymosis after fall on stairs – suspect?"	**→ ligament connecting the second metatarsal base to the medial cuneiform**

"Clinical photo of plantar bruising – diagnosis?"	→ **Lisfranc fracture dislocation** (TO)
"Midfoot pain, nonstanding films normal – next?"	→ **weight-bearing AP radiograph**
"Plantar ecchymosis, X-ray negative – next?"	→ **standing radiographs**
"Negative CT, high suspicion – next?"	→ **weightbearing radiographs of both feet**
"Most likely WB finding?"	→ **dorsal displacement of the 1st metatarsal proximal base on lateral radiograph**
"MRI dorsal capsule torn, plantar ligament intact?"	→ **protected weight-bearing in a CAM boot**
"Hockey puck to medial foot, normal X-rays?"	→ **MRI of the foot**
"Preferred treatment of purely ligamentous Lisfranc?"	→ **medial tarsometatarsal joint arthrodesis** (TO)
"Neglected ligamentous Lisfranc in runner, > 5 mm diastasis?"	→ **Lisfranc injury and arthrodesis of medial column** (TO)
"Equestrian, stirrup injury 1 wk ago?"	→ **arthrodesis of the medial tarsometatarsal joints**
"Alternative to ORIF for ligamentous pattern?"	→ **arthrodesis of medial and middle columns, reduction + stabilisation of lateral column**
"MVC, medial pain with abduction, plantar bruising?"	→ **open reduction and arthrodesis of the medial two TMT joints**
"Primary arthrodesis vs ORIF of 1st–3rd TMT – true?"	→ **improved return to activity with primary arthrodesis**
"Crush injury – true statement?"	→ **primarily ligamentous patterns do better with primary arthrodesis**
"Best outcome for acute purely ligamentous dislocation?"	→ **open reduction and primary arthrodesis with screw fixation**
"ORIF vs PA for ligamentous injury – ORIF has?"	→ **higher overall cost of treatment**
"Arthrodesis vs ORIF (bony) – true?"	→ **decreased rate of hardware removal at 1 year**
"ORIF vs PA outcomes – true?"	→ **higher hardware removal rates with ORIF**
"Prognosis with primary arthrodesis?"	→ **decreased overall reoperation rate**
"Delayed 8 wk, medial + middle columns – ORIF vs fusion?"	→ **functional outcomes equivalent**
"ORIF with rigid implants vs fusion – higher rate of?"	→ **hardware removal**
"All five TMTs fused vs medial fusion + lateral K-wires?"	→ **inferior functional outcomes**
"Twisting injury, displaced Lisfranc (no fusion option)?"	→ **open reduction and internal fixation**
"Standard fixation of TMT fracture-dislocation?"	→ **plating of joints 1, 2 and 3; K-wire fixation of joints 4 and 5**

– – Active space – –

"Soccer Lisfranc – best technique?"	→ **open reduction and internal fixation of medial column** (cortical screws)
"Ligamentous 1st + 2nd TMT gapping on stress view?"	→ **open reduction and rigid internal fixation of the first and second TMT joints**
"WB X-ray > 2 mm, no lateral instability?"	→ **ORIF of the injury** ✗ Wrong options: lateral column stabilisation
"Rugby player, 2nd MT–medial cuneiform diastasis?"	→ **open reduction and internal fixation** ✗ Wrong options: pins
"Dancer with Lisfranc dislocation?"	→ **open reduction and internal fixation**
"Homolateral dislocation after temporising reduction?"	→ **arthrodesis of the medial column and K-wire fixation of the lateral column**
"Soldier, plantar ecchymosis, displaced Lisfranc?"	→ **Lisfranc open reduction and internal fixation**
"Chronic injury, push-off pain, arch loss?"	→ **tarsometatarsal joint arthrodesis**
"6-month-old 1st–3rd TMT dislocation?"	→ **arthrodesis of 1st through 3rd TMT joints**
"6 months, instability + arch collapse?"	→ **open reduction and primary arthrodesis**
"Malunion, pain at push-off – nonsurgical?"	→ **shoe modification with a cushioned heel and rocker sole**
"Factors favouring primary midfoot arthrodesis?"	→ **primarily ligamentous injury + delay in treatment (1 and 3)**
"Midfoot arthrodesis best indicated for?"	→ **55-year-old with chronic ligamentous Lisfranc injury**
"Untreated injury → progressive planovalgus?"	→ **Lisfranc injury**
"Chronic Lisfranc with progressive flatfoot?"	→ **midfoot arthrodesis**
"Most common complication of Lisfranc injury?"	→ **midfoot arthritis**

LOWER EXTREMITY OS

– Foot & Ankle · 31 / 48 –

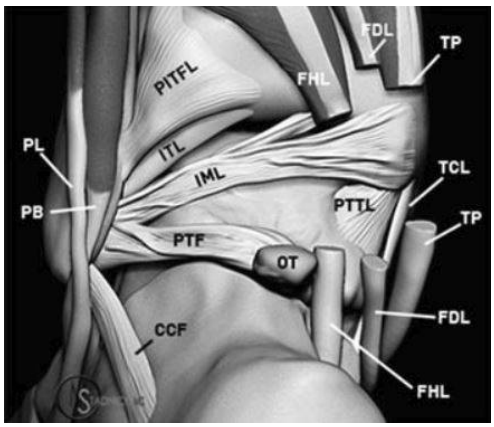
1 Basics

Definitions & epidemiology

Accessory ossicle	**secondary ossification centre that stays separate** from parent bone · may mimic fracture
Sesamoid	**bone incorporated into a tendon**, moves with it
Number	up to **40 accessory ossicles** + multiple sesamoids in foot and ankle
Commonest ossicles	**os trigonum**, **accessory navicular (os tibiale externum)**, os intermetatarseum · os trigonum = **second commonest accessory bone** (OB)
Commonest sesamoids	**os peroneum** (in **peroneus longus**) · **hallux sesamoids** (in **FHB**, under 1st MT head)

Os trigonum – anatomy

Origin	separated **posterolateral (lateral) tubercle of talus** · ossifies **8–13 yr**, normally **fuses within 1 yr** · unfused → **synchondrosis** with talus
Stieda process	elongated lateral tubercle (unfused) that may fracture with repetitive trauma
Relations	os lies **lateral to FHL, tibial nerve, PTT, posterior tibial artery** · FHL runs in fibro-osseous tunnel between **posterolateral (os) + posteromedial talar tubercles**



FHL runs just medial to the os trigonum (OT)

2 Mechanism

Os trigonum syndrome	**repetitive forced plantarflexion** (microtrauma → stress fracture) or **acute forced plantarflexion** (acute fracture) · traumatic disruption of synchondrosis → pain
Impingement	os squeezed between **posterior tibial malleolus + calcaneal tuberosity / superior calcaneus**

– – Active space – –

At risk	**ballet dancers (en pointe)**, basketball, soccer, football, kicking, downhill running, push-off
Os peroneum injury	direct trauma · **sudden inversion + supination** → associated **PL rupture**

3 Presentation

Os trigonum & posterior impingement – diagnosis

Symptom	chronic / recurrent **posterolateral ankle pain** with **forced plantarflexion**, **en pointe**, push-off, jumping
Exam	pain with **passive maximal plantarflexion** · no Achilles insertion tenderness · ± FHL swelling / tenderness if FHL tenosynovitis

Feature	Os trigonum syndrome	FHL tendinitis / stenosing tenosynovitis
Pain site	**Posterolateral**	**Posteromedial** (behind medial malleolus)
Provocation	**Forced plantarflexion**	Hallux motion, push-off, deep squat
Sign	Plantarflexion pain	**Triggering**, crepitus, **great toe locking**
X-ray	Tibio-calcaneal impingement on plantarflexion lateral	May show incidental os

Inflamed structure in os trigonum + posteromedial tenderness	**FHL** (labelled C on axial MRI)
✗ Posterolateral look-alikes	peroneal pathology (instability, pain with passive DF + eversion) · subtalar arthritis / loose body (pain with inversion / eversion) · tarsal tunnel (medial heel, plantar paraesthesia) · PTT tendinitis (posteromedial; **does not abut os trigonum**)

FHL stenosing tenosynovitis – examination

Functional hallux rigidus	**low-lying distal FHL muscle belly** jams in retrotalar fibro-osseous tunnel ("**cork in a bottle**")
Test	**maximal knee extension + ankle dorsiflexion → passive hallux dorsiflexion limited + painful** ✗ Wrong options: forced plantarflexion (os trigonum) · posterolateral tenderness behind peroneals (os trigonum) · DF + eversion (anterior impingement) · calf squeeze (Achilles)
FHL origin	inferior fibula + interosseous membrane
Hamilton	good results after **FHL release** · **presence of os trigonum on X-ray does not mandate excision**

Os subfibulare & os peroneum – presentation

Os subfibulare	often asymptomatic · focal tenderness + swelling · **chronic ankle instability, recurrent sprains** · laxity on anterior drawer + inversion / eversion stress
Os peroneum	painful os peroneum syndrome · fracture · diastasis · PL rupture

4 Diagnosis & Classification

Accessory ossicles and sesamoids

Accessory bone	Prevalence	Clinical significance	Differential
Os trigonum 	**10–25%**	**Posterior ankle impingement, FHL entrapment**	**Shepherd's fracture** (posterolateral process)
Type II accessory navicular	**2–12%**	**PTT dysfunction**	Navicular tuberosity avulsion, type I accessory navicular
Os subfibulare	**2%** (table) · **1–2%** (text)	Painful os subfibulare, instability	**Lateral malleolus (ATFL) avulsion fracture**
Os peroneum	**9–20%** · **bilateral 60%** · **bipartite 30%**	Painful os peroneum, fracture, diastasis, **PL rupture**	Painful os vesalianum, bipartite os peroneum
Os vesalianum	**2%**	Painful os vesalianum	**Avulsion fracture of 5th MT base**
Hallux sesamoids	**~100%**	Fracture, stress fracture	**Bipartite tibial sesamoid**

Key radiology findings

Ossicle	Study	Finding
Os trigonum	**Lateral X-ray in plantarflexion**	Round / oval / triangular ossicle **impinged between posterior tibia + calcaneal tuberosity**
Os trigonum	**Axial CT**	Ossicle **off posterior talus** (✗ Wrong options: posterior malleolus (tibia) · accessory navicular (medial + distal) · coalition (distal) · anterior loose body)
Os trigonum	**MRI**	**Fluid around os + marrow oedema in os**, **↑ T2 signal**, FHL oedema / inflammation · **no talar marrow oedema** (vs acute posterior process fracture)
Os subfibulare	**Weight-bearing AP, lateral, mortise + varus stress view**	Ossicle · **talar tilt** on varus stress → instability · **↑ separation of os from fibular tip**
Os peroneum	X-ray + MRI	Normal os · acute fracture · **proximal migration = PL rupture**

– – Active space – –

Os subfibulare vs type VII transepiphyseal fracture (child)

	Os subfibulare	Type VII transepiphyseal fracture
X-ray	**Smooth-edged, short ossicle** at **inferior pole** of epiphysis	**Long, irregular line in middle third** of distal fibular epiphysis
Nature	Anatomic variant **or ossified ATFL avulsion**	**Sleeve avulsion** through secondary ossification centre into joint (ATFL, CFL, PTFL stronger than physis)
Initial treatment	**Protected weight-bearing (CAM boot)**	**Protected weight-bearing (CAM boot)**

Differentiates them **radiographs**, not clinical picture

Paediatric ankle injuries **~17% of all physeal injuries**



Os subfibulare in a child after inversion injury

5 Management

Symptomatic os trigonum

- First → **NSAIDs, rest, immobilisation (CAM), restricted WB, PT**
- Failed ≥ 3 mo → **excision (open lateral or posterior arthroscopy)**
  - FHL involved → **± FHL tenolysis, debridement or repair**

Os trigonum excision (arthroscopic vs open, sural nerve)

Nonop NSAIDs, rest, immobilisation, restricted weight-bearing, activity modification, PT · PT = short-term relief only · **steroid injection not recommended** here

Surgery after **≥ 3 months failed nonop → excision**, open **or** arthroscopic · **similar return-to-play** both (Orthobullets) · **open excision = high RTP, low complications** (OB) · **arthroscopic = 80% good / excellent at 2–5 yr** (Van Dijk) · **RTP 2–3 months**

✗ Os trigonum traps **fixation / ORIF** (not indicated) · FHL debridement alone (doesn't remove culprit) · PTT debridement · peroneal debridement + SPR repair (no snapping / instability) · retrial of failed rest · Haglund excision · os naviculare excision · biopsy

Posterolateral approach **sural nerve at greatest risk**
✗ Wrong options: SPN (more anterior)

Ballet dancer, failed 1 mo rest + NSAIDs + PT	**arthroscopic os trigonum excision**
Basketball player, chronic posterior pain	**open excision** = highest success ✗ Wrong options: PT · steroid · arthroscopic fixation · ORIF
Soccer player, posterolateral pain with plantarflexion, ↑ T2 around os	**arthroscopic excision** ✗ Wrong options: PB tubularisation · FHL debridement · EMG · subtalar loose body removal

Os subfibulare & os peroneum treatment

Os subfibulare / type VII	**protected weight-bearing first** · **failed nonop → excision** ± ligament re-attachment ✗ Wrong options: initial MRI (only for chronic pain / instability) · acute surgery · mandatory ATFL re-attachment
Os subfibulare excision for functional instability without laxity (children)	better AOFAS than rehab alone

Os peroneum situation	Treatment
Painful os peroneum syndrome, minimally displaced fracture	**NSAIDs, rest, immobilisation, restricted WB**
Refractory syndrome (minimal tendon involvement), **displaced fracture**	**Excision**
With PL tendon rupture	**Excision + PL repair or tenodesis to peroneus brevis**

X vs Y

	Os trigonum	Shepherd's fracture
Edges	Smooth, corticated, synchondrosis	Acute fracture of posterolateral process
MRI	Fluid + oedema around os, **no talar body oedema**	**Talar marrow oedema**

	Os vesalianum	5th MT base avulsion
Edges	Smooth ossicle	Acute fracture line

How it's asked

"Ballet dancer, posterior pain en pointe, failed 1 mo nonop?"	→ **arthroscopic os trigonum excision**
"Basketball player, painful max plantarflexion – highest success?"	→ **open excision**
"CT arrow posterior to talus?"	→ **os trigonum fragment**
"Posterolateral approach for os trigonum – diagnosis + nerve at risk?"	→ **symptomatic os trigonum, sural nerve**

– – Active space – –

"Dancer with os trigonum, posteromedial tenderness — inflamed structure?"	→ **C — FHL**
"Posterolateral pain with plantarflexion, no pain DF + eversion, ↑ T2?"	→ **arthroscopic excision of os trigonum**
"Elite dancer — high RTP, low complications?"	→ **open surgical excision of the os trigonum**
"Child, os subfibulare vs type VII fracture — true?"	→ **both treated initially with protected weightbearing**
"Functional hallux rigidus from low FHL belly — exam?"	→ **maximal knee extension + ankle dorsiflexion, assess passive hallux dorsiflexion**

METATARSAL FRACTURES

– Foot & Ankle · 32 / 48 –

1 Basics

Epidemiology

Adults	**5th MT** most commonly fractured
3rd MT	rarely isolated · **68%** with 2nd or 4th MT fracture
Peak age	2nd–5th decade

Paediatric metatarsal fractures – most common location

Young child	**1st MT (proximal)** most common · **< 4 yr** (Orthobullets) · **< 5 yr** (OB) – read the options
Older child (> 5 yr)	**5th MT**
Course	usually minimally displaced, rarely operative
Ossification (age clue)	lateral cuneiform **1 yr** · middle **3 yr** · medial **4 yr**

Anatomy

1st MT	**widest + shortest** · plantar crista articulates with sesamoids · **bears 30–50% of weight** in gait · **50%** (Orthobullets anatomy) · TO: standing = 1st MT, walking = **2nd MT** – read the options · **no intermetatarsal ligament** support
2nd MT	**longest** · **commonest stress fracture site**
Distal intermetatarsal ligaments (2nd–5th)	maintain length + alignment in **isolated** fractures · implicated in Morton's neuroma · **multiple fractures lose this stability → ↑ displacement**
Muscle balance	extrinsic (EDL, FDL) vs intrinsic (interossei, lumbricals)
Blood supply	dorsal + plantar metatarsal arteries

2 Mechanism

Indirect (commonest)	forefoot fixed, hindfoot / leg rotates
Direct crush	significant soft-tissue injury
Multiple proximal MT fractures	**Lisfranc-equivalent** injury
Stress	repetitive microstress; MTs loaded **perpendicular to long axis** · relatively fixed proximal 2nd + 3rd MTs → diaphyseal stress fractures

3 Presentation

Symptoms	pain, inability to bear weight · **antecedent pain** suggests stress fracture

– – Active space – –

Exam foot alignment (neutral / **cavovarus** / planovalgus) · tenderness · soft tissues in crush · **overlap / malrotation** of toes · Semmes–Weinstein monofilament if neuropathy

4 Diagnosis & Classification

Key radiology findings

Study	Use
AP, lateral, oblique foot	**Required** · ± contralateral, stress / WB views
Repeat X-ray at 2 wk	X-rays **lag symptoms by weeks** → callus / periosteal reaction appears
CT	Not routine · periarticular injury, **rule out Lisfranc** · better shows bone when fracture visible on X-ray
MRI	**Occult / stress fracture** · fluid sequences show **endosteal marrow + periosteal oedema** (earliest) · useful screening tool
Bone scan (Tc-99)	Occult stress fracture when X-rays unclear · **most useful** (OB1095 reference) · finds **noncontiguous simultaneous fractures** (e.g. 2nd MT + tibia)

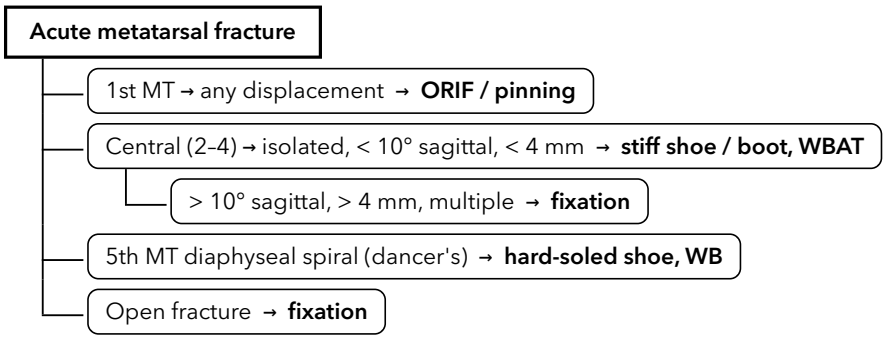


March fracture: periosteal reaction, no fracture line yet

Classification

Descriptor	Record
Location	Base / shaft / neck / head
Pattern	Transverse, spiral, comminuted
Displacement	mm translation
Angulation	Sagittal plane degrees
Articular involvement	Yes / no

5 Management



Operative criteria

Nonoperative (stiff-soled shoe / walking boot, WBAT)	Operative (percutaneous or open)
1st MT non-displaced	**1st MT any displacement** (✗ Wrong options: 3 mm displaced 1st MT nonop)
Central MT isolated, non / minimally displaced	**Central MT sagittal deformity > 10°** (✗ Wrong options: 12°)
–	**> 4 mm translation** (✗ Wrong options: 6 mm)
–	**Multiple MT fractures**
5th MT spiral diaphyseal (dancer's fracture)	**Open fractures**
Stress fractures	Failed nonop stress fracture / nonunion

Shaft fractures – nonoperative indications

5th MT spiral diaphyseal (dancer's) fracture	**high union, low morbidity → hard-soled shoe + WB**
Fixation goals	restore alignment for normal force transmission across MT heads · **maintain length** → avoid **transfer metatarsalgia**
Techniques	antegrade / retrograde pinning · lag screws or mini-fragment plates for length-unstable patterns
Prognosis	majority of isolated MT fractures heal nonop

Stress fractures – second MT, dancers, management & metabolic work-up

March fracture	**2nd, 3rd, 4th MTs**, **2nd commonest** · diaphyseal, callus + good alignment · runners, **military recruits**
Ballet dancers	**base of 2nd MT** · **amenorrhoea / delayed menarche** · pain with push-off + en pointe ✗ Wrong options: Lisfranc injury · cuneiform · 1st MT base · plantar fascia strain
Commonest stress site	**distal 2nd MT shaft** (length + immobility) · "2nd + 3rd most frequent", also 5th (OB)
Risk factors	**long 2nd MT**, **hypermobile 1st ray**, **female**, recent ↑ training, **cavovarus** (recurrent), metabolic bone disease

– – Active space – –

Female athlete triad	repeated stress + **low BMD** + **dietary restraint** → ↑ fracture risk
Metabolic work-up	**serum cortisol, TSH, vitamin D, calcium** · menstrual history · **DEXA** when needed · (hypothyroidism, amenorrhoea) · older patient (55 yr) → **fragility fracture**
Treatment	**protected WB / short walking cast / fracture boot WBAT / hard-soled shoe** + activity modification · rest **4–6 wk** (OB1095) · return over **6–8 wk** (OB218223) – read the options · base fractures sometimes limited WB
✗ Stress-fracture traps	ORIF, K-wires, IM screw, calcium phosphate / BMA (only after failed nonop) · **biopsy / oncology referral** (no red flags) · continue running with orthotic · observation alone · NWB cast · **bone stimulator** (no evidence) · NSAIDs + immediate RTP · bone density in a teenager (normal, no help) · MRI / bone scan once X-ray diagnostic
MRI already shows oedema, X-ray normal	**fracture boot WBAT** (limit miles + repeat X-ray at 2 wk only if no MRI)
Nonunion (dancer, 2nd MT base)	fixation



Second metatarsal base stress fracture in a ballet dancer

Differential	metatarsalgia, MTP synovitis, **Freiberg's infarction**, Morton's neuroma

6 Complications

Malunion	**transfer metatarsalgia** or plantar keratosis → **corrective osteotomy**
Multiple fractures	loss of intermetatarsal ligament stability → displacement



Multiple metatarsal shaft fractures – loss of intermetatarsal ligament support

X vs Y

	1st MT	Central MTs (2–4)	5th MT diaphysis
Ligament support	**None**	Distal intermetatarsal ligaments	–
Load	**30–50%** of weight	–	–
Fix if	**Any displacement**	**> 10° sagittal, > 4 mm, multiple**	Rarely (spiral → nonop)

	Bone scan	MRI
Strength	Occult fracture · **noncontiguous fractures**	Earliest oedema · occult fracture · screening
When not needed	X-ray already diagnostic	X-ray already diagnostic

How it's asked

"Runner, 3rd MT callus – next step?"	→ **protected weightbearing with crutches, slow return to running**
"Ballet dancer, delayed menarche, midfoot pain?"	→ **second metatarsal base stress fracture**
"55-yo ballet, 2nd MT stress fracture on X-ray / scan / MRI?"	→ **walker boot + evaluation for metabolic bone disease**
"Ballerina, 2nd MT shaft, callus on 2-wk X-ray – next?"	→ **protected weight-bearing** ✗ Wrong options: MRI · bone scan · biopsy · observation
"Runner, normal X-ray, MRI oedema of 3rd MT?"	→ **fracture boot, WBAT**
"Recruit, amenorrhoea, distal 2nd MT on MRI – treatment?"	→ **immobilisation in a short walking cast**
"Field-hockey player, 3rd MT stress reaction – next?"	→ **serum cortisol, TSH, vitamin D + calcium levels**
"Young child fell – commonest MT fracture site?"	→ **1st MT** (label 1)
"Best managed nonoperatively?"	→ **5th MT spiral fracture**

– – Active space – –

MIDFOOT ARTHRITIS

– Foot & Ankle · 33 / 48 –

1 Basics

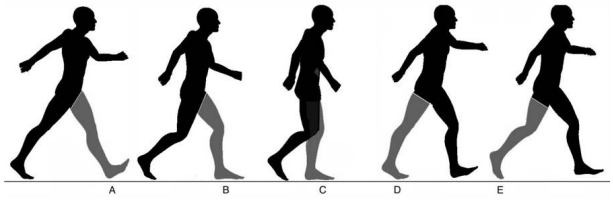
Definition & etiology

Joints	**naviculocuneiform**, **intercuneiform**, **tarsometatarsal (metatarsal–cuneiform)** · 2nd + 3rd TMT commonly narrowed
Commonest cause	**idiopathic / primary OA** (Orthobullets, OB) · **post-traumatic** articular damage (another OB) – read the options
Post-traumatic	missed / untreated **Lisfranc injury** (plantar bruising, "ankle sprain") → progressive flatfoot + midfoot arthrosis
Inflammatory	RA, lupus · **rule out inflammatory + neuropathic causes** before treating
Pathoanatomy	large forces through joints with **limited motion** · supporting soft tissues overloaded → **midfoot collapse** · over time Achilles shortens → **equinus**
RA foot	**forefoot commonest** (dislocations, hallux valgus) · midfoot: symmetric degeneration across **talonavicular + intertarsal** joints

3 Presentation

Post-traumatic vs inflammatory, abduction & gait

Symptom	**midfoot / arch pain with toe-off (push-off)** · dull, achy, worse with weight-bearing · walking distance limited
Gait phase	**toe-off** (load passes hindfoot → midfoot → forefoot) ✗ Wrong options: heel strike · midstance · swing



Gait cycle: midfoot arthritis pain peaks at toe-off (E)

Deformity on WB	Present?
Longitudinal arch collapse	**Yes** (✗ Wrong options: arch maintained)
Midfoot collapse – looks like **PTTI**	**Yes**
Forefoot abduction	**Yes**
Hindfoot valgus	**Yes** (✗ Wrong options: hindfoot varus)
Equinus (Achilles) contracture	**Yes** (✗ Wrong options: peroneus brevis contracture)
Hallux valgus	Yes
Dorsal midfoot prominence	Large **dorsal osteophyte**

Palpation	arch / midfoot tender
Midfoot vs PTTI	**single-limb heel rise possible**, **inversion strength 5+**, supple hindfoot → PTT working → deformity **isolated to midfoot**
Differential	**PTTI** · post-traumatic **Lisfranc** · lateral ankle instability

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

View	Finding
Lateral (WB)	**Break in Meary's line (talus–1st MT)** with **apex at midfoot** · longitudinal arch collapse · **dorsal osteophytes** · normal calcaneal pitch (vs PTTI)
AP (WB)	Midfoot arthritic change · **forefoot abduction** · normal talar head coverage (vs PTTI)
Inflammatory	**Symmetric joint degeneration**, minimal extra bone
Primary OA	**Subchondral cysts, osteophytes**, asymmetric narrowing
Image-guided steroid injection	**Diagnostic + therapeutic**



Lateral radiograph: midfoot collapse with Meary's line break at the midfoot



AP radiograph: midfoot arthritis with forefoot abduction

– – Active space – –

X-ray pattern	Diagnosis
Symmetric degeneration	**Inflammatory (lupus, RA)**
Subchondral cysts + osteophytes	**Primary OA**
Isolated dorsal osteophyte at 1st MTP	**Hallux rigidus**
Navicular sclerosis + collapse	**Köhler disease**
1st–2nd base widening ↑ on stress	**Lisfranc injury**

5 Management

Midfoot arthritis

- First line → **NSAIDs, activity change, stiff rocker-bottom shoe, arch support, image-guided steroid**
- Failed nonop → **midfoot (medial column) arthrodesis**
  - Equinus → **+ Achilles lengthening**
  - Hindfoot valgus → **+ hindfoot realignment**

Nonoperative care – orthoses & bracing

Best orthosis — **stiff-soled rocker-bottom shoe** → transfers load **hindfoot → forefoot** with minimal midfoot stress

Others — **cushioned heel**, **longitudinal arch supports**



Rocker-bottom shoe for midfoot arthritis

Orthosis	Use
Rocker-bottom shoe	**Midfoot (1st–2nd TMT) arthritis**
Arizona brace	**Flexible hindfoot planovalgus**; ankle + hindfoot stability
Lateral heel wedge	**Cavovarus with fixed forefoot pronation**
Lateral wedge + **1st ray cutout**	Flexible **cavovarus**
Morton's carbon-fibre extension	**1st MTP arthritis / hallux rigidus**
Posterior leaf-spring AFO	**Foot drop**
Medial wedge insole, **UCBL**	Flexible **planovalgus**
Metatarsal pad	**Freiberg disease** (2nd MT head AVN)

Midfoot / medial column arthrodesis

Indication	**failed nonoperative management** · midfoot joints are **non-essential** → fusion gives **near-normal foot function**, esp. push-off
Isolated midfoot flatfoot (heel rise OK, flexible hindfoot, mild heel cord contracture)	**medial column arthrodesis + heel cord lengthening** ✗ Wrong options: PTT debridement · tendon transfer + spring ligament / LCL · triple arthrodesis (PTTI / hindfoot procedures)
TMT collapse, inversion 5+, supple hindfoot, normal heel cord	**medial Lisfranc arthrodesis** ✗ Wrong options: LCL + FDL · TN · subtalar · triple (hindfoot procedures)
Atraumatic midfoot OA, push-off pain, failed nonop	**midfoot arthrodesis** ✗ Wrong options: PRP (not studied) · 1st MT opening wedge (hallux valgus) · **dorsal exostectomy – limited relief in advanced OA** · double calcaneal osteotomy (planovalgus abduction)
Iliac crest autograft	**improved fusion rate / ↓ nonunion** ✗ Wrong options: cost · fewer implants · long-term alignment (depends on technique + fixation) · donor morbidity (significant donor pain)
Realignment arthrodesis	**1st ray → 1st TMT** · **2nd / 3rd rays → NC + intercuneiform joints** · **do NOT fuse 4th / 5th TMT** (lateral ray mobility accommodates stance) · select cases: **interposition arthroplasty of 4th / 5th TMT** (keeps lateral column length)
Technique	TMT joints **2–3 cm deep** → prepare fully · screws, staples, midfoot plates · ± **Achilles lengthening**, ± **hindfoot realignment**
Outcome	union linked to satisfactory outcome in **up to 90%** (Kurup) · **union 92%**, few complications at 4.5 yr (Nemec) · inflammatory midfoot: **primary fusion** most reliable

6 Complications

Triple arthrodesis – talonavicular nonunion (TO)

Complication of triple arthrodesis	Fact (TO)
Nonunion	**Commonest complication overall** + **commonest early / midterm** (TO)
Site of nonunion	**Talonavicular** (most difficult joint to fuse) **> calcaneocuboid** (TO)
Ankle arthritis	**40–100%** with long follow-up = **commonest long-term** complication (TO)
Midfoot arthritis	**~50%** (TO)
Hardware irritation, infection, osteonecrosis, malunion, DVT, nerve injury	**Rare** (TO)

– – Active space – –

X vs Y

	Midfoot arthritis	PTTI flatfoot
Single-limb heel rise	**Possible**	Unable
Apex of collapse	**Midfoot** (Meary break at TMT / NC)	Talonavicular / hindfoot
Calcaneal pitch, talar coverage	**Normal**	Abnormal
Surgery	**Medial column / Lisfranc arthrodesis**	Tendon transfer, osteotomies, triple

	Inflammatory	Primary OA
X-ray	**Symmetric degeneration**, little bone formation	**Cysts, osteophytes**, asymmetric

How it's asked

"Lupus, midfoot pain – X-ray finding?"	→ **symmetric joint degeneration**
"Clinical finding with post-traumatic midfoot arthritis?"	→ **forefoot abduction**
"Gait phase that worsens midfoot arthritis?"	→ **E – toe-off**
"1st–3rd TMT OA, arch collapse – initial orthosis?"	→ **Figure E – rocker-bottom shoe**
"Rocker-bottom shoe prescribed – diagnosis?"	→ **first and second tarsometatarsal arthritis**
"Active 70-yo, push-off pain, failed nonop?"	→ **midfoot arthrodesis**
"Flat foot, heel rise OK, flexible hindfoot, tight heel cord?"	→ **medial column arthrodesis and heel cord lengthening**
"TMT collapse, inversion 5+, supple hindfoot?"	→ **medial Lisfranc arthrodesis**
"Why iliac crest autograft in midfoot fusion?"	→ **improved fusion rate**
"Commonest complication of triple arthrodesis?"	→ **talo-navicular nonunion** (TO)
"Commonest pseudoarthrosis after triple arthrodesis?"	→ **talonavicular** (TO)
"Commonest long-term complication of triple arthrodesis?"	→ **ankle arthritis** (TO)

MTP DISLOCATIONS

– Foot & Ankle · 34 / 48 –

1 Basics

Definition **multiplanar instability of a lesser MTP joint → crossover toe** (dorsomedial deviation of 2nd toe)

Commonest joint **2nd MTP**

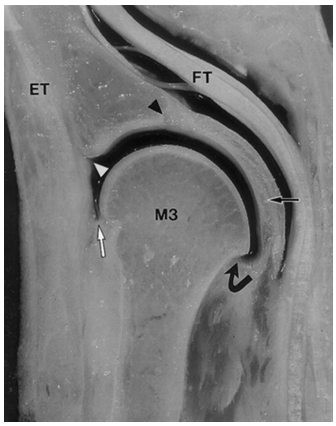
Plantar plate anatomy

Structure broad, thick ligamentous plate on plantar MTP joint

Origin **metatarsal head** via thin synovial attachment, just proximal to articular surface (**weakest point = origin at metatarsal neck**)

Insertion **plantar base of proximal phalanx** (ruptures here and retracts proximally)

Function **resists sagittal tensile load, esp. MTP dorsiflexion** · cushions joint, supports weight-bearing



Plantar plate of a lesser MTP joint

2 Mechanism

Pathoanatomic stages

Stage	Event	Result
1	**Plantar plate disrupted** – **chronic inflammation (commoner)** or traumatic rupture	Dorsal laxity
2	**Lateral collateral ligament fails** (OB: **LCL fails first**, plantar plate second – read the options)	**Medial deviation of 2nd toe**; plantar plate + flexors displace medially
3	**Medial structures contract** – lumbrical, interosseous, MCL, medial capsule	Adduction force
4	**Plantar plate fails completely**	**Hyperextension → dorsal instability / dislocation**

Trauma **hyperextension injury** of toe (push-off, spin) → plantar plate tear

– – Active space – –

Chronic	**repeated MTP dorsiflexion** attenuates plantar aponeurosis, plantar plate, capsule
Accelerated by	**cortisone injection** for misdiagnosed **Morton's neuroma** or MTP synovitis
Once plate fails	intrinsic + extrinsic forces dorsiflex the proximal phalanx · main deforming force = **EDL**

3 Presentation

Symptoms	pain at **toe-off / push-off** · **"walking on a marble / pebble / rock in shoe"** · better in padded shoes than barefoot
Point tenderness	**plantar aspect of 2nd MTP**; pain on hyperextension
Calluses	**plantar under MT head** (fat pad migrates **dorsally**, uncovering head) + **dorsal over PIP** (abuts toe box)
Deformity	**dorsomedial deviation (crossover toe)** · **hammertoe**: MTP extension + PIP flexion · crossover develops when **collateral ligament(s)** also injured
Early instability	before deformity, **mimics Morton's neuroma**



Crossover second toe

4 Diagnosis & Classification

Plantar plate injury & instability – drawer test

MTP drawer / modified Lachman	grip proximal phalanx (neutral) + MT head → **dorsally directed force → ↑ sagittal (dorsal–plantar) translation / dorsal subluxation** · **> 50% subluxation** = marked instability



MTP drawer test

Key radiology findings

Study	Finding
WB AP, oblique, lateral	AP: proximal phalanx dislocated – **medial > lateral** · lateral: **hyperextension + dorsal dislocation** of proximal phalanx
X-ray early	Often **normal** in plantar plate tear
MRI	**Early MTP synovitis / plantar plate tear**; rules out other pathology · not useful once dislocated

Clinical grading of plantar plate rupture (OB)

Grade	Injury	Treatment
I	**Sprain**	**Rest, ice, NSAIDs, stiff-soled shoe, MT pad**
II	**Partial tear**	**Nonop first** (± cast)
III	**Complete tear**	Repair if nonop fails

Surgical grading of plantar plate tears

Grade	Tear pattern
0	Plate / capsular **attenuation** ± discolouration
1	**Transverse distal tear < 50%** (next to phalangeal insertion) ± midsubstance < 50%
2	**Transverse distal tear > 50%** ± midsubstance < 50%
3	**Extensive transverse ± longitudinal tear** (may involve collaterals)
4	**Extensive tear with button-hole (dislocation)**; little plate left to repair

Differentials by exam

Finding	Diagnosis
↑ sagittal translation, + drawer / Lachman, MTP hyperextension pain	**Plantar plate tear** ✓
Squeeze test + / palpable click (Mulder), numbness, **3rd web space**	**Morton's neuroma**
Point tenderness **2nd MT shaft**	**2nd MT stress fracture**
Flexion contracture + dorsal ulceration	**Hammer toe**
AVN, flattening / collapse of 2nd MT head from repetitive microfractures	**Freiberg's infraction**
Intrinsic weakness	**Claw toe**
TMT pain, X-ray arthritis	Midfoot arthritis

– – Active space – –

5 Management

Lesser MTP instability / plantar plate tear

- Sprain, partial tear, first presentation → **nonop (does not correct deformity)**
- Failed nonop, painful instability → **plantar plate repair + Weil osteotomy**
- Severe subluxation / dislocation, fixed → **Weil shortening osteotomy + soft-tissue balancing**
  - Release + tendon lengthening fail → **add MT shortening osteotomy**

Nonop (first line)	**taping**, **shoe modification / extra-depth shoes**, **metatarsal pads**, well-padded liners, **Budin splint**, **stiff-soled shoe**, NSAIDs, Achilles stretching, ± cast · **won't correct deformity**
Crossover toe 1 yr after hallux valgus, 1 month symptoms, > 50% drawer	**orthotics with MT pads + shoe modification + taping + NSAIDs** ✗ Wrong options: steroid injection (attenuates plate/capsule further) · revision HV correction · Weil + plate repair before nonop fails · MTP fusion
Grade II plantar plate tear in athlete	**rest, NSAIDs, stiff-soled shoe** ✗ Wrong options: direct repair · repair + osteotomy · reconstruction (plate usually repaired) · arthrodesis
Plantar plate repair indications	failed nonop · (hallux) **sesamoid retraction, loose fragments** · lesser toe repair is done **with Weil osteotomy** (protects repair) · dorsal or plantar approach, both good

Operative criteria

Situation	Procedure
Significant pain + loss of function, fixed deformity	**Weil (distal oblique shortening) osteotomy**
Mild–moderate deformity	**Capsular release, collateral tightening, tendon lengthening, flexor transfer**
Severe subluxation / dislocation, persists after release	**MT shortening osteotomy** (decompresses joint, lengthens soft tissues)
Incompetent plantar plate	**Plantar plate repair** (sutures distal plate → drill holes in proximal phalanx) + osteotomy
Incompetent lateral structures	**EDB transfer under transverse intermetatarsal ligament** (dynamic stabiliser)
Dynamic flexion support	**FDL → EDL transfer (Girdlestone-Taylor)** – split FDL over proximal phalanx
Salvage	**DuVries metatarsal head resection arthroplasty**

Metatarsal shortening osteotomy (Weil/Helal) & floating toe

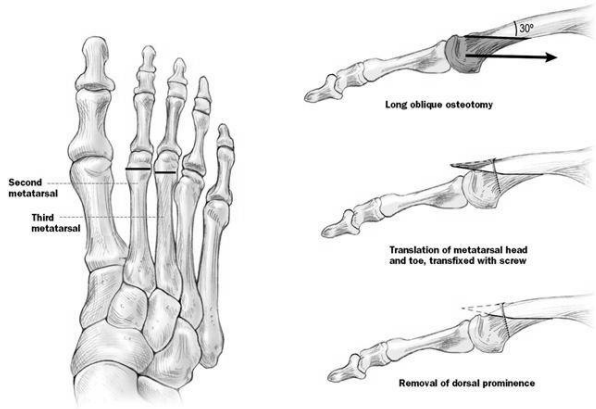
Weil	**intra-articular capital (metatarsal neck / head) osteotomy** · cut **parallel to plantar foot** · **longitudinal decompression by shortening** → joint reduces · **screw perpendicular to osteotomy** · with dorsal + medial capsular release, lateral reefing

Helal **more proximal (shaft)** osteotomy · **no fixation**

	Weil	Helal
Level	**MT neck / head (distal)**	**Shaft (proximal)**
Fixation	**Screw**	**None**
Satisfaction, AOFAS	**Higher**	Lower
Recurrent metatarsalgia, transfer lesions	**Fewer**	More
MTP reduction maintained	**Better**	Worse
Malunion / pseudoarthrosis	**Lower**	**Higher**

Weil vs Helal EXCEPT Weil does **NOT ↑ malunion / pseudoarthrosis**

✗ Next-step traps (subluxation after release) **flexor tendon resection** (floppy toe, no dorsal correction) · **proximal phalangeal osteotomy** (distal to problem) · **lesser MTP arthrodesis** (poorly tolerated) · **distraction osteogenesis of MT** (lengthens → more instability)



Weil osteotomy

6 Complications

Floating toe **commonest complication of Weil osteotomy** · toe elevated in sagittal plane, **doesn't touch floor** standing · inability to flex MTP → **2nd toe dorsiflexion deformity** · from surgery, injury, or shortened MT
✗ Wrong options: dorsal release + pin · silicone implant · excisional arthroplasty · Helal

Toe vascular compromise correcting **chronic dislocation** stretches contracted vessels → **may need to reverse the procedure** to save the digit

– – Active space – –

X vs Y

	Plantar plate rupture	Freiberg's infraction	Morton's neuroma
Pathology	Ligament failure → **dorsal instability**	**AVN of MT head subchondral bone** (repetitive microfracture)	Interdigital nerve
Exam	**+ drawer / Lachman**, ↑ sagittal translation	MT head pain	**Mulder click**, squeeze, numbness
Site	**2nd MTP**	**2nd MT head**	**3rd web space**
X-ray	Normal early	**Flattening / collapse**	Normal

How it's asked

"Golfer, 2nd MTP hyperextension, calluses dorsal PIP + plantar head – all true EXCEPT?"	→ **repetitive microfractures compromising MT subchondral blood supply** (that's Freiberg)
"Plantar callus mechanism?"	→ **dorsal displacement of fatty cushion under MT head**
"MRI 2nd MTP plantar plate tear – exam finding?"	→ **↑ sagittal translation (+ drawer)** ✗ Wrong options: hyperflexion · squeeze pain · shaft tenderness
"Grade II plantar plate tear, footballer – next step?"	→ **rest, NSAIDs, stiff-soled shoe**
"Rock in shoe, + modified Lachman, normal X-ray – diagnosis?"	→ **plantar plate rupture** ✗ Wrong options: Freiberg · Morton's · claw toe
"Joints still subluxate after extensor lengthening + capsular release?"	→ **metatarsal shortening osteotomy**
"Weil vs Helal – all true EXCEPT?"	→ **↑ malunion / pseudoarthrosis**
"Floating toe – commonest complication of?"	→ **Weil (metatarsal neck) osteotomy**
"Crossover toe after HV surgery, 1 month symptoms?"	→ **MT pads, shoe modification, taping, NSAIDs** ✗ Wrong options: steroid injection
"Commonest complication of MTP dislocation correction?"	→ **floating toe**
"Pale toe after correcting chronic dislocation?"	→ **vascular stretch → reverse correction**

– – Active space – –

NERVES OF THE FOOT

– Foot & Ankle · 35 / 48 –

1 Basics

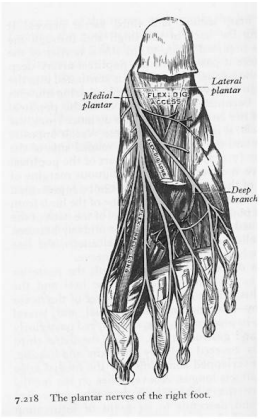
Tibial nerve branches

Tibial nerve ankle **plantarflexion**, **toe flexion**, **plantar sensation** · 2 primary branches **medial + lateral plantar → all intrinsic foot muscles except EHB + EDB**

Branch	Motor	Sensory
Medial calcaneal n.	–	**Plantar medial heel** (lesser role)
Medial plantar n.	**Abductor hallucis**, **FHB**, **FDB**, **lumbricals to 2nd + 3rd toes** (TO: **1st lumbrical**)	Medial plantar foot
Lateral plantar n. (LPN)	**Adductor hallucis**, **quadratus plantae**, **all interossei**, **lateral two lumbricals**, **abductor digiti minimi** (via Baxter's)	**Plantar lateral foot**, lateral 4th toe, 5th toe (TO: **lateral ⅓ plantar foot + 4th + 5th toes**)
Baxter's n.	**1st branch of LPN** = nerve to **abductor digiti quinti (ADM)**	Courses **anterior to medial tuberosity between QP and FDB**

Lateral / medial plantar injury **weak or absent MTP flexion** (**intrinsic minus**)

Medial plantar proper branch at risk in **medial-plantar approach to tibial sesamoid**



Plantar nerves of the foot

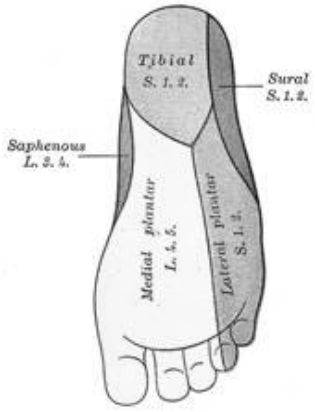
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Dorsal & cutaneous nerves

Nerve	Supplies	Most vulnerable
Sural	Dorsal sensation **4th web space (70–80%)** · lateral foot	**Extensile lateral approach to calcaneus** · **screw for proximal 5th MT stress fracture** · **Achilles repair (esp. percutaneous)** · **gastroc recession** · **peroneal tendon repair (posterolateral incision)**
Deep peroneal (DPN)	Motor **EDB + EHB** · sensation **1st dorsal web space**	Compressed **beneath inferior extensor retinaculum** (anterior tarsal tunnel) · **lateral branch**: **calcaneonavicular bar excision**, **Lisfranc reduction approach** · anterior ankle approach
Superficial peroneal (SPN)	**Most of dorsum of foot**	**Medial branch (dorsomedial cutaneous)**: **bunion surgery** · **lateral cutaneous branch**: **anterolateral arthroscopy portal**, **distal fibula ORIF (esp. Weber C)** · **ankle inversion injury** · anterior ankle approach (medial branch)
Saphenous	**Medial side of foot**	–
Tibial	Plantar foot	**Tarsal tunnel release (posteromedial)**

DPN injury — **weak / absent EDB + EHB** · trauma or **CMT**

SPN exit from deep fascia — **12–15 cm** proximal to lateral malleolus tip (Orthobullets, lateral cutaneous branch) · **10–12 cm** (OB) – read the options · wraps **fibular head**, branch of **common peroneal**



Plantar sensory distribution – tibial, sural, saphenous, medial and lateral plantar

3 Presentation

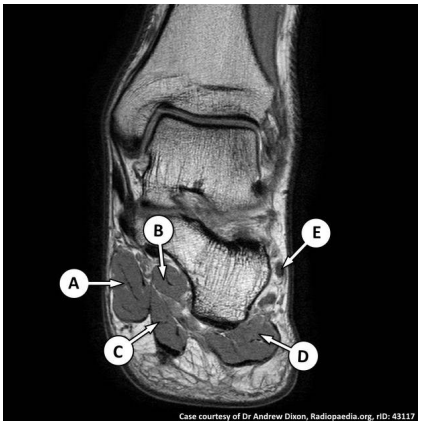
Tarsal tunnel – denervation pattern

Tarsal tunnel syndrome — **commonest tibial nerve compression** · plantar numbness, Tinel **posterior to medial malleolus** · EMG **prolonged sensory + distal motor latencies** · untreated → intrinsic **denervation atrophy on MRI**

Spared muscle **peroneus brevis** (**SPN**) – least likely to atrophy

– – Active space – –

Coronal T1 MRI label	Muscle	Nerve
A	**Abductor hallucis**	Medial plantar
B	**Quadratus plantae** (flexes 2nd–5th toes)	Lateral plantar
C	**Flexor digitorum brevis**	Medial plantar
D	**Abductor digiti minimi**	Lateral plantar (Baxter's)
E	**Peroneus brevis**	**SPN – spared**



Coronal T1 MRI hindfoot muscles A–E

5 Management

Iatrogenic nerve injury by incision

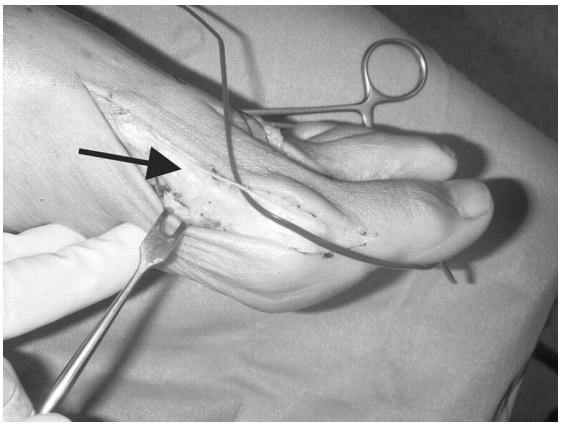
Incision	Nerve at risk
Gastroc recession (posterior calf)	**Sural** ✓
Anterior total ankle approach	**Medial branch of SPN + DPN** (✗ Wrong options: tibial)
Tarsal tunnel release (posteromedial)	**Tibial nerve + branches** (✗ Wrong options: DPN)
Peroneal tendon repair (posterolateral)	**Sural** (✗ Wrong options: SPN)
Calcaneus ORIF	**Sural** (✗ Wrong options: lateral plantar)
Bunion (medial MTP incision)	**Dorsomedial cutaneous n.**

Dorsomedial cutaneous nerve **terminal branch of SPN** · **5–15 mm medial / plantar to EHL** · injury → **reduced sensation over medial aspect of great toe** ± painful neuroma
✗ Wrong options: saphenous · DPN · motor weakness

✗ Motor traps 1st MTP **abduction** weakness = **abductor hallucis (medial plantar)** · IP **extension** weakness = **EHL / EHB (DPN)** · IP **flexion** weakness = **tibial n.**

Dorsomedial cutaneous nerve syndrome severe neuritic pain over medial eminence + 1st toe after bunion surgery · nerve transected / scarred → **resection + burial** if nonop fails · **positive nerve block** before surgery

– – Active space – –



Dorsomedial cutaneous nerve isolated in a bunion incision

Painful neuroma – neurectomy & burial

Chronic SPN injury after inversion sprain	paraesthesia dorsum, **Tinel anterolateral ankle**, no instability · failed 1 yr nonop (steroid, brace, PT) → **neurotomy + transposition to bone** (bone burial > muscle burial for pain) ✗ Wrong options: Semmes-Weinstein (diabetic screening) · PRP (no evidence) · neuroma excision + nerve grafting · Broström-Gould (no instability)
Glass laceration, plantar lateral forefoot numbness	**lateral plantar nerve laceration** (MRI: through abductor hallucis + LPN) ✗ Wrong options: foreign body · Baxter's (Tinel deep to abductor hallucis) · interdigital neuroma (Mulder's click) · digital nerve (more distal)

X vs Y

	SPN	DPN
Sensation	**Dorsum of foot**, dorsomedial hallux	**1st dorsal web space**
Motor (foot)	None	**EDB, EHB**
Classic injury	Bunion (medial branch), anterolateral portal, Weber C, inversion sprain	Anterior tarsal tunnel, CN bar excision, Lisfranc approach

	Medial plantar	Lateral plantar
Motor	AbH, FHB, FDB, lumbricals 2–3	AdH, QP, all interossei, lateral 2 lumbricals, ADM
Sensation	Medial plantar	Plantar lateral, 4th (lateral) + 5th toes

How it's asked

"Nerve looped in medial bunion incision – true?"	**→ terminal branch of the superficial peroneal nerve; reduced sensation over medial aspect of great toe**
"Correct incision–nerve pairing?"	**→ gastroc recession – sural nerve**
"Chronic SPN pain after ankle sprain, failed nonop – next?"	**→ neurotomy and transposition to bone**

"Stepped on glass, plantar lateral numbness, no Mulder's, no Tinel?"	→ **lateral plantar nerve laceration**
"Tarsal tunnel — muscle least likely to atrophy?"	→ **peroneus brevis (E)**
"First branch of LPN?"	→ **Baxter's nerve (to abductor digiti minimi)**
"Sensation to 1st dorsal web space?"	→ **deep peroneal nerve**

– – Active space – –

– – Active space – –

ORTHOTICS

– Foot & Ankle · 36 / 48 –

1 Basics

Uses & principles

Uses	**off-load** high pressure + ↓ shear · **cushion** vulnerable tissue (diabetics) · **correct flexible / accommodate rigid** deformity (planovalgus) · **eliminate painful motion** (hallux rigidus) · **replace lost motion**, improve gait (**ankle fusion**)
Stabilisation	**static** = rigid, holds segment fixed · **dynamic** = mobile, permits motion · or combination
Naming	by joints controlled (ankle + foot = **AFO**)
Patient factors	soft tissue at risk · functional level (orthosis must match) · **pathology**: soft tissue, flexible vs rigid deformity, painful motion, weakness
Design	**three-point pressure** system · aligned at anatomic joint · simple, light, durable, acceptable cosmesis

4 Diagnosis & Classification

Foot posture index (TO)

FPI (TO)	**static** clinical assessment of **standing** foot posture · quantifies supinated / neutral / pronated foot · **relaxed double-limb stance ~2 min** ✗ Wrong options: **dynamic biomechanical assessment** = FALSE
6 items (TO)	**talar head palpation** · curves above + below malleoli · **calcaneal frontal-plane position (inversion / eversion)** · **talonavicular congruence** · **medial longitudinal arch height + congruence** · forefoot abduction / adduction

Item (TO)	−2 (highly supinated)	0 (neutral)	+2 (highly pronated)
Talar head palpation	Lateral side only	Equal medial + lateral	Medial side only
Curves above / below malleolus	Below straight or convex	Roughly equal	Below markedly more concave
Calcaneal inversion / eversion	> 5° inverted (varus)	Vertical	> 5° everted (valgus)
TN congruence	Markedly concave	Flat	Bulging markedly
Medial arch height	High, acutely angled	Normal	Very low, arch contacts ground
Forefoot abd / adduction	No lateral toes visible	Equal	No medial toes visible

Total score (TO)	Foot type
−5 to −12	**Highly supinated**
−1 to −4	Supinated
0 to +5	**Neutral**
+6 to +9	Pronated
≥ +10	**Highly pronated**

Uses (TO) diabetic neuropathic ulcer risk · research inclusion / exclusion · sports injury risk · falls in elderly · age-related foot change · **more valid** than goniometric WB / NWB methods

5 Management

Shoes & shoe modifications

Shoe	Foot
Extra-depth	Room for orthosis / deformity
Stiff, supportive	**Flexible** foot (flexible planovalgus)
Soft, shock-absorbing (running shoe)	**Rigid, bony** foot
Custom	Severe deformity (**Charcot**)
High-top lace-up	**Poor distal proprioception** (proximal feedback, balance)

Internal vs external modification internal = **more mechanically effective**, ↓ shoe space · external = keeps volume, worse cosmesis, **wears faster**

Modification	Indication
Cushioned heel	**Heel pad atrophy**
Lateral flare	Resists **inversion** (ankle instability)
Medial flare	Resists **eversion**
Lateral sole wedge	**Cavovarus with fixed forefoot pronation** (forefoot reaches ground without compensatory varus)
Medial wedge	**Flexible planovalgus (PTTD)** → corrects hindfoot valgus
Lateral heel wedge	Unloads **medial knee** compartment (fixed knee varus)
Heel lift	**Equinus**, **LLD**
Extended shank (carbon / steel)	Splints forefoot / midfoot: **hallux rigidus (Morton's carbon-fibre extension)**, midfoot arthritis
Sole excavation, metatarsal pad, toe crest, scaphoid pad	Bony prominences / painful sites

Lateral wedge for cavovarus with fixed forefoot pronation **lateral sole wedge** (Orthobullets) · **lateral heel wedge** (OB) — read the options

– – Active space – –

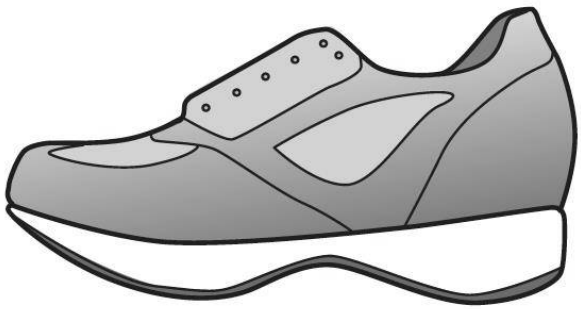
Rocker soles

Effect transfer body weight forward · **can destabilise the knee** → caution with balance / proprioception problems

Rocker	Shape	Use
Mild	Mild angle toe + heel – **commonest**	↓ MT head pressure, forward propulsion
Heel-to-toe (single)	More angled toe + heel	↓ heel-strike pressure + **need for ankle motion**: **ankle / subtalar arthritis or fusion**, midfoot amputation, **calcaneal ulcers**
Toe only	Angled at toe	↑ WB proximal to MT heads
Severe angle	Steep toe	**MT head / toe-tip ulcers**
Negative heel	Heel lower than sole	**Fixed dorsiflexion** deformity
Double	Two rockers over forefoot + hindfoot	↓ midfoot pressure: **midfoot prominence, Charcot**



Heel-to-toe rocker sole



Double rocker sole for midfoot prominence

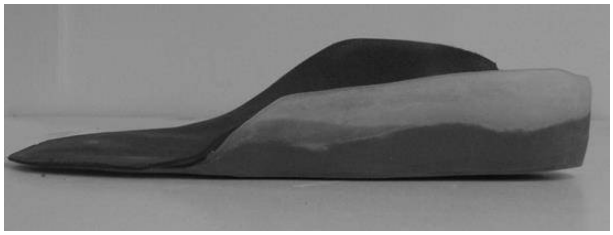
Gait orthosis after ankle arthrodesis

Ankle fusion gait **↑ metabolic energy 9–15%** vs normal

Rocker-bottom sole **mimics ankle DF / PF** · redistributes plantar pressure · improves kinematics + function
✗ Wrong options: **Arizona brace** (immobilises ankle, ankle arthritis) · non-articulating / articulating **AFO** (immobilise, foot drop) · **Morton's extension** (hallux rigidus)

Foot orthoses (inserts)

Insert	Key facts
Heel cup	Rigid plastic; prevents **lateral calcaneal shift** in flexible planovalgus
UCBL	Rigid plastic over cast in **maximum manual correction**; heel + midfoot, medial / lateral / posterior walls; heel **vertical neutral** · **flexible planovalgus only** – **rigid → painful, skin breakdown**
Longitudinal arch support	Prevents subtalar depression, corrects pes planus
Metatarsal pad / bar	**Metatarsalgia** → load moved from MT heads proximally



UCBL orthosis

Ankle orthoses & AFO

Arizona brace — **UCBL + lace-up ankle support** · **flexible planovalgus** (rigid hindfoot support) (Orthobullets) · **immobilises ankle** → nonop **ankle arthritis** (OB) · **PTTI / AAFD**: limits ankle motion → **articulated AFO** preferred in active walkers (OB)

AFO construction — footplate, calf support, calf band · plastic, metal, leather

Ankle → knee — **ankle PF = knee extension force** · **ankle DF = knee flexion force**

	Non-articulating AFO	Articulating AFO
Look	More aesthetic	–
Knee	**Flexion force at weight acceptance** (neutral ankle)	Adjustable stops / assists
Stance	No eccentric PF in early stance · **trim lines set late-stance flexibility** (max / moderate / min DF resistance)	More natural gait; also medial–lateral control

– – Active space – –

AFO design	Key facts	Use
Posterior leaf spring	**Commonest AFO** · narrow trim line **behind malleoli** · resists PF · **no medial–lateral control**	**Foot drop**
Solid	Trim line **anterior to malleoli** · blocks PF + varus / valgus	Rigid control
Hinged	Adjustable stops	Multiplanar instability, ankle pain, **spina bifida (mid-lumbar)**
Ground reaction	Pretibial shell, forefoot trim line · **max PF resistance → knee extension**	**CP with over-lengthened triceps surae + mild crouch**
Patellar tendon bearing	Load on patellar shelf	**↓ WB force through foot**
CAM walker	Off-the-shelf immobilisation, WB allowed	Ankle sprain, stable ankle fracture, Achilles rupture
SMO (supramalleolar)	**Shortest AFO**, ends above malleoli; heel neutral vertical	Flexible pes planus, planovalgus, hyperpronation



Posterior leaf spring AFO

Ankle joint option	Effect	Use
Free motion	Free DF + PF; **medial–lateral stability only**	**Ligamentous ankle instability**
Plantarflexion stop	Blocks PF, free DF → **knee flexion moment** at weight acceptance (✗ Wrong options: **quadriceps weakness**)	**Swing-phase foot drop**, flexible equinus
Dorsiflexion stop	Blocks DF, free PF → **knee extension moment** (prevents buckling)	**Weak plantarflexion in stance** (± weak quads)
Limited motion stop	Blocks both	**Global ankle weakness**
DF assist spring	Posterior coil spring aids swing DF	**DF weakness with preserved motion**
T-strap	Medial strap → lateral upright = **valgus correction** · lateral strap → medial upright = **varus correction**	Varus / valgus

Knee orthoses

KAFO	AFO + uprights + mechanical knee + 2 thigh bands · **quadriceps weakness / paralysis**, flexible genu varum / valgum · harder to don · not with moderate–severe cognitive dysfunction
Double-upright metal KAFO	**commonest** KAFO
Scott-Craig orthosis	**paraplegia (SCI)** standing + ambulation · T-shaped footplate, adjustable ankle stops, pawl-lock knee + bail · **hip hyperextension** puts COG behind hip, in front of locked knee + ankle · **10° ankle DF** alignment → swing-to / swing-through with crutches

X vs Y

	Plantarflexion stop	**Dorsiflexion stop**
Blocks	PF	DF
Knee moment	**Flexion** (avoid if quads weak)	**Extension** (prevents buckling)
Use	**Foot drop**, flexible equinus	**Weak plantarflexors** in stance

	UCBL / medial wedge / Arizona	**Lateral wedge / 1st ray cutout**
Foot	**Flexible planovalgus**	**Cavovarus**

How it's asked

"Gait problems after ankle arthrodesis – device?"	→ **Figure C – rocker-bottom sole**
"FALSE about foot posture index?"	→ **it is a dynamic biomechanical assessment** (TO)
"Toe-tip / MT head ulcer – rocker?"	→ **severe angle rocker**
"Fixed ankle dorsiflexion deformity – rocker?"	→ **negative heel rocker**
"Charcot midfoot prominence – rocker?"	→ **double rocker**
"Commonest AFO, for foot drop?"	→ **posterior leaf spring AFO**
"UCBL in rigid planovalgus?"	→ **painful, skin breakdown**
"CP, over-lengthened triceps surae, mild crouch?"	→ **ground reaction AFO**
"AFO joint to avoid with quadriceps weakness?"	→ **plantarflexion stop**
"Paraplegic standing + swing-through gait?"	→ **Scott-Craig orthosis**

– – Active space – –

OSTEOCHONDRAL LESIONS OF THE TALUS

– Foot & Ankle · 37 / 48 –

1 Basics

Epidemiology

Incidence	**69% of ankle fractures**, **70% of ankle sprains** (Orthobullets, OB) · **2–6% of all ankle injuries** (another OB) – read the options
Unexplained chronic ankle pain	OLT in **81%**
Sprains that fail to settle	**~10%** → think **OLT** or persistent **instability**
Bilateral	**10%**
Medial vs lateral frequency	**medial more common**: ~**68%** (OB) · 50–56% vs lateral 34–44% · paediatric medial **53–73%** · most lesions actually **medial + central** on MRI (classic teaching: posteromedial / anterolateral)
Chronic lateral ankle instability (CLAI)	**32–56%** develop OLTs · repetitive talar tilt + translation → **impaction on medial dome**

Diagnosis & aetiology – medial vs lateral lesions

	Medial talar dome	**Lateral talar dome**
Trauma history	**Usually none** (overuse / microtrauma / ischaemia)	**Usually traumatic** (inversion ± dorsiflexion)
Trauma rate	80% (Flick) · 60% (McGahan)	100% (Flick) · ~90% (McGahan)
Position	**More posterior** (posteromedial)	**More central or anterior** (anterolateral)
Size / depth	**Larger, deeper**	**Smaller, superficial / shallow**
Frequency	**More common**	Less common
Spontaneous healing	Better	**Lower** → harder with nonop
Behaviour	Less symptomatic	**More often displaced + symptomatic**

Anatomy

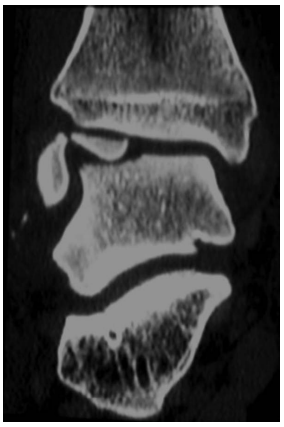
Talus	frustum shape · **anterior broader than posterior** · **no muscular attachments**
Cartilage	covers **70% of talus** · **among thickest in body** · keeps tensile strength longer than femoral head with age
Blood supply	relies on **extra-osseous** supply · **deltoid artery supplies majority of talar body + dome** ✗ Wrong options: release deltoid in medial malleolar osteotomy

Mortise	highly congruent · oriented **15° externally** from midsagittal line

– – Active space – –

2 Mechanism

Lateral OLT	**inversion + dorsiflexion + axial load** → shear of lateral dome
Medial OLT	**inversion + external rotation + plantarflexion + axial load** → shear of medial dome
Pathophysiology after inversion injury	**chondral shearing injury** ✗ Wrong options: failure of embryonic segmentation = coalition · ATFL / CFL sprain · SPR tear
Microtrauma	repetitive microtrauma → ischaemic environment → loss of subchondral integrity → softened, disrupted cartilage
Talar tilt	**inverse relationship** between talar tilt on stress X-ray and OLT (bony constraint) · stable ankle + effusion after sprain → OLT
Associated	**cavus hindfoot** · **CLAI**
LIFT lesion	**lateral inverted osteochondral fracture of talus** after inversion · fragment's joint surface **faces talus, not tibia** · arthroscopy → **open reattachment** (most) or excision · good function at ~9 yr, osteophytes in all



Coronal CT – LIFT lesion: inverted lateral dome fragment

3 Presentation

History	**inversion ankle sprain** · persistent pain after sprain that does not heal
Symptoms	pain over **ankle joint line** · **effusion** · **mechanical catching / locking**
Exam	effusion · **palpation rarely reproduces pain** · cavus · ROM limited by pain / effusion · **stable anterior drawer** (no instability) in classic stem
Check ligaments	**untreated lateral ligament insufficiency at time of OLT repair ↑ failure**

Differential (option)	Why wrong
Tarsal coalition (failure of segmentation)	Sprains but ↓ subtalar motion
ATFL sprain	**Anterior drawer laxity in plantarflexion**
CFL sprain	**Anterior drawer laxity in dorsiflexion** · 2nd commonest sprained
SPR tear	Peroneal instability, **no effusion**

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Key facts
Weight-bearing ankle X-ray	First step · **often normal** · subtle lucency / fragmentation · shows **open physes**
Bone scan	Equivocal X-ray · **94% sensitive, 96% specific**
CT	**Subchondral bone + cysts** · pre-op planning · **less reliable for purely cartilaginous / nondisplaced** lesions
MRI	**Persistent pain after sprain, normal X-ray, stable ankle → next step** · size, oedema (may **overestimate**), stability · **fluid deep to subchondral bone = unstable** · **92% sensitive** for stability

Persistent pain 6 mo after sprain, no instability, normal X-ray

MRI of ankle
✗ Wrong options: continue PT · CAM boot 2 wk · steroid injection · diagnostic arthroscopy



AP ankle – lateral talar dome osteochondral lesion

Juvenile lesions – nonoperative care & prognostic MRI features

Only factor correlated with stability
open physes / skeletal immaturity
✗ Wrong options: absence of effusion · subchondral sclerosis · perilesional marrow oedema · T1 homogeneity · cysts · signal rim · intra-articular body

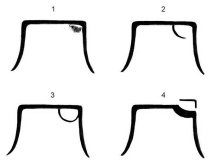
MRI prognostic factor
incomplete cartilage cap with fluid penetration beneath lesion = **unstable, poor prognosis** · intact cap = stable
✗ Wrong options: perilesional stress fracture · ATFL / CFL rupture (not prognostic) · physeal closure (seen on X-ray) · synovial hypertrophy

Open physes vs MRI sign
open physes = only correlate of stability (OB, Patel) · four MRI stability criteria: **cartilage defect, oedema-like signal, T2 bright rim, subchondral cysts** (Hembree) – read the options (physes are judged on **X-ray**)

Juvenile OCD counselling
nonop frequently relieves pain but often no radiographic healing even at 6 months
✗ Wrong options: spontaneous healing common (rare) · healing < 12 wk · hyperbaric O_2 · fusion frequently needed

Paediatric
mean age ~13 yr · stable lesions in skeletally immature → **trial of nonop** · conservative treatment best in children

Berndt and Harty (radiographic)

Stage	Description	Diagram
1	**Small area of subchondral compression**	
2	**Partial fragment detachment**	
3	**Complete detachment, not displaced**	
4	**Displaced fragment**	

Ferkel and Sgaglione (CT)

Stage	Description
1	**Cystic lesion within dome, intact roof** on all views
2a	Cystic lesion **communicating with dome surface**
2b	**Open articular surface** lesion with overlying nondisplaced fragment
3	Nondisplaced lesion **with lucency**
4	**Displaced fragment**

Hepple (MRI)

Stage	Description
1	**Articular cartilage oedema**
2a	Cartilage injury + underlying fracture **+ surrounding bone oedema**
2b	2a **without** bone oedema
3	**Detached, nondisplaced** fragment
4	**Displaced** fragment
5	**Subchondral cyst** formation

5 Management

OLT

- Acute, nondisplaced, incomplete → **immobilise + NWB 6 wk**
- Acute displaced, viable cartilage + bone (> 0.5 cm²) → **surgical fixation**
- Chronic, < 1 cm, or small fragment with little bone → **arthroscopic debridement + marrow stimulation**
- > 1 cm, intact cartilage cap → **retrograde drilling ± bone graft**
- > 1 cm displaced, shoulder, cystic, failed BMS / drilling → **osteochondral graft (OAT, ACI, allograft)**

– – Active space – –

Treatment depends on	**age, activity demands, lesion size, stability** · surgery if nonop fails after **6 months**

Operative criteria

Lesion	Treatment
Acute injury, nondisplaced fragment, incomplete fracture	**Immobilisation + NWB** (cast / boot 6 wk → PT) · **45% good–excellent** (Orthobullets) · 45–50% (OB)
Acute displaced, viable cartilage, adequate bone, > 0.5 cm²	**Surgical fixation** (headless compression screws / absorbable pins)
Chronic, < 1 cm, displaced small fragment with minimal bone	**Arthroscopic removal of loose fragment, debridement + marrow stimulation**
Ideal for BMS	**< 10 mm diameter, < 100 mm² area, < 5 mm depth** · excellent < 1.5 cm², best < 1 cm
> 1 cm with intact cartilage cap	**Retrograde drilling ± bone grafting**
> 1 cm + displaced, shoulder lesions, salvage of failed BMS / drilling	**Osteochondral grafting** (OAT, ACI, bulk allograft)
Contraindications to grafting	**Diffuse ankle arthritis · bipolar kissing lesions · advanced talar dome osteonecrosis**

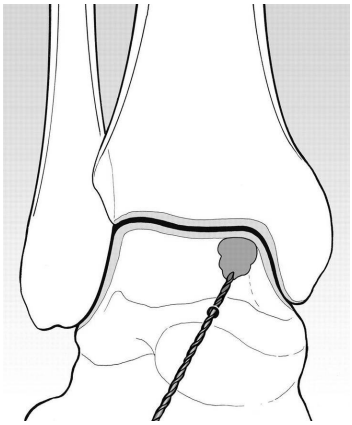
Early surgery favoured by	**displaced fragment no longer congruent with talar dome** ✗ Wrong options: lateral location alone · < 1 cm · presenting within 2 wk · subchondral sclerosis (chronic / contained)
Surgery indicated	chronic lesions · acute displaced fragments · higher grade · failed nonop · **loose bodies**
Success rates	excision + curettage + drilling **85%** · excision + curettage 78% · nonop **45%** · excision alone **38%** · arthroscopic excision + BMS **87%** (OB)

Small lesions – arthroscopic marrow stimulation

Chronic medial OLT < 1 cm, failed nonop	**arthroscopic marrow stimulation** (microfracture) ✗ Wrong options: total contact cast (diabetes) · arthrodesis (no arthritis) · open ACI (large lesions, not first line) · steroid injection (short-lived)
Antegrade microfracture	**subchondral bone violated through tibiotalar joint arthroscopically** ✗ Wrong options: via sinus tarsi (retrograde) · knee autograft (larger) · posterolateral talus autograft (not described)
Technique	debride unstable flaps → stable contained defect · **awl perpendicular**, **2–4 mm deep**, holes **2–3 mm apart** · stop inflow → **fat / blood** from holes = adequate · K-wire via anterior portals or **transmalleolar** for central / posterior lesions
Repair tissue	**fibrocartilage (type I collagen)** · fibrocartilage in 60% at second look, no correlation with outcome ✗ Wrong options: hyaline at 6 months
Outcomes	**85% pain improvement** · 65–90% ↑ PROs · larger lesions worse · **age not a predictor**
Complications	cartilage delamination, graft failure

Intact cartilage cap – retrograde drilling ± grafting

Indication	**> 1 cm (e.g. 15 mm), contained, intact cartilage cap → retrograde drilling and/or bone grafting** (initial surgery)
Best outcomes	**1.2 cm defect + intact cartilage cap + no fragment displacement**
Retrograde drilling not for	displaced fragment (→ fixation or debridement + antegrade drilling) · 0.8 cm, no oedema (→ immobilise) · 0.5 cm displaced (→ debridement + antegrade drilling) · acute loose body (→ arthroscopic removal + debridement + antegrade drilling)
Technique	probe cap (softening, dimpling) · **K-wire from sinus tarsi into defect** (fluoro / guide) · cannulated drill → curette cyst → graft
Complication	**violation of intact cartilage cap**



Retrograde drilling from the sinus tarsi beneath an intact cap

Displaced fragments – fixation

Acute displaced 1 × 1 cm anterolateral, viable cartilage + intact subchondral bone	**surgical fixation** ✗ Wrong options: observation · excision · excision + microfracture (chronic, < 1 cm², poor healing potential) · OAT (failed BMS or > 1.5 cm², cystic)

Large / cystic lesions – osteochondral autograft

1.6 cm² displaced lesion	**osteochondral autograft transplantation of type II collagen** (hyaline) ✗ Wrong options: marrow stimulation (type I fibrocartilage) · NWB 6 wk · retrograde drilling (needs intact cap) · excision
OAT indications	**> 1.0–1.5 cm²** · cystic / uncontained · **failed BMS** · displaced · failed drilling · CLAI with > 50–60 mm²
OAT contraindication	**bipolar kissing lesions** · diffuse arthritis · advanced talar dome osteonecrosis ✗ Wrong options: displaced · failed drilling · failed BMS · > 1 cm (all indications)
OAT technique	donor **ipsilateral knee** · graft depth **12–15 mm** · perpendicular access, **flush** placement · knee cartilage **thinner** than talus → graft looks **prominent on X-ray** though cartilage flush
OAT results	80–87% good / excellent · failure ↑ **> 2.25 cm²** · **donor-site morbidity up to 15%** · no worse after prior BMS · lesion size ∝ worse MOCART

– – Active space – –

Fresh osteochondral allograft	failed prior arthroscopic cartilage procedures · **large cystic component** · contained > 1.5 cm² · uncontained shoulder · **knee OA / prior knee infection** · patient fears knee donor morbidity · comparable to femoral autograft
ACI	**two-stage**: arthroscopic chondrocyte harvest (ankle or knee) → culture → implant via osteotomy under **tibial periosteal patch** (fibrin glue) · > 1 cm², ± cyst, shoulder, primary or revision · **MACI** = equivalent, may avoid osteotomy
Approach	**medial malleolar osteotomy** for **medial + posterior** lesions (2 parallel K-wires aim at lateral extent of OLT; predrill 2 holes) · lateral lesions → **lateral malleolar osteotomy or ATFL / CFL release**

Concomitant instability – ligament reconstruction

7 mm medial OLT + CLAI (> 15° talar tilt, + drawer) for BMS	add **lateral ligamentous complex reconstruction** ✗ Wrong options: peroneal debridement · deltoid reconstruction · OATS · OCA
Unaddressed CLAI	**↑ clinical failure** (AOFAS < 80) · larger lesions (**≥ 150 mm²**, > 1.5 cm) · **bipolar lesions** (medial malleolus tip, tibial plafond)
1.8 cm posteromedial OLT, 12 mm cyst, intact cartilage, CLAI	**OAT via medial malleolar osteotomy + concomitant lateral ligament repair** (+ subchondral bone graft) ✗ Wrong options: microfracture + Broström (> 1.5 cm², cystic) · **staged** ligament surgery 6 mo later · retrograde drilling alone · ACI via anterior arthrotomy (two-stage)

6 Complications

Complication	Cause / procedure
Graft failure	All grafting procedures
Persistent pain	Small % regardless of treatment
Osteotomy delayed / nonunion	Malleolar osteotomy
Talar body ischaemia	**Deltoid released** (deltoid artery)
Ankle impingement	**Graft plug left proud**
Donor-site morbidity	Knee autograft harvest (up to 15%)
Violation of cartilage cap	Retrograde drilling
Progression	Lesion may involve entire ankle

X vs Y

	Antegrade BMS / microfracture	Retrograde drilling	OAT
Size	**< 1 cm**	**> 1 cm**	**> 1–1.5 cm²**, cystic
Cartilage cap	**Absent / unstable**	**Intact**	Displaced / shoulder
Route	**Through tibiotalar joint**	**Through sinus tarsi**	Malleolar osteotomy
Tissue	**Fibrocartilage (type I)**	Own cap preserved	**Hyaline (type II)**
Pitfall	Delamination	**Breach of cap**	Proud plug, donor site

How it's asked

"Dancer, pain 6 mo after sprain, stable ankle, normal X-ray — next step?"	→ **MRI of the ankle**
"Lateral vs medial talar OCD — true?"	→ **increased incidence of traumatic aetiology** ✗ Wrong options: deeper · better spontaneous resolution · more posterior · more common
"Persistent pain after inversion, stable drawer, effusion, 2° talar tilt — pathophysiology?"	→ **chondral shearing injury**
"12-yo talar OCD — counsel parents?"	→ **nonop relieves pain but often no radiographic healing even at 6 months**
"MRI / lesion feature associated with stability and nonop?"	→ **open physes**
"MRI ordered in a 13-yo to assess which prognostic factor?"	→ **incomplete cartilage cap with fluid penetration**
"Chronic medial OLT < 1 cm, failed 6 mo nonop?"	→ **arthroscopic marrow stimulation**
"0.8 cm anteromedial lesion, no cap — best procedure?"	→ **subchondral bone violated through tibiotalar joint arthroscopically**
"15 mm contained lesion, intact cartilage cap — initial surgery?"	→ **retrograde drilling and/or bone grafting**
"Best outcomes after retrograde drilling?"	→ **1.2 cm defect, intact cartilage cap, no displacement**
"Acute displaced 1 × 1 cm anterolateral fragment, viable cartilage + bone?"	→ **surgical fixation**
"Strongest reason for early surgery over immobilisation?"	→ **displaced fragment no longer congruent with talar dome**
"1.6 cm² displaced talar dome lesion?"	→ **osteochondral autograft transplantation (type II collagen)**
"Contraindication to talar OAT?"	→ **bipolar kissing lesions**
"1.8 cm posteromedial OLT + cyst + CLAI?"	→ **OAT via medial malleolar osteotomy + concomitant lateral ligament repair**
"7 mm medial OLT + CLAI — add to BMS to reduce failure?"	→ **lateral ligamentous complex reconstruction**
"Repair tissue after microfracture?"	→ **fibrocartilage (type I collagen)**
"Graft looks proud on post-op X-ray after knee OAT?"	→ **thinner knee cartilage** (surface is flush)

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PERONEAL TENDON TEARS AND INSTABILITY

– Foot & Ankle · 38 / 48 –

1 Basics

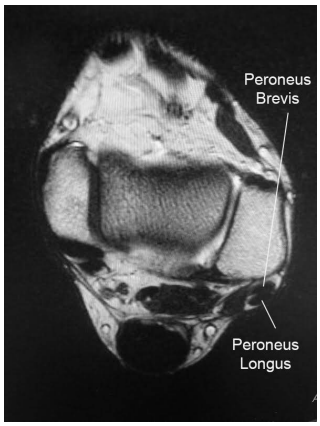
Epidemiology

Spectrum	tenosynovitis → tendinopathy → tears → instability (subluxation / dislocation)
Tears in lateral ankle instability	**23–77%** (Orthobullets) · "up to **77%**" (OB) → often **missed** in lateral ankle pain
Prevalence	tears = **4%** of ankle injuries
Instability	young, athletic · presents late, mistaken for **ankle sprain**
PB vs PL	**PB tears far commoner** (vulnerable between fibula + PL) · PL tears less frequent (TO) ✗ Wrong options: "PL tear more common" (TO)

Anatomy – brevis / longus relationship & course

Level	Peroneus brevis (PB)	Peroneus longus (PL)
Origin	**Distal ⅔ fibula** + intermuscular septum (TO)	**Proximal ⅔ lateral fibula**, septum, **lateral tibial condyle** (TO)
Becomes tendinous	**2–4 cm** above fibular tip (Orthobullets) · **2–3 cm** (TO)	–
Retromalleolar groove	**Directly posterior to fibula**, flat (TO) · **anterior + medial (deep) to PL**	**Posterior to PB**, rounded (TO)
Peroneal tubercle	**Superior / dorsal** to tubercle (at sinus tarsi level)	**Inferior / plantar** → cuboid tunnel
Insertion	**Base of 5th MT** (dorsolateral)	**Plantar base 1st MT + medial cuneiform** (via cuboid tunnel between **long plantar ligament** + cuboid groove – TO)
Function	**Primary evertor** · **63% of eversion power** (TO) · assists PF	**Plantarflexes 1st ray** + **everts foot** · **secondary ankle PF** + **stabilises medial column** (TO) · **transverse arch** · antagonises TA
Nerve	**Superficial peroneal (S1)**	**Superficial peroneal (S1)**

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Axial MRI: PB directly behind fibula, PL posterior to PB

✗ **Insertion traps**	navicular + medial cuneiform = **tibialis posterior** · base of hallux distal phalanx = **FHL** · distal phalanges 2–5 = **FDL** · 1st MT base + medial cuneiform = **PL** (not PB)
✗ **Course traps**	90° turn at medial malleolus → navicular + plantar MT 2–4 = **tib post** · anterior ankle → medial cuneiform + 1st MT base = **tib ant** · between talar tubercles → hallux = **FHL** · inferior to tubercle → medial cuneiform + 1st MT = **PL**
✗ **Ligament traps**	fibula → talar neck = **ATFL** · posterior fibula → posterolateral talar tubercle = **PTFL** · fibula → calcaneus = **CFL**
Common sheath (TO)	**~4 cm above to 1 cm below** lateral malleolus tip · splits into two sheaths at **peroneal tubercle** (TO)
Retromalleolar groove	fibro-osseous tunnel · deepened by **fibrocartilaginous ridge** (lateral) · only **~5 mm** deep · bounded by **SPR, PTFL, CFL, PITFL** · surface ↑ laterally by fibrocartilaginous ridge, medially by osseous groove (TO)
Superior peroneal retinaculum (SPR)	**primary restraint** to peroneal subluxation · from **posterolateral ridge / distal 1–2 cm of fibula** (TO: distal 2 cm) → lateral calcaneus / **Achilles sheath** · secondary restraint to anterolateral ankle instability · single fibro-osseous tunnel
Inferior peroneal retinaculum	at **peroneal tubercle**, forms 2 separate sheaths · not involved in instability · "sheaths thicken to form the inferior extensor retinaculum" (TO wording)
Blood supply	vincula from **anterior + posterior tibial arteries**, **whole tendon vascularised, avascular zones disproven** (Orthobullets) · vincula from **posterior peroneal + medial tarsal arteries** (TO) · "**vascular watershed** just posterior to fibula" (OB) — read the options
Peroneus tertius (TO)	separate muscle in **anterior** compartment (TO)

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Variants – peroneus quartus, os peroneum, low-lying PB

Variant	Facts
Peroneus quartus	Arises from **PB muscle belly, distal ⅓ leg** (✗ Wrong options: **from PL** (TO EXCEPT)) · runs posteromedial to tendons · inserts **retrotrochlear eminence** of calcaneus · incidence **up to 21%** (OB) · quartus or quintus **10–34%** (Orthobullets) · **mass effect** → pain, tears
Low-lying PB belly	Below superior margin of SPR · up to **33%** (TO)
Enlarged peroneal tubercle	**29%** · entraps PL → **PL tenosynovitis / tears** · Harris view
Flat / convex retromalleolar sulcus	**18%** (Orthobullets) · "flat or concave" (TO wording)
Os peroneum	Sesamoid in **PL** near **CC joint / lateral plantar cuboid** · radiographically as few as **5%** (TO) · uni / bi / multipartite · symptomatic = "**stepping on a pebble**"
POPS (TO)	**Painful os peroneum syndrome** → acute fracture … hypertrophied peroneal tubercle entrapping PL / os (TO)

Associated conditions & risk factors

Risk factors	**lateral ligament (ATFL / CFL) instability** · **cavovarus hindfoot** · **Charcot–Marie–Tooth** · low-lying PB · enlarged tubercle · quartus / quintus · flat sulcus · os peroneum · **calcaneal malunion + subfibular impingement**
Not risk factors	**pes planus / PTT insufficiency** (TO, OB) · 5th MT base fracture (↑ with cavovarus but doesn't cause tears) · talocalcaneal coalition · OLT
PL tears associated with	**diabetes**, RA, **hindfoot varus**, enlarged peroneal tubercle

2 Mechanism

SPR tear (TO)	**rapid forced dorsiflexion + inversion → reflex peroneal contraction →** SPR torn → subluxation / dislocation = **most common mechanism** (TO) · e.g. **ski tips caught**, foot caught on astroturf
Mechanism wording	**dorsiflexion + inversion** (Orthobullets, TO, most OB) · "dorsiflexion + eversion" in some OB explanations – read the options ✗ Wrong options: plantarflexion + inversion (line of pull direct, little subluxation)
PB split – two mechanisms (TO)	**subluxation over posterolateral fibular edge** (SPR lax / torn, chronic instability) → attrition · **compression between fibula + PL** in inversion (TO)
PB tear pattern (TO)	**longitudinal split** at **tip of fibula** (tendon changes direction acutely + compression by PL) (TO) ✗ Wrong options: radial · base of 5th MT
PL tears	direct trauma, sports · **cuboid notch / tunnel** = commonest PL site ✗ Wrong options: musculotendinous junction (TO)

3 Presentation

History	**pop** / snap with **dorsiflexion** injury · lateral instability · pain + clicking returns after cast removal

Pain lateral / **posterolateral** ankle, retrofibular · worse with **active eversion**, PF, **passive dorsiflexion**

Inspection swelling behind lateral malleolus · **cavovarus** · "pseudotumour" · voluntary subluxation ± pop · tendons palpable as ridge over lateral fibula

Test	Finding
Apprehension test	Subluxation / apprehension with **active DF + eversion against resistance** (most consistent)
Palpable snapping	**Tendon snapping over fibula during ankle dorsiflexion** (+ eversion) → **SPR injury**
Compression test	Pain / crepitus with **passive DF + eversion**
Active circumduction	Recreates instability
Coleman block	Hindfoot flexibility (cavovarus)
Anterior drawer	Concomitant ATFL laxity

✗ Exam traps +anterior drawer = **ATFL** · +ER stress = **syndesmosis** · anterolateral crepitus = arthritis / OCD / loose body · tenderness at **5th MT base with eversion = peroneal (PB) tear**, not SPR

PL tendinopathy signs pain with **resisted eversion** + **resisted plantarflexion of 1st MT**

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
X-ray (WB)	Often normal · **fleck sign** = cortical avulsion of SPR off **distal / posterior lateral malleolar ridge** (rim fracture) · cavovarus: plantarflexed 1st MT, ↑ Meary angle
Os peroneum migration	**Proximal migration = pathognomonic of PL rupture** · best on **oblique + lateral** · compare contralateral
Harris view	Peroneal tubercle morphology
Ultrasound	**Dynamic subluxation**, intra-sheath subluxation · **90% accurate** for tears · user dependent
CT	Calcaneal malunion, subfibular impingement, groove / tubercle shape
MRI	**Axial, slightly plantarflexed ankle** (↓ **magic angle**) · **83–90% sens, 72–75% spec** · circumferential sheath fluid = tenosynovitis · fatty infiltration · **three structures behind fibula = longitudinal split** · **boomerang sign** (PB wraps around PL) · accessory muscles, low PB belly · ATFL / CFL, OCD

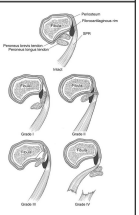
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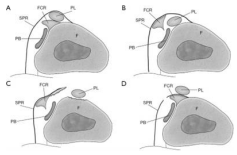
Fleck sign – SPR avulsion off the lateral malleolus

✗ PL rupture traps avulsion at 5th MT base (PB) · lateral tenderness (sprain) · **inability to plantarflex** (gastrocsoleus still works) · **pes planovalgus** (PTTD)

Anatomic classification of SPR tears (Orthobullets)

Grade	Description	Image
1	SPR **partially elevated** off fibula, **fibrocartilaginous ridge intact** → both tendons sublux	
2	SPR separated from **cartilofibrous ridge** → tendons sublux between SPR and ridge	
3	**Cortical avulsion** of SPR off fibula (fleck) → tendons under fragment	
4	SPR torn from **calcaneus**, not fibula	

Eckert & Oden classification of peroneal dislocation (TO)

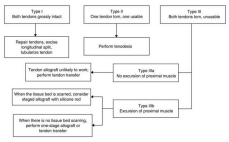
Type	Pathoanatomy	Frequency	Image
1	**SPR detaches from FCR + fibula with the anterior fibular periosteum** → false pouch	**51% – most common**	
2	**FCR + SPR + periosteum avulsed together** from anterior fibula	**33%**	
3	**Bony avulsion** of anterior fibula with FCR + SPR	**13%**	
4	**SPR itself ruptured** (added by Oden)	Rare, unknown	

Raikin classification – intra-sheath subluxation

Type	Description
1	PL lies **deep** to PB
2	PL subluxes **through a PB tear**

Redfern & Myerson tear classification

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Type	Tendons	Treatment	Algorithm
I	**Both intact, partial tears**	**Debride + tubularise**	
II	**One intact, other majority torn**	**Tenodesis** to intact tendon	
IIIa	**Both torn (unusable), no muscle excursion**	**FHL / FDL transfer**	
IIIb	**Both torn, muscle excursion present**	**Allograft** (one-stage) · scarred bed → **staged silicone rod**	

Other OB wording "Type I = < 50% one or both tendons, Type II = > 50% complex" — read the options

Brandes–Smith zones (TO)

Zone	Site	Typical tear
A	Under **SPR**	Subluxation, mass effect of **quartus / low-lying PB**
B	**Inferior peroneal retinaculum**	**Peroneal tubercle hypertrophy**
C	**Cuboid notch**	**Commonest site of PL pathology**

5 Management

Peroneal instability (SPR tear)

- Acute, nonprofessional → **reduce + SLC 6 wk, slight PF + inversion**
- Acute, high-level athlete → **SPR repair ± groove deepening**
- Chronic / failed nonop → **SPR repair + groove deepening**
- Recurrent, bony abnormality → **bone block / rerouting (salvage)**
  - hindfoot varus → **add calcaneal osteotomy**

Peroneal tendon tear (after failed rest, boot, PT)

- < 50% (50–60%) of width → **debride + tubularise**
- > 50% (> 60%), one tendon usable → **excise + tenodesis**
- Both > 50%, muscle excursion → **interposition allo / autograft**
- Both > 50%, no excursion → **FHL / FDL transfer**

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Operative criteria

Situation	Procedure
Tenosynovitis / tendinopathy, recalcitrant, < 50–60% to resect	**Tenosynovectomy + debridement** (tendoscopic)
Tear < 50% (Orthobullets 50–60%)	**Debridement + tubularisation**
Complex PB tear > 50% (multiple splits, tendinosis; OB: **> 60%**)	**Excise + proximal + distal tenodesis PB → PL** · 70–80% success, RTS 12 wk
Complex PL tear > 50%	**Tenodesis PL → PB**
Both > 50%, excursion preserved (IIIb)	**Interposition graft** (hamstring auto / allo)
Both > 50%, no excursion (IIIa)	**FHL / FDL transfer**
Acute dislocation, high-level athlete wanting quick return · longitudinal tear	**SPR repair + groove deepening**
Chronic / recurrent with bony abnormality / incompetent SPR	**Groove deepening + soft-tissue transfer / bone block** (salvage)
Rigid hindfoot varus / valgus	**Add corrective osteotomy**

Peroneal subluxation – SPR repair ± groove deepening

Acute, nonop	**tendons reduced** + **short-leg cast 6 wk**, protected WB, **slight plantarflexion + inversion** (TO, Orthobullets) · **irreducible acute dislocation → surgery** · some advise acute repair in athletes (high nonop failure) (TO) · **U- or J-shaped pad** · effective in **~50%** (OB) · **> 50% failure for chronic** · up to **83% failure for tears**
No PT / rehab for unstable tendons	may further damage them · no more casting or stirrup splint after failed cast
Failed nonop / chronic subluxation	**SPR repair ± fibular groove deepening** ✗ Wrong options: ATFL repair · CFL repair · Broström–Gould with solid drawer end point · fibular shortening osteotomy · walker boot in PF
Fleck sign + persistent popping, solid drawer	**peroneal retinacular reconstruction**
Dislocated tendons + convex fibular groove	**débridement + fibular groove deepening + SPR repair** ✗ Wrong options: tear repair alone (won't stop dislocation) · tenodesis alone
Commonest procedure for chronic dislocation (TO)	**SPR repair + groove deepening** (TO) ✗ Wrong options: CFL rerouting · distal fibular osteotomy + CFL rerouting · complete fibular osteotomy / bone block
SPR repair + groove deepening	**RTS 83–100%** (higher than SPR repair alone) · excellent PROMs

Surgical categories (TO) SPR repair / replacement · groove deepening · bone block / osteotomy · soft-tissue rerouting · tendon graft (TO)

Technique (TO)	Detail
Direct SPR repair	Chronic SPR redundant → **pants-over-vest** (anterior flap over posterior) · direct, bone tunnels or anchors
Groove deepening – direct	Elevate cartilaginous floor → burr / curette cancellous bone → tamp floor back
Groove deepening – indirect	**3.5 mm drill retrograde from fibular tip** parallel to groove → tamp floor into tunnel (preserves floor)
Bone block (Kelly)	Partial / complete fibular osteotomy, posterior displacement → barrier · firmer, but **fracture, nonunion, tendon irritation / adherence**
CFL rerouting (Platzgummer)	Tendons placed under CFL · salvage
Tendon graft	Achilles, PB, **plantaris** reinforce SPR via fibular tunnel

Bone block + rerouting (TO) nonunion, tendon adherence, **sural neuropathy**, instability → **salvage / revision only** (TO)

Approach longitudinal incision over tendons, **avoid sural nerve** · SPR split longitudinally leaving cuff, or transected from fibula

Tendon tears – debridement / tubularisation vs tenodesis vs transfer

First line rest, activity modification, **boot then PT** (boot stops when rest pain resolves) · orthosis for varus · NSAIDs, bracing

Tubularise (TO) after excising thin split, **> 50% of PB remains → tubularise** · **< 50% remains → tenodesis to PL** (TO)

Severe degeneration (TO) **muscle excursion adequate → hamstring auto / allograft** · **scarred muscle → FHL / FDL transfer** · **fibrotic sheath → staged silastic rod → pseudosheath → repair** (TO)

Single longitudinal PB tear, no tendinosis **core repair + tubularisation**

PB: multiple splits, > 70% tendinosis over 3 cm **excise + proximal + distal tenodesis to PL**
✗ Wrong options: core repair · excision without transfer · groove deepening · arthroscopic debridement

PB tenodesis levels proximal **~3 cm above** fibular tip · distal **~2 cm below** tip, side-to-side

30% PB + PL tear **debridement + tubularisation only**
✗ Wrong options: 30% PB + shallow groove → add **groove deepening** · 60% PB → tenodesis · 60% both + excursion → graft · 60% both, no excursion → transfer

SPR tear + complex PB 65% **SPR repair + groove deepening + PB → PL tenodesis**
✗ Wrong options: tendinitis → tenosynovectomy · 30% → tubularise · both 70% → graft vs transfer

PL tendinopathy (cuboid tunnel), PB normal **debride PL + tenodesis PL → PB** · proximal tenodesis just above fibular groove, distal just proximal to cuboid
✗ Wrong options: PL / PB repair (chronic, no excursion) · PB allograft

PL tear (MRI) **peroneus longus repair**
✗ Wrong options: SPR repair + groove deepening (only if intact + dislocated)

– – Active space – –

Failed PB repair, both tendons torn	**FDL transfer to 5th MT** → balance + hindfoot mobility ✗ Wrong options: **subtalar fusion** (salvage) · **PTT transfer** (loses inversion + arch) · **excise both** → imbalance → **forefoot varus**
Persistent pain after boot + PT, eversion weakness	**SPR repair + peroneal tendon repair / tenodesis** ✗ Wrong options: recast · syndesmotic fixation · OLT excision · biopsy
Tendoscopy	viewing portal **2 cm distal**, working portal **3 cm proximal** to fibular tip

Hindfoot varus – correction to prevent failure

Hindfoot varus	**risk factor for failure / recurrence** of peroneal stabilisation + repair → correct it ✗ Wrong options: bunion · tarsal coalition · accessory navicular · talar OCD
Correction	**calcaneal osteotomy (lateral closing wedge, Dwyer)** or **lateral closing-wedge subtalar arthrodesis** (rigid / subtalar arthritis)
PB tear + cavovarus NOT correcting on Coleman block	**calcaneal osteotomy + 1st ray dorsiflexion osteotomy + PL → PB transfer** ✗ Wrong options: 1st ray osteotomy alone (Coleman non-correcting) · PB repair alone · **PRP** · **amniotic augmentation**
Supple hindfoot, chronic PB tear	contributor = **hindfoot alignment (cavovarus)**

Calcaneal malunion – subfibular impingement

Anterior ankle pain + ↓ dorsiflexion	horizontal talus impinges on tibia → **subtalar bone-block arthrodesis**
Lateral pain	peroneal dislocation, **subfibular impingement**, subtalar arthritis → **lateral wall exostectomy**
Malunion + subtalar arthrosis + exostosis + height loss	**subtalar bone-block arthrodesis + lateral wall exostectomy ± Achilles lengthening** ✗ Wrong options: PTT transfer · Achilles lengthening + exostectomy alone · SPR repair ± sural neurolysis (only for isolated incompetent SPR) · TTC nail (needs tibiotalar arthritis)

Stephens–Sanders type	Treatment (OB)
I	**Lateral wall exostectomy + peroneal tenolysis**
II	+ **subtalar bone-block arthrodesis** (bone graft)
III	+ **closing-wedge calcaneal osteotomy** for varus, or triple arthrodesis

6 Complications

– – Active space – –

Complication	Fact
Sural neuroma	**Most common** complication after surgery
Recurrent instability	**Unaddressed malalignment (hindfoot varus)** → corrective osteotomy ± soft-tissue reconstruction
Persistent pain	**Over-tightened SPR** → tendon stenosis
Tibial nerve compression	After **FHL transfer** → release FHL–NV bundle adhesions
FHL / FDL transfer	Residual **eversion weakness**
Bone block	Fracture, nonunion, tendon irritation / adherence

X vs Y

	Peroneal instability (SPR)	Peroneal tear
Exam	Snapping / subluxation with **DF + eversion**, circumduction	Retrofibular tenderness, pain on resisted eversion, **no instability**
X-ray	**Fleck sign**	Usually normal (PL: os peroneum migration)
Surgery	**SPR repair ± groove deepening**	Tubularise / tenodesis / graft / transfer

	SPR	Inferior peroneal retinaculum
Attachment	Distal posterior fibula → Achilles sheath / lateral calcaneus	**Peroneal tubercle** of calcaneus
Role	**Primary restraint** to subluxation	Separates tendons into 2 sheaths
In instability	**Torn – repair**	Rarely injured, no repair

	Debride + tubularise	Tenodesis	Graft	FHL / FDL transfer
Tear size	**< 50%**	**> 50%** one tendon	Both > 50%	Both > 50%
Muscle excursion	–	–	**Present**	**Absent**

How it's asked

"Peroneal muscles – all true EXCEPT?"	→ **peroneus quartus originates from the peroneus longus belly** (it arises from PB) (TO)
"PB relation to PL in retromalleolar groove?"	→ **anterior**
"MRI longitudinal tear, tendon just behind fibula – insertion?"	→ **fifth metatarsal base**
"MRI shows PL tear – management?"	→ **peroneus longus repair**
"Structure A in peroneal sheath at surgery?"	→ **peroneus quartus**

– – Active space – –

"Tendon to plantar 1st MT base — name + function?"	→ **peroneus longus; eversion of foot**
"Snap while walking — most indicative of PL rupture?"	→ **proximal migration of the os peroneum**
"Chronic PB tear on MRI — true of the structure?"	→ **dorsal to the peroneal tubercle**
"Torn tendon course to insertion?"	→ **superior to peroneal tubercle at sinus tarsi → base of fifth metatarsal**
"Commonest pathoanatomy of peroneal dislocation?"	→ **SPR detached from FCR + fibula with anterior periosteum** (Eckert–Oden 1) (TO)
"Commonest procedure for chronic dislocation?"	→ **SPR repair + groove deepening** (TO)
"Mechanism causing SPR disruption?"	→ **dorsiflexion + inversion**
"Skier, snapping behind lateral malleolus — mechanism?"	→ **forced dorsiflexion + inversion**
"Popping, tendons sublux on active eversion — injured structure?"	→ **superior peroneal retinaculum**
"Chronic subluxation after failed PT?"	→ **fibular groove deepening + SPR repair**
"Exam finding of SPR injury?"	→ **palpable tendon snapping over the fibula during ankle dorsiflexion**
"Failed boot + physio, posterolateral pain, eversion weakness?"	→ **SPR repair + peroneal tendon repair / tenodesis**
"Popping + weakness after remote injury — deficient structure?"	→ **superior peroneal retinaculum**
"MRI dislocated tendons — repair what to stop popping?"	→ **superior peroneal retinaculum**
"Fleck sign, solid drawer end point — treatment?"	→ **peroneal retinacular reconstruction**
"Skier, clicking after cast, convex fibular groove?"	→ **débridement + fibular groove deepening + SPR repair**
"Axial T2 MRI after rolling ankle — diagnosis?"	→ **PL dislocation at the level of the SPR**
"Football player, labelled figure — injured structure?"	→ **SPR (label 4)**
"Skier, instability on circumduction — repair which structure?"	→ **SPR (structure B)**
"Commonest pattern + site of peroneal tear?"	→ **longitudinal tear at tip of fibula** (TO)
"True about peroneal tendon tear?"	→ **rapid forced dorsiflexion + inversion is the commonest mechanism** (TO)
"Diabetic, +Coleman block, PB normal, PL scarred?"	→ **debridement of PL + tenodesis of PL to PB**
"PB multiple splits, > 70% tendinosis?"	→ **excision with proximal + distal transfer to PL**

"Failed PB repair, PL + PB tears?"	→ **FDL transfer to the fifth metatarsal**
"Indication for debridement + tubularisation only?"	→ **30% longitudinal PB tear + PL tear**
"Soccer player, stable ankle, PB tear on MRI?"	→ **symptomatic PB tear, tubularise if < 50%**
"MRI finding indicating SPR repair + groove deepening + PB→PL tenodesis?"	→ **complex PB tear of 65%**
"PB tear + cavovarus not correcting on Coleman block?"	→ **calcaneal osteotomy + 1st ray dorsiflexion osteotomy + PL to PB transfer**
"Failure to correct what jeopardises peroneal stabilisation?"	→ **hindfoot varus**
"Supple hindfoot, chronic PB tear – contributor?"	→ **hindfoot alignment**
"Nonop calcaneus malunion – anterior + lateral pain, ↓ DF?"	→ **subtalar bone-block arthrodesis + lateral wall exostectomy ± Achilles lengthening**

– – Active space – –

PLANTAR FASCIITIS

– Foot & Ankle · 39 / 48 –

1 Basics

Epidemiology

Demographics	**men = women** · typical age **40–70** (OB) · common **bilaterally** · **posteromedial heel**
Risk factors	**obesity (↑ BMI)** · **↓ ankle dorsiflexion** (tight calf / foot musculature) in non-athletes · **weight-bearing endurance activity** (running, dancing) · steep ↑ training volume

Diagnosis & anatomy

Plantar fascia	nonelastic multilayered **connective tissue** originating on the **medial calcaneal tubercle** → inserts on **metatarsal heads** · **supports the arch** · **windlass mechanism** in late stance ✗ Wrong options: abductor hallucis origin · Baxter's nerve · Achilles insertion on calcaneal tuberosity · tibialis posterior on navicular tuberosity
Shared origin on medial calcaneal tubercle	**abductor hallucis, FDB, quadratus plantae** (may be inflamed too)
Pathology	**degenerative**, chronic overuse → **microtears at fascia origin** · recurrent inflammation + periostitis
Heel spur	may coexist · **not the cause**

Associated conditions

Heel pain triad	**plantar fasciitis** + **posterior tibial tendon dysfunction** + **tarsal tunnel syndrome**
Others	**calcaneal apophysitis** · **gastrocnemius–soleus contracture** · femoral anteversion · pes cavus · pes planus

3 Presentation

"Start-up" pain	insidious **inferior heel pain on first steps out of bed** / after rest · may walk on toes first · **eases with ambulation** · worse at **end of day** after prolonged standing
Exam	**tender medial calcaneal tuberosity** (plantar medial / anteromedial heel) · **↑ with dorsiflexion of toes + foot** · **↓ ankle DF** (tight Achilles · gastrocnemius contracture on **Silfverskiöld**)
Abductor hallucis origin tenderness	small subset → **Baxter's nerve** (first branch of lateral plantar nerve = **inferior calcaneal nerve**) entrapment · **pain persists after activity** rather than improving

4 Diagnosis & Classification

Diagnosis **clinical**; normal radiographs
✗ Wrong options: navicular stress fracture · Freiberg · Baxter's entrapment · anterior tarsal tunnel

Key radiology findings

Study	Use
Radiographs	**Not necessary on first visit** · often normal · may show **plantar heel spur** · optional WB axial + lateral hindfoot
MRI	**Surgical planning**
Bone scan	Quantifies inflammation · **rules out stress fracture**
Labs	Not routine · if **inflammatory arthritis / infection** suspected
EMG	**Rule out nerve entrapment**



Lateral radiograph – plantar calcaneal spur



Sagittal MRI – plantar fascia at its calcaneal origin

– – Active space – –

5 Management

Plantar fasciitis

- First line → **PF-specific + Achilles stretching, prefabricated insert, NSAIDs**
- Persists > 6 wk → **night splint (± steroid injection, cast)**
- > 6 mo, failed → **ESWT (FDA-approved)**
- Failed 6–12 mo (9 mo) → **partial plantar fasciotomy (medial ⅓)**
  - + tarsal tunnel compression → **open release + distal tarsal tunnel decompression**

First-line treatment – stretching, night splints & orthoses

Best initial	**Achilles + plantar fascia stretching** with **prefabricated shoe insert** / heel cups ✗ Wrong options: short leg cast · steroid injection · custom orthotic · surgical release
Plantar medial heel pain + calcaneal spur	**Achilles stretching** ✗ Wrong options: walker boot · steroid · 50% release · **heel spur resection**
Highest satisfaction at 8 wk	**plantar fascia–specific stretching** (recreates **windlass**) ✗ Wrong options: isolated Achilles stretching · steroid · ESWT · tarsal tunnel decompression + release

NWB plantar fascia–specific stretching > WB Achilles stretching · all stretching programs **equal satisfaction at 2 yr**

Prefabricated vs custom	**prefabricated more effective** (with stretching) + cheaper (Orthobullets, OB) · "Level III evidence supports custom orthotics" (OB) – read the options
Chronic plantar fasciitis orthosis	**dorsiflexion night splint** = **only orthosis proven effective** ✗ Wrong options: gel heel cushion · full-length prefabricated / accommodative insert · **Morton's extension** (1st MTP pathology) · **recessed first ray / cavus orthotic** (cavus foot) · semi-rigid medial post insert



Dorsiflexion night splint

Plateau at 6 wk with NSAIDs + PT	add **night splints**, continue PT ✗ Wrong options: stop PT for custom orthotics · ESWT · endoscopic fasciotomy · bipolar radiofrequency
Second tier (> 6 wk)	**night splints**, repeat **corticosteroid injection**, **cast immobilisation** (short leg cast **8–10 wk**, Orthobullets)

Symptom duration up to **6–12 months** to resolve

– – Active space – –

Shock wave therapy

ESWT **second line** · **chronic heel pain > 6 months** failing other treatment · **FDA approved** · painful · effective at 6-month follow-up · alternative to surgery once nonop exhausted

Operative criteria

Procedure	Indication
Plantar fasciotomy (open or arthroscopic / endoscopic)	**Persistent pain > 9 months** of failed conservative care (Orthobullets, AAOS) · **6–12 months** (OB) · after **6 months of second-tier** modalities (OB)
Release + distal tarsal tunnel decompression	**Concomitant tibial nerve compression** in tarsal tunnel → **open** procedure · **70–90% success**
Gastrocnemius recession	**No clear indications** established
Baxter's nerve release	Release deep fascia of **abductor hallucis** → may improve outcomes

Release extent **medial ⅓ to ⅔** of fascia only (OB: **medial third**) → protects **medial longitudinal arch**

Avoid complete release arch destabilisation · **lateral column overload** · **dorsolateral foot pain**

Fasciotomy outcomes complications common · protracted recovery

6 Complications

Complication	Fact
Lateral plantar nerve injury	Surgical
Complete release	**Medial longitudinal arch collapse** · ↑ stress **dorsolateral midfoot**
Chronic pain	–
Heel pad (fat pad) atrophy	**Steroid injection** → shoe padding
Plantar fascia rupture	**Athletes**, **minimalist runners**, **steroid injection**

Plantar fascia rupture – steroid injection & immobilisation

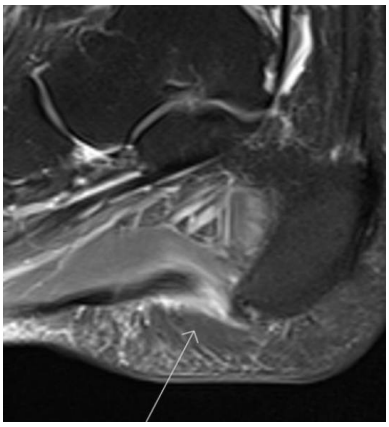
Steroid mechanism **inhibit heel pad fibroblast proliferation → fat pad atrophy**
✗ Wrong options: decrease inflammation (PF is degenerative) · no rupture association · good medium / long-term relief · US guidance prolongs relief (not needed)

Steroid evidence ↓ **short-term** pain only · **no high-quality evidence** · risks: fat pad atrophy, **PF rupture**, local reaction · used rarely / second line

Rupture presentation acute **pop** in runner after steroid shot → can't bear weight · **loss of medial arch → pes planus**
✗ Wrong options: os trigonum (posterior pain) · pes cavus + peroneal subluxation · PTTI (no heel pain) · lateral talar process (snowboarders)

– – Active space – –

Rupture treatment **cast immobilisation 3–6 wk** (OB) · **CAM boot** sufficient (OB) · stretching **after** immobilisation
✗ Wrong options: urgent / delayed primary repair (no evidence) · stretching first · steroid



Plantar fascia rupture on MRI

X vs Y

	Plantar fasciitis	Baxter's nerve entrapment	Calcaneal stress fracture
Pain pattern	**Start-up**, eases with walking	**Persists after activity**	Worse with WB
Tender	Medial calcaneal tubercle	**Abductor hallucis origin**	**Heel squeeze**
Test	Clinical	**EMG**	**Bone scan** / MRI

Orthosis	Use
Dorsiflexion night splint	**Chronic plantar fasciitis** (only proven)
Morton's extension	1st MTP pathology
Recessed first ray	Cavus foot
Gel heel cushion / prefabricated insert	Acute PF with stretching; not proven in chronic

How it's asked

"Start-up heel pain, BMI 35, gastrocnemius contracture, normal X-ray?"	→ **plantar fasciitis**
"Structure causing morning plantar heel pain?"	→ **connective tissue originating on medial calcaneal tubercle**
"Runner, anteromedial heel pain – best initial?"	→ **Achilles + plantar fascia stretching with prefabricated insert**
"Plantar heel pain with calcaneal spur – initial?"	→ **Achilles stretching exercises** ✗ Wrong options: spur resection
"Highest satisfaction at 8 weeks?"	→ **plantar fascia–specific stretching program**
"Chronic PF, failed stretching – which orthosis?"	→ **dorsiflexion night splint**
"Orthosis proven in chronic PF?"	→ **night splint**

"Improved 40% then plateaued at 6 weeks?"	→ **night splints**
"ESWT indicated when?"	→ **> 6 months failed nonop** (FDA)
"Fasciotomy — how much to release?"	→ **medial third** (avoid arch collapse)
"PF + tarsal tunnel compression — surgery?"	→ **open plantar fascia release + distal tarsal tunnel decompression**
"Steroid injection — mechanism / efficacy?"	→ **inhibits heel pad fibroblast proliferation**
"Pop after steroid shot in marathoner — exam / diagnosis?"	→ **pes planus, plantar fascia rupture**
"First line for plantar fascia rupture?"	→ **cast immobilisation for 3–6 weeks**

– – Active space – –

– – Active space – –

POLIOMYELITIS

– Foot & Ankle · 40 / 48 –

1 Basics

Post-polio syndrome & epidemiology

Pathology	viral destruction of **anterior horn cells** (spinal cord) + **brain stem motor nuclei** · **dorsal columns spared** · up to **95% of anterior horn cells** may be infected in acute stage (TO)
Virus	**enterovirus (poliovirus)** · **faecal–oral** spread
Household contacts	**seroconversion > 90% (90–100%)** · 4–7 secondary infections per case
Outcome of infection	**~75% asymptomatic** / mild flu-like · **paralysis in ~1%** · respiratory failure **5–15%** of paralytic cases
Age at risk	**younger children (< 5 yr)** ✗ Wrong options: older > 5 yr
Course	**transient infection → lifelong motor dysfunction** in most paralytic cases ✗ Wrong options: paralysis from active / chronic infection

Eradicated in the USA (vaccine) · late sequelae common in **developing countries** (TO)

3 Presentation

Acute polio presentation

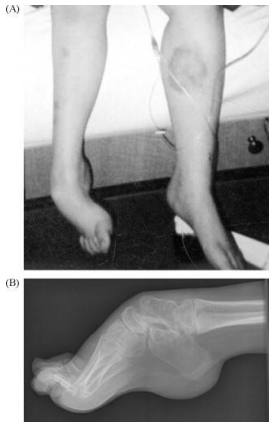
Acute polio (TO)	**flaccid paralysis, DTR absent, muscle atrophy** (TO) ✗ Wrong options: spastic paralysis · DTR ++
Probable case definition (TO)	**acute flaccid paralysis of ≥ 1 limb** + **↓ / absent tendon reflexes** · no other cause · **no sensory or cognitive loss** (TO)
Acute stage (TO)	lasts **7–10 days** · malaise → encephalomyelitis · **asymmetrical paralysis** (TO)
Younger children (TO)	listlessness, sore throat, low fever → recurrence with hyperaesthesia / paraesthesia, headache, vomiting, **nuchal rigidity**, back pain, **limited SLR** (TO)
Older children + adults (TO)	low fever, **flushing**, apprehension, **tender muscles** · **superficial reflexes lost first**, **DTRs disappear when muscle group paralysed** (TO)
Paralytic polio	**proximal, asymmetric weakness**, back pain + stiffness, **no sensory deficit** (after diarrhoeal / flu-like illness)
Hallmark (Orthobullets)	**muscle weakness with normal sensation**

Polio foot deformities

Causes (TO)	**muscle imbalance** · abnormal posture · gravity · walking on the deformed foot (TO)
Tibialis anterior (TO)	**one of the most frequently completely paralysed muscles → foot drop → fixed equinus** (TO)

Commonest deformity	**equinocavovarus** (OB) · TA weakness (↓ dorsiflexion), **cavus**, **claw toes**, **equinus**
Range (TO)	equinus · calcaneus · hindfoot varus / valgus · pes cavus / planus · combinations (TO)
Neglected deformity effects (TO)	abnormal gait · pressure areas + **callosities** · ↑ energy expenditure · arthritis (TO)

– – Active space – –



Polio equinocavovarus foot — clinical and lateral X-ray

4 Diagnosis & Classification

Diagnostic criteria – post-polio syndrome (TO)

Criterion (TO)	Detail
Prior paralytic polio	Motor neuron loss — history, residual weakness + atrophy, **EMG denervation**
Recovery + stable interval	Partial / complete recovery → **stable function usually ≥ 15 yr**
New weakness	**Gradual** (rarely abrupt), progressive, persistent weakness or **↓ endurance** ± fatigue, atrophy, muscle / joint pain · may follow trauma, surgery, inactivity
Duration	Symptoms **≥ 1 year**
Exclusion	Other neuromuscular, medical, orthopaedic causes

5 Management

Polio foot deformity — goal: painless, stable, plantigrade foot

- First → **ORTHOTICS, physio, stretching, bracing**
- Flexible, failed nonop → **SOFT TISSUE: release, lengthening, tendon transfer**
- Fixed / rigid → **BONY: osteotomy, wedge resection, arthrodesis**
  - Adult, flail (TA + TP grade 2), rigid equinus → **PANTALAR ARTHRODESIS**

– – Active space – –

Operative criteria

Factor	Rule (TO unless stated)
Flexible deformity	**Soft-tissue procedures** – release, lengthening, **tendon transfer**
Fixed / rigid deformity	**Osteotomy + arthrodesis**; wedge resection **extra-articular** where possible
Arthrodesis timing	**Reserved until skeletal maturity**
Tendon transfer	Needs **supple joints** + donor power **grade IV–V (normal / near normal)** · document **manual muscle testing** first
Younger child	Release / lengthen contracted tendons · **older child → bony surgery**
Orthoses fail to give satisfactory standard of living (Orthobullets)	**Contracture release, tendon transfer, arthrodesis**

Lightweight orthoses **first line** for polio foot deformity · keep patient functionally independent

Equinocavovarus correction

Rigid equinus, 18 yr, TA + TP grade 2 (TO) **pantalar arthrodesis** (TO)
✗ Wrong options: tendon transfer (rigid + weak donors) · ankle arthrodesis alone · triple (second best)

Pantalar vs triple (TO) pantalar = **triple + tibiotalar arthrodesis** → better stability · triple without muscle balance → **late secondary ankle deformity** (unbalanced forces on mobile joint proximal to fusion) · **pantalar > triple** (TO)

Flexible equinocavovarus, failed nonop **Achilles tendon lengthening + tibialis posterior tendon transfer (to dorsum) + plantar fascia release** · **FDL → EDL** for claw toes

Component	Procedure
Equinus	**Achilles lengthening**
↓ Dorsiflexion (TA weak)	**Tibialis posterior transfer to dorsum**
Cavus	**Plantar fascia release**
Claw toes	**FDL → EDL transfer**

✗ Flexible-foot traps **triple arthrodesis** / subtalar arthrodesis + medial cuneiform osteotomy (for **fixed** deformity) · **medializing calcaneal osteotomy + FDL transfer** (tibialis posterior insufficiency) · **lateralizing calcaneal osteotomy + tibialis anterior transfer** (TA usually weak)



Polio foot deformity

Post-polio syndrome – management

Nonop (first line)	**limited exercise with rest periods** ± lightweight orthosis · **exercise to sub-exhaustion levels** → tone muscle without breakdown · maintain, don't overuse
Operative	tendon transfers, contracture releases, arthrodesis to optimise function
Drugs (TO)	**no effective treatment** · lamotrigine, **IVIg** some success · analgesics / antidepressants (amitriptyline, duloxetine, codeine) – may worsen fatigue (TO)
Team (TO)	physio, OT, speech, respiratory, podiatry, psychology, dietetics, pain, orthotics (TO)

6 Complications

Post-polio syndrome (PPS)

Nature	**aging phenomenon**: more nerve cells inactive with time · **NOT viral reactivation** / chronic infection
Incidence	**up to 50%** of polio cases (Orthobullets, OB) · **20–85%** depending on criteria (TO) – read the options
Onset	**≥ 15 yr after initial infection** (TO) · **20–40 yr after** (OB) · **after middle age** (Orthobullets) · **not 5–10 yr after paralytic polio** (TO)
Severity (TO)	**more severe after acute / severe initial disease**; minimal initial symptoms → mild PPS (TO)
Risk factors (TO)	**female sex** + **respiratory symptoms** = most important · permanent disability, severe residual weakness, aging, muscle overuse / disuse, immune reaction (TO)
Motor features	**new weakness**, ↓ endurance, atrophy, **myalgia**, fasciculations, **fatigue**, cold intolerance, dysarthria, **dysphagia**, respiratory compromise · ↑ difficulty with ADLs
Sensory symptoms (TO)	**frank sensory deficits + paraesthesias occur in PPS** (sensory evoked potential changes ↮ cord atrophy on MRI) · contrast: acute polio = **normal sensation** (Orthobullets) (TO)
Cognitive (TO)	word-finding difficulty, poor concentration, memory impairment, mood change → social isolation (TO)

– – Active space – –

Counselling a recovered child — **may experience muscle weakness and fatigue as an adult**
✗ Wrong options: no risk · safe after skeletal maturity · certain paralysis + avoid sport · wheelchair by 25

X vs Y

	Acute polio	Post-polio syndrome
Cause	Active poliovirus infection	**Aging of depleted motor units** (✗ Wrong options: reactivation)
Timing	7–10 day acute stage	**≥ 15 yr** (TO) / **20–40 yr** (OB) later
Weakness	**Flaccid, asymmetric, proximal**	New, gradual, previously affected muscles
Reflexes	**Absent DTRs**	–
Sensation	**Normal**	**Sensory symptoms possible** (TO)

	Flexible deformity	Fixed deformity
Surgery	**Soft tissue** – TAL, PT transfer, plantar fascia release	**Bony** – osteotomy, **triple / pantalar arthrodesis**
Age	Younger	Older child / **after maturity**

How it's asked

"Post-polio syndrome – all true EXCEPT?"	→ **seen 5–10 years after the onset of paralytic polio** (TO)
"Recovered child from Liberia – counselling?"	→ **may experience muscle weakness and fatigue as an adult**
"Enterovirus flaccid paralysis – true?"	→ **seroconversion rate > 90% of household contacts**
"Rigid equinus, 18 yr, TA + TP grade 2 – best treatment?"	→ **pantalar arthrodesis** (TO)
"Flexible equinocavovarus in polio, failed nonop?"	→ **Achilles tendon lengthening, tibialis posterior tendon transfer, plantar fascia release**
"Acute poliomyelitis presentation?"	→ **flaccid paralysis, DTR absent, muscle atrophy** (TO)
"Polio involves which cells?"	→ **anterior horn cells + brain stem motor nuclei** ✗ Wrong options: dorsal column
"Exercise advice in PPS?"	→ **sub-exhaustion levels**

POSTERIOR TIBIAL TENDON INSUFFICIENCY (PTTI)

– Foot & Ankle · 41 / 48 –

1 Basics

Epidemiology

Commonest cause	**adult-acquired flatfoot deformity (AAFD)** · renamed **progressive collapsing foot deformity (PCFD)** (Myerson consensus) = hindfoot valgus + forefoot abduction + midfoot varus
Who	**women**, **6th decade** (50s–60s), obese, middle-aged
Risk factors	**obesity**, **hypertension**, **diabetes**, ↑ age, **corticosteroid use** (incl. repeated injections), **seronegative inflammatory** disorders
Associated conditions	**inflammatory arthropathy** · **tarsal coalition** (young, rigid pes planus ± recurrent ankle sprains) · acute onset after **pronation–external rotation** ankle fracture

Anatomy & biomechanics of the PTT

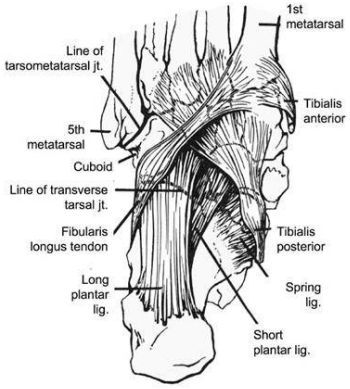
Muscle	origin **posterior fibula, tibia + interosseous membrane** · **tibial nerve (L4–5)**
Course	**posterior to medial malleolus** → divides into **3 limbs**

Limb	Insertion
Anterior	**Navicular tuberosity + 1st (medial) cuneiform**
Middle	**2nd + 3rd cuneiforms, cuboid, MT 2–4**
Posterior	**Sustentaculum tali** (anteriorly)

Blood supply	**posterior tibial artery branches** distally (Orthobullets) · branches of sural, peroneal + posterior tibial arteries (OB)
Watershed	**2–6 cm proximal to navicular insertion** (navicular → distal medial malleolus) = main site of **synovitis + rupture**
Excursion	**2 cm** (normal)
Axis	**posterior to tibiotalar** + **medial to subtalar** joint axis
Functions	**primary dynamic arch support** · **hindfoot invertor** · **adducts + supinates forefoot** in stance · **secondary ankle plantarflexor** · with FDL + FHL dynamically restrains arch
Primary antagonist	**peroneus brevis** (evertor) ✗ Wrong options: peroneus longus (antagonist of tibialis anterior) · tibialis anterior (antagonist of PL) · gastroc-soleus (antagonised by ankle dorsiflexors) · EHL (antagonised by FHL)
Locking mechanism (TO)	midstance: **TP initiates subtalar inversion → locks the transverse tarsal (TN + CC) joints → rigid lever arm for toe-off** · adducts + plantarflexes navicular on talar head → buttresses medial arch (TO)

– – Active space – –

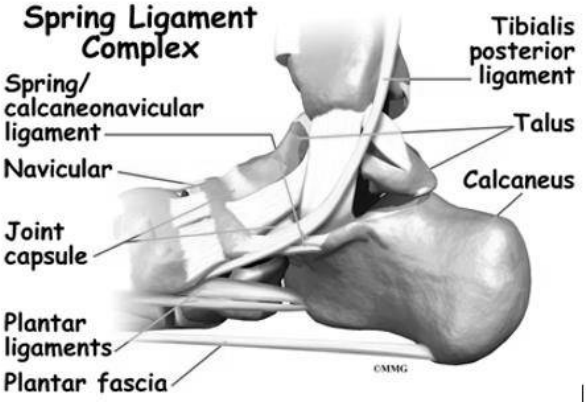
Dysfunction (TO)	**failure of inversion of the hindfoot that locks the transverse tarsal joints** → heel rise difficult, **heel remains in valgus** (TO) ✗ Wrong options: eversion · "unlocks"
Calf fatigue	unlocked transverse tarsal joint → no rigid lever → ↓ triceps surae mechanical advantage → **medial + distal calf pain, fatigue** → tendinopathy primarily affects the **transverse tarsal joint** ✗ Wrong options: 1st TMT (dorsiflexed, not the cause) · subtalar (altered, no extra triceps work) · Lisfranc (unaffected) · tibiotalar (stage IV only)



PTT plantar insertions with spring, long and short plantar ligaments and peroneus longus

Spring ligament & related anatomy

Spring ligament complex	**superomedial calcaneonavicular (CNL)** + **inferior calcaneonavicular** ligaments · **anterior sustentaculum tali → plantar-medial navicular** · supports **plantar-medial talar head** · **primary static stabiliser of the TN joint**
Torn / attenuated in AAFD	**superomedial calcaneonavicular** (most often; 18 / 22 PTT ruptures had spring injury) ✗ Wrong options: dorsal cuneonavicular · dorsal intercuneiform · dorsal calcaneocuboid · anterior tibiofibular
Deltoid link	distal (superficial) deltoid **blends with superomedial spring ligament** = **tibiocalcaneonavicular ligament (TCNL)** · also attenuated · deltoid-spring reconstruction for advanced medial peritalar instability
Static arch stabilisers	**spring ligament complex**, **plantar fascia**, **plantar ligaments** (long + short) · TO: long + short plantar, spring + bifurcate ligaments kept tight by TP tension → lost tension → plantar ligament failure, esp. **calcaneonavicular** (TO)
Failure	**plantar + medial migration of the talar head** → arch flattens → **loss of medial peritalar stability**
Medial labelled figure	**A = spring (superomedial CN) ligament** · B = joint capsule · C = plantar ligaments · D = plantar fascia · E = tibialis posterior tendon
Plantar labelled figure	**structure D = spring (plantar calcaneonavicular) ligament** · A = peroneus longus · B = long plantar ligament · C = short plantar ligament · E = tibialis anterior



Spring ligament complex, medial view

✗ Ligament–condition traps	Lisfranc injury = **Lisfranc ligament** (plantar-lateral medial cuneiform → plantar-medial 2nd MT base) · peroneal dislocation = **SPR** · pes cavus = muscle imbalance · 2nd toe crossover = **plantar plate + collateral ligament**

2 Mechanism

Aetiology	exact cause **unknown** · acute injury vs **long-standing tendon degeneration**
Sequence	**tenosynovitis → tendinosis / elongation → loss of dynamic arch stabilisation → attritional failure of static stabilisers** (spring ligament, plantar fascia, plantar ligaments) → fixed degenerative joint changes
Structure order	**PTT → spring + talocalcaneal ligaments → deltoid** (last = stage IV)
Deformity	**pes planus** · **hindfoot valgus** · **forefoot abduction** · **forefoot varus / supination** (compensatory → keeps foot plantigrade)
Equinus cycle	hindfoot valgus → **Achilles lateral to subtalar axis → acts as evertor** · valgus shortens gastrocnemius → **fixed equinus, Achilles / gastroc contracture** → worse arch collapse → worse equinus
Stage IV	**deltoid incompetence → valgus talar tilt in mortise** → tibiotalar arthritis

3 Presentation

Diagnosis & examination (too-many-toes, single-limb heel rise, equinus)

Early	**medial ankle / foot pain + weakness** · tenderness + swelling **just posterior to tip of medial malleolus** / along PTT · ↓ endurance
Progressive	**loss of arch** ("lost her arch", needs shoe inserts), foot "looks wider"

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Late — **lateral (sinus tarsi) pain from subfibular impingement** (calcaneus / talus against fibula tip)

Present in AAFD	Not a feature
Hindfoot valgus	Hindfoot varus
Forefoot abduction ("too many toes")	Forefoot adduction
Forefoot varus / supination (compensatory)	Forefoot valgus / pronation (cavovarus)
Pes planus	Cavus, plantarflexed 1st ray (cavovarus compensation)
Equinus / Achilles / gastrocnemius contracture	Claw toes, hallux varus, plantar fasciitis
–	**Hallux valgus** (no ↑ rate in PTTI)

Single-limb heel rise — **hallmark** · patient **unable to raise the heel off the ground and stand on the toes while supported on one leg** · **stages II–IV can't** · stage I can (TO: grossly normal single raise but **fails repeated / successive raises**)

Heel on double-leg heel rise — normal / flexible: heel **inverts**, arch **reconstitutes** (stage II) · rigid: heel **stays valgus** (stage III)

Too-many-toes sign — viewed from behind · **forefoot abduction** · **> 40% TN uncoverage** (stage IIB)



Too-many-toes sign: forefoot abduction seen from behind

PTT power — foot **plantarflexed + fully inverted** → can't hold against examiner's eversion force · pain with resisted inversion

Forefoot varus check — hold flexible heel **neutral** → fixed varus = **lateral border more plantarflexed than medial** · flexible = MT heads perpendicular to long axis of tibia + calcaneus

Flexible vs rigid — flexible = passively correctable to plantigrade (**stage II**) · rigid = not correctable (**III, IV**) → **decides joint-sparing vs fusion**

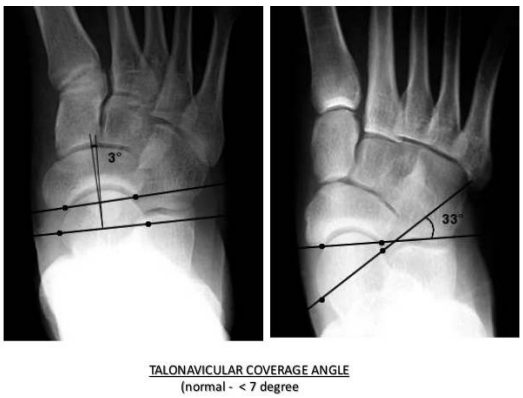
Equinus (must be evaluated + addressed at surgery) — **Silfverskiöld** with hindfoot / TN held **neutral**: **increased ankle dorsiflexion with knee flexion = gastrocnemius contracture** · no change = gastroc + soleus (Achilles) · TO: equinus = **< 10° DF with knee extended** (TO)

✗ Exam traps — anterior drawer in 20° PF = **ATFL** · **Coleman block** = cavovarus (correction = flexible, forefoot-driven cavovarus) · **piano-key** = Lisfranc · **Thompson** = Achilles rupture · inability to flex hallux = **FHL**

4 Diagnosis & Classification

Key radiology findings

Study	Finding
Views	**WB AP + lateral foot**, **ankle mortise**
AP foot	**↑ TN uncoverage** (TN coverage angle normal **< 7°**) · **↑ AP talo-1st MT angle (Simmon)** – stages II–IV
WB lateral	**↑ lateral talo-1st MT angle (Meary) > 4° = pes planus** (II–IV) · **↓ calcaneal pitch** (normal **17–32°**) · **↓ medial cuneiform–floor height** · **dorsal navicular subluxation on talus**
Lateral (TO)	**Midfoot break at talo-MT angle**, abnormal talocalcaneal angle, **↓ cuneiform height** (stage 2) (TO)
Hindfoot	**Subtalar arthritis** – stages **III–IV**
Mortise	**Talar tilt (valgus)** = **deltoid insufficiency** – **stage IV**
US	Stage 1 / 2a **tenosynovitis – target sign** · degenerating tendon **hyper- → hypo-echoic** · stage 2b **empty groove sign** = ruptured tendon gap (TO) · ↑ role
MRI	Tendinosis, tenosynovitis, tearing · **elongated, thinned tendon** (TO) · TN, subtalar, tibiotalar arthritis



Talonavicular coverage angle: normal under 7°, flatfoot 33°

Meary in stems 8°, 12°, 30° = pes planus · collapse sometimes reported as **"negative" Meary angle**

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Diagnostic criteria

Finding	Threshold / meaning
Pes planus	**Meary (lateral talo-1st MT) > 4°**
Calcaneal pitch	Normal **17–32°** → ↓ = loss of arch height
Stage IIB / LCL	**TN uncoverage > 40%** (Orthobullets, OB, Thordarson consensus) · **> 30%** (OB; Myerson IIa < 30% vs IIb > 30%) · "> 40% or > 30°" (OB) – read the options
Gastroc contracture	**DF ↑ with knee flexion** (Silfverskiöld +)
Equinus (TO)	**< 10° DF with knee extended**
Stage I vs II	**Can vs can't single-leg heel rise**
Stage II vs III	**Flexible vs rigid** hindfoot · rigid = stage III **irrespective of arthritis on X-ray**
Stage III vs IV	**Deltoid compromise / talar tilt / ankle arthritis**
1st TMT instability	**Plantar gapping medial cuneiform–1st MT base**

Differential

Midfoot-driven flatfoot	midfoot OA / **chronic Lisfranc**; apex at **naviculocuneiform** = **cuneiform subluxes dorsally, TN alignment maintained** (vs PTTI = navicular subluxes dorsally on talus) → **midfoot fusion + realignment**
Isolated spring ligament incompetence (no PTT pathology)	standard flatfoot reconstruction **+ adjunctive spring ligament reconstruction**
Tarsal coalition	young, rigid pes planus, recurrent sprains
Adolescent flexible flatfoot	ligamentous laxity → **conservative**

Staging (Johnson & Strom, Myerson modification)

Johnson and Strom classification for **tibialis posterior** (TO) · stage IV added later (Myerson)
✗ Wrong options: tibialis anterior · peroneus longus · FHL

Stage	Deformity	Exam	X-ray	Treatment
I	**Tenosynovitis, no deformity**	**(+) single heel raise**	Normal	**Cast / boot 3–4 mo** → tenosynovectomy
IIA	**Flexible** hindfoot valgus, **normal forefoot**	**(–) single-leg heel raise**, mild sinus tarsi pain	Arch collapse	**FDL + MDCO** ± TAL / gastroc
IIB	Flexible + **flexible forefoot abduction** (too many toes, **> 40% TN uncoverage**)	(–) heel raise, mild sinus tarsi pain	Arch collapse	**+ lateral column lengthening**
IIC	Flexible hindfoot + **rigid forefoot abduction / varus**	(–) heel raise, **moderate** sinus tarsi pain	Arch collapse, **gapping of 1st TMT**	**+ Cotton** or **medial column fusion**
III	**Rigid hindfoot valgus**, rigid forefoot abduction	(–) heel raise, **severe sinus tarsi pain**	**Subtalar arthritis**	**Triple (hindfoot) arthrodesis**
IV	III + **deltoid ligament compromise**	+ **ankle pain**	**Talar tilt in ankle mortise**	Correctable → **triple + deltoid recon** · rigid + arthritis → **TTC**

Johnson & Strom stages (TO)

Stage	TO description
1	Tenosynovitis, **no deformity** · tendon pain, normal function, **able to repeat tiptoes**
2	**Flexible deformity** · tendon rupture, ↓ strength, **unable to repeat tiptoes**
3	**Rigid flat foot**, valgus heel, **loss of subtalar motion**, forefoot varus, absent heel raise, **anterolateral impingement**, secondary OA
4	**Ankle joint involvement – tibiotalar degeneration**

Stage 1 (TO) medial pain, ↓ endurance · erythema, swelling, warmth, tenderness along TP · arch + alignment normal (TO)

Stage 2 (TO) inflammatory signs + **flexible hindfoot valgus** · flattened arch **returns on single-leg raise** · forefoot **abducted + supinated** · yet usually **can't do an unsupported single heel rise** (TO) – Orthobullets: (–) single-leg heel raise – read the options

Stage 3–4 (TO) **subtalar arthropathy**, deformity fixed · stage 4 = **tibiotalar degeneration** · X-rays diagnostic · treatment = **arthrodesis** (TO)

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Four-stage AAFD with surgical treatment (Myerson)

Stage	Deformity	Surgery
I	No AAFD deformity (may have pre-existing flatfoot)	**Tenosynovectomy**, possible tendon transfer and / or medial slide osteotomy
IIa	Mild / moderate flexible, **< 30% TN uncoverage**	Tendon transfer + medial slide osteotomy ± **Cotton**
IIb	Severe flexible, abduction through TN, **> 30% TN uncoverage**	Transfer + medial slide ± **LCL** or hindfoot fusion (subtalar or TN + CC) · Cotton / TMT fusion to lower 1st ray
III	**Fixed** (triple-joint complex)	**Hindfoot fusion, usually triple** – all three joints
IV	Foot deformity + **lateral talar tilt**	Correct foot ± **deltoid reconstruction** · severe arthritis → **ankle fusion or TAA** + foot correction
IVa / IVb	Flexible / fixed foot	Foot corrected as **IIb** / as **III**

Bluman classification (detailed Myerson)

Stage	Hallmark	Treatment
I A / B / C	Normal anatomy, PTT tenderness (A ± systemic disease · C slight HF valgus)	Immobilisation, NSAIDs, cryotherapy, orthoses, tenosynovectomy
II A1	Supple HF valgus, **flexible forefoot varus**	Orthoses · MDCO · **TAL / Strayer + FDL** if deformity corrects only with ankle PF
II A2	Supple HF valgus, **fixed forefoot varus**	MDCO + FDL + **Cotton**
II B	**Forefoot abduction**, TN uncovering	MDCO + FDL + **LCL**
II C	**Fixed forefoot varus + medial column instability**, **1st TMT plantar gapping**, 1st ray dorsiflexes with HF correction	MDCO + FDL + **Cotton or medial column fusion**
III A	Rigid HF valgus, sinus tarsi pain, subtalar space loss, angle of Gissane sclerosis	Custom bracing if not surgical · **triple**
III B	+ forefoot abduction	**Triple ± LCL**
IV A	**Supple tibiotalar valgus**	Correct HF + **deltoid reconstruction**
IV B	**Rigid tibiotalar valgus**	**TTC or pantalar fusion**

RAM classification (rearfoot · ankle · midfoot)

Stage	Rearfoot	Ankle	Midfoot
Ia	PTT tenosynovitis	Neutral	Neutral
Ib	PTT tendonitis, no deformity	Mild valgus **< 5°**	Mild flexible supination
IIa	Flexible planovalgus: **< 40% talar uncoverage, < 30° Meary, incongruency 20–45°**	Valgus + deltoid insufficiency, no arthritis	Supination, no instability
IIb	Flexible: **> 40%, > 30° Meary, incongruency > 45°**	Valgus + deltoid insufficiency + **tibiotalar arthritis**	Supination + **midfoot instability**, no arthritis
IIIa	Fixed / arthritic, < 40% · < 30°	Valgus from **lateral plafond bone loss**, deltoid normal	Arthritis **isolated to medial column** (NC or 1st TMT)
IIIb	Fixed, > 40% · > 30° – **not correctable through triple**	Plafond bone loss **+ deltoid insufficiency**	**Medial + middle column** arthritis

5 Management

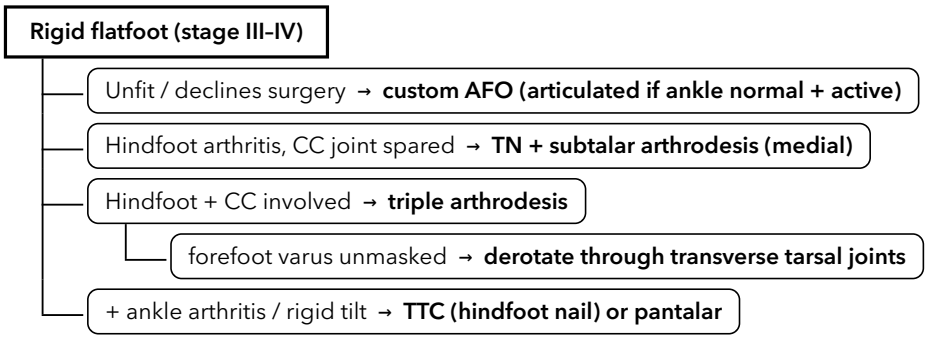
PTTI – nonoperative first at every stage

- I tenosynovitis, heel rise OK → **walking cast / boot 3–4 mo**
  - fails → **tenosynovectomy / PTT debridement**
- II flexible → **UCBL / AFO with medial post**
  - fails → **FDL transfer + MDCO (+ adjuncts)**
- III rigid hindfoot ± subtalar arthritis → **triple arthrodesis**
- IV deltoid failure, talar tilt → **decide by ankle**
  - ankle valgus correctable → **triple + deltoid reconstruction**
  - rigid valgus + tibiotalar arthritis → **TTC / pantalar**

Stage II reconstruction – base FDL → navicular + MDCO

- Forefoot abduction, TN uncoverage > 40% (IIB) → **+ lateral column lengthening**
- Residual forefoot varus, stable medial column → **+ Cotton osteotomy**
- 1st TMT plantar gapping / hypermobility / arthritis → **+ 1st TMT fusion (Lapidus)**
- DF ↑ with knee flexion (gastroc only) → **+ gastrocnemius recession**
- No change with knee flexion (gastroc + soleus) → **+ TAL**

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Operative criteria

Procedure	Indication	Contraindication
Tenosynovectomy / PTT debridement	**Stage I after failed immobilisation**	Stage II+ alone (no deformity correction)
FDL transfer → navicular	**Stage II** (FDL synergistic with PT)	**Subtalar rigidity (< 15° motion)** · **fixed forefoot varus > 10–12°**
Stage II reconstruction	Failed nonop, flexible	**Hypermobility** · **neuromuscular** · **severe subtalar arthritis** · obesity (relative) · **age > 60–70** (relative)
MDCO	**Flexible hindfoot valgus** (IIA; all stage II)	Rigid valgus
Lateral column lengthening (Evans)	**Forefoot abduction / TN uncoverage > 40%** (IIB)	Stage IIA · normal calcaneal pitch or mild valgus (Chi)
Cotton	**Residual forefoot varus / supination, stable medial column**	Medial column unstable / arthritic
1st TMT (Lapidus) / NC fusion	**Stage II with 1st TMT hypermobility, instability or arthritis** · NC sag	–
Isolated subtalar arthrodesis	**No fixed forefoot deformity**	**Fixed forefoot supination / varus** (→ lateral border overload) · hypermobility
Triple / hindfoot arthrodesis	**Stage III** · **stage II with severe subtalar arthritis**	Ankle valgus / arthritis not addressed
Triple + TAL + deltoid reconstruction	**Stage IV, passively correctable ankle valgus**	Ankle arthritis
TTC arthrodesis	**Stage IV: rigid hindfoot, valgus talus, tibiotalar + subtalar arthritis**	–

Nonoperative care – orthoses (UCBL, AFO, medial posting)

Stage I — **immobilisation in walking cast / boot 3–4 months = first line** → custom-molded in-shoe orthosis after

Stage I (TO) — immobilisation, ice, NSAID, rest · **CAM walker boot → less muscle atrophy than plaster** · orthotics eliminate forefoot pronation, rest TP · PT once inflammation settles · recalcitrant → **arthroscopic or open debridement + synovectomy** (TO)

AFO indications — **initial treatment for stage II, III + IV** · non-operative candidates (**sedentary / low demand, age > 60–70**, comorbid)

Orthosis	Key facts
Custom in-shoe orthosis	Stage I after immobilisation, stage II · **medial heel lift + longitudinal arch support** · **+ medial forefoot post if fixed forefoot varus / supination**
UCBL with medial posting	Plastic over cast in **maximal manual correction** · **flexible** deformity · first-line orthotic for stage II · not for rigid → uncomfortable, **skin breakdown**
AFO family (Arizona, molded, articulating)	**Most effective** · **acquired flexible flatfoot deformity** · medial post supports valgus collapse
Short articulated AFO	**Rigid flatfoot (stage III), active patient, normal stable ankle** → supports deformity, **keeps ankle motion**
Arizona brace	**Molded leather lace-up gauntlet** (UCBL + laced ankle support) → stabilises tibiotalar, hindfoot + arch · **significantly limits ankle motion** · TO: custom rigid AFO above malleoli accommodates rigid flatfoot (TO)
Non-articulated AFO	**Rigid planovalgus** (OB) · **stage IV** – deltoid incompetence + talar tilt (Elftman; stage III → articulated AFO) – read the options
Custom-molded AFO	Advanced rigid AAFD, **not an operative candidate** (fitted in situ)
Rocker-bottom shoe	**No role in PTTI** (OB – ankle / subtalar arthritis, calcaneal ulcers) · palliative for rigid flatfoot (TO) – read the options
PTB brace (TO)	Offloads foot in severe rigid flatfoot (TO)

Medial post — **promotes hindfoot inversion** · elevates medial heel → ↓ lateral hindfoot impingement pressure · fits any shoe
✗ Wrong options: medial post for eversion · lateral post for inversion · lateral post for forefoot varus / adduction

Rigid flatfoot nonop (TO) — palliative only, accommodates deformity · activity modification, NSAIDs, **steroid injection into subtalar / TN joint** · for elderly, low-demand, medically unfit or surgery-averse (TO)

Best nonop responders — **elderly patient with a sedentary lifestyle**
✗ Wrong options: young active · inflammatory systemic disorder · balance / ambulation problems: **Parkinson's**, severe ipsilateral **hip / knee arthritis**

Outcomes — AFO + PT: **83% improved at 4 mo, 89% satisfied** (Alvarez) · molded AFO (rigid) / UCBL (flexible): **67%** good–excellent (Chao) · **~90%** satisfaction nonop stage II (Deland) · eccentric exercise group improved most (Kulig)

✗ Orthotic traps — **lateral hindfoot posting + 1st ray cutout** = cavovarus (worsens PTTI) · full-length semi-rigid insole + 1st ray depression + lateral wedge = cavovarus · **total contact cast** = insensate / Charcot · **Morton's extension** / rigid forefoot plate limiting MTP DF = hallux rigidus · wide toe box = hallux valgus · rigid sole + rocker = midfoot arthritis · pressure-dissipating insoles + accommodative laced shoes = diabetic neuropathy · AFO = mild equinovarus · **CROW** = neuropathic severe deformity

Stage II, fixed forefoot supination, first visit — **medial heel lift, longitudinal arch support + medial forefoot posting**
✗ Wrong options: Lapidus + LCL + spring repair (1st TMT hypermobility, after failed orthosis) · short boot + lateral heel wedge (stage I) · triple · FDL + MDCO (after failed nonop)

Stage II, untreated, no abduction — **physical therapy and orthotics with medial hindfoot posting**
✗ Wrong options: FDL + calcaneal osteotomy + Cotton + gastroc (only after failed nonop) · triple (rigid)

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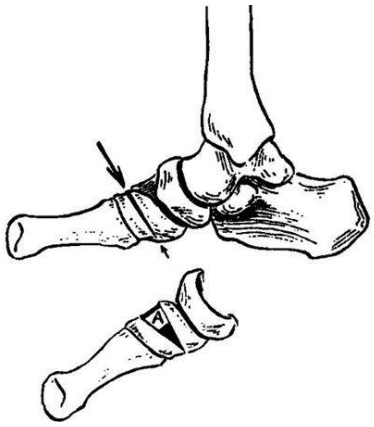
Stage II, improved with 6 wk PT	custom orthotic with **medial post + arch support** → hindfoot inversion
Rigid stage III, avid walker, normal ankle, declines surgery	**short articulated AFO** ✗ Wrong options: short leg cast NWB (early tendon "rest") · CAM boot + PTT strengthening + Achilles stretch (immobilises normal ankle) · UCBL + medial heel wedge (stage I–II flexible) · Arizona (limits ankle motion, affects gait)

Flexible deformity – FDL transfer, calcaneal osteotomy, lateral column lengthening, Cotton, gastrocnemius recession

Classic stage II	**medial displacement calcaneal osteotomy + posterior tibial augmentation with flexor digitorum longus tendon transfer** to navicular ✗ Wrong options: PTT debridement alone · triple · ankle fusion · Lapidus
Why MDCO	corrects hindfoot valgus (**static**) · **shifts Achilles pull medially** · **offloads / protects the transferred FDL** · moves heel WB axis toward tibial axis (TO) · **transfer alone → early failure** ✗ Wrong options: isolated FDL transfer
FDL (dynamic)	**synergistic with PT** → augments a deficient PT · found at **knot of Henry** · inserted into **navicular** near PT insertion
Knot of Henry	at ankle FDL anterior to FHL ("Tom, Dick and Harry") · **in the midfoot FHL crosses dorsal (deep) to FDL** → structure immediately deep to FDL at harvest = **flexor hallucis longus** ✗ Wrong options: Achilles (not in exposure) · FHB (anterolateral to FHL) · adductor hallucis (too distal) · abductor hallucis (never crosses FDL)
FDL vs FHL	**FDL is standard** (Orthobullets) · FHL harder to mobilise, no better results · FHL + MDCO: arch not reconstituted, good satisfaction (OB) · **TO answer = FHL transfer** (FDL not offered) – read the options
Stage II (TO)	bilateral planovalgus, heel valgus **passively correctable non-WB**, can't single heel rise → **medial displacement calcaneal osteotomy, tibialis posterior tendon debridement and FHL transfer** (TO) ✗ Wrong options: posterior tibial tenotomy + lengthening · PTT debridement + medial arch support orthotics · subtalar arthrodesis + MDCO + FHL (no arthritis)
Outcome	FDL + MDCO: **87% arch reconstitution, 84% normal shoes without orthotics** (Myerson)
IIA (flexible, no forefoot abduction)	**medial calcaneus displacement osteotomy + FDL transfer to navicular** alone ✗ Wrong options: lateral calcaneal displacement (worsens valgus) · peroneus longus → brevis (cavovarus) · **split PTT transfer** (spastic equinovarus)
IIB (too many toes)	**+ lateral column lengthening (Evans)**: opening wedge of **anterior calcaneus** or through **CC joint** · **LCL corrects forefoot abduction** · aims: adduct TN joint, plantarflex midfoot, derotate calcaneus out of valgus · may need MDCO too ✗ Wrong options: hindfoot valgus (MDCO) · hindfoot varus, forefoot adduction, cavus (not PTTI features)
LCL titration	overlengthening → correct with 1st TMT fusion or medial cuneiform osteotomy · **overcorrection (> 5–10 mm) → lateral column overload**

IIB package	**medial calcaneal displacement osteotomy, lateral column lengthening, FDL tendon transfer and tendoachilles lengthening** (or gastroc recession) ± **Cotton** ✗ Wrong options: tenosynovectomy + UCBL (stage I) · **Dwyer closing wedge + 1st MT closing wedge + plantar fascia release** (cavovarus) · triple (stage III) · **lateral** calcaneal displacement
Flexible planovalgus, failed AFO	**lateral column lengthening, medializing calcaneal osteotomy, FDL transfer to the navicular** ✗ Wrong options: isolated FDL · 1st ray dorsiflexion osteotomy + PL → PB (cavus) · lateralizing calcaneal osteotomy · triple
Large talar head uncoverage, painful heel rise that corrects valgus	**medializing calcaneal osteotomy + lateral column lengthening + flexor digitorum longus transfer to the medial navicular** ✗ Wrong options: PTT débridement (stage I only) · FDL alone · MCO + FDL without LCL · triple (supple, no arthritis)
Equinus not correcting with knee flexed, 50% uncoverage, no arthritis	**stage IIB: transfer of FDL to medial navicular + MCO + lateral column lengthening + tendo-Achilles lengthening (TAL)** ✗ Wrong options: stage I debridement · IIA (has abduction) · triple / TTC
Gastroc-only equinus	DF **−5° knee extended → +10° flexed → open gastrocnemius muscle lengthening** (recession / Strayer) · gastroc + soleus both tight → **TAL** (open Achilles lengthening for significant deformity, percutaneous) · recession = less weakness, better cosmesis, less powerful correction ✗ Wrong options: posterior ankle capsule release (ankle can DF) · PL → PB
No equinus	Silfverskiöld **+5° / +5°** → no lengthening needed
Forefoot varus / supination	compensatory; **unmasked once hindfoot corrected** intra-op → must be corrected or residual deformity
Cotton osteotomy	**plantarflexion (dorsal) opening wedge medial cuneiform osteotomy** = **plantarflexion osteotomy to correct residual forefoot varus** · requires **stable medial column** (navicular colinear with 1st MT) ✗ Wrong options: medial opening wedge for abduction · for forefoot valgus · for hindfoot valgus
Cotton corrects	**forefoot supination / varus** ✗ Wrong options: forefoot abduction (LCL) · forefoot pronation (cavovarus) · hindfoot equinus (gastroc / Achilles release) · hindfoot valgus (MCO static + FDL dynamic)
After FDL + MCO + TAL, forefoot still supinated	**plantarflexion opening wedge medial cuneiform osteotomy** ✗ Wrong options: dorsiflexion closing wedge (worsens) · in-situ 1st–3rd TMT fusion (Lisfranc) · lateral column closing-wedge shortening · subtalar fusion (subtalar arthritis)
Cotton vs 1st MT–cuneiform fusion	Cotton: **predictable union, preserves 1st-ray mobility, easily varied correction**

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Cotton osteotomy: plantarflexion opening wedge of the medial cuneiform

Unstable medial column (plantar sag at 1st TMT and / or NC)	**medial column fusion**: isolated 1st TMT, NC, or both
1st TMT instability + abduction, flexible hindfoot	**first tarsal-metatarsal arthrodesis, medializing calcaneal osteotomy, flexor digitorum longus tendon transfer and lateral column lengthening** ✗ Wrong options: Cotton + MCO + FDL (no 1st TMT instability) · triple + Achilles lengthening (rigid) · TTC + gastroc (stage IV) · 1st TMT + subtalar fusion + PTT debridement ± LCL (IIB with severe subtalar arthritis; subtalar fusion + LCL rarely combined → CC fusion instead; PTT debridement no proven benefit)
Modified Lapidus	indication = **plantar gapping between medial cuneiform and first metatarsal base** (1st TMT instability) · = 1st TMT fusion + distal soft tissue (modified McBride) ± medial cuneiform osteotomy · also 1st TMT arthritis, hypermobility, neuromuscular ✗ Wrong options: flexible hindfoot valgus · Meary 12° · TN uncoverage 50° → LCL · + Silfverskiöld → gastroc recession
Stage IIB + gastroc tightness, no TMT instability	**flexor digitorum longus transfer, medial calcaneal slide osteotomy, lateral column lengthening, Cotton osteotomy, gastrocnemius recession** ✗ Wrong options: PTT synovectomy + FDL (no bony correction) · lateral calcaneal slide (varus feet) · 1st TMT arthrodesis (no instability) · triple
Contraindication to FDL + osteotomies + LCL + Cotton	**rigid hindfoot valgus** → arthrodesis ✗ Wrong options: failed heel raise · forefoot varus · forefoot abduction · pes planus (all expected in stage II)

Procedure (Chi)	Indication	Contraindication
Lateral column lengthening	TN sag, forefoot abduction, **valgus calcaneus**	**Normal calcaneal pitch or mild valgus** · can overcorrect → forefoot varus
Medial column stabilisation	NC / cuneiform–1st MT **sag**, forefoot abduction, **neutral calcaneus**	Valgus calcaneus with flat pitch
Combined medial + lateral	NC / cuneiform–1st MT sag **+ valgus calcaneus**	–
Medial column + MCO	**Valgus calcaneus with normal pitch**	–

MIS vs open calcaneal osteotomy	**traditional open technique associated with higher wound complications + longer surgical time compared to MIS technique** · MIS: smaller incision, **fewer wound complications**, **↓ nerve injury**, **shorter time**, **same radiographic + patient outcomes** · radiation **↓ with MIS** (Orthobullets) · **no difference** (OB) — read the options
Exam ↔ procedure	**flexible forefoot abduction and hindfoot valgus deformities + gastrocnemius contracture** ↔ FDL + MCO + LCL + gastroc lengthening · flexible HF valgus + gastroc contracture ↔ MCO + gastroc lengthening · isolated PTT tenderness ↔ tenosynovectomy after failed immobilisation · rigid forefoot + hindfoot ↔ triple + gastroc · + tibiotalar arthritis ↔ tibiotalar fusion ± triple
FDL + MCO alone	female with pain, **inability to perform single leg heel raise + flexible hindfoot valgus deformity**, no abduction = **IIA** ✗ Wrong options: rigid → triple · + deltoid compromise → triple + deltoid / TTC · retained heel raise → stage I, maximise nonop then debridement · + forefoot abduction → IIB, needs LCL ± spring reconstruction

Rigid deformity – triple, hindfoot & TTC arthrodesis

Rigid adult flatfoot causes (TO)	**stage III / IV PTTD**, post-traumatic deformity, inflammatory arthropathy (TO)
Surgery indication (TO)	persistent pain + functional limit despite adequate nonop, fit + willing (TO)
Stage III gold standard	**triple arthrodesis (talocalcaneal, talonavicular + calcaneocuboid fusion) with appropriate structural realignments** (TO) (TO) ✗ Wrong options: LCL (Evans) + Cotton · isolated PTT reconstruction with FDL transfer + spring repair · isolated subtalar fusion without addressing the midtarsal complex
Triple technique (TO)	one or two incisions (medial + lateral) · realign valgus, abducted, dorsiflexed hindfoot → **neutral / slight valgus (5°), adducted, plantigrade** · **multiple cannulated screws** · bone graft (resected surfaces / iliac crest) · **NWB 6–8 wk** → boot until union **12–16 wk** (TO) · fusion **98%** at 2 yr (Pell)
Principles (TO)	**plantigrade, braceable foot** > perfect angles · **correct all components** (valgus + abduction + arch; subtalar alone leaves TN subluxation) · **equinus → Strayer or percutaneous TAL** concurrently (deforming force into pronation) · **forefoot varus → Cotton** (dorsal opening wedge medial cuneiform, plantarflexes 1st ray) or **lateral border overload** (TO)
Preferred stage III surgery	**arthrodesis of calcaneocuboid, talonavicular and subtalar joints** ✗ Wrong options: FDL + MDCO + TAL (flexible) · pantalar (stage IV) · FDL + LCL through anterior calcaneus · PTT debridement + tenodesis to FDL
Forefoot derotation in triple	after subtalar realigned, **forefoot varus unmasked → derotation of the forefoot through the transverse tarsal joints** (TN + CC) → plantigrade, no **lateral border foot pain** · alternatives: Cotton, 1st TMT fusion, NC fusion ✗ Wrong options: derotation through CC alone (can't fix abduction) · dorsiflexion osteotomy medial cuneiform · isolated TN fusion (subtalar degenerate too) · no further step
Double arthrodesis	**talonavicular + subtalar** via **single medial incision** when **CC not substantially arthritic** (OB; medial approach: no wound complications) · TO: double = **TN + CC**, spares subtalar when subtalar arthrosis minimal + deformity at TN (TO) — read the options

– – Active space – –

TN + subtalar case	rigid hindfoot + TN not passively correctable, pain on resisted inversion, failed bracing → **talonavicular and subtalar arthrodesis** ✗ Wrong options: CROW (neuropathic) · MCO + PTT debridement ± FDL ± LCL (joint-sparing fails in rigid: no arthritic pain relief, loss of correction)
Isolated subtalar fusion	**stage III** with mainly subtalar symptoms / arthritis (OB) · no fixed forefoot deformity (Orthobullets) · **rarely sufficient** for rigid flatfoot – TN usually degenerate → isolated subtalar coalition or post-traumatic subtalar arthritis (TO) · obese elderly with Chopart arthritis → subtalar alone leaves **TN + CC symptomatic** → triple (OB) – read the options
Rigid = stage III examples	70-year-old, **hindfoot not passively correctable** → **triple arthrodesis** (✗ Wrong options: talocalcaneal arthrodesis · FDL + calcaneal slide ± LCL through cuboid) · obese 75-year-old → **triple** (✗ Wrong options: double calcaneal osteotomy · subtalar alone) · obese 62-year-old, **plantar-medial callus = talar head**, TN dislocation, no hindfoot motion → **triple** (✗ Wrong options: exostectomy (destabilises) · LCL · MCO · TN arthrodesis)
Heel rise test for rigidity	**hindfoot valgus persisting with double-leg heel rise** = rigid → **triple** even with only 25% TN uncoverage ✗ Wrong options: heel inverts to varus on heel rise = flexible IIA / IIB → reconstruction · 14-year-old flexible flatfoot → conservative
Hindfoot arthrodesis	**stage III** (usually triple) · **stage II with severe subtalar arthritis** ✗ Wrong options: walking cast / boot 3–4 mo (stage I) · FDL + calcaneal osteotomy + TAL ± forefoot osteotomy (stage II) · 1st TMT fusion + calcaneal osteotomy + TAL ± LCL (stage II + 1st TMT hypermobility) · TTC (no tibiotalar arthritis / tilt)
Stage IV	**passively correctable ankle valgus** (cartilage preserved) → **triple arthrodesis + deltoid ligament reconstruction** (+ TAL) · **rigid ankle valgus + tibiotalar arthritis** → **tibiotalocalcaneal (TTC) arthrodesis** with **retrograde hindfoot nail** · end-stage ankle arthritis → **pantalar (triple + ankle) fusion** – eliminates all hindfoot + ankle motion (TO; OB: "stage IV → pantalar")
Stage IV, Meary 30°, ankle osteophytes, rigid subtalar + tibiotalar valgus	**tibiotalocalcaneal arthrodesis** ✗ Wrong options: AFO (functionally limiting deformity) · MCO + FDL ± deltoid repair (rigid) · triple (ignores ankle)
TTC caveat	salvage only; **tibiotalar joint provides most DF / PF**
Unfit for surgery	stage IV + **severe symptomatic aortic stenosis** → **custom-molded ankle-foot orthosis (AFO)** ✗ Wrong options: TTC · triple (stage III) · triple + deltoid (stage IV correctable) · in-shoe orthosis with medial forefoot post (stage I–II with forefoot varus)
Midfoot (NC) flatfoot	flexible hindfoot, full strength, collapse at NC joint on X-ray → **navicular-cuneiform arthrodesis** ✗ Wrong options: lateral calcaneal slide (worsens, cavovarus) · MCO / LCL (stage II PTTI) · TN arthrodesis (part of stage III)

6 Complications

Complication	Cause / fact
Lateral column overload → 5th MT stress fracture, lateral column stiffness	**LCL (anterior calcaneal opening wedge) for stage IIb** · overcorrection > 5–10 mm
CC arthritis	After LCL – **14%** radiographic, rarely symptomatic (Mosier-LaClair)
Lateral border foot pain	**Forefoot varus not corrected** after hindfoot realignment / isolated subtalar fusion with fixed forefoot varus
Early transfer failure	FDL transfer **without** hindfoot valgus correction
Lateral plantar nerve	**Retrograde hindfoot (TTC) nail start point** (mid plantar heel, centred on talus + tibia) → **numbness lateral plantar midfoot** → careful dissection to bone for guidewire + nail
FHL impingement	**Long lateral → medial calcaneal interlocking screw** by **sustentaculum** → ↓ hallux flexion
Loss of correction	Joint-sparing surgery in rigid deformity
Progression untreated	Flexible → rigid (peritalar arthritis) → deltoid failure → **tibiotalar valgus**, arthritis, near-total loss of ankle ROM

Lateral column overload & hindfoot-nail nerve injury

5th MT stress fracture after correction of
stage IIb PTTI (LCL)
✗ Wrong options: ligamentous Lisfranc (planovalgus malreduction, PTOA) · moderate hallux valgus (iatrogenic hallux varus, transfer metatarsalgia) · stage I (tenosynovectomy) · stage IIa (normal TN coverage, no LCL)

Numbness after retrograde hindfoot nail
lateral plantar midfoot (lateral plantar nerve, branching medially from tibial nerve)
✗ Wrong options: medial plantar midfoot (medial plantar n.) · lateral hindfoot (sural) · medial hindfoot (saphenous) · plantar hindfoot (medial calcaneal n.)

Long distal interlocking screw by sustentaculum
flexor hallucis longus
✗ Wrong options: lateral plantar nerve (nail start point, not screw) · tibialis anterior · saphenous vein (superficial) · tibial nerve (anterior to FHL, not by sustentaculum)

X vs Y

	MDCO	LCL (Evans)	Cotton	1st TMT fusion (Lapidus)
Corrects	**Hindfoot valgus**	**Forefoot abduction / TN uncoverage**	**Forefoot varus / supination**	Forefoot varus + **unstable / arthritic 1st TMT**
Typical stage	All stage II (IIA alone with FDL)	**IIB**	IIA2 · IIB · IIC (stable medial column)	IIC (unstable medial column)
Pitfall	Lateral slide = wrong (cavovarus)	**Lateral column overload, 5th MT stress fracture**	Dorsiflexion wedge = wrong	–

– – Active space – –

	Gastrocnemius recession	TAL
Silfverskiöld	**DF ↑ with knee flexion**	**No change** (gastroc + soleus)
Trade-off	Less weakness, better cosmesis, less power	More powerful

	PTTI flatfoot	Midfoot (NC / Lisfranc) flatfoot
Apex	TN – navicular dorsal on talus	NC / TMT – cuneiform dorsal, TN aligned
Treatment	FDL + MCO ± LCL → triple	**NC / midfoot fusion + realignment**

	Flexible planovalgus orthotic	Cavovarus orthotic
Post	**Medial hindfoot post** (± medial forefoot post)	**Lateral post + 1st ray cutout / depression**

	Articulated AFO	Arizona / non-articulated AFO
Ankle motion	**Preserved**	**Limited**
Use	Active patient, rigid stage III, normal ankle	Stage IV / talar tilt · rigid planovalgus

	Lateral plantar nerve	FHL
Hindfoot nail injury from	**Start point** in plantar heel	**Long lateral → medial interlocking screw**
Sign	Numb lateral plantar midfoot	↓ hallux flexion

How it's asked

"Heel rise difficult, heel stays in valgus – failure of?"	→ **inversion of the hindfoot that locks the transverse tarsal joints** (TO)
"Advanced PTTI – feature to evaluate + address at surgery?"	→ **equinus contracture**
"Obese 46-year-old, flatfoot X-rays – likely physical finding?"	→ **Achilles tendon contracture**
"Primary antagonist of tibialis posterior (MRI arrow)?"	→ **peroneus brevis**
"Too many toes, calf pain + fatigue – tendinopathy affects which joint?"	→ **transverse tarsal joint**
"Exam finding of disease causing attritional spring ligament failure?"	→ **unable to raise the heel off the ground and stand on the toes on one leg**
"Medial pain, needs arch inserts – exam finding?"	→ **increased ankle dorsiflexion with knee flexion**
"Can't single-limb heel rise + arch loss – other finding?"	→ **hindfoot valgus**
"Johnson and Strom classification is for which tendon?"	→ **tibialis posterior** (TO)
"Rigid valgus hindfoot, subtalar DJD, no talar tilt – stage?"	→ **III**
"Stage II (Meary 8°, too many toes) – procedures EXCEPT?"	→ **deltoid ligament reconstruction**

"Triple arthrodesis performed — stage?"	→ **stage III**
"Supple, 50% uncoverage, equinus unchanged with knee flexed — stage + treatment?"	→ **stage IIB: FDL → medial navicular + MCO + LCL + TAL**
"Isolated subtalar arthrodesis shown — stage?"	→ **stage III**
"Triple arthrodesis shown in figure — likely pre-op state?"	→ **radiograph with subtalar arthritis, inability to perform single-leg heel raise, severe sinus tarsi pain**
"Heel not reducible under tibia, > 40% uncoverage, no tilt — procedure (figure D)?"	→ **double (TN + subtalar) or triple arthrodesis**
"Superomedial calcaneonavicular ligament tear most associated with?"	→ **adult-acquired flatfoot deformity**
"Ligament attenuated in AAFD?"	→ **superomedial calcaneonavicular**
"Incompetent spring ligament on labelled medial figure?"	→ **A**
"Structure causing loss of medial peritalar stability (plantar figure)?"	→ **structure D — spring ligament**
"Stage II, fixed forefoot supination — next step?"	→ **medial heel lift, longitudinal arch support + medial forefoot posting**
"Custom AFO is most effective nonop for?"	→ **acquired flexible flatfoot deformity**
"Medial hindfoot posting treats?"	→ **flexible pes planovalgus**
"AFO / UCBL — highest satisfaction in?"	→ **elderly patient with a sedentary lifestyle**
"Sedentary 65-year-old, flexible, can't heel raise — initial?"	→ **orthotic management with the UCBL (figure C)**
"Rigid flatfoot, normal ankle, avid walker, declines surgery?"	→ **short articulated ankle-foot orthosis (AFO)**
"Custom orthotic after successful PT for stage II — feature?"	→ **medial post to promote hindfoot inversion**
"Flexible, no abduction, untreated — long-term next step?"	→ **physical therapy and orthotics with medial hindfoot posting**
"Talar tilt + subtalar arthritis, severe aortic stenosis?"	→ **custom-molded ankle-foot orthosis (AFO)**
"Correctable bilateral planovalgus, can't heel rise (FNB)?"	→ **MDCO + tibialis posterior tendon debridement + FHL transfer** (TO)
"Structure immediately deep (dorsal) to FDL in the midfoot?"	→ **flexor hallucis longus (FHL)**
"Passively correctable valgus, failed orthotics?"	→ **MDCO + posterior tibial augmentation with FDL transfer**
"Flexible valgus, abducted forefoot, equinus — surgery?"	→ **medial calcaneal displacement osteotomy + LCL + FDL transfer + tendoachilles lengthening**

– – Active space – –

"Cotton osteotomy is?"	→ plantarflexion osteotomy to correct residual forefoot varus
"Flexible planovalgus, failed AFO + PT – surgery?"	→ LCL + medializing calcaneal osteotomy + FDL transfer to the navicular
"After FDL + MCO + TAL, forefoot still supinated – next?"	→ plantarflexion opening wedge medial cuneiform osteotomy
"Triple for stage 3, forefoot varus after subtalar aligned – prevent lateral border pain?"	→ derotation of the forefoot through the transverse tarsal joints
"Flexible hindfoot, abduction, 1st TMT instability?"	→ first TMT arthrodesis + MCO + FDL transfer + LCL
"DF –5° knee extended, +10° flexed – add?"	→ open gastrocnemius muscle lengthening
"Large talar uncoverage, correctable hindfoot, no arthritis?"	→ MCO + LCL + FDL transfer to the medial navicular
"Evans procedure corrects?"	→ forefoot abduction
"Full subtalar motion, flexible forefoot supination, no abduction?"	→ medial calcaneus displacement osteotomy + FDL transfer to navicular
"Indication for modified Lapidus?"	→ plantar gapping between medial cuneiform and first metatarsal base
"Contraindication to FDL + slide + LCL + Cotton?"	→ rigid hindfoot valgus
"IIB with gastroc tightness, no TMT instability?"	→ FDL transfer, medial calcaneal slide osteotomy, LCL, Cotton osteotomy, gastrocnemius recession
"5th MT stress fracture from lateral column overload after surgery for?"	→ stage IIb PTTI
"FDL + MCO alone suits which patient?"	→ 57-year-old female, pain, inability to perform a single leg heel raise, flexible hindfoot valgus deformity
"MIS vs open calcaneal osteotomy – true?"	→ traditional open technique = higher wound complications + longer surgical time
"Opening wedge medial cuneiform osteotomy addresses?"	→ forefoot supination
"FDL + MCO + LCL + gastroc lengthening matches which exam?"	→ flexible forefoot abduction and hindfoot valgus deformities, gastrocnemius contracture
"Stage III AAFD, rigid eversion, subfibular impingement – reconstruction?"	→ triple arthrodesis with appropriate structural realignments (TO)
"Preferred surgery for stage III?"	→ arthrodesis of calcaneocuboid, talonavicular and subtalar joints
"70-year-old, uncorrectable hindfoot, orthotics fail?"	→ triple arthrodesis

"Obese, no hindfoot motion, plantar-medial callus over talar head?"	→ **triple arthrodesis**
"Obese 75-year-old, rigid acquired flatfoot?"	→ **triple arthrodesis**
"Rigid hindfoot + TN, resisted inversion painful, CC spared?"	→ **talonavicular and subtalar arthrodesis**
"Rigid subtalar + tibiotalar valgus, ankle osteophytes?"	→ **tibiotalocalcaneal arthrodesis**
"Stage IV with passively correctable ankle valgus?"	→ **triple arthrodesis + deltoid ligament reconstruction**
"Numbness after retrograde hindfoot nail — where?"	→ **lateral plantar midfoot**
"Long distal interlocking screw by sustentaculum — structure?"	→ **flexor hallucis longus**
"Flexible hindfoot, collapse at naviculocuneiform joint?"	→ **navicular-cuneiform arthrodesis**
"Most likely to benefit from triple arthrodesis?"	→ **57-year-old male, hindfoot valgus persisting with double-leg heel rise, 25% talonavicular uncoverage on AP radiographs**
"Rigid forefoot abduction + rigid hindfoot valgus, severe sinus tarsi pain?"	→ **hindfoot arthrodesis**

– – Active space – –

SESAMOID INJURIES OF THE HALLUX

– Foot & Ankle · 42 / 48 –

1 Basics

Spectrum	fracture (hyperextension + axial load) · dislocation · sprain (**turf toe**) · sesamoiditis (trauma / infection) · chondromalacia · OCD · FHB tendonitis

Epidemiology

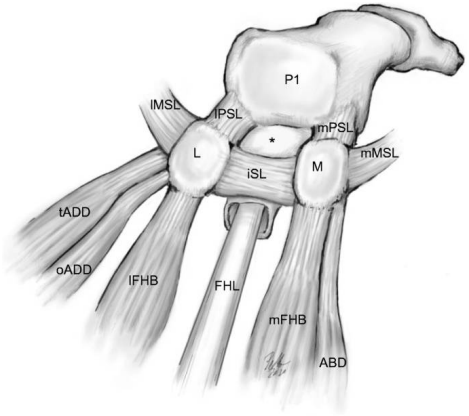
Tibial (medial) sesamoid	**more commonly injured**: **greater weight-bearing** + **larger** than lateral
Stress fracture	**commonest sesamoid problem** (OB) · **medial (tibial) sesamoid** · **mainly females** (OB)
Bipartite sesamoid	**10–25%** · **97% tibial** · **25% bilateral**

Anatomy – intersesamoid ligament

Held together by	**intersesamoid ligament + plantar plate** ✗ Wrong options: intermetatarsal ligament (bases of 2nd–5th MTs) · spring ligament (supports talar head / arch) · plantar fascia · FHL (inserts distal phalanx)
Plantar plate	inserts **base of proximal phalanx** of hallux
Embedded in	tendons of **FHB** (short flexor); FHB attaches to **both** sesamoids · both lie **within 1st MTP capsule**, plantar to MT head (TO)

Sesamoid	Muscle attached	Flexor-tendon complex (TO)
Tibial (medial)	**Abductor hallucis**	**Medial FHB head + abductor hallucis** → medial proximal phalanx
Fibular (lateral)	**Adductor hallucis**	**Lateral FHB head + adductor hallucis** → lateral proximal phalanx

Functions	**absorb weight-bearing** pressure · ↓ **friction** at MT head · **protect FHL** (glides **between** sesamoids) · **fulcrum for FHB →** ↑ **MTP flexion power**
Analogy	**patella**: ↑ mechanical advantage of FHB



Sesamoid complex: M/L sesamoids, iSL intersesamoid ligament, FHL between, FHB heads, ABD medial, ADD lateral

Metatarsal load distribution (TO)

Most weight during walking (TO)	**2nd MT** · **standing → 1st MT** (TO) · foot anatomy note (Orthobullets): **1st MT 50% in gait** – read the options
1st MTP joint (TO)	carries **~119% body weight** each step
Sesamoid complex (TO)	transmits **50% body weight** · **> 300% during push-off**

2 Mechanism

Commonest	**forced dorsiflexion (hyperextension) of 1st MTP** → plantar plate avulsion off **base of proximal phalanx** → **proximal migration of sesamoids**
Fracture	**hyperextension + axial loading**
Chronic overuse	dancers, players: **grade 3 1st MTP sprain → chronic sesamoiditis** (TO)
Complete plantar plate tear (TO)	may cause **sesamoid fracture** · **separation of bipartite sesamoid** · rarely **proximal migration**
Bilateral sesamoiditis	red flag → **Reiter's disease** (urethritis, conjunctivitis / iritis, IBD) · **psoriatic arthritis** · **seronegative RA**

3 Presentation

Symptoms	generalised big-toe pain · **worse in terminal stance** / hyperextension
Exam	possible **plantar-flexed MTP with cavus foot**

Sesamoiditis in dancers (TO)

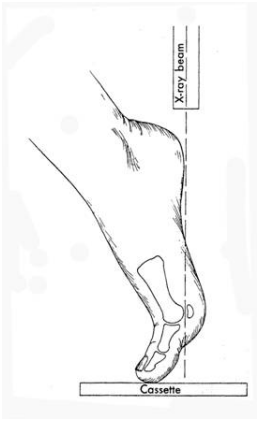
Dancer, walks on lateral margin, painful great-toe MTP plantar flexion (TO)	**sesamoiditis of first MTP joint** (TO) ✗ Wrong options: 1st MTP joint arthritis · stress fracture of 1st MT head · AVN of 1st MT head
Features (TO)	**transfer of weight-bearing to lateral surface** · pain + restricted MTPJ movement · **painful push-off** of great toe
DDx (TO)	sesamoid stress fracture · AVN · osteochondral fracture · **hallux rigidus** (1st MTP arthritis)
Treatment (TO)	orthotics + immobilisation → sesamoidectomy if operative

– – Active space – –

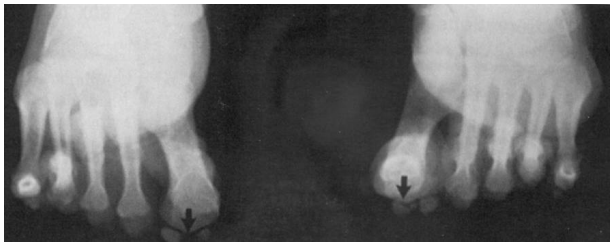
4 Diagnosis & Classification

Key radiology findings

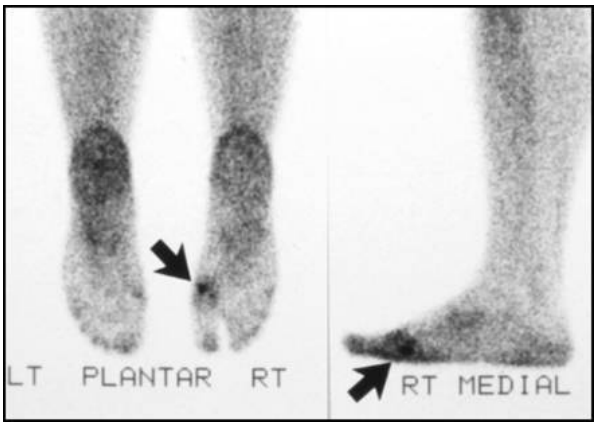
Study	Use / finding
X-ray	**AP + lateral foot** · **medial oblique (sesamoid view)** · **axial sesamoid view** · (TO) start with radiographs incl. **axial + lateromedial** sesamoid views
X-ray finding	**Proximal migration of sesamoids** → suspect **intrinsic-minus hallux** (plantar plate avulsion)
Early X-ray	**Often negative** in stress fracture (OB)
Bone scan	**Bipartite vs fracture** (↑ uptake = fracture) · **25–30% of asymptomatic** show ↑ uptake → **compare with uninjured side**
Bone scan + CT	**More reliable** for confirming **stress fracture** (OB)
MRI	**Confirms** diagnosis · indicated if radiographs **inconclusive** (TO)



Axial sesamoid view technique



Axial view showing fractured sesamoid (arrows)



Bone scan: ↑ uptake in right tibial sesamoid

5 Management

Sesamoid injury

- First line → **NSAIDs, ↓ WB, activity change, orthoses**
  (acute fracture: SLC with toe extension – controversial)
- Keratotic lesion, normal alignment → **shave plantar half of sesamoid**
- Plantar-flexed 1st ray → **dorsiflexion osteotomy**
- Failed nonop 3–12 mo → **partial / complete sesamoidectomy**
  - Nonunion, 2 large fragments, no osteonecrosis → **ORIF / autologous bone graft**

Operative criteria

Procedure	Indication
Partial / complete sesamoidectomy	**Failed nonop 3–12 months** (Orthobullets) · **≥ 3 months** (OB)
Autologous bone grafting	**Nonunion** or fracture
ORIF of sesamoid	**Two large fragments without osteonecrosis** (OB)
Dorsiflexion osteotomy	**Plantar-flexed first ray** with sesamoid injury
Sesamoid shaving	**Keratotic lesion** ↑ pressure on sesamoid · **contraindicated with plantar-flexed 1st MT**
Both sesamoids	**Never excise both**

Stress fracture/osteonecrosis – excision indications & outcomes

Initial treatment almost always **accommodative orthosis** (OB)

Medial sesamoid excision indicated **recalcitrant medial sesamoid stress fracture with fragmentation**
✗ Wrong options: with Lapidus · with 1st MTP arthrodesis (neither needs sesamoidectomy) · acute fracture (nonop first) · Morton's neuroma (neurectomy, bury in intrinsic muscle)

Excision technique **proximal fragment** in some · **entire sesamoid** in others

– – Active space – –

3-yr medial sesamoid pain, fragmentation + ↑ density, failed orthotics	**medial sesamoid excision** ✗ Wrong options: 1st MTP arthrodesis (doesn't unload sesamoid) · ORIF (only 2 large fragments without osteonecrosis) · shoe / heel-height change (already tried) · steroid injection (no long-term relief)
Triathlete, failed nonop tibial sesamoid fracture – fastest reliable RTS	**tibial sesamoidectomy** ✗ Wrong options: ORIF + autogenous calcaneal bone graft (valid but **slower** RTS) · plantar plate reconstruction (turf toe) · MTP arthroscopy + debridement (mild arthritis) · chevron + Akin (bunion)

Intractable plantar keratosis – sesamoid shaving

IPK under 1st MT head, normal standing alignment, adequate passive DF	**planing / excising superficial (plantar) half of medial sesamoid** → preserves function, ↓ pressure
✗ IPK traps	**complete medial sesamoid excision** (→ hallux valgus) · **FHL transfer** (no cock-up toe) · **dorsiflexion osteotomy** (no deformity) · **gastrocnemius recession** (passive DF adequate → more instability) · Keller bunionectomy

Turf toe (TO)

Definition (TO)	**hyperextension injury to plantar plate + sesamoid complex** of hallux MTP · mechanism **hyperextension + axial loading**
Plantar plate (TO)	fibrocartilaginous: **thickened plantar capsule + medial and lateral flexor-tendon complexes** · 1st MT head → proximal phalanx · stabilisers allow **~50° DF + 30° PF**
Plantar plate insufficiency (TO)	**primary pathology of MTP instability** → main focus of surgery
Clinical (TO)	pain, swelling, tenderness at MTP · **push-off painfully weak** · **negative plantar (palmar) grip** · **positive vertical Lachman / drawer test**
Treatment (TO)	**mostly nonop: ice, rest, taping, stiff-sole shoe / rocker-bottom sole** · **plantar plate repair only in a few** → "treatment is plantar plate repair" is the **false** statement (TO)
Surgery indications (TO)	failed nonop · **retraction or fracture of sesamoids with diastasis** · **loose fragments** in joint · **hallux deformity**

Approaches – sesamoidectomy

Sesamoid	Approach	Nerve at risk
Tibial (medial)	**Medial-plantar**, at junction of **glabrous skin**	**Proper branch of medial plantar nerve** = **plantar-medial cutaneous nerve of hallux** (directly under incision)
Fibular (lateral)	**Plantar**	**Proper branch to lateral hallux** + **1st common digital branch to 1st web space**

Medial sesamoidectomy – most vulnerable	**plantar-medial cutaneous nerve** → painful neuroma, shoe-wear problems ✗ Wrong options: abductor hallucis tendon (**dorsal** to incision) · intermetatarsal ligament · dorsomedial cutaneous nerve · crista

6 Complications

Complications of sesamoidectomy – cock-up toe, hallux malalignment & nerve injury

Excision	Complication
Both sesamoids	**Cock-up toe deformity** (commonest)
Tibial (medial)	**Hallux valgus**
Fibular (lateral)	**Hallux varus**
Either	Pain, **nerve injury (neuroma)**, infection, deformity

Cock-up toe
↓ **FHB moment arm** / FHB weakening → **MTP extension** + relative flexor overpull → **IP flexion** · prevent by **meticulous FHB repair**; **avoid excising both**

Hallux valgus after tibial sesamoidectomy
commonest reported deformity · from **inadequate medial capsular repair / reefing** (OB)

HV 1 yr after "removing a painful bone"
medial (tibial) sesamoidectomy
✗ Wrong options: accessory navicular excision · dorsal cheilectomy · lateral sesamoidectomy → varus · os vesalianum excision



Hallux valgus after medial sesamoidectomy

X vs Y

	Tibial (medial) sesamoid	Fibular (lateral) sesamoid
Injury frequency	**More common** (larger, ↑ WB)	Less
Bipartite	**97%**	Rare
Muscle	Abductor hallucis	Adductor hallucis
Approach	Medial-plantar	Plantar
Excision →	**Hallux valgus**	**Hallux varus**

	Bipartite sesamoid	Sesamoid fracture
Bone scan	No ↑ uptake vs other side	**↑ uptake**
Laterality	25% bilateral	Usually unilateral

– – Active space – –

	Sesamoiditis (TO)	Turf toe (TO)
Cause	**Chronic overuse** (dancers)	**Acute hyperextension + axial load**
Sign	Walks on **lateral margin**, painful push-off	**+ drawer / vertical Lachman**, negative plantar grip
Treatment	Orthotics, immobilise → sesamoidectomy	**Nonop** (taping, stiff sole) · repair in few

How it's asked

"Dancer bears weight on lateral margin, painful hallux MTP plantar flexion?"	→ **sesamoiditis of first MTP joint** (TO)
"Turf toe – all true EXCEPT?"	→ **treatment is plantar plate repair** (TO)
"Turf toe – exam?"	→ **positive drawer test, negative plantar grip** (TO)
"Metatarsal bearing most weight during walking?"	→ **2nd** (TO)
"Hallucal sesamoids held together by?"	→ **intersesamoid ligament**
"Medial sesamoid excision indicated for?"	→ **recalcitrant stress fracture with fragmentation**
"Triathlete, failed nonop tibial sesamoid fracture – fastest RTS?"	→ **tibial sesamoidectomy**
"Fragmented medial sesamoid, failed orthotics?"	→ **medial sesamoid excision**
"Commonest complication of excising both sesamoids?"	→ **toe cock-up deformity**
"Structure most at risk in medial sesamoidectomy?"	→ **plantar-medial cutaneous nerve of hallux**
"HV after removal of a painful bone – which procedure?"	→ **medial (tibial) sesamoidectomy**
"Lateral sesamoidectomy causes?"	→ **hallux varus**
"IPK under 1st MT head, normal alignment, failed nonop?"	→ **plane superficial half of medial sesamoid**
"Bipartite vs fractured sesamoid?"	→ **bone scan** (compare with uninjured side)
"Bilateral sesamoiditis – think of?"	→ **Reiter's**, psoriatic arthritis, seronegative RA
"Proximal migration of sesamoids on X-ray?"	→ **plantar plate avulsion / intrinsic-minus hallux**

SYNOVITIS OF 2ND MTP

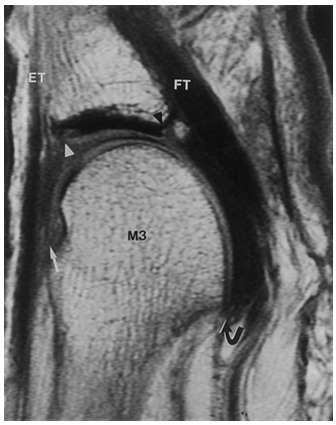
– Foot & Ankle · 43 / 48 –

1 Basics

Definition	**commonest monoarticular synovitis of the MTP joints**
Risk factors	**elongated 2nd MT relative to 1st (Morton foot)** · **hallux valgus**

Plantar plate anatomy

Role	weight-bearing pushes proximal phalanx **dorsally** · **plantar plate + intrinsics (interossei, lumbricals)** resist → hold phalanx neutral at MTP
Stabilisers of 2nd MTP	**plantar plate + collateral ligaments**



Sagittal MRI of lesser MTP plantar plate (FT flexor tendon, ET extensor tendon)

2 Mechanism

Cascade	**synovitis → capsuloligamentous stretch → instability → deformity**
Plantar plate attenuation	MTP **extension** · **sagittal-plane deformity** · → **crossover toe**
MTP instability	→ **dorsal dislocation of MTP** · predisposes to **hammer toe**

Plantar plate deficiency – crossover toe deformity

Plantar plate deficiency – commonest association	**crossover toe deformity** ✗ Wrong options: Freiberg's infraction (no plantar plate pathology) · Lisfranc injury (acute trauma) · claw toe (weak intrinsics vs stronger EDL on extensor sling) · curly toe (FDL + FDB contracture, unknown cause)
High-grade 2nd MTP plantar plate rupture – greatest risk	**second toe crossover** ✗ Wrong options: hallux varus · hallux valgus (may contribute to or follow crossover but not from rupture) · Freiberg's (microtrauma → blood supply loss → 2nd MT head collapse) · 2nd mallet toe (DIP flexion + lateral deviation, FDL contracture)
Crossover toe	2nd toe crosses **over or under the hallux** · requires **plantar plate + collateral ligament** failure

– – Active space – –

Order of failure	**LCL fails first → medial deviation**, plantar plate **second → dorsal displacement** of proximal phalanx (OB) · (Orthobullets, MTP dislocations): **plantar plate first (stage 1), LCL second (stage 2)** – read the options



Crossover second toe

3 Presentation

Symptoms	pain · **warm** joint · **fullness** of joint · "walking on marbles" (OB)
Tenderness	**plantar over plantar plate** or **over dorsal capsule** · may be in **2nd web space** (inflammation or pressure on interdigital nerve)
Motion	pain with ROM, **↓ with traction** · **↓ plantar flexion** · global MTP swelling
Deformity	**passively correctable** in predislocation stages · progresses to **crossover** (varus or valgus) when collateral + plantar plate disrupted
Dorsal drawer test	demonstrates instability · **dorsal subluxation possible = plantar plate attenuated**

Diagnosis – pain worse barefoot

Synovitis finding	**more pain when barefoot** (✗ Wrong options: less pain barefoot (neuroma)) · **toe ROM painful** (✗ Wrong options: painless ROM (neuroma))
✗ Wrong options (these are neuroma signs)	**pain + click with medial–lateral forefoot compression** (Mulder) · **pain with 2nd–3rd web-space compression**
Site	Morton's neuroma usually **3rd interspace** · pain in **2nd interspace → think 2nd MTP synovitis / metatarsalgia**
Why it matters	**steroid injection** helps neuroma but **weakens capsuloligamentous structures** → instability → **dorsal subluxation, crossover toe**

Key radiology findings

Study	Finding
Weight-bearing AP + lateral foot	**Widening / medial–lateral joint-space imbalance** of 2nd MTP
X-ray	**Dorsal subluxation** – looks like **joint-space narrowing** or **proximal phalanx overlapping MT head**
X-ray	Varus or valgus toe deformity
MRI	If **diagnosis unclear** · **quantifies plantar plate / ligament disruption**



Dorsal dislocation of the 2nd MTP joint – phalanx overlaps MT head

Causes of metatarsalgia (TO)

Metatarsalgia (TO) pain in forefoot **under one or more MT heads** · groups: **primary, secondary, iatrogenic** (after forefoot surgery)

Primary (TO)	Secondary (TO)
Plantar soft-tissue lesions	**Iatrogenic** – 1st ray shortening, MT head resection
Abnormal metatarsal parabola	**Neurogenic** – referred pain, tarsal tunnel syndrome
Morton's foot	OA · RA
Neuroma	**Atrophied plantar fat pad** · stress fracture
Trauma · brachymetatarsia	Capsulitis / bursitis / tendinitis · **Freiberg's infraction**
Bone tumours	Pes cavus / planus · inappropriate shoegear

Rocker (TO)	Phase	Metatarsalgia from
1st	Heel contact → foot flat	Congenital deformity, **tight heel cord**, severe cavus
2nd	Tibia over talus; triceps surae activated at end	**Limited ankle DF**, **plantarflexed metatarsals**
3rd	Foot turns over dorsiflexed MTP joints	**MTP joint deformities**

Work-up (TO) exam of plantar foot + **weight-bearing radiographs**

– – Active space – –

5 Management

- **2nd MTP synovitis**
  - First line → **shoe / activity change, NSAIDs, taping 10–12 wk**
  - No deformity, failed nonop → **synovectomy**
  - Fixed deformity
    - Long 2nd MT → **Weil distal oblique shortening osteotomy**
    - No long 2nd MT, sagittal → Girdlestone-Taylor FDL → **EDL / capsular release + EDL lengthening**

Operative criteria

Situation	Treatment
First line	Activity / shoe change, NSAIDs, external MTP support (**crossover taping, Budin-type splint**) for **10–12 weeks**
No deformity + failed nonop	**Synovectomy**
Fixed deformity + long 2nd MT	**Weil (distal oblique shortening MT osteotomy)**
Fixed deformity, NO long 2nd MT, sagittal deformity	**FDL → EDL transfer (Girdlestone-Taylor)** or **MTP capsular release + extensor lengthening**
No	**Steroid injection** – contraindicated (weakens capsule)

Weil osteotomy technique

Weil used for (TO) — **metatarsalgia** (TO)
✗ Wrong options: Perthes · SCFE · genu valgum

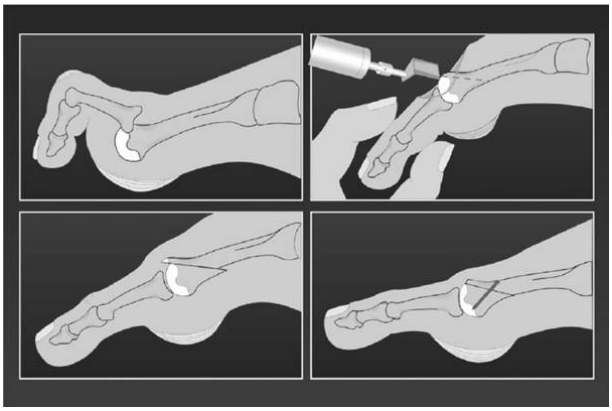
Nonop metatarsalgia first (TO) — physio, orthotics, shoe modification, injections

Definition (TO) — **distal oblique osteotomy of MT head** · corrects **transverse + sagittal** MTP pathology · **decompresses MTP joint → relaxes soft tissues**

Orthobullets — preserves joint · **rebalances MT cascade** · relaxes plantar plate

Indications (TO) — **long or plantarflexed metatarsal** · **transverse-plane MTP deformity** · **MTP subluxation / dislocation** · **recalcitrant generalised metatarsalgia**

Technique (TO) — start at **dorsal articular surface** of MT head · cut **dorsal-distal → proximal-plantar**, **30° to shaft**, **parallel to weight-bearing surface** · **screw** fixation



Weil osteotomy: dorsal-distal to proximal-plantar cut, head slides proximally, screw

Long 2nd MT + fixed dorsiflexion deformity, "walking on marbles" after 4 mo orthotics	Weil · avoid commonest complication (**floating toe**) by **osteotomy parallel to plantar surface of foot** (✓)
✗ Weil traps	protect **collateral ligaments** (prevents **AVN** — much rarer than floating toe) · **inject plantar to intermetatarsal ligament** (steroid ↑ instability) · **shorten ≥ 4–7 mm** (over-shortening → plantar displacement → floating toe; **shorten ≤ 3 mm**) · **strict immobilisation 4 wk** (↑ stiffness; use **protected WB in surgical shoe + aggressive ROM**)
Other floating-toe prevention	remove a **wafer of bone** (shorten without plantar migration) · combine soft-tissue procedures

6 Complications

Complication	Fact
Floating toe	**Commonest after Weil** (OB) — **plantar shift of MT head centre of rotation** → intrinsics pass **dorsal** to axis → act as **MTP dorsiflexors**
Weil (TO)	**Dorsiflexed toe** · pain on MTP DF · **↓ MTP ROM** · **PIP contracture** (TO)
AVN	Collateral ligament injury · uncommon
Stiffness	Prolonged immobilisation
Vascular compromise	Correcting **chronic dislocation** → contracted vessels stretched → **may need to reverse procedure** to save digit

X vs Y

	2nd MTP synovitis	Morton's neuroma
Interspace	**2nd**	**3rd** (usually)
Barefoot	**Worse**	Less pain
Toe ROM	**Painful** (↓ with traction)	Painless
Compression	–	**Click** (medial–lateral), web-space pain
Drawer	**+** if unstable	–
Steroid	**worsens instability**	Appropriate

Deformity	Cause
Crossover toe	**Plantar plate + collateral ligament** failure
Claw toe	Weak intrinsics, EDL overpull on extensor sling
Curly toe	FDL + FDB contracture
Mallet toe	FDL contracture — DIP flexion
Freiberg's	2nd MT head collapse (blood supply)

– – Active space – –

How it's asked

"Finding most likely in 2nd MTP synovitis?"	→ **more pain when barefoot**
"Click on medial–lateral forefoot compression?"	→ **Morton's neuroma** ✗ Wrong options: synovitis
"Plantar plate deficiency most associated with?"	→ **crossover toe deformity**
"High-grade 2nd MTP plantar plate rupture — progressive deformity?"	→ **second toe crossover**
"Weil's osteotomy is used for?"	→ **metatarsalgia** (TO)
"Long 2nd MT, fixed deformity — avoid commonest Weil complication?"	→ **osteotomy parallel to plantar surface of foot**
"Commonest complication of Weil?"	→ **floating toe**
"Maximum Weil shortening?"	→ **≤ 3 mm** ✗ Wrong options: 4–7 mm
"Fixed deformity without long 2nd MT?"	→ **FDL → EDL transfer / capsular release + extensor lengthening**
"No deformity, failed nonop?"	→ **synovectomy**
"Why avoid steroid injection?"	→ **weakens capsule → instability, crossover**
"Correcting chronic dislocation — feared complication?"	→ **vascular compromise → reverse procedure**

– – Active space – –

TARSAL NAVICULAR FRACTURES

– Foot & Ankle · 44 / 48 –

1 Basics

Rare midfoot fracture	**traumatic** (avulsion, tuberosity, body) or **stress** (repetitive microstress)
Articulations	**talus, calcaneus, cuboid, cuneiforms** · key to **inversion–eversion** motion
Medial column	navicular supports **medial longitudinal arch** · only **displaced fracture / navicular dislocation** → arch instability (OB)
PTT	inserts on navicular (tuberosity)

Stress fracture epidemiology

Who	**athletes running on hard surfaces** · **baseball players** (predilection; e.g. catcher jumping from crouch) · gymnasts, soccer, marathon runners (OB stems)
High-risk stress fracture	**risk of AVN** · commonest complications **delayed union + nonunion**
At presentation	**~83% incomplete** · complete ones usually **nondisplaced** (OB)

2 Mechanism

Type	Mechanism
Avulsion	**Plantarflexion** or **eversion / inversion** · TN or naviculocuneiform ligaments
Tuberosity	**Eversion + simultaneous PTT contraction** · may be **acute diastasis of an accessory navicular**
Body	**Axial loading**
Stress	**Chronic overuse** · hard surfaces
Mueller-Weiss	Spontaneous adult **navicular AVN**

3 Presentation

Symptoms	**vague midfoot pain** + swelling · **insidious** in stress fracture → high index of suspicion
Exam	midfoot (**dorsomedial / medial**) swelling + **tenderness** · **ankle + subtalar ROM usually full** · full DF / PF / inversion strength
Not	anterior tibial tendinitis (no evidence → no cortisone injection into tibialis anterior sheath)

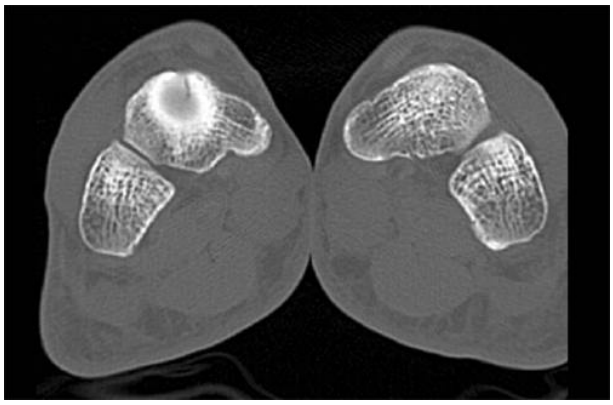
– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Finding / role
X-ray	**AP, lateral, oblique 45°** · **often missed / read as normal** · **45° oblique best for tuberosity fractures**
CT	**More sensitive than X-ray** · shows **nondisplaced fracture line** through navicular · may be **false-negative** without frank cortical disruption
MRI	**T2 high signal** (oedema / inflammation) · **best test to confirm** when X-ray negative + high suspicion or failed restricted WB (OB) · stress reaction = **oedema, no fracture line**
Mueller-Weiss X-ray	**Narrow navicular + lateral sclerosis** · **"double navicular sign"** (split navicular) (OB)

Best imaging **CT more sensitive than X-ray** (Orthobullets, OB) · **MRI best to confirm** (OB, CT false negatives) – read the options



Axial CT: navicular stress fracture

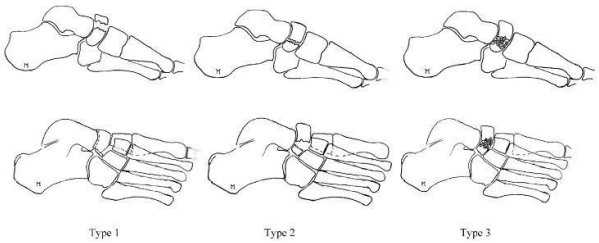


Sagittal T2 MRI: navicular stress fracture (arrow)

Sangeorzan classification (navicular body)

Type	Fracture	Deformity
I	**Transverse**, **dorsal fragment < 50%** of bone	**None**
II	**Oblique**, usually **dorsolateral → plantar-medial**	**Forefoot ADduction**
III	**Central or lateral comminution**	**Forefoot ABduction**

– – Active space – –



Sangeorzan types 1–3

5 Management

Navicular stress fracture

- Any type, first presentation → **NWB short-leg cast 6–8 wk**
- Stress reaction (oedema, no line) → **restricted WB short-leg cast**
- Failed cast / nonunion / recurrence → **ORIF + autologous bone graft**
  - High-level / pro athlete → **ORIF (OB, Orthobullets)**

Traumatic navicular fracture

- Avulsion, most tuberosity, undisplaced type I–II → **NWB cast**
  - Failed avulsion / symptomatic tuberosity nonunion → **fragment excision**
- Displaced / intra-articular type I–II → **ORIF (medial approach), lag screws**
- Type III or AVN → **ORIF + ex-fix vs primary fusion**

Operative criteria

Injury	Operate when
Avulsion	**> 25% of articular surface** → ORIF · failed nonop → **excision**
Tuberosity	**> 5 mm diastasis** or **large intra-articular fragment** → ORIF · **symptomatic nonunion → excision**
Body type I–II	**Displaced or intra-articular** → ORIF (**medial approach**)
Body type III	**ORIF + ex-fix vs primary fusion** · maintain **lateral column length**
Stress fracture	**Fracture line across 2 cortices**, or **1 cortex + displacement or cystic change** (OB) · **failed NWB cast** · **nonunion** · **high-level athletes**
Navicular AVN	**TN + naviculocuneiform fusion**

Stress fractures – non-weight-bearing cast first-line

First line (any type) — **short-leg cast + NWB 6–8 weeks** (± walking boot + crutches) → **high success**

45-yo runner — **short leg cast + non-weight bearing**
✗ Wrong options: hard-sole shoe (insufficient immobilisation) · partial WB boot / cast (risks progression)

– – Active space – –

21-yo baseball player, normal X-ray, MRI T2 signal	**NWB cast for planned 6–8 weeks** ✗ Wrong options: observation · steroid into anterior tibial sheath · partial WB boot · ORIF first line
Gymnast, CT nondisplaced fracture	**NWB + cast** · **NWB cast more effective than surgery** ✗ Wrong options: WBAT hard-soled shoe or Aircast boot · partial WB cast (inferior outcomes) · ORIF first
15-yo soccer, 4 mo pain, MRI oedema, no fracture line (stress reaction)	**restricted weight bearing in short-leg cast** (± boot) ✗ Wrong options: ORIF · WBAT boot (may **displace + injure TN joint**) · orthotic + bone stimulator · carbon-fibre orthotic
High-level athletes	**NWB cast works even in high-level athletes** · operative first line not supported (OB) · vs **ORIF for pro athletes** who can't tolerate long recovery, to prevent painful nonunion (OB, Orthobullets) — read the stem

Failed or complete stress fractures – ORIF with bone graft

Marathoner, recurrent chronic stress fracture with diastasis, failed immobilisation, intra-articular	**ORIF with autologous calcaneus bone graft** ✗ Wrong options: WBAT hard-soled shoe · repeat NWB cast · fragment excision + PTT advancement (not described) · CRPP (not described)
18-yo gymnast, 1 yr pain, failed 6 wk NWB cast	**ORIF with autologous bone grafting** ✗ Wrong options: ORIF alone · more NWB cast · PWB walking cast · no treatment

Acute fracture fixation

Acute displaced body fracture with diastasis (basketball)	**open reduction + lag screw fixation** → interfragmentary compression, restores **medial column length** + articular congruity
✗ Acute traps	**NWB cast 6–8 wk** (nondisplaced body fracture / stress reaction only) · **CRPP** (insufficient for displacement) · **ex-fix** (**highly comminuted** / substantial **medial column shortening**) · **primary TN fusion** (pre-existing **TN arthritis** or non-reconstructable)
Column length	Type III: maintain **lateral column length** (Orthobullets) · ex-fix for **medial column shortening** (OB) — read the options

Navicular arthrosis – talonavicular & naviculocuneiform fusion

56-yo, longstanding midfoot pain, no trauma, failed semirigid insoles	Mueller-Weiss → **fuse talonavicular + naviculocuneiform joints** ✗ Wrong options: TN only (misses NC joint) · any fusion including **subtalar – spared** in this disease
Pain source	abnormal motion at **TN + NC joints**



Mueller-Weiss: collapsed navicular on lateral X-ray

Mueller-Weiss syndrome (adult navicular osteonecrosis)

Definition (TO)	**spontaneous adult-onset tarsal navicular osteonecrosis** · aka **Brailsford disease** (TO) · related to **Köhler's** (Köhler = paediatric) (OB)
Who (TO)	**40–60 yr** · **women** · **frequently bilateral** (TO) · 4th–5th decade (OB) · middle-aged (Orthobullets)
Aetiology (TO)	uneven load → **compression of lateral half of navicular** · preconditions: **subtalar varus**, **1st ray brachymetatarsia**, **mild clubfoot** → lateral shift of force, **lateral incompetence of acetabulum pedis**
Clinical (TO)	chronic **mid + hindfoot pain** · navicular collapse → secondary OA → **pes planovarus** · tibia forced into **external rotation → knee OA**
Exam (TO)	dorsomedial midfoot swelling + tenderness · **heel varus** · prominent navicular tuberosity → **pseudo hindfoot valgus** · flattened medial arch
Paradoxical pes planovarus (TO)	**pseudo hindfoot valgus + pes planus** · "**pseudo hindfoot varus**" = the **false** option (TO)
Treatment	**semirigid insoles first**, fusion if failed (OB) · **mostly surgical** (TO) – read the options · fuse symptomatic joints + restore arches by correcting **Meary-Tomeno line** · **talonavicular-cuneiform arthrodesis with bone grafting** = best results (TO)

6 Complications

Nonunion risk

Complication	Fact
Nonunion + delayed union	**Two commonest** after both nonop + surgery
AVN	Why stress fracture is **high risk**
Malunion	**Very rare** (nondisplaced)
Infection	Rare unless open
Arch instability	Only **displaced fracture / dislocation**
Neuroma	Rare (stress fracture, nonunion)

Most common complication (stress fracture image)	**non-union** ✗ Wrong options: malunion · infection · longitudinal arch instability · neurovascular injury

– – Active space – –

Pro baseball catcher, 5 mo dorsomedial pain, failed nonop, MRI stress fracture – most at risk

nonunion
✗ Wrong options: arch collapse / acquired flatfoot (chronic **PTTI** sequela) · **cavovarus** · PTTI (chronic, not from stress fracture) · transfer metatarsalgia (after 1st ray shortening)

X vs Y

	Navicular stress fracture	Mueller-Weiss (adult AVN)
Age	Young athletes	**40–60**, women
Cause	Overuse, hard surfaces	Spontaneous; lateral compression
X-ray	Often **normal**	**Narrow, lateral sclerosis, double navicular**
Treatment	**NWB cast 6–8 wk** → ORIF + graft	Insoles → **TN + NC fusion** (subtalar spared)

	Mueller-Weiss	Köhler's
Age	**Adult (4th–5th decade)**	**Paediatric**

Sangeorzan	Deformity
II (oblique)	**ADduction**
III (comminuted)	**ABduction**

How it's asked

"Navicular stress fracture – initial immobilisation + WB?"	**→ short leg cast + non-weight bearing**
"Baseball player, normal X-ray, MRI signal in navicular?"	**→ NWB cast for planned 6–8 weeks**
"Gymnast, CT navicular stress fracture – next step?"	**→ non-weightbearing with cast immobilization**
"Adolescent navicular stress reaction on MRI?"	**→ restricted weight bearing in short-leg cast**
"Surgery for navicular stress fracture when?"	**→ line across 2 cortices, or 1 cortex + displacement / cysts**
"Chronic recurrent stress fracture, failed casting?"	**→ ORIF with autologous calcaneus bone graft**
"Gymnast, failed 6 wk NWB cast?"	**→ ORIF with autologous bone grafting**
"Most common complication of navicular stress fracture?"	**→ non-union**
"Pro catcher with navicular stress fracture – most at risk for?"	**→ nonunion**
"Acute displaced navicular body fracture?"	**→ open reduction and lag screw fixation**
"Best view for tuberosity fracture?"	**→ 45° oblique**
"Avulsion ORIF threshold?"	**→ > 25% articular surface**

"Tuberosity ORIF threshold?"	→ **> 5 mm diastasis**
"Sangeorzan III deformity?"	→ **forefoot abduction**
"Mueller-Weiss – fuse which joints?"	→ **talonavicular and naviculocuneiform**
"Mueller-Weiss – all true EXCEPT?"	→ **pseudo hindfoot varus** (TO)
"Paradoxical pes planovarus = ?"	→ **pseudo hindfoot valgus + pes planus** (TO)

– – Active space – –

TARSAL TUNNEL SYNDROME

– Foot & Ankle · 45 / 48 –

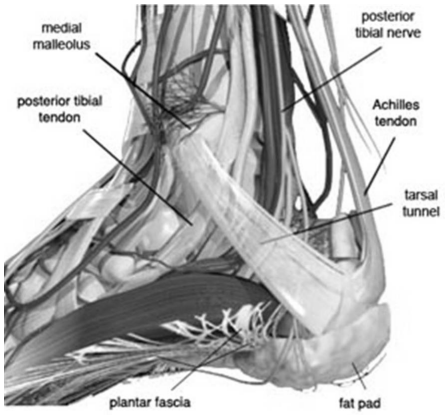
1 Basics

Definition — **compressive neuropathy of the tibial nerve** at the tarsal tunnel → pain + paraesthesia of the **plantar foot**

Diagnosis — **clinical**: burning plantar pain + **Tinel's over tibial nerve** · EMG / NCS can help confirm · **history = most useful diagnostic aid**

Tarsal tunnel – anatomy, causes (flatfoot, space-occupying lesion) & symptoms

Posterior tarsal tunnel	Structure
Roof (superficial)	Flexor retinaculum (laciniate ligament)
Medial wall (deep)	Calcaneus + talus
Inferior	Abductor hallucis
Contents	Tibial nerve · posterior tibial artery · FHL · FDL · tibialis posterior



Posterior tarsal tunnel – flexor retinaculum over tibial nerve, PTA and flexor tendons

Tibial nerve course — along fascia of **tibialis posterior in deep posterior compartment** → gives off **medial calcaneal nerve just proximal to ankle** → dives **deep to flexor retinaculum** → most often **bifurcates in the tunnel** into medial + lateral plantar nerves

3 distal branches — **medial plantar · lateral plantar · medial calcaneal** · bifurcation **proximal to tunnel in 5%**

Plantar nerves pass through **two separate tunnels within and deep to abductor hallucis** (superficial + deep fascial layers) · can be compressed in their own sheath **distal** to the tarsal tunnel

Intrinsic causes	Extrinsic causes
Ganglion cyst (classic)	Shoes
Tendinopathy, tenosynovitis	Trauma
Lipoma / tumour	Deformity – **tarsal coalition, valgus hindfoot**
Perineural fibrosis	Post-surgical scarring
Osteophytes	Systemic inflammatory disease, lower-limb oedema

Identifiable cause **80%** of cases · commonest = **space-occupying lesion in the tunnel proper** (OB)

Heel pain triad **posterior tibial tendon deficiency (adult-acquired flatfoot) + plantar fasciitis + tarsal tunnel syndrome** · loss of **static + dynamic stabilisers of medial arch → traction neuropathy** of tibial nerve · **86% improvement** after surgery (Labib)

Not associated with TTS insertional Achilles tendinitis · hypermobile first ray · metatarsal stress fracture · gastrocnemius contracture

Baxter's nerve (heel pain) (TO)

Tibial nerve divisions (TO) **medial calcaneal nerve (MCN) · medial plantar nerve (MPN) · lateral plantar nerve (LPN) · first branch of LPN** · tibial = larger of the two sciatic divisions

Nerve in heel pain syndrome (TO) **1st branch of lateral plantar nerve**
✗ Wrong options: medial plantar nerve · medial calcaneal nerve · 2nd branch of sural nerve

First branch of LPN (TO) = **nerve to abductor digiti minimi** · sensory branches to **calcaneal periosteum** · **supplies no skin**

Commonest neural cause of plantar heel pain (TO) **entrapment of first branch of LPN**

Entrapment sites of first branch (TO) **sharp edge of deep fascia of abductor hallucis** · **just distal to medial edge of calcaneus**

Second commonest (TO) **medial calcaneal nerve**

LPN (TO) supplies **most foot muscles** · skin of **lateral ⅓ of plantar foot + 4th and 5th toes** · entrapped **between abductor hallucis and quadratus plantae**

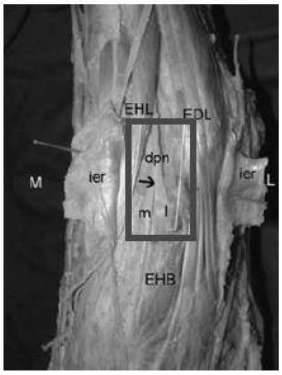
MPN (TO) motor **abductor hallucis, FHB, FDB, 1st lumbrical** · skin **medial ⅔ of plantar foot** · entrapment **less common** than the others

– – Active space – –

Anterior tarsal tunnel (deep peroneal nerve)

Anterior tarsal tunnel	Orthobullets	OB
Roof (most superficial)	**Inferior extensor retinaculum**	**Inferior extensor retinaculum**
Floor (deep)	Fascia over **talus + navicular**	**Talonavicular capsule**
Medial / lateral	–	**Medial malleolus / lateral malleolus**
Contents	**DPN + branches, EHL, EDL, peroneus tertius, dorsalis pedis a.**	Same **+ tibialis anterior + dorsalis pedis vein**

EHB — **not in the tunnel** (originates **distal** to it) · DPN may also be trapped **under the EHB tendon** (OB)



Anterior tarsal tunnel – DPN under the inferior extensor retinaculum, EHL, EDL, EHB

Commonest DPN entrapment site — **under the inferior extensor retinaculum = anterior tarsal tunnel syndrome**
✗ Wrong options: EHB tendon · TN osteophytes · os intermetatarseum · base of 5th MT

Other DPN compressors (runners) — **dorsal talonavicular osteophytes** · **os intermetatarseum** (between 1st + 2nd MT bases)

Causes — contusion of dorsum · **tight shoe laces / compressive shoes** (commonest) · TN osteophytes · ganglion · **pes cavus** · scar · synovial cyst · hypertrophic extensor muscles · **dorsalis pedis aneurysm** · **dorsal navicular avulsion fracture**

Peroneal neuropathy — **commonest compressive neuropathy of lower limb** · CPN at **fibular prominence** · SPN at **exit from lateral compartment** · DPN **under extensor retinaculum**

Frequency — anterior TTS **5% of foot numbness**, often misdiagnosed

3 Presentation

Pain — **prolonged standing / walking** · **vague, misleading medial foot pain** · **sharp, burning** · may radiate proximally **up into the calf**

Numbness — **intermittent paraesthesia + numbness of plantar foot** · lateral foot burning = **lateral plantar nerve fibres** · history of trauma or surgery

Exam — **Tinel's over tibial nerve** · sensory exam equivocal · **pes planus** · **intrinsic wasting: abductor digiti quinti or abductor hallucis**

Provocative test	Manoeuvre
Compression test	**Plantar flexion + inversion + digital pressure over tunnel → highly sensitive + specific**
Dorsiflexion–eversion test	**Ankle dorsiflexion + heel eversion + toe dorsiflexion** → reproduces pain / paraesthesia
Simple stretch	Pain with **dorsiflexion + eversion**

Ganglion in tunnel — **burning + numbness on the bottom of the foot** · pain medial heel + plantar foot · affects **toe flexion**
✗ Wrong options: loss of toe extension

Nerve — **tibial nerve**
✗ Wrong options: sural · saphenous · SPN · DPN (not in the triad)

Anterior TTS — **dull ache / vague dorsomedial pain, numbness, paraesthesia of dorsum → 1st web space** · ↑ with **plantar flexion**, inactivity, **tight shoes** · **EDB atrophy** · **Tinel's over anterior ankle** radiating between 1st + 2nd toes · ballet dancers, marathon runners, hikers

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
Weight-bearing X-ray	Osseous structure (deformity, coalition, osteophytes)
MRI	Rules out **accessory muscle** or **soft-tissue tumour** · **ganglion cyst in tunnel** (sagittal: bright fluid lesion)
Ultrasound	Complements clinical diagnosis
Lateral foot X-ray (anterior TTS)	**Dorsal navicular avulsion fragment** / dorsal osteophytes compressing DPN



Sagittal MRI – ganglion cyst in the tarsal tunnel

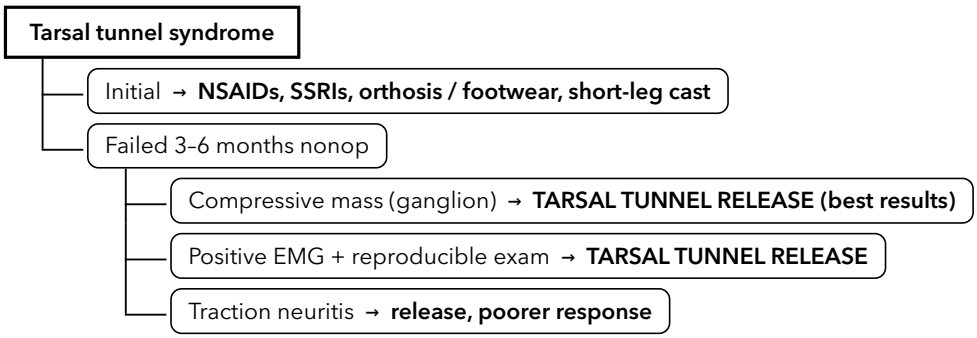
– – Active space – –

Diagnostic criteria

EMG / NCS (posterior TTS)	Positive finding
Distal motor latency	**≥ 7.0 ms**
Sensory latency	**> 2.3 ms**
Motor amplitude	↓ to **abductor hallucis** or **abductor digiti minimi**
Which first	**Sensory (SAP) abnormal before / more often than motor**

Electrodiagnostics **not very sensitive or specific**, **false negatives** in symptomatic patients · still helpful to **localise compression** (nerve root vs tarsal tunnel vs more distal) · **EMG confirms anterior TTS**

5 Management



Nonoperative anti-inflammatories · **SSRIs** · orthosis / footwear to correct **hindfoot alignment** · period of **short-leg cast** · usually **ineffective**

Operative criteria

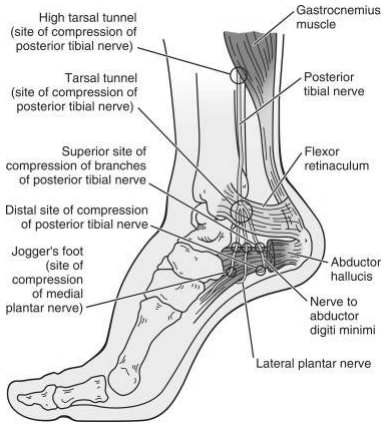
Release indicated after 3–6 months failed nonop and	Outcome
Compressive mass (ganglion cyst) identified	**Best results** – compressing anatomic structure removed
Positive EMG	–
Reproducible physical findings	**Positive Tinel's pre-op** → good results
Traction neuritis	Responds less well

Success **50–90%** (Orthobullets) · **44–96%** (OB, Ahmad) – read the options · **early** operation to prevent fibrosis

Release sites & causes of recurrence

Technique **identify the nerve proximally** → release all layers

Must release	Not a TTS site
Flexor retinaculum	**Medial band of the plantar fascia**
Deep investing fascia of the leg	–
Superficial fascia of abductor hallucis	–
Deep fascia of abductor hallucis	–



Compression sites of the tibial nerve — high tarsal tunnel, tarsal tunnel, distal sites, jogger's foot

Anterior TTS nonop: **activity modification, remove compressive shoes** → surgical release · **endoscopic** for **osteophytes / isolated ganglion** · **open** for **synovitis / unknown cause**

6 Complications

Recurrence

Commonest cause **incomplete / inadequate release**
✗ Wrong options: < 5% recurrence · revision = primary outcome · motor latencies more sensitive

Other causes adhesive neuritis · **neuroma** · incorrect initial diagnosis · infection · **double crush** · bleeding + scarring · anatomic variation · pre-existing nerve damage

Recurrence rate **5–40%**

Revision **less successful than index operation** · **repeat release not recommended** (Orthobullets; some authors) · barrier materials ↓ adherence · **excise neuromas** · then nerve stimulators, systemic medication

Worse prognosis **double crush** · post-operative scarring · revision surgery

– – Active space – –

X vs Y

	Posterior TTS	Anterior TTS	Baxter's nerve (TO)
Nerve	**Tibial**	**Deep peroneal**	**1st branch LPN**
Roof	**Flexor retinaculum**	**Inferior extensor retinaculum**	**Deep fascia of abductor hallucis**
Symptoms	**Burning plantar foot**	**Dorsum → 1st web space**	Plantar heel pain
Motor	AbH / ADM wasting	**EDB atrophy**	**Abductor digiti minimi**

Heel / foot pain	Clue
Plantar fasciitis	**Worst first thing in the morning** · commonest misdiagnosis of TTS
Tarsal tunnel	Burning plantar numbness, Tinel's, ganglion on MRI
Malignant tumour (e.g. synovial sarcoma)	**Night pain** · aggressive imaging
Os trigonum	Impingement, **no burning / tingling**
Missed foot compartment syndrome	**Intrinsic-minus (claw) toes** from interosseous + lumbrical fibrosis · patchy paresis
L5/S1 radiculopathy	**Dermatomal** numbness, radiating pain
Morton's neuroma	**2nd / 3rd interdigital nerve** at transverse intermetatarsal ligament
CRPS	Diffuse burning pain, touch / movement intolerable

How it's asked

"Flatfoot, burning lateral foot, medial heel tenderness – nerve?"	→ **tibial nerve** ✗ Wrong options: sural · saphenous · SPN · DPN
"Associated with tarsal tunnel syndrome?"	→ **adult-acquired flatfoot** (heel pain triad)
"Plantar burning after failed stretching + night splints, sagittal MRI cyst?"	→ **tarsal tunnel syndrome** ✗ Wrong options: recalcitrant plantar fasciitis
"Ganglion in tarsal tunnel – symptom?"	→ **burning and numbness on the bottom of the foot**
"Release sites EXCEPT?"	→ **medial band of the plantar fascia**
"Recurrent TTS – true?"	→ **commonest cause is incomplete release**
"Revision release outcome?"	→ **worse than primary**
"Sensory vs motor latencies?"	→ **sensory prolonged first**
"Best surgical results when?"	→ **compressing structure (ganglion) identified + removed**
"Ballet dancer, dorsomedial pain to 1st web space, EDB atrophy, navicular avulsion?"	→ **anterior tarsal tunnel syndrome**

"Commonest DPN entrapment site?"	→ **inferior extensor retinaculum**
"Most superficial boundary of the space (dorsal pain to 1st–2nd toes)?"	→ **inferior extensor retinaculum** ✗ Wrong options: EHL · EDL · peroneus tertius (inside) · EHB (distal)
"Nerve involved in heel pain syndrome?"	→ **1st branch of lateral plantar nerve** (TO)

– – Active space – –

– – Active space – –

TIBIOTALAR IMPINGEMENT

– Foot & Ankle · 46 / 48 –

1 Basics

Definition	anterior ankle pain, **most often osteophyte impingement** in anterior tibiotalar joint · **"footballer's ankle"** (high incidence in soccer players)
Who	athletes on **turf / grass** or needing **maximum dorsiflexion**: **rugby, football, soccer, dancers (ballet), wrestlers**
Associated (OB)	**lateral ankle instability**, **osteochondral lesions**, **gastrocnemius contracture** → look for + treat concomitantly

2 Mechanism

Causes	**repetitive overuse** · trauma · degenerative sequelae · excess **anterolateral soft tissue** · posterior soft-tissue / osseous abnormalities
Pain generator (OB)	spurs of **talar neck + anterior tibia rarely touch** (cadaver) → **anterior synovial + capsular tissue impingement** is the source

Subtype (OB)	Cause
Anteromedial	**Intra-articular exostoses of anterior tibia + talar neck** pinch interposed synovium / capsule → chronic inflammation
Anterolateral	**AITFL catching on talus** · or **after ATFL sprain** → **fibrotic tissue in anterolateral gutter**

Sources of anterior impingement pain	**AITFL** · **tibial osteophyte** · **inflamed synovium + capsule** · **talar neck exostosis** ✗ Wrong options: **os trigonum** = accessory ossification near **posterolateral talar process** → **posterior** impingement

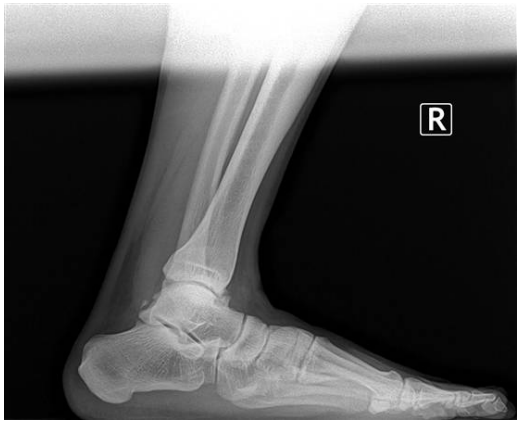
3 Presentation

Symptoms	**anterior ankle pain** · worse **crouching** / squatting
Exam	**pain with forced dorsiflexion** · **limited dorsiflexion** (e.g. passive DF to **10°** reproduces pain) · **anterior joint-line point tenderness** · soft-tissue swelling, effusion · **subtalar joint pain-free**
Anterolateral soft-tissue impingement	old **ankle sprain** · pain **anterolaterally at end-arc passive dorsiflexion** · **normal talar drawer** · **no** tenderness at fibula, ATFL, CFL · no posterior pain with passive PF

4 Diagnosis & Classification

Key radiology findings

Study	Finding
X-ray: AP, lateral, oblique	**Spurs on anterior distal tibia** + **dorsal talus (talar neck)** · dorsiflexed lateral shows contact
Oblique view	Reveals **anteromedial talar spurs** · better than standard series (Tol & van Dijk)
CT	**Extent of bony osteophytes**
MRI	**Spurring + joint fluid** · fluid-sensitive: **anterior capsular thickening + tibial osteophyte**
MR arthrogram	**Anterolateral gutter soft-tissue mass** · MRI **78.9% accuracy** for anterolateral soft-tissue impingement (Ferkel)



Dorsiflexed lateral: anterior tibial spur abutting talar neck



Axial MR arthrogram: anterolateral gutter soft-tissue impingement

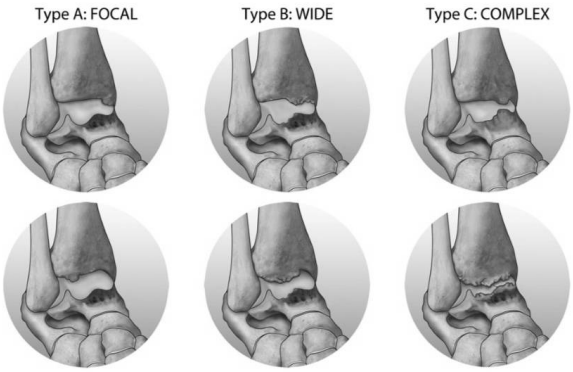
Diagnosis **meticulous physical exam** + oblique radiographs (Tol & van Dijk)

Parma classification (anterior bony impingement)

Type	Pattern
A	**Focal**
B	**Wide**
C	**Complex**

– – Active space – –

Prognosis (Parma)	**isolated tibial spurs do better · concomitant OCD, older age, previous ankle fracture → worse**



Parma classification: type A focal, B wide, C complex

5 Management

Anterior ankle impingement

- First line → **PT, activity / lifestyle change, NSAIDs, bracing**
  (steroid injection → soft-tissue impingement + synovitis pain)
- Failed nonop, bony spurs ± mild arthritis → **arthroscopic debridement / cheilectomy**
- Failed nonop, anterolateral soft tissue → **arthroscopy + synovial debridement**
- Severe arthritis → **ankle arthrodesis (gold standard)**

Anterior bony impingement – arthroscopic debridement

Rugby player, painful limited DF, tibial spur on dorsiflexed X-ray	**arthroscopic tibial debridement** ✗ Wrong options: chondral drilling · modified Broström · os trigonum excision · tibiotalar arthrodesis
Impingement + mild arthritis	**anterior ankle cheilectomy / debridement** (arthroscopic or open) · many still need **open approach** (Talusan)
Wrestler, failed PT + NSAIDs + bracing, MRI capsular thickening + spur	**arthroscopic debridement** ✗ Wrong options: os trigonum excision (posterior) · percutaneous Achilles repair · more proprioceptive rehab (already failed) · loose body removal (none seen)
Arthroscopy	debride **bony abnormalities + inflamed synovium** · effective alternative to open · anteromedial: **97% return to prior sport** (Murawski & Kennedy)
Technique	**supine**, **external traction**, leg on padded bump · **knife cuts skin only → hemostat spread** for portals · **adequate view before burring** anterior tibia (protect dorsal NV bundle)
Severe arthritis	**ankle arthrodesis = gold standard** · **93–94% fusion with cancellous screws** (74% overall, 93% low risk) (OB) · **~90% union** (Orthobullets, ankle arthritis) – read the options · **TAA** = technically demanding alternative

Anterolateral soft-tissue impingement

Military officer, old sprain, anterolateral pain at end-DF, stable drawer, failed nonop	**ankle arthroscopy with synovial debridement** ✗ Wrong options: Broström-Gould (anatomic ATFL + CFL shortening / reinsertion + inferior extensor retinaculum) · Chrisman-Snook (non-anatomic split peroneus brevis) · syndesmosis screw (diastasis) · loose body removal
Outcome	partial synovectomy + lateral-gutter scar debridement → **good–excellent at 2 yr** (Ferkel)

Evidence-based indications for ankle arthroscopy

Evidence (Glazebrook)	Indication
Fair → recommended	**Osteochondral lesions · ankle arthrodesis**
Indicated	**Anterior ankle impingement** (isolated bony)
Limited	Instability · septic arthritis · arthrofibrosis · **loose body removal**
Not effective → not recommended	**Debridement of diffuse degenerative cartilage (ankle arthritis)**

6 Complications

Portal / step	Structure injured
Anterolateral portal	**Superficial peroneal nerve**
Anteromedial portal	**Saphenous vein**
Spur burring of anterior tibia	**Dorsal neurovascular bundle**

X vs Y

	Anterior impingement	Posterior impingement
Painful motion	**Forced dorsiflexion**	**Passive plantarflexion**
Bony cause	Tibial / talar neck spurs	**Os trigonum** (posterolateral talar process)
Sport	Soccer, rugby, wrestling, ballet	–

	Anteromedial	Anterolateral
Cause	**Tibia + talar neck exostoses**	**AITFL** on talus / **post-ATFL-sprain gutter fibrosis**
Imaging	Oblique X-ray spurs	MR arthrogram gutter tissue
Surgery	Arthroscopic spur + synovial debridement	**Arthroscopic synovial debridement**

How it's asked

"Rugby player, painful limited DF, anterior tibial spur — surgery?"	→ **arthroscopic tibial debridement**
"Wrestler, failed nonop, MRI capsular thickening + spur?"	→ **arthroscopic debridement**

– – Active space – –

"NOT a source of anterior impingement pain?"	**→ accessory ossification near posterolateral talar process (os trigonum)**
"Old sprain, anterolateral pain at end-DF, stable drawer?"	**→ ankle arthroscopy with synovial debridement**
"Ankle arthroscopy indications EXCEPT?"	**→ debridement of diffuse degenerative ankle cartilage**
"Best plain view for anteromedial talar spurs?"	**→ oblique**
"Portal injuring SPN?"	**→ anterolateral**
"Real pain generator in bony impingement?"	**→ impinged synovium + capsule** (spurs rarely touch)

TOTAL ANKLE ARTHROPLASTY

– Foot & Ankle · 47 / 48 –

1 Basics

Role — **alternative to ankle arthrodesis for end-stage ankle OA** · aims to **preserve functional ROM** sacrificed by fusion

Ankle OA — typically **post-traumatic** · less frequent than hip / knee OA · nonop: gauntlet brace, PT, steroid injection, NSAIDs

Implant generations

Generation	Design	Problem / feature
1st	Highly **constrained / semi-constrained, two-component**, **cemented** both sides	**High loosening**, wide osteolysis, subsidence, mechanical failure
2nd	**Two-component fixed-bearing**, **polyethylene bearing** built into talar or tibial component	Conservative bone cuts · **cementless press-fit, porous coating** for ingrowth
3rd	Adds a third component: **independent polyethylene mobile-bearing meniscus**	Ligaments for stability · anatomic balancing · minimal resection

Three vs two components — **no evidence three-component designs are better**

Two-component	Three-component (mobile bearing)
Agility (cemented era; tibia titanium, central fins)	**STAR** (Scandinavian Total Ankle Replacement)
INBONE (tibial **IM stem**)	Mobility · Buechel-Pappas
Salto Talaris (**no modular polyethylene liner**, OB) · Eclipse	HINTEGRA · BOX

FDA-approved (Orthobullets) — **STAR · INBONE · Agility · Salto Talaris**

4 Diagnosis & Classification

Key radiology findings

Study	Use
WB AP + lateral ankle	Pre-op: **extent of arthritis** · non-WB views **underestimate joint space narrowing**
MRI (pre-op)	**Osteonecrosis** (amount), bone loss, **subchondral cyst size**
CT with metal artifact-reducing protocol	**Best study for extent of osteolysis** / loosening / subsidence in painful TAA
Bone scan	Infection **or** aseptic loosening but **cannot differentiate** · high false-positive + false-negative
SPECT	**Not indicated** for PJI
Flexion–extension X-rays + gutter exam	Impingement, arthrofibrosis



Painful TAA – balloon osteolysis around the tibial component with talar subsidence

Diagnostic criteria – painful TAA & infection

Painful TAA – first step — **CBC, ESR, CRP** (rule out chronic infection) → aspiration + synovial fluid to confirm → then **CT (metal artifact-reducing)**

Causes of failure — **aseptic loosening**, **subsidence**, **impingement**, **arthrofibrosis**, **infection** · persistent pain after "successful" TAA is common

Parameter	Aseptic (OB)	PJI (OB)
ESR / CRP	**19 mm/h / 1.1 mg/L** (normal)	**Elevated**
Serum WBC	**9K**	–
Synovial nucleated cells	**900 / mL, 44% PMN**, culture negative	**25,000 / µL, 85% PMN** · or **70K**, **gram-positive cocci in clusters**
Intra-op	–	**Purulence · > 5 PMN / hpf**
Next	**CT metal artifact-reducing**	**Explant + antibiotic spacer**

PJI timing	Definition	Treatment
Acute	**< 4 wk** from index	**I&D + polyethylene exchange** (option)
Chronic	**> 4 wk**	**Explant + antibiotic spacer** (biofilm on metal)

Heterotopic ossification classification (Brooker-based) (TO)

Class	Finding (TO)
0	**No HO**
I	**Islands of bone in soft tissue** about the ankle
II	**Bone spurs from tibia or talus, posterior joint space ↓ < 50%**
III	Spurs **↓ posterior joint space > 50%**, **non-bridging**
IV	**Bridging bone** continuous **tibia ↔ talus**

Periprosthetic fracture classification (Manegold, OB)

Fracture type	Location	Prosthesis stability
1 Intraoperative	**A Medial malleolus**	**S Stable**
2 Postoperative trauma	B Lateral malleolus	**U Unstable**
3 Postoperative stress	C Tibia · D Talus	–

5 Management

Operative criteria – patient selection

	Orthobullets	TO
Indication	**Unilateral or bilateral end-stage ankle OA**	Degenerative, **inflammatory** or post-traumatic arthritis
Ideal patient	**Older (middle-old), low demand**, reasonably mobile, no major comorbidity · **normal / low BMI** · **well-aligned, stable hindfoot** · good soft tissues	**Older, thin, low-demand**, **minimal deformity**, **retained ankle ROM**
Contraindications	**Active infection** · **peripheral vascular disease** · inadequate soft-tissue envelope · **Charcot arthropathy** · insufficient bone stock · **severe osteoporosis** · **osteonecrosis of the talus** · **peripheral neuropathy**	**Age < 50** · poor compliance · **heavy industrial labourer** · heavy smoker · **uncontrolled diabetes with neuropathy** · **significant ankle instability** · **angular deformity > 10–15°** · vascular insufficiency · **obesity > 250 lb** · significant bone loss · **osteonecrosis (AVN talus)** · **active or previous infection**

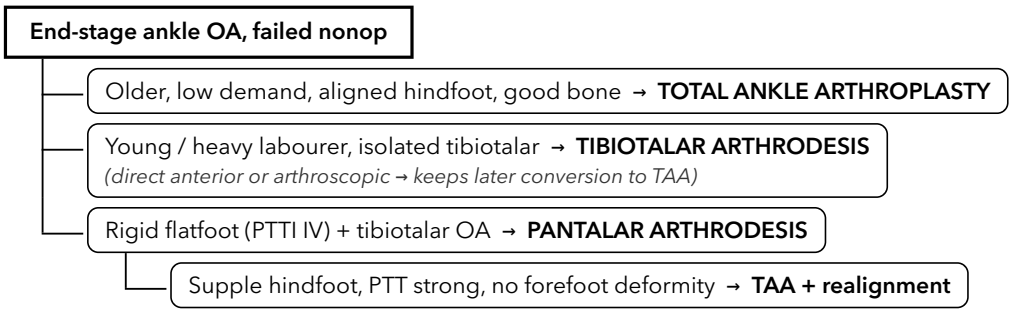
RA – NOT a contraindication (TO) arthrodesis used to be standard · arthroplasty now chosen to **preserve motion + ↓ stress on midfoot and subtalar joints**
✗ Wrong options: AVN talus · ankle instability · heavy labourer · young age

AVN talus (TO) **avascular, fragmented, collapsed talar body → not a candidate → arthrodesis**

Diagnosis vs outcome **best for primary OA** · **worse with post-traumatic + inflammatory arthritis** (prior surgery, tissue laxity) (OB) · TO lists **inflammatory** among ideal indications – read the options

– – Active space – –

Coronal deformity	**> 10–15° = contraindication** (TO) · **> 15° varus / valgus** did as well as neutral when corrected, third-generation implants (OB, Queen) — read the options
Young, active, heavy job	**high stress on prosthesis + bone–implant interface → ↑ failure** → fuse



Patient selection & outcomes vs arthrodesis (deformity, flatfoot)

49-yo construction worker, isolated tibiotalar OA, no subtalar pain	**tibiotalar arthrodesis through direct anterior or arthroscopic approach** → allows **later conversion to TAA** ✗ Wrong options: TAA (young, high demand) · TTC (no subtalar arthritis) · **medial malleolar osteotomy / transfibular approach** (prevents later conversion, ↑ wound complications after conversion) · repeat steroid
Arthroscopic arthrodesis	**< 15° coronal deformity** · **100% fusion** without graft (OB, Duan)
PTTI stage IV, rigid hindfoot, tibiotalar OA, flatter despite Arizona brace	**pantalar arthrodesis** (addresses arthritic joint + forefoot deformity)

✗ Option in stage IV rigid + ankle OA	Why wrong
Medial slide calcaneal osteotomy + FDL transfer + LCL + Cotton + spring / deltoid recon	Early disease; ignores tibiotalar OA
Supramalleolar osteotomy	Only when tilt comes from **distal tibia**; stage IV tilt = **deltoid insufficiency**
TAA	Deformity must be **reconstructed first** (staged or concurrent)
Triple arthrodesis + deltoid reconstruction	Only if **no ankle arthritis** (< 10° valgus)

Stage IV (Bluman)	**rigid hindfoot → TTC or pantalar fusion** · **supple → realignment + deltoid reconstruction**
Isolated TAA in valgus ankle with deltoid insufficiency (collapsing flatfoot, minimally correctable)	**lateral polyethylene edge-loading + progressive lateral talar tilt** ✗ Wrong options: durable neutral alignment + symmetric wear · medial gutter impingement from deltoid overconstraint · subtalar nonunion (no fusion done) · peroneal rupture from varus overcorrection

Adjuncts to get plantigrade foot — **deltoid reconstruction**, osteotomies, arthrodeses · **pain relief after tibiotalar injection = ankle is pain generator**

Outcome	TAA vs arthrodesis
Pain relief	**Equivalent**
Stairs (up + down)	**TAA better** ✓ · better on **uneven surfaces**
ROM	**TAA greater** (✗ Wrong options: decreased ROM)
Gait	**TAA ↑ stride length**, gait velocity, **cadence** (✗ Wrong options: decreased stride length)
Daily step count	**Neither improves**
AOFAS / SF-36	No difference (meta-analysis)
Reoperation / surgical complications	**Higher with TAA**
Adjacent joints	Arthrodesis → **premature OA of subtalar, TN, CC, NC, TMT, 1st MTP** (✗ Wrong options: knee) at 22 yr

Post-op ROM — **pre-op ROM = best predictor** · minimal extra ROM gained · little high-quality TAA vs fusion evidence

Survivorship — **70–90% at 10 yr** · **> 80% at 5 yr if < 50 yo** (Orthobullets) · **70–98% at 3–6 yr, 80–95% at 8–12 yr** (OB, Easley) · **5-yr survival up to 60%, reoperation up to 30%** – worse than TKA / THA (OB) – read the options

Surgical technique

Approach — **anterior, between tibialis anterior + EHL**

Goal — restore **mechanical alignment** (guides for tibia, talus ± fibula cuts) · **ligament balancing** · **stable, neutral, plantigrade** ankle + hindfoot · may need ligament reconstruction, tendon transfer, osteotomy, **heel cord lengthening**, arthrodesis

Soft tissue — **long incision** (↓ skin tension) · **thick flaps** · minimise retractors · **avoid tibialis anterior sheath** (prevents **bowstringing** + wound problems)

Technical errors — prosthesis **too lateral** · **too small → subsides** · uncorrected varus / valgus · replacing an ankle **too anteriorly subluxated**

Failure – loosening/subsidence work-up & salvage (TTC arthrodesis)

- **Failed TAA**
  - Infection (erythema, drainage, ↑ ESR/CRP, pus) → STAGED: SPACER + IV ABX → **FUSION**
  - Severe talar loss / arthritic STJ / subsidence into STJ → **TTC FUSION**
    - Large defect → **FEMORAL HEAD ALLOGRAFT + nail / plate / nail-plate**
  - Good residual talus + healthy STJ → **ISOLATED ANKLE FUSION**
    - Small defect → **shortening / cancellous graft + anterior plate**
    - > 2 cm defect → **structural iliac crest / distal tibial allograft**

4 key decisions (Berkowitz) — **rule out infection → ankle only vs ankle + subtalar → which bone graft → which fixation**

– – Active space – –

Include subtalar joint (TTC) if **insufficient talar bone stock · talar subsidence crosses STJ · pre-existing STJ arthrosis**

Bone graft	Use
Particulate cancellous	**< 2 cm** talar bone loss
Bulk / structural allograft	**> 2 cm** bone loss
Femoral head allograft	**Graft of choice for large talar loss / TTC**
Campbell technique	**Tricortical iliac crest wedges**
Fresh-frozen distal tibial allograft	Structural option

Fixation **nail · plate · nail + plate**

9 yr, talar subsidence into STJ, infection negative **TTC arthrodesis with structural (femoral head) allograft**
✗ Wrong options: revise talar component (subsided not loose, no bone stock) · revise tibial (not loose) · revise both (early failure, poor talar purchase) · TTC without / cancellous graft (only < 2 cm)



Talar component subsidence into the subtalar joint with talar bone loss

Fusion outcomes isolated tibiotalar fusion: **1 nonunion** vs **5** with ankle–hindfoot fusion · **subtalar nonunion = commonest complication** of ankle–hindfoot fusion → separate STJ approach · **arthrodesis after failed TAA worse than primary arthrodesis**

Revision TAA preferred when **sufficient bone stock**, adequate soft tissue, **preserved ROM**, compliance → **one-stage revision** (long-stem)

Salvage choice	When
One-stage revision	Preserved ROM + **sufficient bone stock**
Two-stage revision	**Established infection**
Tibiotalar arthrodesis	Extensive bone loss but **preserved talar stock + no subtalar arthritis**
Tibiotalar + subtalar (TTC)	**Insufficient talar stock** or **subtalar arthritis**
Below-knee / transtibial amputation	**Non-salvageable** only

Painful TAA, labs + aspirate negative, osteolysis **CT with metal artifact-reducing protocol**
✗ Wrong options: bone scan · repeat aspiration · routine follow-up (subsidence worsens) · two-stage revision

Persistent worsening pain since TAA, bone loss on CT	**CBC, ESR, CRP** first ✗ Wrong options: two-stage · one-stage · tibiotalar arthrodesis · tibiotalar + subtalar arthrodesis

Periprosthetic infection

Deep infection	**0.5–3.5%** · **risk factors: prior ankle surgery, diabetes, wound healing problems > 14 days**, low pre-op scores
Gold standard	**two-stage revision of all components: explantation + cement spacer** → reimplantation or arthrodesis
Reimplantation	**rarely successful**: **0% early prosthesis salvage**, **16% reimplantation** (Myerson) · persistent infection, bone loss from debridement, recurrence risk
Myerson series (19 infections)	3 revision arthroplasty · 6 arthrodesis · **7 permanent spacer** · 3 transtibial amputation
Diabetic, 2 yr post-op, 5 wk symptoms, 70K WBC, GPC clusters	**antibiotic spacer; reimplantation unlikely** ✗ Wrong options: IV antibiotics + retention (only for suppression) · **I&D + poly exchange — biofilm** · spacer with successful reimplantation · amputation first
Salto Talaris, 18 mo, ↑ CRP / ESR, 25,000 WBC 85% PMN	**two-component explantation + antibiotic spacer** ✗ Wrong options: oral antibiotics 6 wk · SPECT · poly exchange (no modular liner) · below-knee amputation
Salvage order	preserve joint → then limb: **permanent spacer, fusion, reimplantation** before **amputation**

6 Complications

Complication	Fact
Delayed wound healing	**Most common** (Orthobullets) · **4–17%**
Superficial wound infection	–
Deep infection	**0.5–3.5%** → explant, antibiotics, reimplant or **salvage arthrodesis**
Sensory deficit	**Up to 21%** · anterior incision near **SPN + DPN**
Intra-op fracture	**Medial > lateral malleolus**
Component loosening	**Talar fails more than tibial**
Subsidence	→ convert to fusion
Osteolysis	**Polyethylene wear** → large expansive cysts in tibia / talus · **CT metal artifact reduction**
Heterotopic ossification	**3.8–82%** (TO)

Intraoperative malleolar fracture – prevention

Most frequent intra-op complication (TO)	**fracture of the medial or lateral malleolus** ✗ Wrong options: fracture of the talus dome
Commonest cause of revision (TO)	**aseptic loosening** (most series) · **arthrofibrosis / impingement** (some series) · **debridement of heterotopic bone** = commonest reoperation (some series)

– – Active space – –

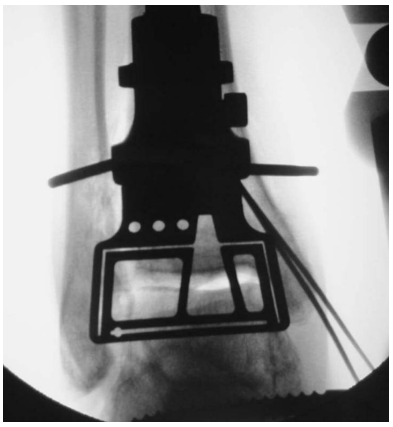
Major reported complications (TO) — **wound dehiscence + heterotopic ossification** · TO marks "**wound dehiscence is the most common postoperative complication**" as false and "**HO is rare**" as false · Orthobullets: **delayed wound healing most common** – read the options

Rate — **4% periprosthetic fracture** (2% intra-op, 2% post-op) · **66% medial malleolus** · most occur **intra-operatively**

Site — **narrow bone bridge** between ankle joint and outer cortex of tibia / fibula

Cause of malleolar fracture	Effect
Plafond cut overextended medially / laterally (more medial cut)	↑ medial malleolus fracture
Tibial cut too proximal	Narrows bone bridge
Oversized tibial component	High compressive load on medial malleolus
Ex-fix over-distraction / early removal	Sudden bending moments on malleoli
Uncontrolled saw-blade excursion · tibial component too medial / lateral	Also risk factors (OB)

Prevention — **prophylactic K-wire (or screw) pinning of the medial malleolus before the osteotomy cut** ✓



Prophylactic pinning of the medial malleolus with the cutting jig in place



Unrecognised intra-operative medial malleolar fracture at 6 weeks

Postoperative heterotopic ossification grading (TO)

Onset (TO) — **as early as 3 months** · incidence **3.8–82%** · ↓ ROM, poor function

Effect (TO)	osteophytes → **malleolar impingement** → **pain on uneven terrain** → may need repeat surgery
Risk (TO)	**males 2× females** · large soft-tissue dissection · **bone debris left in field** · post-op haematoma · prolonged surgery · **wrong-sized components** · mispositioned components
Spurs, posterior joint space ↓ < 50% (TO)	**Grade 2** ✗ Wrong options: 1 (islands) · 3 (> 50%) · 4 (bridging)
Management (TO)	**early active + active-assisted ROM** once confirmed → **surgical resection of mature HO** if symptomatic · exclude **component malalignment** (else recurs → **TAA revision**) · **indomethacin** prophylaxis works for THR, **doubtful for TAA**

X vs Y

	TAA	Ankle arthrodesis
Patient	**Older, low demand, thin**	**Young, heavy labourer**
ROM / gait	**Preserved**, ↑ stride, cadence	↓
Stairs, uneven ground	**Better**	Worse
Pain relief	Equivalent	Equivalent
Reoperation	**Higher**	Lower
Adjacent joints	Protected	**Premature OA (subtalar → 1st MTP)**
Talar AVN	**Contraindicated**	Option

	Acute PJI (< 4 wk)	Chronic PJI (> 4 wk)
Treatment	**I&D + polyethylene exchange**	**Explant + antibiotic spacer**
Why	No mature biofilm	**Biofilm on metal**

	Isolated tibiotalar fusion	TTC fusion
Talus	**Good residual bone**	**Severe loss / subsidence into STJ**
Subtalar	**Healthy**	**Arthritic**
Graft	Cancellous (< 2 cm) / iliac crest	**Femoral head allograft**
Nonunion	Lower	**Higher (subtalar)**

How it's asked

"9 yr after TAA, talar subsidence into STJ, infection negative – next?"	**→ TTC arthrodesis with structural allograft**
"Painful TAA, normal ESR/CRP, 900 cells, culture negative, osteolysis – next?"	**→ CT with metal artifact-reducing protocol**
"Worsening pain since TAA, bone loss – next step?"	**→ CBC, ESR and CRP**
"Chronic PJI, 70K WBC, GPC – treatment + outcome?"	**→ antibiotic spacer; reimplantation unlikely**

– – Active space – –

"Infected Salto TAA, 25,000 WBC — next?"	→ **two-component explantation and antibiotic spacer**
"True about TAA complications?"	→ **commonest intra-op complication = medial or lateral malleolus fracture** (TO)
"Reduces risk of intra-op medial malleolar fracture?"	→ **medial malleolar pinning prior to osteotomy cut**
"NOT a contraindication to TAA?"	→ **rheumatoid arthritis** (TO)
"Contraindication to TAA (TO list)?"	→ **AVN talus · ankle instability · heavy industrial labourer · young age**
"Rigid stage IV PTTI with tibiotalar OA?"	→ **pantalar arthrodesis**
"TAA vs arthrodesis function?"	→ **improved ability to go up and down stairs**
"49-yo construction worker, isolated tibiotalar OA?"	→ **tibiotalar arthrodesis through a direct anterior or arthroscopic approach**
"Isolated TAA in uncorrected valgus + deltoid insufficiency?"	→ **lateral polyethylene edge-loading and progressive lateral talar tilt**
"HO with spurs, posterior joint space ↓ < 50%?"	→ **Grade 2** (TO)
"Commonest complication of TAA?"	→ **delayed wound healing** (Orthobullets)
"Which component loosens more?"	→ **talar**

TURF TOE

– Foot & Ankle · 48 / 48 –

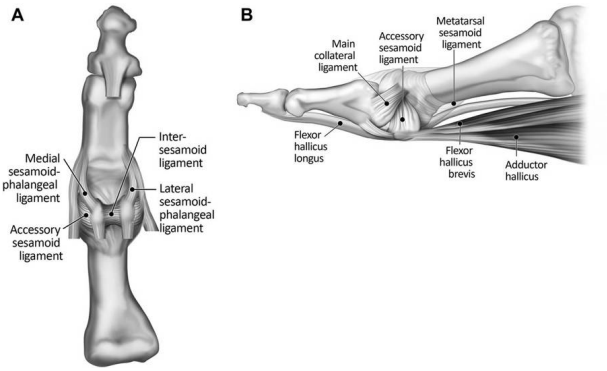
1 Basics

Definition	**hyperextension (hyperdorsiflexion) injury to the plantar plate + sesamoid complex** of the **hallux MTP joint**
Epidemiology	**contact sports on rigid / artificial turf** · **flexible cleats** · **football** (first described **1976**, collegiate players) · **up to 45% of NFL players** · **running backs + quarterbacks** · games > practice
Surgery needed	**< 2%** · **1.7%** of turf toe injuries → **< 5% need surgery** ✓
Lesser toes	great toe = commonest, but **plantar plate tears also occur in lesser toes → crossover toe**

Anatomy – hallux MTP stabilisers

Stabiliser	Components
Osseous	MT head – proximal phalanx articulation
Tendons	**FHB** (contains **tibial + fibular sesamoids**) · **abductor hallucis → medial sesamoid** · **adductor hallucis → lateral sesamoid**
Ligaments	**Medial + lateral collateral** · intermetatarsal · **intersesamoid** · sesamoid-phalangeal · accessory sesamoid
Plantar plate	Joint capsule · attaches to **transverse head of adductor hallucis, flexor tendon sheath, deep transverse intermetatarsal ligament**

Plantar plate — **main stabiliser of the MTP joint**



Hallux MTP ligaments — medial / lateral sesamoid-phalangeal, intersesamoid, accessory sesamoid, collateral ligaments

2 Mechanism

Mechanism	**forefoot fixed to ground** + **hallux MTP hyperextended** + **axial load on the heel** (ankle / hindfoot plantarflexed) → attenuation or tear of plantar capsuloligamentous complex
Two patterns	**axial load on a dorsiflexed great toe (35%)** · **landing on another player's foot → hyperdorsiflexion (33%)** · most after contact with surface or another player

– – Active space – –

Tear site capsuloligamentous-sesamoid complex tears **off the proximal phalanx (phalangeal end), not the metatarsal**



Turf toe mechanism — axial load on the heel with the hallux hyperextended

Associated injury	Note
Varus / valgus hallux MTP injury	**Traumatic hallux valgus** = turf toe variant (**MCL** rupture)
Sesamoid fracture	± diastasis
Proximal sesamoid migration	Grade III
Cartilage injury / loose body	Debride
Proximal phalanx stress fracture	Often overlooked
Hallux rigidus	**Late sequela**

3 Presentation

Symptoms acute pain, stiffness, swelling · **inability to push off** · reduced agility · **pain at toe-off** · difficulty with **cutting drills**

Exam **plantar swelling + ecchymosis** · **pain with passive toe extension / dorsiflexion** · **inability to hyperextend without pain** · gait: shortened time after heel rise

Vertical Lachman **↑ dorsal / plantar laxity vs contralateral side** = plantar plate injury

Lateral stress → excessive lateral deviation + lateral sesamoid subluxation **medial collateral ligament** rupture (label B)
✗ Wrong options: A = IP joint · C = accessory sesamoid ligament · D = intersesamoid ligament · E = lateral sesamoid-phalangeal ligament

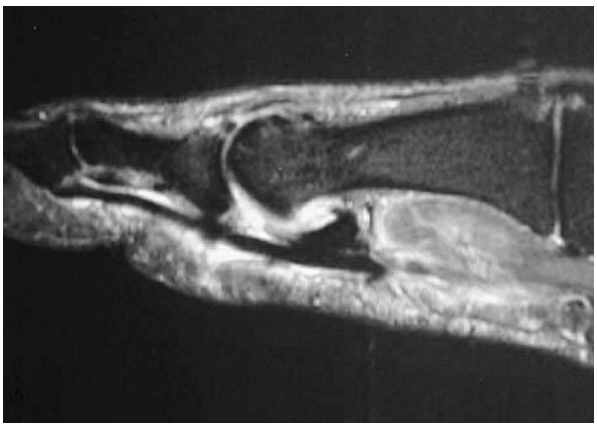
4 Diagnosis & Classification

Key radiology findings

Study	Finding / indication
WB AP, lateral, oblique foot	Usually **concentric** MTP (no dislocation seen)
Sesamoid axial + forced dorsiflexion views	**Sesamoid-to-joint distance vs contralateral** · **medial sesamoid displaced proximally** · sesamoid fracture
Bone scan	Negative X-ray + persistent pain, swelling, weak push-off → **↑ uptake at 1st MTP** · **proximal phalanx stress fracture**
MRI	**Most information** on plantar structures + cartilage · indicated for **abnormal X-ray**, **clinical partial / complete plantar plate tear**, + bone scan, persistent symptoms · **not routine** · rules out phalanx stress fracture

MRI in severe tear

plantar plate / intersesamoid ligament rupture · osteochondral injury + contusion **plantar medial MT head** next to distal medial sesamoid · **dorsal capsule avulsion → slight plantar subluxation of proximal phalanx**



Sagittal MRI – disruption of the hallux plantar plate

Grade III X-ray

proximal retraction of sesamoids vs normal side · MRI: **complete plantar plate disruption distal to the sesamoid**



Grade III turf toe – left sesamoids retracted proximally vs normal right

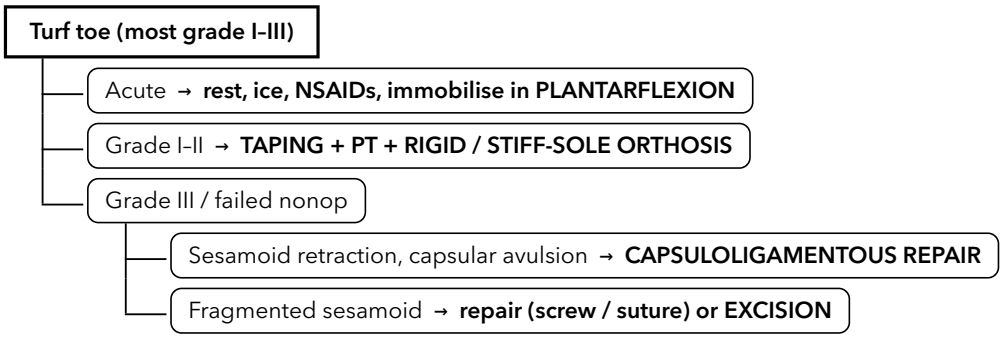
– – Active space – –

Classification

Grade	Orthobullets	OB (clinical)	Treatment (Anderson)
I	**Sprain** of plantar plate	**Attenuation**, localised swelling, **minimal ecchymosis**	**Taping + early rehab**, early RTP
II	**Partial tear**	Partial tear, **moderate swelling**, **motion restricted by pain**	**Turf-toe plate / carbon-fibre orthosis** limiting extension, taping, RTP **~2 wk** (often time off)
III	**Complete tear**	Complete disruption, **marked swelling / ecchymosis**, **hallux flexion weakness**, **frank MTP instability**	Nonop possible (**splint in plantarflexion**) · **direct primary repair** good in elite athletes, RTP **6–12 mo**

Evidence **fair** for nonop grade I–II · **insufficient** for operative grade III (Kadakia)

5 Management



Nonoperative (most injuries, grade I–III) rest, ice, **NSAIDs**, **taping**, **stiff-sole / rocker-bottom shoe**, walking boot · severe → **boot or short-leg cast 2–6 wk** · progressive motion once stable

✗ Nonop traps **taping in acute phase** (vascular compromise with swelling) · **splinting in 5–10° dorsiflexion** (immobilise in **plantarflexion**) · **early motion in grade II–III**

Operative criteria

Indication for surgery (usually grade III)	Source
Failed conservative treatment	Orthobullets + OB
Retraction of sesamoids	Orthobullets + OB
Sesamoid fracture with diastasis · **diastasis of bipartite sesamoid**	Orthobullets + OB
Traumatic bunion / traumatic hallux valgus	Orthobullets + OB
Loose fragments / bodies, chondral injury	Orthobullets + OB
Large capsular avulsion · **vertical instability**	OB
Hallux toe deformity	Orthobullets

✗ Surgery traps **surgical repair as best initial treatment** of complete tear · **sesamoid excision as best initial treatment** (only failed nonop, depends on fragment size)

Treatment – nonoperative vs capsuloligamentous repair

Grade III, sesamoid retraction, failed nonop	**capsuloligamentous repair** ✗ Wrong options: **resection** (sesamoid fracture / fragmentation) · **tendon repair / tenotomy** (no tendon disruption or contracture) · **internal fixation** (no fracture)
Running back, 1 wk, sesamoids aligned, MRI plate intact with signal (low–mid grade)	**taping the first MTP + physical therapy + rigid orthotic** ✗ Wrong options: bone scan (MRI already done) · **1st MTP arthrodesis** / **lateral closing wedge osteotomy** (hallux varus options) · **sesamoidectomy** (sesamoid fracture)
Technique	**medial plantar incision** · **repair or excision of sesamoid** by fragmentation · **headless screw or suture** for sesamoid fracture · synovitis / OCD → **debridement or cheilectomy** · **abductor hallucis transfer** if plantar plate / flexors irreparable
Post-op	immediate **NWB** → progressive ROM + PT · **RTP 3–4 months** (Orthobullets) · **6–12 months** for grade III repair (OB, Anderson) — read the options
Traumatic hallux valgus	**74% return to pre-injury level at ~3.4 months** · **¼ never return**

Lesser-toe plantar plate injury (crossover toe)

Hyperextension injury 1 yr ago → 2nd toe crossover (varus)	exam: **↑ laxity of MTP with dorsal / plantar directed force (vertical Lachman)** ✗ Wrong options: pain improves with hyperextension (it worsens) · valgus instability = MCL deficiency
Pathology	**transverse plantar plate tears** in all crossover specimens · severe → **mid-substance + collateral ligament tears**, complete disruption (Coughlin anatomical grading)
Treatment	**activity + shoe-wear modification** first → **direct plantar plate repair** (plantar or dorsal approach) or **advancement through bone tunnels**

6 Complications

Complication	Fact
Hallux rigidus	**Late sequela → cheilectomy vs arthrodesis** by severity
Proximal phalanx stress fracture	**May be overlooked** → bone scan / MRI
Career impact	**Devastating** for professional athletes

X vs Y

Lesser MTP pain	Provocation
Plantar plate tear	**+ vertical Lachman**, pain **worse with hyperextension**
Early Freiberg's	Pain with **distraction**
Late Freiberg's	Pain with **axial loading / compaction**
MCL deficiency	**Valgus instability**

– – Active space – –

	Turf toe (plantar plate)	Sesamoid fracture
Mechanism	Hyperextension + axial load	Same, ± diastasis
Imaging	Sesamoids **retracted proximally**, plate torn	Fracture line / fragments
Surgery	**Capsuloligamentous repair**	**Screw / suture repair or sesamoidectomy**

How it's asked

"Hyperdorsiflexion, plantar ecchymosis, pain at toe-off, MRI plate torn – diagnosis?"	→ **rupture of the plantar plate** ✗ Wrong options: FHL tear · FHB tear · medial sesamoid fracture · MT head fracture
"Lateral deviation on stress + lateral sesamoid subluxation – structure?"	→ **medial collateral ligament (B)**
"Crossover 2nd toe after old hyperextension – exam?"	→ **increased laxity of the MTP joint with dorsal / plantar directed force**
"Grade III with sesamoid retraction, failed nonop – next?"	→ **capsuloligamentous repair**
"Low-mid grade turf toe, MRI plate intact – next?"	→ **taping the first MTP, physical therapy, rigid orthotic**
"Turf toe treatment – true?"	→ **less than 5% of injuries require surgery**